# Observers Are All You Need

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**Abstract**

Observer-Patch Holography (OPH) starts with finite observers that keep records, compare what both sides can read on an overlap, and repair disagreements. Public physics is the stable record left by that process. Finite repair gives normal forms and repeatable measurement records. Modular screen geometry supplies the three-dimensional observer chart, Lorentz kinematics, and a conditional Einstein branch. Compact sector reconstruction under the stated matter assumptions gives the Standard Model quotient, exact charge assignments, three colors, and three generations.

The primary quantitative closure asks one screen cell to reproduce its own electromagnetic resolution,

$$P = \varphi + \frac{\sqrt{\pi}}{A_T(P)}.$$

Here $A_T(P)$ is the Thomson-limit inverse electromagnetic coupling emitted by the trial cell. The fixed-point existence and uniqueness theorem used by this equation, together with outward-rounded interval certificates, gives one root for each declared map. The secondary global extension $N = \log M_0(\mathfrak{U}_N)$ asks a trial universe to reproduce its logarithmic correctable-record capacity. Its direct physical capacity packet, unique finite-size selector, horizon and Higgs identifications, particle pole masses, and the common-domain Einstein construction are work in progress.

**The paper's thesis is that public physical structure is the fixed point of finite observer records, overlap comparison, and repair; spacetime and matter are then recovered as observer-facing readouts of that settled structure.**

**Keywords:** observer patch holography; quantum foundations; emergent spacetime; fixed-point consensus; Standard Model structure; quantum gravity

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## What This Paper Contributes

This synthesis paper owns the observer interpretation and the two quantitative closures. The local equation

$$P = \varphi + \frac{\sqrt{\pi}}{A_T(P)}$$

is the stronger result: its fixed-point uniqueness schema and outward-rounded interval certificates give one root for each declared map on the full analytic domain. The global equation $N = \log M_0(\mathfrak{U}_N)$ asks a trial universe to reproduce its logarithmic correctable-record capacity. The finite consensus and capacity algebra is established separately. The source-derived physical hadronic transport for $P$, the physical public-record producer and unique finite-size selector for $N$, the horizon/record identification, and the common screen/electroweak load carrier are work in progress. Numerical comparisons and falsifiers are stated in their owning sections.

OPH combines familiar mathematics under one operational constraint. Finite observers compare the records visible on overlaps, repair mismatches by declared recovery moves, and accept as public the normal form that survives those comparisons. The finite patch is the primitive object. Spacetime, fields, and particles enter only through additional physical readout maps.

The consensus paper proves the same-source finite normal-form result. The substrate-neutral treatment in Ref. [1] separates it from cross-source boundary identification, schedule-independent liveness, and refinement compatibility. The compact SM/GR paper owns the controlled modular route to Lorentz and Einstein structure and the compact-gauge route to the Standard Model quotient selected by Minimal Admissible Realization (MAR). The particle paper owns the downstream hierarchy, mass, and flavor calculations; the screen paper owns the finite carrier and record model. The common-domain Einstein tower and the physical closure packets required for $P$ and $N$ are work in progress.

This paper supplies the shared interpretation and closure language. Its claims remain typed: finite theorem, controlled scaling result, conditional physical branch, numerical diagnostic, and interpretation are not interchangeable. A counterexample to a load-bearing theorem, a nonunique or target-leaking closure, failure of the physical readout premises, or a carrier-dependent change in the observable quotient falsifies the corresponding OPH claim.

## Canonical Observer Language

An abstract observer patch is the algebraic object

$$\mathsf{O}_i = (\mathcal{A}_i, \rho_i, \mathcal{R}_i, \{(\mathcal{I}_e, \pi_{i,e})\}_{e \ni i}, \mathcal{U}_i, \mathrm{Chk}_i)\,.$$

Here $\mathcal{A}_i$ is the finite or regulated accessible algebra, $\rho_i$ the accessible state, $\mathcal{R}_i$ the exact or approximate record algebra, $\mathcal{I}_e$ the overlap-visible interface algebra, $\pi_{i,e}$ the visible restriction, $\mathcal{U}_i$ the allowed update and repair instruments, and $\mathrm{Chk}_i$ the checkpoint data used for continuation.

A support patch, such as a cap $P_i \subset S^2$ or a causal diamond, is a geometric chart of the abstract patch on a geometric branch. A carrier patch is a physical or digital implementation that realizes the same visible interface and record statistics within a declared error. Echosahedral hardware is therefore a reference carrier architecture for a fixed-cutoff implementation surface. The physical claim lives at the quotient of visible records, interface statistics, repair maps, and checkpoint continuation.

The observer patch is therefore the formal access structure used by the proof. A human observer, a horizon cap, and a hardware module can realize or chart it, but none of those examples is the definition. Physical predictions attach to quotient data: visible interface statistics, record readout, repair maps, and checkpoint continuation. A carrier implementation is admissible only when those data are invariant within its declared error model.

**Definition 0.1** (Semantic observer history)**.** *For an observer token $O$, the observer-readable history is a labeled causal poset*

$$\mathsf{H}_O = (E_O, \preceq_O, \ell_O).$$

*The elements of $E_O$ are semantic record events, $\preceq_O$ is physical or informational dependence, and $\ell_O$ is the observer-visible event label. Event identity is derived from the canonical semantic payload, observer token, visible footprint, and semantic causal parents. It is not derived from worker ID, repair iteration, queue position, retry counter, timestamp, or packet latency metadata. Latency is execution gauge only when it does not alter semantic causality, delivered data, timeout-sensitive operations, or operational clock records.*

**Definition 0.2** (Observer registry groupoid and namespaces)**.** *Observer identities live in a registry groupoid, not in a flat UUID table. Objects are observer tokens*

$$\mathrm{RID}(O) = (\mathrm{kind}(O), [\mathrm{birth}(O)]_{\mathsf{H}}, \mathrm{lineage}(O)),$$

*where* $\mathrm{kind}(O)$ *lies in the disjoint namespace*

$$\{\textit{patch observer},\ \textit{cap observer},\ \textit{future observer}\}.$$

*The birth event is a semantic event class in $\mathsf{H}$, and lineage is carried by explicit continuation, split, and merge arrows. Local registries on overlapping presentations must preserve observer kind, birth event, lineage arrows, visible observer structure, and checkpoint cuts. The global registry is the descent groupoid or colimit of those local registries when the cocycle and trivial-monodromy checks pass. A split creates child tokens with parent links; it does not silently assign one identity to multiple descendants.*

**Definition 0.3** (Operational observer clock)**.** *The modular parameter of an observer-facing cap pair is not, by itself, an arbitrary operational clock reading. A clock claim supplies an observer-accessible instrument*

$$\Theta_O : \mathrm{Chk}_O(D) \to \mathrm{Prob}(\mathbb{R})$$

*and a declared affine calibration law. For a corresponding observer $O'$, the required finite or exact certificate has the form*

$$\Theta_{O'}(\Phi_{O*}\rho) \approx (A_{a,b})_*\Theta_O(\rho), \qquad A_{a,b}(\tau) = a\tau + b,$$

*with an explicit residual bound from state mismatch, calibration error, and record-readout error. Execution counters are provenance; observer record order, modular parameter, and clock uncertainty are separate semantic or operational fields.*

**Remark 0.4** (Observer-clock naturality gate)**.** *This paper proves observer-first quotient and read-back claims only where the cited theorem surface supplies the required data. Scheduler-independent observer history and clock naturality require a history-augmented quotient relation with history-coherent diamonds, registry descent, state-preserving observer-algebra extraction, support-cap chart naturality, and the operational clock instrument above. A public evidence bundle can falsify the claim by showing counter-dependent semantic event keys, duplicate or namespace-colliding registry identities, broken registry cocycles, non-isomorphic semantic history DAGs across executors, or clock residuals exceeding the declared bound. Implementation field names alone are not the proof.*

This definition leaves the primitive observer size open. The axioms require finite or regulated access at fixed cutoff and permit connected support regions with different finite or regulated algebras. A primitive observer may therefore be a variable finite federation of screen cells. The stronger claim that elementary carriers are isomorphic fixed-capacity cells would require a separate branch: homogeneous UV cellulation, isomorphic local cell algebras, equivariant overlap maps, refinement preservation of the cell type, and a proof identifying one cell or a fixed block of cells with the primitive carrier.

## Failure Tests

OPH is wrong as stated if a load-bearing theorem admits a counterexample under its declared assumptions, if the controlled modular/Einstein branch cannot be instantiated under physically realizable conditions, if MAR fails to select the stated realized package on the declared admissible class, if a quantitative closure is nonunique or target-leaking, or if a physical implementation shows branch-essential dependence on hidden carrier presentation after visible quotient data are held fixed. Concrete falsifiers include failure of a displayed Maxwell, pure-Yang–Mills, or pure-Einstein quadratic kernel to have its stated transverse or transverse-traceless (TT) classical mode under all of that theorem's action/background/phase hypotheses; failure of a claimed quantum pole after its full quantization and spectral receipt is supplied; an observed low-energy $X/Y$-type $(3, 2, \pm 5/6)$ gauge generator on the realized product branch; a fourth light matter generation in the realized low-energy branch; a smaller admissible one-Higgs Standard Model package; nonunique $P_\star$ or $N_{\mathrm{CRC}}$ closure; a failed fixed-cap entropy-to-Einstein tensor upgrade; or carrier-dependent observables after the visible interface is held fixed. A failed continuation does not erase the recovered core unless that continuation is one of the stated parents of the core claim.

## Contents

# 1 Introduction

Observer-Patch Holography asks how much of physics is forced once the world is described from the inside. No observer has access to a view from nowhere. In the idealized description there is a shared screen net, often charted by $S^2$, but each finite observer has access only to a local patch $P_O$ of that net. The observer carries records inside that patch and can compare only the observables shared with neighboring patches. OPH takes that comparison rule literally: where descriptions overlap, they must agree. Thus an "observer screen" is an operational access cut, not a private physical sphere assigned to the observer.

The formalism used here is deliberately quantum-algebraic. Patches carry algebras and states, record events use the trace/Born rule on their declared operator surface, and the gravity branch uses generalized entropy. OPH's theory-of-everything claim is that the observed effective universe can be recovered from this axiom set and its stated branch conditions if the framework is correct. The reconstruction may use declared standard mathematics and branch hypotheses without treating those ingredients as hidden empirical constants.

The computational interpretation has a sharp boundary. A conventional simulation evolves surrogate universe states,

$$U_t \longrightarrow U_{t+1} \longrightarrow U_{t+2} \longrightarrow \cdots .$$

OPH uses a fixed-point computation, but the equation alone is not the definition of simulation. In the fundamental description there is no global timeline on which the contents of spacetime are updated. The readback-and-repair operator acts on observer-readable world candidates and selects a stable solution

$$\mathcal{T}(W) = W.$$

The companion consensus paper [3] defines an OPH simulation by five independent clauses: recovery-derived endogenous update, nontrivial quotient-readable records, strict overlap-repair descent with a schedule-independent normal form, elimination of a proper candidate basin in the selected boundary/sector fiber, and implementation/clock closure. Its simulation theorem proves that the selected finite OPH packet meets those clauses under the stated boundary and record hypotheses and then derives the fixed-point equation. The same theorem gives an identity-map/constant-functional counterexample, so an ordinary equilibrium is insufficient as a simulation.

The local readout of the selected normal form supplies history for observers who subjectively experience time. Time is therefore internal record order instead of an external update parameter. In OPH, "simulation" means this certified observer-facing self-consistency structure, not a bare fixed point and not a frame renderer. The executable OPH-FPE simulator and its finite-run receipt bundles are a separate public evidence surface for finite OPH experiments [10]; they do not replace the theorem-level definition above.

The effective bulk is reconstructed from compatible screen data. Locality comes from the way collars control recovery across patch boundaries. Gauge freedom is the freedom to change the local description without changing any shared observable. Records are the stable parts of a patch that can be checked by other patches. Matter is the stable charge and excitation content that survives transport, fusion, and minimal admissible realization.

The first major result is spacetime kinematics. Modular flow on screen caps becomes geometric, and the compact cap-normal theorem identifies a sky direction with $q(\Omega) = (1, \Omega)$, represents an oriented round cap by $n_C = (\cot\alpha, \csc\alpha\,\mathbf{c})$, proves $\eta(n_C, q(\Omega)) = 0$ exactly on the cap boundary with the interior/exterior sign rule, and gives $n_{gC} = \Lambda_g n_C$. Future unit observer frames then form $H^3 \simeq \mathrm{SO}^+(3,1)/\mathrm{SO}(3)$, exactly three-dimensional. A cap determines an $H^3$ plane/half-space, not an event position or populated bulk. On the conditional frame-response branch, quotient-visible record tokens whose calibrated modular cap responses factor through frame-local $H^3$ data admit exact frame identification and stable finite-noise frame estimates. The finite frame radius is $R_H[(L/\alpha)\varepsilon + (2/\alpha)\sigma]$ for exact net minimization, with a positive residual gap required before a unique finite frame value is reported. Event location requires the separate affine translation and ancestry receipts of the event-manifold branch. Generalized entropy at fixed cap then gives a tensor first-variation relation in the large-scale regime. The absolute Einstein equation additionally uses one source-derived common-domain tower with uniform asymptotics, universal coupling, a vacuum reference, and independent scale readouts; construction and certification of that tower are work in progress. The proposed cosmological capacity relation is

$$\Lambda_{\mathrm{CRC}} = \frac{3\pi}{G N_{\mathrm{CRC}}}.$$

The Gibbons–Hawking entropy and the Planck-2018 late-time de Sitter benchmark give a concrete normalization [24, 25]. With $R_{\mathrm{dS}} \simeq 1.66 \times 10^{26}$ m and $\ell_P \simeq 1.616 \times 10^{-35}$ m, the bare horizon ratio is

$$N_{\mathrm{patch}} = \left(\frac{R_{\mathrm{dS}}}{\ell_P}\right)^2 \simeq 1.05 \times 10^{122}.$$

The entropy capacity is

$$N_{\mathrm{scr}} = S_{\mathrm{dS}} = \frac{A_{\mathrm{dS}}}{4\ell_P^2} = \pi N_{\mathrm{patch}} \simeq 3.31 \times 10^{122},$$

so $\Lambda \ell_P^2 \simeq 2.85 \times 10^{-122}$. OPH uses $N_{\mathrm{scr}}$ for the Gibbons–Hawking entropy capacity and $N_{\mathrm{patch}}$ for the bare horizon area ratio. The proposed input-free closure begins at finite cutoff with a frozen carrier dimension $D$, the unclosed terminal quotient fiber, local and interface record atoms, compatible public global sections, endogenous reachability, a frozen publicness policy, and the globally coupled semantic checkpoint kernels. Their compound confusability graph $G_q$ gives the exact finite readback

$$M_0(q) = \alpha(G_q).$$

This correctable-code capacity includes joint checkpoint compatibility and confusability. A cyclic checkpoint can preserve every label while fixing only constant functions, and equal local marginals can hide different joint capacities. The first producer is the set

$$\mathfrak{F}_{r,\varepsilon}(D) = \{M_\varepsilon(q) : q \in \widetilde{\Omega}_{r,D}\}.$$

A scalar exists only when the whole nonempty terminal fiber agrees. A faithful capacity-carrier representation gives $M_\varepsilon \leq D$; exact equality means a complete rank-one public record basis. If the scalar exact map is total, monotone, and deflationary on a declared finite chain, top-down iteration reaches its greatest fixed point. The official universe-level cosmic record-closure equation and its selector-free finite implementation are

$$N = \log M_0(\mathfrak{U}_N), \qquad \mathfrak{F}_{r,0}(D_\star) = \{D_\star\}, \qquad N_{\mathrm{CRC}} = \log D_\star.$$

The finite public-section, correctable-code, boundary, scalarization, order, extension, refinement, and sewing implications can be proved from the stated data. They do not select a cosmic value: $M(D) = D$ and $M(D) = 1$ are both compatible with monotone deflation. The remaining source constructions are the record-atom restrictions, endogenous reachability, publicness policy, global checkpoint coupling, capacity-carrier representation, whole-fiber scalar producer, confusability-reflecting extension and refinement packets, and an exact finite-size slack law

$$s(D) = \log D - \log M_0(D)$$

with one physical zero. Two physical identifications remain afterward: the correctable-record carrier must be identified with the de Sitter horizon record for the de Sitter area law, and the screen load must be identified with the electroweak load by a positive, unital, refinement-natural map for the electroweak bridge. An independently produced operational scale is a commuting-square test of the code capacity, not its definition. The electroweak bridge supplies a mathematical candidate of about $3.532 \times 10^{122}$ only after the common-load receipt. The Planck–$\Lambda$ central read-off is about $3.313 \times 10^{122}$, which is about 6.6 percent below the bridge candidate. The joint Planck posterior, propagated through the $\Lambda$-to-$N$ map across three degeneracy routes, places the gap at 2.4 to 2.5 one-dimensional standard deviations under the consumed likelihood combination and 3.8 to 3.9 under Planck+BAO, so the combination freeze is part of any future test registration; a final statement requires the closed premises. Thus neither the physical fixed-point identity nor the cosmological display $\Lambda_{\mathrm{CRC}} = 3\pi/(GN_{\mathrm{CRC}})$ is established by the bridge calculation.

The second result is the Standard Model structure. On a cofinal tail carrying the explicit compact-gauge refinement receipt (including coherent surjective charge pullbacks, finite block embeddings, tensor realizations, $3+1$D symmetry, and compatible forgetful fibers), edge sectors reconstruct the bosonic compact-gauge branch. On the realized low-energy branch with the explicit one-generation/one-Higgs matter package, minimal admissible realization then selects the compact gauge quotient

$$\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)/\mathbb{Z}_6.$$

The realized low-energy branch then fixes exact hypercharge, three generations, three colors, integer charge for color singlets, and the absence of the simple-GUT $X/Y$ gauge channel. On a declared trace-balanced block carrier $V = C \oplus W$, with dimensions 3+2 and hypercharges $(-1/3, 1/2)$, the exact exterior package $\Lambda^2 V \oplus \Lambda^4 V$ branches as one fifteen-state Standard Model generation, supports the three one-Higgs invariant lines, cancels all five listed perturbative gauge and mixed anomalies, and has weak-doublet multiplicity $3 + 1 = 4$ with even Witten parity. This is a conditional representation witness. A screen-forced Standard Model requires physical current, determinant, spin-lift, deck-descent, carrier-selection, extra-sector-exclusion, and family-attachment receipts. The selected exterior package is not the full even Clifford module, because $\Lambda^0 V$ is omitted. On separately stated Maxwell, perturbative/deconfined Yang–Mills, and pure-Einstein action/phase branches, the quadratic reductions have the corresponding classical null carrier modes. Photon, gluon, or graviton particle claims require the additional physical-Hilbert-space and spectral-pole receipts.

The third result is quantitative and sits outside the recovered core. Its primary equation closes one local observation cell on its own electromagnetic readout:

$$P = \varphi + \frac{\sqrt{\pi}}{A_T(P)}, \qquad A_T(P) = \alpha_{\mathrm{em}}^{-1}(0; P).$$

The outside reading is the normalized pixel detuning $(P - \varphi)/\sqrt{\pi}$. The inside reading is the Thomson-limit coupling emitted by the same trial cell. The contraction-based theorem gives existence and uniqueness on a closed interval. Outward-rounded interval certificates verify the concrete hypotheses, enclose one root for each declared map, and exclude a second root across the full analytic domain. No measured value is inserted into either solve. The source-derived same-scheme hadronic transport is still required before a declared-map root can be identified with the physical Thomson endpoint.

The secondary global extension uses the exact finite correctable-record capacity $M_0(q) = \alpha(G_q)$ and the universe-level equation $N = \log M_0(\mathfrak{U}_N)$. Its finite public-section, carrier-bound, scalarization, order, and refinement implications are conditional theorems. The physical checkpoint packet, whole-fiber scalar producer, unique finite-size slack zero, horizon identification, and common screen/electroweak carrier are work in progress. The Planck–$\Lambda$ central capacity $3.313 \times 10^{122}$

and the conditional electroweak value $3.532 \times 10^{122}$ are comparison coordinates. Their 6.6-percent difference does not enter the construction of the direct capacity map.

The local pixel coordinate and independent scale certificate are

$$P := \frac{a_{\text{cell}}}{\ell_\star^2}, \qquad \gamma_\star = \frac{\ell_\star \nu_{\text{Cs}}}{c}, \qquad B_\star = \frac{3\pi}{\ell_\star^2}.$$

If the two declared-map outputs acquire their missing physical identifications, $P_\star$ and $N_{\text{CRC}}$ fix $\Lambda_\star \ell_\star^2 = 3\pi/N_{\text{CRC}}$ and $\Lambda_\star a_{\text{cell}} = 3\pi P_\star/N_{\text{CRC}}$. They fix the dimensionless geometry. The selected scale certificate supplies the SI scale product $B_\star = \Lambda_\star N_{\text{CRC}} = 3\pi/\ell_\star^2$. The selected scale certificate does not construct the cosmic readback map or remove the 6.6-percent central-value mismatch. Under those same identifications, the pixel $P_\star$ together with the $\Lambda$-located working capacity $N_\Lambda = 3.31 \times 10^{122}$ fixes a canonical geometric pixel area and a total geometric cell count on an equal-area screen chart,

$$K_{\text{cell}} = \frac{A_{\text{screen}}}{a_{\text{cell}}} = \frac{4N_\Lambda}{P_\star} \simeq 8.12 \times 10^{122}.$$

This count is a measured-side comparison display and does not enter either closure. At the conditional bridge capacity the same chart count reads $8.66 \times 10^{122}$. The result supplies a geometric chart count. It does not choose a primitive observer capacity. The quantity $P_\star/4 \simeq 0.408$ nats, or about 0.588 bits, should not be read as $\log \dim \mathcal{H}_{\text{cell}}$ for an autonomous tensor factor. OPH capacity lives on shared cuts, edge centers, and constrained overlap data; adjacent geometric pixels are not assumed to factor into independent Hilbert components. No fixed primitive observer algebra follows from $P_\star$ and $N_{\text{CRC}}$ alone. The no-$G$ burden of the scale proof record is the clock hierarchy. The first load-bearing object must be a strict source-root proof for $\alpha_U(P_\star)$; the source-audit branch is only a witness. After that, the cesium gap $\varepsilon_{\text{Cs}} \sim 10^{-33}$ must be emitted by source-side electroweak transmutation, QCD/hadronic data, flavor and nuclear data, and a cesium hyperfine spectral readout, with no dependency path from measured gravity or any calibrated electroweak or gravity scale. The public $\alpha_U$ record is a one-dimensional comparison-branch fixed-point certificate:

$$\begin{aligned} \alpha_U(P_\star) &= 0.0411243361956304\,95, \\ \frac{v}{E_\star} &= 2.0199803239725553 \times 10^{-17}, \end{aligned}$$

and the Krawczyk image of $[0.041123336195630494,\ 0.041125336195630496]$ lies strictly inside that interval. The source-side diagnostic branch excludes the public Thomson endpoint upstream. Its certified forward pixel is $P_{\text{fwd}} = 1.630972095858897\ldots$. That source branch is a witness, not a full endpoint proof; it does not supply the strict interval root for the Ward-projected self-map. The dimensionless formula is the strongest clean hierarchy statement. It does not supply an independently physical $E_\star$, a mass in GeV, or a complex propagator pole. The full SI gravity row requires the full set of cesium-clock source records. The QCD and hadronic weakpoint is narrow. The low-energy Thomson endpoint and the cesium-clock display require confined-quark and nuclear source payloads that the first-principles trunk in this paper does not emit. For a source-only Thomson endpoint, the missing object is the source-derived hadronic spectral hadronic backend: source QCD parameters, a quotient ensemble, finite Euclidean vacuum transfer, Ward-normalized electromagnetic currents, the two-current spectral marginal, same-scheme remainder, systematics, and no-target-leak receipts.

Four fine-structure coordinates must be kept separate. The source/root audit records the undressed declared-map witness $\alpha_{\text{root}}^{-1} = 136.994835177413\ldots$ (interval-certified unique fixed point of that numerical map, with enclosure width $7.2 \times 10^{-24}$). The declared map is incomplete: no derivation identifies this root with a physical fine-structure endpoint. Adding the finite-screen unified gauge-width contribution at the CODATA-derived comparison pixel gives the mixed-provenance, no-hadron diagnostic

$$A^{\mathrm{fp}}_{\alpha_U} = 137.0359595136\ldots.$$

The certified self-consistent gauge-width fixed point is $\alpha^{-1} = 137.035660136946577\ldots$. The mixed diagnostic $137.0359595136\ldots$ combines the inner value at $P_{\mathrm{fwd}}$ with $\alpha_U$ at the CODATA-derived comparison pixel, is not a fixed point of any single declared map, and is excluded from the physical output ledger. An executed empirical closure, using external $e^+e^- \to$ hadrons spectral data, records

$$\Delta\alpha^{(5)}_{\mathrm{had}}(M_Z) = 0.0267591200 \pm 0.0007524955, \qquad \alpha^{-1}_{\mathrm{emp}}(0) = 136.2636589431$$

with interval $[136.1583832834, 136.3690975240]$. The measured endpoint is $\alpha^{-1}(0) = 137.035999177(21)$, outside that interval. The resulting same-scheme anchor correction is $[0.6485541111, 0.8547920666]$ inverse-alpha units. This empirical payload tests the endpoint map and exposes the missing contribution; it does not supply a source-only OPH hadron law. The two-current spectral measure covers the running-$\alpha$/HVP marginal. HLbL and rare-decay rows require higher-point and transition spectral sectors from the same source law. For that reason the compact quantitative stress test is the hierarchy row above: it avoids the public Thomson endpoint, $G$, $\Lambda$, $W/Z$, Higgs mass, hadronic payloads, and the cesium clock.

The same hierarchy exponent has a local/global capacity form:

$$\frac{v}{E_{\mathrm{cell}}} = \left(\frac{N^{\mathrm{EW}}_{\mathrm{CRC}}}{\pi}\right)^{-P_\star/12}, \qquad N^{\mathrm{EW}}_{\mathrm{CRC}} = \pi \exp\left[\frac{6\pi}{P_\star \alpha_U(P_\star)}\right].$$

This is the electroweak bridge capacity, about $3.53235 \times 10^{122}$ on the declared comparison branch. Identifying it with the cosmic capacity requires the source-derived correctable-public-record producer and the common screen/electroweak load-carrier identification. The Planck-$\Lambda$ central capacity, about $3.313 \times 10^{122}$, is lower by about 6.6 percent. The factor 12 has a conditional geometric reading. Euler gives $\sum_v (6 - \deg v) = 12$. A feasible strict unit-splitting cost selects twelve unit defects, and a source-derived D-optimal tomography rule or strictly completely monotonic pair cost selects the icosahedral vertex orbit. Edge-center collars expose the defects as central ports. Reversible write/check orientation gives an oriented 24-slot screen register. Independently, the product adjoint has count $m_{\mathrm{rep}} = 2(8+3+1) = 24$. Their equality is bookkeeping and supplies no load or clock. The exterior matter package gives the exact four-copy weak multiplicity; attaching that multiplicity to the physical screen load remains an explicit hypothesis.

The oriented coefficient module is

$$P_{12} \cong_{A_5} \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{3}' \oplus \mathbf{5}.$$

For the six antipodal axes, the exact frame maps split the even and odd modes, and the outward faces orient the complementary triplet. Pulling the block commutator back through

$$\Theta(b + d_G + d_W) = \left(\kappa_E(\tfrac{1}{2} U d_G) + i\Phi(b), \kappa_W(d_W)\right)$$

constructs

$$(P_{12}, [\ ,\ ]_\Theta) \cong \mathfrak{u}(1) \oplus \mathfrak{su}(3) \oplus \mathfrak{su}(2)$$

with center $\mathbf{1}$, derived dimension eleven, and a noncentral $\mathbf{5}$. This bracket acts on coefficient fluctuations; the central record projectors commute. Compact classification gives the same Lie type for every full-rank, compact skew-adjoint, commutator-closed physical current lift with an inner $A_5$ action. Source production of that lift is work in progress.

A chosen trace balance completes the map to $\mathfrak{s}(\mathfrak{u}(3) \oplus \mathfrak{u}(2))$, whose connected group is

$$S(U(3) \times U(2)) \cong \frac{SU(3) \times SU(2) \times U(1)}{\mathbb{Z}_6}.$$

The six-axis lattice independently has quotient $\mathbb{Z}_6$. These exact constructions require determinant balance, a physical spin lift, and refinement-natural deck-holonomy descent before they identify a physical global group. Hypercharge, chiral content, and the selected quotient are on the MAR route. One sufficient branch contract combines source-derived screen tomography with a reversible physical current; its realization from the recovered core is work in progress.

The finite algebra establishes the positive twelve-term sum-to-twelve unit split, abstract matrix commutator, dimension $9+3$, a matrix noncentrality witness, the six-axis lattice quotient, and the $B_6$ arithmetic. Euler charge, the cost, and the selector remain open. The port-frame map, $A_5$-equivariance, classification, trace-balanced group, and physical descent are paper proofs or open receipts. The face-$C_3$ representation theorem gives a three-dimensional minimal extension; physical family attachment is work in progress. See Ref. [2].

The compact proof packages this quantitative surface as a source-certified compression test. A row counts in the accidental-hit estimate only when its declared source map has no dependency path from the measured target or a calibrated proxy. If $p_i$ bounds the conditional accidental-hit probability for row $i$ after the preceding accepted rows, then

$$P_{\mathrm{acc}} \leq \prod_i p_i.$$

The closure ledger records the qualifying rows, including the structural gauge-boson zeros and the clause-conditional hypercharge lattice. Any compression claim cites those declared settings and landed basins. The proof load is the fixed-point maps, uniqueness certificates, source records, and falsifiers.

The declared map for $P$ gives the same screen cell two readings. From outside, the cell sits a little above the exact self-similar entropy balance point $\varphi = (1+\sqrt{5})/2$. From inside, the same cell appears as the smallest electromagnetic observation step available in the encoded world. The unique source-map root defines the local closure coordinate. The measured fine-structure constant tests the resulting electromagnetic endpoint and does not enter the solve. The physical low-energy relation requires the unconstructed same-scheme hadronic spectral transport.

The named $P$ branches organize the electroweak carrier chart, the low-energy electromagnetic endpoint, the conditional Higgs/top surface, the rejected reciprocal-ray quark candidate and its six-coordinate physical interface, a target-informed weighted-cycle neutrino comparison candidate, and the local cell/edge consistency used by the gravity readout. The hierarchy theorem fixes $v/E_\star$. An independent physical $E_\star$ and the separate pole receipt govern any mass in GeV. The gravity normalization itself comes from $\ell_\star^2$, and $P$ cancels in the Newton area-law readout. The support table records the distinct electroweak chart, conditional law, and inverse target adapter, the target-informed charged-lepton candidate, the NuFIT 6.1 rejection of the neutrino candidate, its tautological declared-basis transport and missing physical charged basis, and the hadron construction [11].

The fourth result is implementation. The screen-microphysics construction makes patches, overlaps, edge observables, records, repair maps, and observer checkpoints explicit at finite cutoff through a federated patch-carrier architecture. OPH has a regulated observer-and-record surface on which the basic consistency machinery can be written directly.

**Axioms in words.** The theory uses five axioms. A screen net assigns algebras to connected screen patches. Overlap consistency requires agreement on shared observables. Local MaxEnt selects the least biased local state compatible with the finite constraint data and keeps the realized construction stable under refinement. Recoverable generalized entropy supplies the collar-recovery and focusing structure used by the gravity lane. Minimal admissible realization selects the simplest low-energy sector compatible with the consistency constraints. These axioms define a quantum-algebraic starting point. Quantum mechanics and quantum field theory are treated as effective descriptions carried by the deeper observer-patch architecture, not as structures whose use here depends on first deriving every mathematical ingredient from operational records alone.

**The quantitative selection chain.** Self-reading, overlap consensus, modular geometry, transportable charge, entropic gravity, and stable records define the structural branch. They do not prove that the branch is inhabited, force the icosahedral screen-to-Standard-Model attachment, or construct the global capacity packet. The two closure equations test independently produced local and global readback maps. If both equations hold with the declared selection rules, the branch admits at most one coordinate pair $(P, N)$. This statement does not prove that exactly one physical universe exists. The local uniqueness certificate gives existence and uniqueness on its certified interval, and the domain-global certificate bounds $\sup |g'| < 1$ on every piece of a 256-piece subdivision of the declared numerical domain ($\alpha^{-1} \in [100, 200]$, both readout maps, empty exceptional set), so each declared readout map has exactly one fixed point on that domain. This certificate does not derive a physical endpoint relation for the incomplete declared map. The global uniqueness claim has a proved conditional implication layer and a physical execution that is work in progress. Public global sections, zero-error graph capacity, approximate stability, the carrier bound, whole-fiber scalarization, greatest-fixed-point order theory, and exact refinement stabilization are finite implication theorems. Execution needs the source-derived public checkpoint packet, capacity-carrier representation, confusability-reflecting embeddings, and an exact finite-size slack law with one physical zero. The de Sitter and electroweak consequences then require the horizon–record identification and the common screen/electroweak load-carrier identification, respectively. The operational resolution experiment is an independent test, not the producer. The certified value $3.5321315434 \times 10^{122}$ is a solution of the declared bridge equation, not of the unconstructed direct public-record producer; it is about 6.6 percent above the Planck–$\Lambda$ central coordinate $3.313 \times 10^{122}$. The two electromagnetic endpoint residuals have no valid target-blind score: the exposed target conflates total and residual coordinates, and no active decision threshold exists. A valid test requires a complete method frozen against genuinely withheld data or an audited clean-room producer.

The synthesis surface gives the operational picture first, then places the technical proof machinery after the reader has the construction in hand.

## 2 Main Results

The OPH papers support the following unified picture.

1. **Observers and records become physical objects.** Finite patches, shared overlaps, record algebras, repair maps, and observer checkpoints are represented on the screen as part of the construction.

2. **Spacetime is reconstructed.** On the controlled Einstein-branch geometry readout, the finite cap-normal Bisognano–Wichmann (BW) certificate (support-order faithfulness, regularized mixed-GNS modular transport, BW framing, held-out oriented cross-ratio rigidity, and

independently normalized geometric $2\pi$-KMS (Kubo–Martin–Schwinger) convergence) makes the support-visible cap modular automorphism geometric and supplies Lorentz kinematics. Recoverable generalized entropy together with the null modular bridge, bounded-interval transport, fixed-volume small-ball area variation, admissible fixed-cap stationarity, and timelike tensor upgrade then supplies the Einstein relation on one source-derived common-domain tower with certified tails and independent physical identifications. The finite artifacts test algebra, evaluator logic, manifests, deletion rules, and synthetic controls. They do not certify such a tower. Its construction is work in progress. Bare finite consensus supplies only the upstream quotient-normal-form input. The screen-capacity branch proposes the dimensionless cosmological-constant relation below. The identification of its cosmic record-closure value with the electroweak bridge requires the common screen/electroweak load-carrier identification, and the physical public checkpoint packet is work in progress:

$$\mathfrak{F}_{r,0}(D_\star) = \{D_\star\}, \qquad N_{\mathrm{CRC}} = \log D_\star, \qquad \Lambda_{\mathrm{CRC}}\ell_\star^2 = \frac{3\pi}{N_{\mathrm{CRC}}}, \qquad \Lambda_{\mathrm{CRC}} = \frac{3\pi}{GN_{\mathrm{CRC}}}.$$

The last display uses the selected scale certificate $G_{\mathrm{geom}} = \ell_\star^2$. The bridge formula has a certified mathematical value near $3.532 \times 10^{122}$; the Planck-$\Lambda$ central value near $3.313 \times 10^{122}$ is lower by about 6.6 percent. A uniqueness certificate for the bridge formula does not establish the unique zero of the public-record slack or construct its physical checkpoint packet.

3. **The Standard Model quotient is selected by MAR.** Under the stated coherent transportable-sector refinement conditions, edge sectors reconstruct a bosonic compact-gauge branch. The overlap/holonomy calculus classifies transportable sectors; it does not by itself choose the Standard Model. The realized low-energy branch with the explicit one-generation/one-Higgs matter package and MAR then selects the compact gauge quotient, exact hypercharge, three generations, three colors, and the structural absence of the simple-GUT $X/Y$ gauge channel. General proton stability is not implied. On the ordinary electromagnetic branch, the unbroken $\mathrm{U}(1)_Q$ factor reduces the compact-gauge curvature to $F_Q = dA_Q$; after separately assuming the low-energy Maxwell action and its nonzero kinetic term, variation gives $dF_Q = 0$ and $d{*}F_Q = g_Q^2{*}J_Q$, i.e. Maxwell's equations after canonical electromagnetic normalization.

4. **The measured endpoint tests the local closure.** The comparison pixel $P_C \simeq 1.6309682094$ is inferred from the measured fine-structure constant solely for comparison: $\alpha^{-1}(0) = 137.035999177(21)$, $\alpha(0) \simeq 0.00729735256433$. The forward model with the source anchor and Ward-projected electromagnetic transport measures the loop-closure residual: $3.0 \times 10^{-4}$ relative on the source chain and $2.5 \times 10^{-6}$ relative for the certified numerical gauge-width fixed point $\alpha^{-1} = 137.035660136946577\ldots$, a certified fixed point of the declared map with the Ward-projected hadronic transport open. The mixed-provenance diagnostic $137.0359595136\ldots$ is a fixed point of no single declared map and is excluded from the physical output ledger. The separate root $136.994835177413\ldots$ is an interval-certified output of an incomplete declared map; no derivation relates it to the physical fine-structure endpoint. The claim table records the source-side diagnostic trunk and endpoint residual.

5. **The framework has a regulated implementation surface.** The microphysics paper turns the observer language into explicit finite patch carriers, edge observables, records, and repair cycles.

6. **Edge sectors also admit two-dimensional Yang–Mills and worldsheet-style effective descriptions.** The heat-kernel edge partition function reorganizes the same boundary data

into the two-dimensional Yang–Mills form, and on the declared large-edge branch this yields a controlled worldsheet effective description. This 2D bridge sits beside the compact paper's controlled compact-gauge repair-gap theorem. The string-vacuum selector uses this bridge only as the effective edge-language input; critical worldsheet closure, Bouchard-Donagi cohomology, operator-safety realization, threshold matching, and moduli locking are separate gates. The selector's rank certificate gives proxy rank zero on the published pre-completion slice and supplies no completed physical slice or Jacobian. The BD row is not a selected string witness, and the recovered OPH core is unchanged.

7. **The compact-gauge branch carries the 4D Euclidean Yang–Mills form and isolates the gap theorem.** On the declared controlled compact-gauge branch, with the four-dimensional scaling chart, reflection-positive ordinary vacuum, no additional gauge-invariant relevant dimension-four pure-gauge operator on that branch besides curvature squared, repair completeness, and the compact paper's explicit finite cylinder system in force, OPH proves projective weak-* / GNS extraction of a support-visible cylinder cluster family. Identification of that family with the Euclidean Yang–Mills action additionally uses the renormalized four-dimensional regularity/universality receipt. Weighted observation-fiber resampling is the finite conditional-expectation projector, with the noncircular matrix-recognition criterion of Ref. [1]. A concrete active repair law is identified with that projector only after its independently extracted transition matrix passes the support, equal-fiber-row, and weighted detailed-balance receipt. Identification of Euclidean transfer with the repair generator, vacuum persistence, and passage of a uniform finite-stage gap use the separate finite ground-state-transform, transfer/vacuum, OS-regularity/noncollapse, and uniform-gap receipts. The checked 244-type Ising collar table is an exact calibration of that finite receipt format, not evidence for the physical compact-gauge source family. Only on that certified branch is $\Delta_{\mathrm{YM}} = \Delta_{\mathrm{rep}}$. Weak-* extraction alone is not the Clay-facing axiomatic construction.

# 3 Physical Implementation and $A_5/E_8$ Significance

The implementation question is what must exist below the observer-level axioms so they have a physical carrier. OPH's answer is a fixed-cutoff federation of finite patch carriers. Each carrier exposes overlap ports, record registers, repair maps, and checkpoint interfaces. The screen is the observer-facing geometry chart. It is not the carrier itself in the sense of one literal smooth ball. The implementation is the finite algebraic machinery that makes compare, write, repair, and re-read operations available.

The term "sphere folding" has a restricted technical meaning in this stack. It is the observer-facing spherical presentation of the repaired quotient normal form:

$$\mathrm{Fold}_{S,r}(s) = \chi_{S,r}(n_r(\pi_r(s)))\,.$$

The quotient map removes hidden carrier presentation, the normal-form map performs accepted repair, and the spherical chart displays caps, collars, cuts, edge sectors, and boundary records. Folding is therefore the geometric face of overlap repair on the screen chart. On the consensus surface this repair is defined first as the quotient operator $\mathrm{Rep}_\lambda : Q \to Q$, with representative-level maps only serving as lifts, so hidden carrier labels do not become additional physical data.

Spherical geometry carries several pieces of the construction at once. An observer-accessible cut has a closed two-dimensional angular chart. Caps and collars on that chart supply the cut data used by modular flow, entropy variation, and overlap comparison. The orientation-preserving conformal group of the same $S^2$ chart is the celestial-sphere realization of $\mathrm{SO}^+(3,1)$, so the sphere is the kinematic bridge between finite screen cuts and the emergent $3+1$-dimensional Lorentz branch once the controlled cap-pair theorem is satisfied. Finite cellulations of the chart provide the regulator surface on which ports, edge sectors, and local comparison data can be made explicit; they are not by themselves a Lorentz invariant continuum.

The echosahedral carrier is the concrete local realization of this chart data. Its multi-port interface supplies discrete boundary directions, face and edge incidence, exposed readout channels, record slots, and repair channels. Recurrent toroidal subchannels supply local winding and memory structure. The smooth spherical chart describes the observer-facing cut; the echosahedral carrier supplies the fixed-cutoff local machine whose quotient data can present that cut.

The $A_5$ and $E_8$ labels mark different roles in that machinery. $A_5$-icosahedral symmetry is the local finite-screen language used when a patch carrier is made concrete: it organizes ports, faces, and overlap data with enough discrete symmetry to stabilize observer-facing cuts. $E_8$-type language is the exceptional closure language used by the high-symmetry branch. It names the root-lattice and affine exceptional structure in which the icosahedral data can sit. The term is representation-closure language for the branches that call for it.

One finite certificate lives on the $E_8$-side of that split. The $E_8/\mathrm{Spin}(8)$ triality sidecar gives exact matrices for an $A_8$ root subsystem inside $E_8$, the subgroup $\mathrm{Alt}(9) \subset W(A_8) \cong \mathrm{Sym}(9)$, its nonsplit $2{\cdot}\mathrm{Alt}(9)$ lift in $\mathrm{Spin}(8)$, and a positive half-spin image preserving an $E_8$ lattice. The vector and half-spin presentations have distinct $E_8/2E_8$ orbit fingerprints, $\{9, 36, 84, 126\}$ and $\{120, 135\}$, before $\mathrm{Spin}(8)$ triality fuses them. This is a finite exceptional representation-closure receipt. It is not a recovered-core theorem, a new particle or force, a heterotic edge-CFT proof, or hardware evidence.

This distinction matters for the synthesis paper. The physical claim lives at the observer-visible quotient: finite patches must expose the same records and shared observables after allowed implementation-hiding changes. The microphysics paper gives the concrete reference architecture for that quotient surface, including the federated patch-carrier model, fixed-cutoff edge heat-kernel/Casimir law, central-record/Born–Lüders measurement interface, Bell/CHSH event surface, and checkpoint/restoration package. See Ref. [4], available as the numbered GitHub reference in the bibliography.

### Famous equations and structural outputs recovered

The table below maps headline formulas by logical role. Some rows are core theorem results, some are short corollaries of recovered actions, and some are standard limits once the parent OPH branch is established.

| Recovered result | OPH role | Famous display | Support boundary |
|---|---|---|---|
| Three-dimensional observer-frame chart, Lorentz kinematics, and invariant light cone | core theorem | $q(\Omega) = (1, \Omega)$, $n_C = (\cot\alpha, \csc\alpha\, \mathbf{c})$, $n_{gC} = \Lambda_g n_C$, $H^3 \simeq \mathrm{SO}^+(3,1)/\mathrm{SO}(3)$ | scaling-limit geometric subnet; chart population and neutral bulk are separate gates |

| Recovered result | OPH role | Famous display | Support boundary |
|---|---|---|---|
| Record-conditioned $H^3$ frame estimate | conditional theorem | $R_i(C,t,O) \to F(X_i)+e$, $\alpha \bar{d} \le \lvert F(X)-F(Y)\rvert_W$, $S_i(t) = B_H(\widehat{X}_i, r_i)$ | frame-local calibrated cap responses, compact frame domain, bounded error, quantitative cap frame, and positive gap for unique finite output; no event position is inferred |
| Einstein equation | conditional composition theorem | $G_{ab} + \Lambda g_{ab} = 8\pi G \langle T_{ab} \rangle$ | one source-derived common-domain tower with uniform asymptotics, universal coupling, a vacuum reference, and independent scale readouts; tower construction and certification are work in progress |
| Newton-Poisson gravity | inherited weak-field limit | $\nabla^2\Phi = 4\pi G\rho$, $\ddot{\mathbf{x}} = -\nabla\Phi$ | absolute Einstein-branch premises, weak field, and slow motion |
| Maxwell equations | electromagnetic corollary | $dF_Q = 0, \quad d{*}F_Q = g_Q^2 {*}J_Q$ | ordinary $\mathrm{U}(1)_Q$ branch plus explicit Maxwell action/current hypothesis |
| Yang–Mills equations | compact-gauge corollary | $DF = 0, \quad D{*}F = g^2 {*}J$ | compact-gauge connection branch plus explicit Yang–Mills action/current hypothesis |
| Bekenstein–Hawking / de Sitter area law | global capacity branch | $N_{\mathrm{scr}} = A/(4\ell_P^2) = 3\pi/(G\Lambda)$ | screen-capacity closure |
| Born rule and Lüders update | fixed-cutoff theorem | $\mathbb{P}(E) = \mathrm{Tr}(\rho P_E)$, $\rho\vert_E = P_E \rho P_E / \mathrm{Tr}(\rho P_E)$ | central record algebra |
| Tsirelson bound | fixed-cutoff theorem | $\lvert S_{\mathrm{CHSH}}\rvert \le 2\sqrt{2}$ | Bell event surface |
| Charge quantization | core theorem | $Q = T_3 + Y$, color singlets have $Q \in \mathbb{Z}$ | quarks are confined fractional-charge fields |
| Classical carrier-mode poles | conditional action theorem | $K_X^{\mathrm{phys}} \propto (\omega^2 - c_\star^2 \lvert\mathbf{k}\rvert^2)\Pi_X$ | Maxwell, perturbative pure-Yang–Mills, or pure-Einstein action/phase receipt; quantum particle gate separate |

| Recovered result | OPH role | Famous display | Support boundary |
|---|---|---|---|
| Coherent-matter scalar susceptibility | conditional scalar theorem | $\chi_\nu^{\mathrm{can}} = 1 - P_\chi/24 = 0.9320429912748350\ldots$ | nondegenerate OPH-coherent material source, Scalar Edge-Center Exhaustion, and protected-reserve collar closure; force additionally requires repair charge, an external gradient, and field-momentum closure |

## Particle Support Status

The particle paper treats the closure matrix explicitly. The non-hadron bundle has the following support levels:

Full $W/Z$ promotion requires a strict source root, an independently physical $E_\star$, the finite quotient-transport certificate, a frozen RG/matching/scheme packet, branch rigidity, target-independent precommitment, and complex-pole and uncertainty receipts.

**Conditional and audit coordinates.** The source-audit zero-selector law outputs the running/chart coordinates $(80.330, 91.119)$ GeV, with a residual freedom of one discrete two-law choice (9.3 MeV on the $W$ coordinate and 5.1 MeV on the $Z$ coordinate). These values are not commensurate with the PDG mass-dependent-width Breit–Wigner masses or with complex-pole masses, so their numerical proximity supplies no physical mass evidence. The selected carrier, conditional value law, and reference-fitted inverse adapter give the downstream audit coordinates

$$\begin{aligned} \text{selected carrier} \quad & (80.3862916924, 91.1829044467)\ \text{GeV}, \\ \text{conditional value law} \quad & (80.3770000154, 91.1879780779)\ \text{GeV}, \\ \text{reference-fitted inverse adapter} \quad & (80.3625, 91.1879)\ \text{GeV}. \end{aligned}$$

The two-loop audit returned $(79.115335, 89.802735)$ GeV in its designated primary cell. Its prescription adds an SM two-loop increment to an MSSM one-loop baseline, making it an inconsistent MSSM-1L+SM-2L hybrid prescription test rather than a consistent two-loop calculation. The pole audit returned approximately $(79.53284, 89.71232)$ GeV under a partial PRTS/Feynman-gauge prescription. The definition of $v$, tadpole treatment, field-content and threshold matching, scale selection, and higher-order terms are open. These audits establish neither a unique scheme conversion nor a unique 1–2% defect, and they do not exhaust the physical prescription family. As a stated diagnostic under the complex-pole conversion of the PDG-2026 reference values, the chart $W$ coordinate differs from the converted $W$ pole 80.3340 GeV by 0.5 propagated experimental standard deviations, and the chart $Z$ coordinate from its converted pole by about 17; both figures carry no OPH theory covariance and acquire physical meaning only when the physical readout contract closes. The Higgs/top pair is read on the double-criticality branch, a zero-continuous-parameter family whose comparison-exposed boundary-scale candidate $E_\star e^{-\pi} P^{-1/6}$ gives $(m_H, m_t) = (125.77, 172.63)$ GeV at two loops, with $m_H = 125.72$ GeV on the fit-free curve at the measured top. The declared calibration surface $(125.1995304097, 172.3523553288)$ GeV is a data-anchored fit that validates the formula stack and never predicts. For the quark audit at $M_Z$,

$$(\rho_u, \rho_d) = (1.1108888543, 0.7519410008), \qquad \rho_u\rho_d = 0.8353228768,$$

where the reciprocal ray requires one. Granting four endpoints leaves a 21.556% held-out minimax error. The generic exact interface is

$$(\mu_u, \sigma_u, \rho_u, \mu_d, \sigma_d, \rho_d).$$

The 0.2946% and 0.29436% target-anchored residuals use mixed conventions, and a lower-order ablation fits that chart better. These conditional and audit coordinates are excluded from the particle-output ledger. Ref. [5] contains the detailed receipt audit.

## Stack-Wide Claim Boundary

The paper stack uses one no-relabeling rule throughout. A finite diagnostic or calibration field does not become a physical observable merely because it is renamed. Capacity bookkeeping needs an independent energy readout before it can count as mass; an archive needs a physical source, propagation, and detector channel before it can count as radiation; a finite repair spectrum needs a continuum operator/readout bridge before it can count as a physical spectrum; and a finite reconstruction threshold needs physical entropy and clock data before it can count as a Page-time claim. The compact paper records this as the source-separation rule behind the support-level table. The $\chi_\nu$ scalar-channel identity is theorem-grade for nondegenerate OPH-coherent material sources once Scalar Edge-Center Exhaustion is imported. Device-scale force, cosmological abundance, and finite covariant dark stress require their own receipt packages.

**Open evidence boundary.** Four generating objects and one selector delimit the open rows; the underlying prescription families are not proved exhaustive. G1 is the Ward-projected hadronic transport for the two electromagnetic endpoint residuals. Its broad internal-mass bracket $S_{\text{hadronic}} \in [0.5578, 1.0543]$ contains the zero-electroweak point diagnostic 0.8954, but it spans about $1.16 \times 10^8$ declared tolerance widths. The value is compare-only, and the evaluation pixel differs from the comparison pixel; the bracket carries no promotion. G2 requires construction of the direct correctable-public-record producer. Its removable conditions are record-atom restriction maps, endogenous public reachability, a frozen publicness policy, the global checkpoint coupling, exact or approximate correctable-code certification across the terminal fiber, the capacity-carrier representation, confusability-reflecting capacity extension and refinement packets, and a finite-size slack law with one physical zero. The de Sitter and electroweak identifications separately require the horizon–record identification and the common screen/electroweak load-carrier identification. The interval certificate at $3.5321315434 \times 10^{122}$ applies to the conditional electroweak bridge; it does not construct $F$, and it differs from the Planck–$\Lambda$ central capacity by about 6.6 percent. G3 carries the forward electroweak chain. Its preregistered 96-entry one-loop sweep excludes entries inside that frozen menu only; higher-order, matching, threshold, tadpole, scale, and field-content prescriptions are open. G4 supplies numerical solver hygiene for the declared map. Twelve preregistered flavor candidates were also excluded, which excludes those candidates rather than the broader selector family.

A third-party audit of forty-two findings identifies no false theorem in the recovered mathematical core. Fail-closed gates and adversarial regression tests reject the hadronic target-coordinate algebra defect and the finite transition-clock eligibility defect; the printed epistemic statuses include those gates. The evidence ledger contains zero discriminating precommitted target-blind hits. The electroweak checks freeze analysis choices against known targets, so they are audits rather than target-blind predictions. Other positive rows are theorem-level, conditional, target-exposed, or non-discriminating.

### Companion Papers

The detailed depth surfaces used throughout this synthesis are:

1. *Observation-Determined Normal Forms: Stability, Obstructions, and Refinement in Constraint and Rewrite Systems* [1], which gives the substrate-neutral same-source/cross-source quantifier split, endpoint and refinement bounds, collar-repair obstructions, and the finite conditional-resampling receipt used by the repair papers.

2. *Recovering Relativity and the Standard Model from Observer Overlap Consistency* [2], which gives the compact technical core for relativity, gravity, receipt-conditional combined-zero-obstruction compact-gauge reconstruction (central or higher-associator strictification plus an allowed trivial-holonomy strict representative), and MAR-selected Standard Model structure.

3. *Reality as a Consensus Protocol: The Fixed-Point Computation That Implements Physics* [3], which carries the finite patch-net fixed-point, defect, quotient, and operator-record consensus package, with the default overlap code treated as a finite constraint code, termination separated from the local-diamond and repair-completeness conditions needed for confluence, and QECC/min-cut, spectral, BFT, and hardware-speedup claims gated by separate certificates.

4. *Federated Echosahedral Screen Microphysics: Patch Hardware, Records, and Observer Synchronization in OPH* [4], which carries the regulated federated patch-carrier architecture, the fixed-cutoff edge heat-kernel / Casimir theorem, and the measurement and observer checkpoint/restoration packages.

5. *Deriving the Particle Zoo from Observer Consistency* [5], which carries the particle derivations, masses, couplings, and sector calculations.

6. *Theoretical Bounds on $\chi_\nu$ in Observer-Patch Holography* and *Observer-Patch Holography and Dark Matter* [6, 7], which carry the coherent-matter channel theorem and the conditional repair-charge condensate action with its current, stress, normal-phase, deep-galaxy, and device-force consequences.

7. *Explaining the Yang–Mills Mass Gap with Observer-Patch Repair Dynamics* [8], which isolates the controlled compact-gauge repair mechanism and states the continuum certificate needed before the finite repair gap becomes a Clay-facing four-dimensional Yang–Mills gap.

8. *Observer-Patch Holography as a String-Vacuum Selector* [9], which treats string theory as an effective edge language and applies OPH gates to critical-string candidates; its rank certificate classifies the Bouchard-Donagi row as a structural benchmark rather than a selected witness.

The sections that follow give the synthesis view. Detailed proof burdens sit with the companion papers and with the late dependency checklist in this paper.

## 4 Recovered Core: Foundations and Structural Branches

Observer-Patch Holography is a reconstruction program for fundamental physics. Physical data live on a horizon screen $S^2$, and the effective world is encoded by the mutual consistency of overlapping patch descriptions. This recovered-core section develops the structural branch: fixed-cutoff overlap repair, the separated cofinal refinement-limit consensus bridge, collar recovery, the separate geometry-readout route to Lorentz and Einstein structure, receipt-conditional compact gauge reconstruction in the bosonic branch, and the realized-branch Standard Model quotient.

## 4.1 Overview

The recovered core has a simple conceptual order. First, overlap consistency turns local patch states into a constrained gluing problem. The fixed-cutoff branch gives collar decompositions, edge centers, and finite repair dynamics on the declared physical quotient. Second, the continuum geometric branch first reads quotient normal forms as cap, stress, diamond, tetrad, and scale data; it then turns modular flow on caps into Lorentz kinematics, a three-dimensional observer-frame hyperboloid, and finally into the Einstein relation through fixed-cap generalized-entropy stationarity plus the null-stress, small-ball, and tensor-upgrade clauses. Bare finite consensus stops at quotient normal forms. Third, the transportable edge-sector branch first classifies fixed-stage zero-obstruction transportable sectors and, on a cofinal tail carrying the compact-gauge refinement receipt, reconstructs some compact internal gauge group in the bosonic branch; Minimal Admissible Realization (MAR), anomaly constraints, and the explicit CKM/weak-sector clauses then select the realized Standard Model quotient with exact hypercharge, structural electroweak force content, the realized color triplet $N_c = 3$, and the generation count $N_g = 3$ on the explicit realized one-generation/one-Higgs package. This branch does not by itself supply a super-Tannakian internalization of fermions/chirality.

The local pixel ratio $P \equiv a_{\mathrm{cell}}/\ell_\star^2$, the cosmic record capacity $N_\star = \log D_\star$, and the selected no-$G$ scale certificate enter only on their declared quantitative branches. $P$ is the root of the outer/inner declared map in Section 7. The global-capacity branch uses the correctable public-record readback

$$M_0(q) = \alpha(G_q), \qquad \mathfrak{F}_{r,0}(D) = \{M_0(q) : q \in \widetilde{\Omega}_{r,D}\},$$

and stable closure requires $\mathfrak{F}_{r,0}(D_\star) = \{D_\star\}$ together with one physical zero of $s(D) = \log D - \log M_0(D)$. The exact reversible branch reduces to $M_0(q) = |X_{\mathrm{reach}}(q)|$. The horizon–record identification and the common screen/electroweak load-carrier identification are independent downstream hypotheses. The selected scale certificate emits $\gamma_\star = \ell_\star \nu_{\mathrm{Cs}}/c$, equivalently $B_\star = 3\pi/\ell_\star^2$, and then $G_{\mathrm{SI}} = c^3\ell_\star^2/\hbar$; the pixel $P$ cancels in the Newton area-law readout. The late classification section records the formal dependency checklist and separates recovered-core claims from quantitative closures and continuations.

## 4.2 Model and Axioms

### 4.2.1 Observers and access model

An observer $O$ is a tuple $(P_O, \mathcal{A}(P_O), \rho_O, R_O)$ where:

- $P_O \subset S^2$ is a connected screen patch (the observer's access region).
- $\mathcal{A}(P_O)$ is the von Neumann algebra associated to $P_O$.
- $\rho_O$ is the local state, obtained by restricting the global state to $\mathcal{A}(P_O)$.
- $R_O$ is a finite record algebra generated by stable internal record projectors within $P_O$.

Observers are internal patterns in the global state. Different observers correspond to different patches and their compatible marginals.

### 4.2.2 Screen, patches, and algebra net

We work in a single static patch with a horizon screen $S^2$. Each connected subregion $P \subset S^2$ is assigned a von Neumann algebra $\mathcal{A}(P)$. The net satisfies isotony:

$$P \subset Q \implies \mathcal{A}(P) \subset \mathcal{A}(Q).$$

A global state $\omega$ is a positive linear functional on the inductive-limit algebra. Overlap consistency is imposed algebraically: for overlaps $P_1 \cap P_2$, $\omega$ restricted to $\mathcal{A}(P_1 \cap P_2)$ is the same from either side.

#### 4.2.3 Five OPH axioms

This paper uses five OPH axioms. The support-visible BW scaling theorem below handles the scaling/BW step. Transportability and the fixed-cutoff bosonic sector category are theorem-level results cited directly where they are used. Refinement/fiber descent and the cofinal MAR-admissible compact-gauge witness are conditional on the explicit compact-gauge refinement receipt stated in Section 6. Other branch-local hypotheses are written out in prose where needed instead of being assigned a separate label family.

**1. Screen Net.** A horizon screen $S^2$ carries a net of algebras $P \mapsto \mathcal{A}(P)$.

**2. Overlap Consistency.** Local states agree on shared observables for any overlap.

**3. Local MaxEnt and Refinement Stability.** At the UV scale the realized state maximizes entropy subject to a finite family of gauge-invariant local constraints, and the resulting family of states is stable under coarse-graining so symmetry-allowed relevant operators are not held at zero by unexplained fine tuning.

**4. Recoverable Generalized Entropy.** A generalized entropy functional exists on caps, obeys quantum focusing on null generators, and comes with the recoverability structure used throughout the manuscript: collar tripartitions have small conditional mutual information (CMI) with controlled recovery maps, with stronger collar/null-strip hypotheses stated explicitly where needed.

**5. Minimal Admissible Realization (MAR).** On the admissible low-energy branch, the realized sector package is the lexicographically minimal one under the complexity vector $C(\mathfrak{S})$. MAR is an explicit structural-economy axiom on admissible realized branches, not a theorem derived from the preceding four axioms.

#### 4.2.4 Closure values and theorem-local technical data

The quantitative branches discussed here use one incomplete declared-map coordinate, one proposed capacity coordinate, and one selected no-$G$ scale certificate:

$$P \equiv a_{\mathrm{cell}}/\ell_\star^2, \qquad N_{\mathrm{CRC}} \equiv \log\dim\mathcal{H}_{\mathrm{tot}}, \qquad \gamma_\star \equiv \frac{\ell_\star \nu_{\mathrm{Cs}}}{c}, \qquad B_\star \equiv \frac{3\pi}{\ell_\star^2}.$$

Here $P = P_\star$ is the local UV-area ratio selected by the unique root of the declared outer/inner self-read map in Section 7. No measured coupling enters that solve. Its physical Thomson-limit interpretation requires the same-scheme hadronic spectral transport. The global coordinate is defined by the source-only correctable public-record packet and the stable whole-fiber saturation equation $\mathfrak{F}_{r,0}(D_\star) = \{D_\star\}$, with $N_\star = \log D_\star$. The independent weak/Higgs common-load bridge gives $N_{\mathrm{bridge}} = \pi\exp[6\pi/(P\alpha_U(P))]$. That bridge is $N_{\mathrm{bridge}} \simeq 3.532 \times 10^{122}$, while the Planck-$\Lambda$ central capacity is $N_\Lambda \simeq 3.313 \times 10^{122}$, a mismatch of about 6.6 percent. Capacity-level neutrino side estimates tied to that branch value are comparison-only bookkeeping terms separate from the rejected weighted-cycle comparison lane. The selected scale certificate supplies $\gamma_\star$, with $B_\star$ as the SI curvature display of the same branch. Once supplied, $\ell_\star^2 = 3\pi/B_\star$ becomes the usual Planck-area display after $G_{\mathrm{SI}} = c^3\ell_\star^2/\hbar$. These are quantitative-branch quantities, not extra axioms.

Observation enters as an external test of the source-defined branches. In the OPH reverse-engineering reading, the universe is a closed mathematical fixed structure, and each readback object is specified independently of its measured target. For N, this object is set-valued before whole-fiber scalarization and its uniqueness theorem is the exact finite-size slack law, not a Banach contraction. Measurements may define comparison coordinates; they construct neither closure. The local map has a certified unique root. Global uniqueness requires the physical checkpoint packet and the regulator-stable condition $s(D_\star) = 0 < s(D)$ elsewhere.

The compact proof uses the same convention as a compression estimate. A quantitative row enters that estimate only when the declared source map has no dependency path from the measured target or a calibrated proxy. If $p_i$ upper-bounds the conditional accidental hit probability of row $i$ after previous accepted rows, then

$$P_{\rm acc} \le \prod_i p_i.$$

The evidence ledger contains no registered discriminating prediction. The electroweak checks use known targets and therefore have comparison-audit status. Other positive rows have theorem, conditional, comparison-only, or non-discriminating status.

If both missing physical identifications are supplied, the two dimensionless coordinates determine dimensionless geometry, not an SI scale by themselves. On the de Sitter branch,

$$N_\star = \frac{3\pi}{\Lambda_\star \ell_\star^2}, \qquad P_\star = \frac{a_{\rm cell}}{\ell_\star^2},$$

so

$$\Lambda_\star \ell_\star^2 = \frac{3\pi}{N_\star}, \qquad \Lambda_\star a_{\rm cell} = \frac{3\pi P_\star}{N_\star}.$$

Equivalently, for the scale product $\mathcal{B}_\ell := \Lambda_\star N_\star$,

$$\mathcal{B}_\ell \ell_\star^2 = 3\pi, \qquad \mathcal{B}_\ell a_{\rm cell} = 3\pi P_\star.$$

These invariants survive the rescaling $\ell_\star^2 \mapsto \lambda^2 \ell_\star^2$, $a_{\rm cell} \mapsto \lambda^2 a_{\rm cell}$, and $\Lambda_\star \mapsto \lambda^{-2}\Lambda_\star$, while $\mathcal{B}_\ell \mapsto \lambda^{-2}\mathcal{B}_\ell$. Thus the SI value of $\mathcal{B}_\ell$, the SI area $\ell_\star^2$, and $G_{\rm SI}$ require the selected scale certificate; they are not derived from $P_\star$ and $N_\star$ alone. If the selected no-$G$ scale certificate supplies $\gamma_\star$, equivalently $B_\star$, then

$$\gamma_\star = \frac{\ell_\star \nu_{\rm Cs}}{c}, \qquad B_\star = \mathcal{B}_\ell = \Lambda_\star N_\star, \qquad \ell_\star^2 = \frac{3\pi}{B_\star}.$$

The branch certificate used for the SI gravity row is

$$B_\star = 3.60787391468\ldots \times 10^{70}\,\mathrm{m}^{-2},$$

hence

$$\ell_\star^2 = 2.61228030237\ldots \times 10^{-70}\,\mathrm{m}^2.$$

Rounded capacity displays such as $N_\Lambda \simeq 3.313 \times 10^{122}$ are useful cosmological labels, but they are not high-precision inputs for the Newton row; with the Planck-2018 curvature benchmark, using only the rounded capacity would shift the Newton row at the $8.4 \times 10^{-4}$ level.

The capacity normalization uses the de Sitter static-patch entropy [24]. If

$$N_{\rm patch} = \left(\frac{r_{\rm dS}}{\ell_P}\right)^2$$

denotes the bare horizon area ratio, then

$$N_{\rm scr} = S_{\rm dS} = \frac{A_{\rm dS}}{4\ell_P^2} = \pi N_{\rm patch} = \frac{3\pi}{\Lambda \ell_P^2}.$$

Using the Planck-2018 late-time de Sitter scale [25] gives $N_{\text{patch}} \simeq 1.05 \times 10^{122}$, $N_{\text{scr}} \simeq 3.313 \times 10^{122}$, and $\Lambda \ell_P^2 \simeq 2.845 \times 10^{-122}$.

For the recovered core, the controlled scaling/BW step for Lorentz/null-modular/Einstein statements is Theorem 4.2, the support-visible BW scaling theorem. Transportability, the fixed-cutoff bosonic category, fixed-cutoff regulator bookkeeping, quasi-local propagation, endpoint-Lipschitz control, collar decomposition, and related branch-internal statements are cited directly from axioms and preceding theorems. Refinement/fiber descent and the cofinal compact-gauge witness additionally require the Section 6 compact-gauge refinement receipt.

**Local MaxEnt at the regulator scale.** At the regulator scale $\ell_{\text{UV}}$, the global state $\omega$ maximizes von Neumann entropy subject to:

1. A finite set $\{O_a\}$ of gauge-invariant local operator densities, each supported on a ball of radius $\leq r_0 = O(\ell_{\text{UV}})$.
2. Constraint equations $\langle \sum_x O_a(x) \rangle = C_a$, one homogeneous global sum per density label $a$, where $x$ runs over the cells of the UV lattice. The constraints are the $N_{\text{con}}$ global sums, not one independent constraint per cell, so the number of independent constraints and of Lagrange multipliers is $N_{\text{con}}$ (plus the optional global constraints below) on every lattice, independent of the cell count.
3. Optionally, a finite number of global constraints (total energy, charge).

This is the minimal specification needed to derive the local Gibbs form of Lemma 2.6 from Axiom 3 with a cutoff-independent multiplier count.

**Clarification (MaxEnt $\neq$ thermal equilibrium).** MaxEnt here is **local state selection** given constraints, not "the universe is in thermal equilibrium." Non-equilibrium physics is modeled in the enlarged *local-multiplier* family obtained by replacing the homogeneous multipliers $\lambda_a$ with slowly varying fields $\lambda_a(x)$, equivalently by imposing per-cell constraints $\langle O_a(x) \rangle = c_a(x)$. That enlarged family has dimension proportional to the number of UV cells, so it is a function-space approximation regime with gradient error terms, not the finite-dimensional family of the refinement argument: the refinement-stable branch and the induced maps $R_{\ell \to L}$ below are always statements about the homogeneous $N_{\text{con}}$-parameter branch. Controlled violations of exact Markov additivity appear under the explicit mixing hypotheses stated later in Section 2.3. Equilibrium is an approximation regime with explicit error terms.

**Rotationally invariant constraint branch.** Constraint sets are SO(3)-invariant on $S^2$.

**Gauge as overlap redundancy at fixed cutoff.** Overlap identifications are not unique; the freedom that leaves overlap observables invariant forms a local groupoid.

**Central-defect subbranch.** Assume the overlap centers are identified with one fixed abelian unitary coefficient group $Z_\Sigma$ on which overlap transport acts trivially. Choose unitary overlap implementers with $U_{ji} = U_{ij}^{-1}$. On triple overlaps, when the failure of strict coherence is central, define

$$\Omega_{ijk} := U_{ij} U_{jk} U_{ki} = z_{ijk}, \qquad z_{ijk} \in Z_\Sigma,$$

equivalently $U_{ij} U_{jk} = z_{ijk} U_{ik}$. The defect is data of the implementer lift; writing only $\mathrm{Ad}(z_{ijk})$ would erase it because conjugation by a central element is the identity.

**Collar double-scaling hypothesis.** There is one refinement family with $\delta \to 0$, $\ell_{\text{UV}} \to 0$, and uniform finite-range and strong conditional Gibbs mixing constants such that

$$I(A_\delta : D_\delta \mid B_\delta)_\omega \leq c |\partial C|_{\text{UV}} e^{-\delta/\xi}, \qquad \frac{\delta}{\xi} - \log |\partial C|_{\text{UV}} \longrightarrow +\infty.$$

The second condition controls the complete boundary-prefactored envelope. The weaker ratio $\delta/\ell_{\mathrm{UV}} \to \infty$ does not imply it. Section 2.3 gives the finite-range theorem, explicit constants, a sufficient logarithmic schedule, and the finite-receipt fields. On the exact central-interface branch the CMI is zero without this mixing premise.

**Geometric-subnet BW lift.** The continuum Lorentz statement is asked only on the support-visible extracted geometric subnet for caps, together with the controlled collar/refinement limit that carries the Markov and recovery remainders explicitly. The finite cap regulators are type-I algebras, but the scaling-limit observer algebra may leave that class, so the theorem is automorphism-level on the emitted geometric cap pair and the special type-I generator form is $K_C = 2\pi B_C + Z_C$ with $Z_C$ central. The compact paper proves that any cofinal finite cap-normal branch satisfying the support-order, held-out oriented cross-ratio, mixed-modular-matrix, and independently normalized geometric $2\pi$-KMS certificate converges to the support-visible BW cap automorphism. It also proves the cap-normal $H^3$ chart: $q(\Omega) = (1, \Omega)$, $n_C = (\cot\alpha, \csc\alpha\, \mathbf{c})$, $n_{gC} = \Lambda_g n_C$, and $H^3 \simeq \mathrm{SO}^+(3,1)/\mathrm{SO}(3)$, with cap half-spaces but no populated-bulk or preferred-observer-point consequence. The compact paper's geometry-producer packet produces that certificate from repair normal forms on a computable-receipt branch (spherical incidence, disk/mesh, cross-ratio, and $2\pi$-normalization receipts) and proves that bare confluence underdetermines topology, dimension, framing, and normalization, so the receipts are the irreducible Einstein branch-entry content. Its null-net-standardness and event-manifold packets supply derived positive translations and a conditional Lorentzian event manifold, and its closure packets construct the local conserved stress tensor, the corrected generalized-entropy first law, the uniform small-diamond limit, and the composed branch-entry theorem. The composition requires one source-derived common-domain tower with certified tails, universal coupling, a source-derived vacuum reference, and independent scale readouts. Construction and certification of the tower are work in progress (Theorems 4.3c–4.3h).

**Derived quasi-local propagation and modular-locality control.** At scale $\ell_{\mathrm{UV}}$, the automorphism group generated by the MaxEnt generator of Lemma 2.6, or more generally by any branch generator lying in the same bounded-support algebraic closure, has a finite Lieb–Robinson velocity $v_{\mathrm{LR}}$: for local operators $A, B$ supported on regions separated by distance $d$,

$$\|[A(t), B]\| \le c\|A\|\|B\| \min(|R_A|, |R_B|) e^{-(d - v_{\mathrm{LR}}|t|)/\xi}$$

for $d > v_{\mathrm{LR}}|t|$, where $\xi = O(\ell_{\mathrm{UV}})$.

This is the branch-internal control statement that turns the quasi-local structure of the local-Gibbs branch into explicit support control for time-evolved operators. It is part of the third-axiom branch stated at regulator scale.

**Approximate modular covariance on the fixed-cutoff realized branch.** Under the fixed-cutoff realized presentation, Lemma 2.6 (local Gibbs), the derived quasi-local propagation bound above, and the collar double-scaling plus mixing hypotheses of Section 2.3, the modular flow $\sigma_t^{\omega,C}$ maps $\mathcal{A}(R)$ into a slightly thickened region algebra:

$$\sigma_t^{\omega,C}(\mathcal{A}(R)) \subseteq \mathcal{A}(R^{+v_{\mathrm{mod}}|t|}) \quad \text{up to error } \eta(d - v_{\mathrm{mod}}|t|),$$

where $R^{+s}$ denotes the $s$-neighborhood thickening, $v_{\mathrm{mod}}$ is a "modular propagation velocity" controlled by $v_{\mathrm{LR}}$ and local norm bounds, and $d$ is the distance from $R$ to $\partial C$.

**Heuristic motivation.** On the reduced-modular-locality branch of Lemma 4.1a-b, modular additivity localizes the nonadditive defect of $K_C = -\log\rho_C$ to the collar. The associated propagation receipt then bounds support spreading under $e^{iK_C t}$. In the rate-controlled collar limit of Section 2.3, the thickening vanishes in macroscopic units. This remains heuristic support for the tangent-limit package; a Lieb–Robinson bound for the global Gibbs generator alone does not prove locality of the reduced modular Hamiltonian.

**Continuum-limit heuristic.** Define the induced region flow

$$f_t^C(R) := \lim_{\ell_{\mathrm{UV}}\to 0} R^{+v_{\mathrm{mod}}|t|}.$$

Then $\sigma_t^{\omega,C}(\mathcal{A}(R)) = \mathcal{A}(f_t^C(R))$ becomes exact in the continuum limit, with error controlled by

$$\eta(\delta) \lesssim 2\sqrt{c|\partial C|_{\mathrm{UV}}}\, e^{-\delta/(2\xi)}$$

when CMI is measured in nats, with the full rate condition imposed on the prefactor.

The discussion above is heuristic support for the tangent-limit package used in Theorem 4.2.

**Refinement-stable MaxEnt branch.** Choose any family of coarse-graining channels $\Phi_{\ell\to L}$ between UV scale $\ell$ and IR scale $L$ that is compatible with the third OPH axiom, including its refinement-closure clause. Then on the realized branch one may write

$$\Phi_{\ell\to L}(\omega_\ell(\lambda)) = \omega_L(R_{\ell\to L}(\lambda)),$$

for an induced map $R_{\ell\to L}$ on the common finite-dimensional homogeneous multiplier family. The existence of $R_{\ell\to L}$ as a self-map of that same finite-dimensional family is exactly the closure clause, not a bookkeeping consequence of reusing the labels $a$: coarse-graining a finite-range Gibbs family generically generates interactions outside the retained density list, and the axiom asserts that on the realized branch those generated directions vanish or are absorbed into the retained family; the closure defect, the moment-matching I-projection realizing $R_{\ell\to L}$, and its trace-norm residual bound are constructed in Definition 2.6a and Lemma 2.6b, and what remains assumed rather than proved is that the defect vanishes along the realized branch. Granting closure, the regulator states lie in that shared multiplier family rather than in unrelated state spaces at different cutoffs, and the realized branch is the persistent trajectory or invariant subset selected inside it under the induced refinement maps. This is a state-side persistence statement; it does not by itself upgrade fixed-cutoff edge labels to a transportable refinement-persistent sector colimit.

**Fixed-cutoff realized presentation.** At a UV scale $\ell_{\mathrm{UV}}$, local patch algebras are type-I with finite-dimensional Hilbert spaces, and gauge-as-gluing is realized as a boundary action on a lifted finite-dimensional presentation. Section 2.3 proves the resulting fixed-cutoff collar package directly from overlap consistency on this realized presentation, while distinguishing the lifted fixed-point algebra from the sector-preserving collar algebra used in the edge-center (EC) theorem.

External mathematical inputs: SSA and recovery theorems (Petz 1986, 1988; Fawzi and Renner 2015), the fixed-volume area-variation identity for small geodesic balls used in the local Einstein rest-frame relation, and the local quadratic-polarization argument used in the tensor upgrade. The null modular bridge and the small-ball kernel are carried internally on the compact surface rather than imported from a separate EFT small-ball law. For SM contact we also use the Doplicher-Roberts reconstruction (Doplicher and Roberts 1989, 1990) once localized zero-obstruction sectors are assembled in the small-region limit. Full citations appear in the References.

**Recovered-core theorem boundary.** The Lorentz/null-modular/Einstein recovered core runs through the support-visible automorphism theorem stated in Section 4.2. Fixed-cutoff statements remain regulator statements.

### 4.2.5 Notation

- $\rho_C$: reduced state on cap $C$.
- $K_C := -\log \rho_C^\omega$: modular Hamiltonian of the reference state.
- $B_C$: geometric generator of the cap-preserving conformal dilation.
- $S_{\text{gen}}(C)$: generalized entropy on a cap.
- $\ell_{\text{UV}}$: UV length scale of the refined screen net.
- $\delta$: collar width around a cap boundary.

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## 4.3 Information-Theoretic Tools

### 4.3.1 Strong subadditivity and Markov states

For any tripartite state $\rho_{ABC}$,

$$I(A : C \mid B) := S(AB) + S(BC) - S(B) - S(ABC) \geq 0.$$

Exact Markov states satisfy $I(A : C \mid B) = 0$ and admit a recovery map:

$$\rho_{ABC} = (\text{id}_A \otimes \mathcal{R}_{B\to BC})(\rho_{AB}).$$

### 4.3.2 Approximate recovery

If $I(A : C \mid B) \leq \varepsilon$ in nats, there exists a CPTP recovery map $\mathcal{R}$ with

$$\|\rho_{ABC} - (\text{id}_A \otimes \mathcal{R})(\rho_{AB})\|_1 \leq 2\sqrt{1 - e^{-\varepsilon}} \leq 2\sqrt{\varepsilon}.$$

### 4.3.3 Collar refinement and sufficient mechanisms

Fix a cap $C \subset S^2$ with boundary circle. For collar width $\delta$ define

$$B_\delta := \{x \in S^2 : \text{dist}(x, \partial C) \leq \delta\}, \qquad A_\delta := C \setminus B_\delta, \qquad D_\delta := (S^2 \setminus C) \setminus B_\delta.$$

Then $S^2 = A_\delta \cup B_\delta \cup D_\delta$ with $A_\delta$ and $D_\delta$ interacting only through $B_\delta$. The collar double-scaling hypothesis is the requirement that $I(A_\delta : D_\delta \mid B_\delta) \to 0$ in the refinement limit. At fixed regulator scale, the same finite-dimensional realized presentation supplies the patch-net and overlap-gluing data used below. We therefore isolate that realized presentation first, then separate the exact-Markov and quantitative routes.

**Finite quotient ensemble theorem surface.** Fix a finite regulator $r$. The physical presentation space is $\Sigma_r$, the presentation redundancy groupoid is $\Gamma_r$, and the finite physical quotient is

$$Q_r = \Sigma_r/\Gamma_r, \qquad \pi_r : \Sigma_r \to Q_r.$$

The quotient removes nonphysical presentation data: gauge representatives, port relabelings, mesh labels, shard or worker identifiers, queue order, repair schedule identifiers, retry counters, timestamps unless declared semantic, hidden carrier coordinates, and inert ancillary labels. If only settled configurations carry probability, the probability space is the normal-form subset

$$N_r = n_r(Q_r).$$

The map $n_r$ is a normal-form map, not a probability law. Any promoted physical branch inherits this firewall: it must declare the quotient-intrinsic source law or action before a normal form can be read as a selection or prediction claim.

**Observable algebras and reference states.** In the finite classical case the quotient observable algebra is

$$\mathcal{O}_r = \ell^\infty(Q_r).$$

In the finite quantum case the physical algebra is a declared quotient algebra $\mathcal{A}_r^{\mathrm{phys}}$ with state

$$\omega_r(A) = \mathrm{Tr}(\rho_r A).$$

When the reference object is obtained from a finite lifted carrier, the load-bearing data are not an abstract groupoid cardinality alone. They are a tracially pointed quotient

$$\left(\mathcal{A}_{r,b}^{\mathrm{phys}}, \tau_{r,b}^0\right), \qquad \mathcal{A}_{r,b}^{\mathrm{phys}} = z_{r,b} B(\widetilde{\mathcal{H}}_r)^{G_r} z_{r,b},$$

where $U_r : G_r \to U(\widetilde{\mathcal{H}}_r)$ is the compact gauge action and $z_{r,b}$ is the central projection for the declared boundary or superselection sector. The reference trace is

$$\tau_{r,b}^0(A) = \frac{\mathrm{Tr}_{\widetilde{\mathcal{H}}_r}(A)}{\mathrm{Tr}_{\widetilde{\mathcal{H}}_r}(z_{r,b})}.$$

If

$$z_{r,b}\widetilde{\mathcal{H}}_r \cong \bigoplus_\alpha V_\alpha \otimes M_\alpha,$$

with $d_\alpha = \dim V_\alpha$ and $m_\alpha = \dim M_\alpha$, then

$$\mathcal{A}_{r,b}^{\mathrm{phys}} \cong \bigoplus_\alpha I_{V_\alpha} \otimes B(M_\alpha), \qquad p_{r,\alpha} = \frac{d_\alpha m_\alpha}{\sum_\beta d_\beta m_\beta}.$$

These are the induced central-sector weights only after the carrier representation and boundary sector have been fixed.

**OPH quotient ensemble.** An OPH quotient ensemble is specified by a quotient-intrinsic base weight and action

$$m_r : Q_r \to \mathbb{R}_{>0}, \qquad S_r : Q_r \to \mathbb{R} \cup \{+\infty\},$$

and

$$w_r(q) = m_r(q) e^{-S_r(q)}, \qquad Z_r = \sum_{q \in Q_r} w_r(q), \qquad \mu_r(q) = Z_r^{-1} w_r(q).$$

Equivalently, one may state an intrinsic projective prior $\nu_r$ on $Q_r$ and set $\mu_r = (n_r)_\# \nu_r$. Uniform quotient counting, uniform representative counting pushed to the quotient, groupoid weights, and tracial central-sector weights are different physical claims. The paper must declare which one is being used.

**Normal-form projector non-selection.** For any retraction $N : Q \to Q_{\mathrm{nf}}$ onto a subset $Q_{\mathrm{nf}} \subseteq Q$, that is, any map whose restriction to $Q_{\mathrm{nf}}$ is the identity, the induced map on laws

$$\mathcal{C}_Q(\mu) = N_\# \mu$$

is idempotent:

$$\mathcal{C}_Q^2 = \mathcal{C}_Q.$$

Every law supported on $Q_{\mathrm{nf}}$ is fixed; both statements use the retraction property. Therefore settlement or canonicalization never selects a unique physical probability law by itself.

**Selection-gap corollary.** Let $X \subseteq Q_{\text{nf}}$ be a finite set of quotient-normal candidates distinguished by visible invariants. Normal-form data determine $X$ and its quotient-visible invariants, but they do not choose a member of $X$. If two laws $\mu, \nu$ are supported on $X$ and concentrate on different candidates, both are fixed by $\mathcal{C}_Q$. Unique sector selection therefore requires source data: an intrinsic action with a unique minimizer, a declared physical ensemble, or a refinement-stable gap certificate. A defect or holonomy classification can classify possible sectors without choosing the physical sector, and a contraction or repair generator can certify convergence toward a declared target without creating the target law.

**Finite MaxEnt quotient ensemble.** For finite $Q$, positive $m$, and quotient observables $F_1, \dots, F_k$, maximizing

$$\mathcal{H}_m(\nu) = -\sum_q \nu(q) \log \frac{\nu(q)}{m(q)}$$

subject to

$$\sum_q \nu(q) = 1, \qquad \sum_q \nu(q) F_a(q) = c_a$$

has the full-support solution, when the feasible full-support surface is nonempty,

$$\mu(q) = \frac{m(q) \exp[-\sum_a \theta_a F_a(q)]}{Z(\theta)}.$$

Boundary optima obey the same formula after restricting to their support. On a finite noncommutative quotient algebra with faithful reference state $\sigma_r$,

$$\rho_r = \frac{\exp(\log \sigma_r - \sum_a \theta_a F_{r,a})}{\operatorname{Tr} \exp(\log \sigma_r - \sum_a \theta_a F_{r,a})}.$$

The finite constraint ledger must name every $F_{r,a}$, its units and support, the target expectation and source, sector or zero-mode treatment, refinement transformation, and proof that no run output or observational output entered the source definition.

**Refinement compatibility and RG closure.** For $s \succeq r$, let $c_{sr} : Q_s \to Q_r$ be the physical coarse map. Exact compatibility of weighted ensembles is equivalent to the fiber-sum identity

$$\sum_{q' : c_{sr}(q') = q} m_s(q') e^{-S_s(q')} = \alpha_{sr} m_r(q) e^{-S_r(q)}$$

for a constant $\alpha_{sr} > 0$ independent of $q$. Then

$$(c_{sr})_\# \mu_s = \mu_r.$$

If the one-step defects are

$$\delta_{k+1,k} = \|(c_{k+1,k})_\# \mu_{k+1} - \mu_k\|_{\text{TV}},$$

then

$$\|(c_{nr})_\# \mu_n - \mu_r\|_{\text{TV}} \le \sum_{k=r}^{n-1} \delta_{k+1,k}.$$

For exponential-family refinement, exact closure requires the fine conditional free energy

$$G_{sr,\theta}(q_r) = -\log \mathbb{E}_{m_s^0} [\exp[-\theta \cdot F_s(Q_s)] \mid c_{sr}(Q_s) = q_r]$$

to equal $\kappa_{sr}(\theta) + R_{sr}(\theta) \cdot F_r(q_r)$. If the residual is uniformly bounded by $\varepsilon$, the induced total-variation defect is bounded by $\tanh \varepsilon$.

**Implementation invariance and representative lifting.** If implementations $A, B$ have quotient bijections $h_r : Q_r^A \to Q_r^B$ satisfying

$$m_r^B(h_r q) = m_r^A(q), \qquad S_r^B(h_r q) = S_r^A(q), \qquad h_r \circ c_{sr}^A = c_{sr}^B \circ h_s,$$

then

$$(h_r)_\# \mu_r^A = \mu_r^B.$$

For tracially pointed quantum quotients the corresponding equivalence is a trace-preserving quotient equivalence. It is invariant under unitary intertwiners preserving the gauge action and sector, and under inert trivial ancillas $A \mapsto A \otimes I_{\mathrm{anc}}$. It is not invariant under arbitrary changes of gauge-representation multiplicities.

If an implementation stores representatives, a representative-level law must be a conditional lift

$$\widetilde{\mu}_r(x) = \mu_r(\pi_r x)\kappa_r(x \mid \pi_r x), \qquad \sum_{x:\pi_r(x)=q} \kappa_r(x \mid q) = 1.$$

Then $(\pi_r)_\# \widetilde{\mu}_r = \mu_r$. Uniform representative sampling yields orbit-size weights and is physical only if representative counting is the declared base measure.

**Quotient-lumpable kernels and sampler correctness.** A representative kernel $\widetilde{P}(x, y)$ descends to $Q_r$ only when

$$P_Q(q, q') = \sum_{y:\pi(y)=q'} \widetilde{P}(x, y)$$

is independent of the chosen representative $x \in \pi^{-1}(q)$. For $w(q) = m(q)e^{-S(q)}$ and proposal $R(q, q')$ with reciprocal support, the Metropolis–Hastings acceptance rule

$$a(q, q') = \min\left\{1, \frac{w(q')R(q', q)}{w(q)R(q, q')}\right\}$$

gives detailed balance

$$\mu(q)R(q, q')a(q, q') = \mu(q')R(q', q)a(q', q).$$

Repair-informed proposals must include the Hastings asymmetry term; otherwise the stationary law is generically changed.

**Repair generators are not selectors.** A repair generator of the form

$$L_{\mathrm{rep}} = \sum_C c_C(I - E_C)$$

is a relaxation or sampling object after a law has been selected. Conditional expectations $E_C$ are defined on $L^2(X_r, \pi_r)$, so the reference law $\pi_r$ is input. On overlapping collars the expectations need not commute. The correct finite gap certificate is the Poincare constant

$$\kappa_r = \inf_{f \perp 1} \frac{\sum_C \|(I - E_C)f\|^2}{\|f\|^2}.$$

If local fiber rates have a positive lower bound $\gamma_*$, then

$$L_{\mathrm{rep}} \geq \gamma_* \kappa_r (I - P_0).$$

Finite repair completeness gives $\kappa_r > 0$ at fixed regulator. A uniform refinement lower bound $\inf_r \kappa_r > 0$ is a separate theorem or receipt.

**Finite evidence accuracy.** For bounded coarse observables $O$, if

$$\|\widehat{\mu}_s - \mu_s\|_{\mathrm{TV}} \le \epsilon_{\mathrm{samp}}$$

and the refinement defects sum to $\epsilon_{\mathrm{ref}}$, then

$$\left|\mathbb{E}_{\widehat{\mu}_s}[O \circ c_{sr}] - \mathbb{E}_{\mu_r}[O]\right| \le 2\|O\|_\infty(\epsilon_{\mathrm{samp}} + \epsilon_{\mathrm{ref}}).$$

Continuum-facing observables require a realization map and correlation Cauchy bound in addition to a finite histogram.

**Vacuum promotion gate.** A stationary sampler is not a physical vacuum. For any faithful target law one can build a positive transfer operator with that law as ground state, so positivity alone is not a selector. Vacuum promotion requires source Euclidean slab data

$$\mathfrak{S}_r^E = (Q_r, m_r^0, J_r, V_r, a_{t,r})$$

whose conductance $J_r(q, q') = J_r(q', q) \ge 0$, local potential $V_r$, and slab thickness $a_{t,r}$ are derived without using the target law or sampler output. With connected event graph,

$$(H_r^E f)(q) = \frac{1}{m_r^0(q)} \sum_{q'} J_r(q, q')(f(q) - f(q')) + V_r(q)f(q)$$

is self-adjoint and bounded below on $L^2(Q_r, m_r^0)$; its finite Feynman–Kac semigroup is positivity improving. Perron–Frobenius gives a unique positive normalized ground state $\Omega_r$, and the finite vacuum law is

$$\mu_r^{\mathrm{vac}}(q) = |\Omega_r(q)|^2 m_r^0(q).$$

For $T_r = e^{-a_{t,r}(H_r^E - E_{0,r})}$, the Doob kernel is stochastic and detailed-balanced with $\mu_r^{\mathrm{vac}}$. Continuum promotion additionally requires reflection positivity or equivalent reconstruction plus refinement compatibility of the transfer family.

**Primordial and cosmological prediction firewall.** A screen covariance contains incomplete radial information. The complete one-shell map

$$C_\ell = 4\pi \int_0^\infty \frac{dk}{k} \Delta_\zeta^2(k) j_\ell^2(k\chi_\star)$$

has an infinite-dimensional kernel that persists under positivity. OPH primordial promotion requires the source-only stress, single-clock, entropy-repair, curvature-evolution, adiabatic-mode, phase-coherence, physical mode, radial-null-space, and forward-projection receipts together with a scale-natural physical dilation intertwiner or complete radial cross-covariance tomography. A finite radial prior produces a conditional continuation. Observable CMB comparison also requires declared source, solver, dataset, covariance, nuisance, data-use, and pooled-reducer provenance.

**Claim tiers and required receipts.** Every ensemble-facing run records its ensemble id, claim tier, regulator, representative schema, gauge action, canonicalizer, base measure, action coefficients, coarse maps, zero-mode projector, amplitude convention, sampler, smoothing policy, source provenance, and explicit nonclaims. The seed belongs to the run receipt rather than the ensemble definition. The claim tiers are

$$\begin{aligned} E0 &: \quad \text{seed noise, proposal noise, repair jitter,} \\ E1 &: \quad \text{conventional reference ensemble,} \\ E2 &: \quad \text{OPH-native quotient ensemble,} \\ E3 &: \quad \text{OPH vacuum,} \\ E4 &: \quad \text{OPH primordial field,} \\ E5 &: \quad \text{observable cosmological prediction.} \end{aligned}$$

The evidence bundle must keep separate receipts for stationary-law schedule invariance, detailed balance of the aggregate kernel, and pathwise partition invariance. Deterministic replay of semantic random streams or a canonical serial chain is useful, but it is not pathwise partition invariance. Smoothing must preserve raw coefficients, raw spectra, smoothing kernels, smoothed coefficients, smoothed spectra, and hashes of each stage; it is not part of $S_r$ unless explicitly declared.

**Derived regulator realization.** Choose a finite UV cellulation at scale $\ell_{\mathrm{UV}}$ and let $R$ be a finite union of cells. Hilbertize each cell's finite local data to a finite-dimensional space $\tilde{\mathcal{H}}_i \cong \mathbb{C}^{n_i}$. The extended algebra before quotienting by overlap redundancy is

$$\widetilde{\mathcal{A}}(R) = \mathcal{B}(\tilde{\mathcal{H}}_R), \qquad \tilde{\mathcal{H}}_R = \bigotimes_{i \subset R} \tilde{\mathcal{H}}_i.$$

Overlap-preserving changes of local trivialization act only on cut data. On each finite-dimensional chart, their unitary image has compact closure, so one may represent the boundary gluing redundancy by a compact group $G_{\partial R} \subset U(\tilde{\mathcal{H}}_R)$. The lifted boundary-invariant endomorphism algebra is

$$A_{\mathrm{inv}}(R) = \widetilde{\mathcal{A}}(R)^{G_{\partial R}} = \mathcal{B}(\tilde{\mathcal{H}}_R)^{G_{\partial R}}.$$

This is the fixed-cutoff realized presentation used throughout the collar analysis. The EC theorem works on the invariant-state realization and uses the sector-preserving induced collar algebra there; the entire fixed-point algebra on the unreduced tensor product is outside that theorem surface. When the consensus paper speaks of quotient repair, the physical repair law is defined on this fixed-point / overlap-invariant data and any representative-level map is only a lift of that quotient update. Descent to the gauge quotient is therefore built into the physical-algebra formulation as quotient-level repair: the local map is $\mathrm{locRep}_\lambda : Q \to Q$, and the global map is the finite quotient normal-form operator

$$\mathrm{Rep}_\lambda = \overline{\mathrm{nf}}_\lambda : Q \to Q.$$

On the declared fixed-cutoff collar branch, the local repair step itself is read from exact Markov splice or a declared Petz/Fawzi–Renner recovery channel. The declared fixed-cutoff branch adds repair completeness, together with control on the declared Petz domain where that branch is used. The consensus paper proves those clauses on a nontrivial rooted-tree packet-net export: finite packet labels live on a rooted tree, hidden labels are quotient-only, weighted parent-copy repair strictly lowers $\Phi$, normal forms are exactly consistent packet assignments, and the classical full-support Petz channel is CPTP and trace-norm contractive on its positive-support-gap domain. Gauge-invariant observables factor through the quotient normal form on that verified branch. The declared touched-overlap acceptance contract makes $\Phi$ the finite-patch Lyapunov functional for primitive repair proposals on the scalar branch, but a proposal becomes a physical step only through the transactional acceptance layer: snapshot-current read sets, unchanged-outside-write validation, preserved boundary/sector/holonomy data, and exact well-founded descent. The read set is validation-complete: it contains every support of every measure term that can change under the write. Overlapping primitive proposals are aggregated by connected conflict component; the restriction-compatible union-collar glue supplies the canonical aggregate payload, while primitive members of that component do not commit independently. On the same fixed-cutoff quantum lift, the terminal expectation functional on each declared physical observable algebra is therefore unique even when microscopic representative lifts differ by gauge labels globally or by sector labels inside one quotient-local glued state. The unqualified coding statement at this stage is only a finite constraint-code statement: the overlap-net codewords are the globally consistent states $C = \Phi^{-1}(0)$. A bare graph does not determine code distance: the same graph can carry constant readouts with distance 1 or repetition constraints with distance $|V|$. Topological-code distance/min-cut, Knill–Laflamme resilience, spectral-gap convergence, BFT wall-clock liveness, and hardware search-work reduction therefore require their own certificates and are not imported by the core overlap graph. Exact descent gives termination. Confluence follows from an independently checked conflict-component diamond on the physical quotient, protected-support and protected-conflict completeness, and repair completeness. The theorem therefore gives one schedule-independent terminal quotient normal form from a fixed initial quotient state. A stronger same-boundary conclusion requires a preserved boundary/sector map and at most one consistent quotient extension in that boundary fiber. The layered functional carrier in the consensus paper proves the finite multi-edge, multi-step witness for $H_B \wedge H_{\mathrm{fib}}$; the functional selected-fiber theorem gives the rooted obstruction-check form by constructing the single candidate from the boundary/root data and then checking all non-tree, sector, and holonomy obstructions. On that branch, multiple same-boundary candidate interiors can be present, inconsistent candidates are eliminated, and all surviving candidates share one quotient normal form. Normal-form hashes are public equality receipts after quotienting, not selectors among physically distinct minimizing endpoints. If accepted schedules terminate at different observer-facing quotient normal forms without a declared holonomy or higher-gauge obstruction, that repair law is outside the OPH consensus theorem. The neutral result of Ref. [1] packages this quantifier split exactly: under observation preservation and completeness, agreement modulo a silent equivalence of normal endpoints reached from all same-boundary sources is equivalent to injectivity of the induced boundary map on the consistent quotient. This cross-source criterion, same-source confluence, weak normalization, and all-schedule liveness are separate obligations. The named layered and functional carriers discharge the cross-source premise only on their stated domains. For separated cofinal refinement systems, the consensus paper adds an inverse-limit bridge: if finite-stage restriction maps commute with the quotient normal-form maps and holonomy maps, and if compatible families are visibly separated on cofinal finite stages, then finite normal forms and holonomy obstructions assemble into unique refinement-limit classes with finite-stage witnesses for nonzero holonomy. It also adds the RG-facing comparison: when a chosen coarse-graining channel shadows the finite-stage normal forms and obstruction maps with declared defects $\varepsilon^n$ and $\varepsilon^h$, reconciling first and then coarse-graining gives the same macroscopic law readout as coarse-graining first and then reconciling, up to $\max\{\varepsilon^n, \varepsilon^h\}$; exact naturality is the zero-defect case. Exact normal-form naturality is discharged by a repair-morphism witness, and holonomy naturality by the corresponding cochain map. For distributed implementations, the same paper proves that a worker run presents one finite universe only when it starts from one global carrier and each event projects to a legal monolithic repair path, a physical stutter, or a certified rollback to an earlier committed projection. The required certificate names the global graph, initial state, partition, cut interfaces, observer registry, run/config/code hashes, linearized commits, restart roots, final monolithic normal form, and recomputed readout. Executor order is separate from observer history. Observer-clock naturality additionally requires a history-augmented quotient with semantic event keys independent of worker ID, repair iteration, queue position, timestamp, and retry metadata; a global observer registry descended from local registry groupoids with disjoint patch/cap/future namespaces and explicit lineage arrows; a state-preserving observer-algebra extraction functor from terminal augmented normal forms; a support-visible cap chart map natural with that extraction; and an operational clock instrument whose readout is calibrated up to the declared affine reparameterization and residual bound. Until those objects or proof-producing finite certificates are supplied, modular-time/readback naturality is a named gate, not a consequence of implementation bookkeeping. The same quotient boundary applies to neutral-bulk readout. For each shard $s$, raw record rows $\Sigma_s$ must first be quotient by the declared presentation groupoid $\Gamma_s$ and then sent through the terminal normal-form map $n_s$, producing

$$X_s = n_s(\Sigma_s/\Gamma_s).$$

Interface maps $\tau_{ts} : U_{st} \to U_{ts}$ between these terminal charts must satisfy the identity, inverse, cocycle, and zero-cycle-holonomy checks before rows from different shards are compared. A global neutral readout is the quotient

$$Q_{\text{vis}} = \left(\bigsqcup_s X_s\right) / \sim_\tau,$$

not the concatenation of shard-local tables. Channel features $F_{s,c}$ descend to $Q_{\text{vis}}$ only when their domains and values are transported by the same atlas. For a complete declared channel set $\mathcal{C}_\star$, the neutral distance is

$$d_{\text{neu}}(x, y) = \left(\sum_{c\in\mathcal{C}_\star} w_c\, d_c(F_c(x), F_c(y))^p\right)^{1/p}.$$

This is always stated first as a quotient-visible pseudometric. It becomes a metric only after quotienting zero-distance feature-collision classes or proving that the channels jointly separate points. Pairwise available-channel comparison is explicitly excluded: missingness is handled only by complete cases, a fixed quotient-visible missing symbol, or a train-only imputation map labelled as an imputed-representation metric. Presentation changes (gauge representatives, port relabelings, observer order, schedules, and shard partitions) preserve the metric only when they induce a bijection of $Q_{\text{vis}}$ together with channel isometries; refinement claims additionally require a cofinally vanishing distance tail modulus. Finite Euclidean claims require the double-centered Gram certificate

$$B = -\frac{1}{2}H(D^{\circ 2})H \succeq 0$$

and noisy runs must report negative spectral mass, rank/effective-rank data, held-out stress, and planted/shuffled controls. Statistical certificates split independent generative shard batches before preprocessing; chart alignment, scaling, imputation, weights, graph construction, dimension choice, and thresholds are train/validation objects, and a test batch, seed, boundary condition, trajectory family, duplicate, or descendant appearing in more than one split blocks the claim. These theorems establish at most a common finite quotient metric or pseudometric. Identifying it with a physical Riemannian bulk or with Lorentzian/Einstein spacetime (via the conditional event-manifold packet of Theorem 4.3e, under its population, chart, cone, and reachability receipts) requires the separate modular/geometric branch below. The same consensus surface is classified explicitly as a finite-state exact theorem package plus this controlled inverse-limit bridge: decidable normal-form computation with the Lyapunov step bound, automatic approximate stability only through the collar-local splice and record controls, a conditional fair-block contraction branch for long-run noisy approximate consensus, and a verified rooted-tree packet-net subdomain. Computational universality for growing patch-net families belongs to the consensus paper's expressive-power boundary and is not a dependency for the recovered-core branches.

**Fixed-cutoff packet closure map and invariant simplex.** On any declared fixed-cutoff packet quotient $Q$ whose repair relation is terminating and confluent, the normal-form map

$$N : Q \to Q_{\mathrm{nf}}$$

is well-defined and schedule-independent. It induces an affine OPH closure map on the probability simplex over packet states,

$$\mathcal{C}_Q : \Delta(Q) \to \Delta(Q), \qquad \mathcal{C}_Q\left(\sum_{q\in Q} p_q \delta_q\right) = \sum_{q\in Q} p_q \delta_{N(q)}.$$

Since $Q$ is finite, $\Delta(Q)$ is a nonempty compact convex set and $\mathcal{C}_Q$ is a continuous affine self-map. Its image $\Delta(Q_{\mathrm{nf}})$ is invariant, and $\mathcal{C}_Q^2 = \mathcal{C}_Q$. This is an actual closure map on the fixed-cutoff packet-closed quotient branch. It is not the full Appendix/habitat closure-map theorem for arbitrary OPH state-and-law data; the first operation not internalized at that level is the general habitat lift from this finite packet quotient to the full compact-convex observer-supporting sector.

**Finite quotient ensemble theorem surface.** The closure map above is an idempotent normal-form projector. It sends any input law to terminal quotient normal forms, and every law supported on those normal forms is a fixed point. Hence the physical probability law at regulator $r$ must be supplied as an additional quotient-intrinsic object. Let

$$Q_r = \Sigma_r / \Gamma_r$$

be the finite physical quotient, or let $N_r = n_r(Q_r)$ be the quotient-normal-form subset when only settled configurations are intended to carry probability. The declared OPH quotient ensemble is specified by an intrinsic base weight and action

$$m_r : Q_r \to \mathbb{R}_{>0}, \qquad S_r : Q_r \to \mathbb{R} \cup \{+\infty\},$$

and

$$w_r(q) = m_r(q) e^{-S_r(q)}, \qquad Z_r = \sum_{q\in Q_r} w_r(q), \qquad \mu_r(q) = Z_r^{-1} w_r(q).$$

Equivalently, one may state an intrinsic projective prior $\nu_r$ on $Q_r$ and set $\mu_r = (n_r)_{\#}\nu_r$. The choice of $m_r$, $S_r$, or $\nu_r$ is load-bearing: uniform quotient counting, uniform representative counting pushed to the quotient, finite-groupoid weights, and trace weights of a declared quotient algebra are generally different.

**Finite MaxEnt form.** For a finite $Q$, positive $m$, and quotient observables $F_1, \ldots, F_k$, maximizing

$$\mathcal{H}_m(\nu) = -\sum_q \nu(q) \log \frac{\nu(q)}{m(q)}$$

over the affine constraint surface

$$\sum_q \nu(q) = 1, \qquad \sum_q \nu(q) F_a(q) = c_a$$

has a unique full-support solution, when the feasible set has one, of the form

$$\mu(q) = \frac{m(q) \exp[-\sum_a \theta_a F_a(q)]}{Z(\theta)}.$$

This is the finite-dimensional Lagrange-multiplier argument with strict concavity of $\mathcal{H}_m$; boundary optima obey the same formula after restricting to their support. On a finite noncommutative quotient algebra the corresponding density operator with faithful reference state $\sigma_r$ is

$$\rho_r = \frac{\exp(\log \sigma_r - \sum_a \theta_a F_{r,a})}{\operatorname{Tr} \exp(\log \sigma_r - \sum_a \theta_a F_{r,a})}.$$

**Refinement and projective compatibility.** For $s \succeq r$, let $c_{sr} : Q_s \to Q_r$ be the physical coarse-graining map. Exact compatibility of the weighted ensembles is equivalent to the fiber-sum identity

$$\sum_{q' : c_{sr}(q')=q} m_s(q') e^{-S_s(q')} = \alpha_{sr} m_r(q) e^{-S_r(q)}$$

for a constant $\alpha_{sr} > 0$ independent of $q$; necessarily $\alpha_{sr} = Z_s/Z_r$. The proof is just the equality

$$(c_{sr})_{\#}\mu_s(q) = Z_s^{-1} \sum_{q' : c_{sr}(q')=q} m_s(q') e^{-S_s(q')}$$

compared with $Z_r^{-1} m_r(q) e^{-S_r(q)}$. For the selected-prior route, if

$$c_{sr} \circ n_s = n_r \circ c_{sr}, \qquad (c_{sr})_{\#}\nu_s = \nu_r,$$

then functoriality gives $(c_{sr})_{\#}(n_s)_{\#}\nu_s = (n_r)_{\#}\nu_r$. Along a chain, finite-stage compatibility defines a unique projective-limit probability measure on the cylinder $\sigma$-algebra. If the one-step defects are

$$\delta_{k+1,k} = \|(c_{k+1,k})_{\#}\mu_{k+1} - \mu_k\|_{\mathrm{TV}},$$

then total-variation contraction gives

$$\|(c_{nr})_{\#}\mu_n - \mu_r\|_{\mathrm{TV}} \le \sum_{k=r}^{n-1} \delta_{k+1,k}.$$

**Implementation invariance and representative lifting.** If two implementations $A, B$ carry quotient bijections $h_r : Q_r^A \to Q_r^B$ with

$$m_r^B(h_r q) = m_r^A(q), \qquad S_r^B(h_r q) = S_r^A(q), \qquad h_r \circ c_{sr}^A = c_{sr}^B \circ h_s,$$

then $(h_r)_{\#}\mu_r^A = \mu_r^B$. If an implementation stores representatives $\pi_r : \Sigma_r \to Q_r$, a representative-level law must be a conditional lift

$$\widetilde{\mu}_r(x) = \mu_r(\pi_r x)\kappa_r(x \mid \pi_r x), \qquad \sum_{x:\pi_r(x)=q} \kappa_r(x \mid q) = 1.$$

Then $(\pi_r)_{\#}\widetilde{\mu}_r = \mu_r$. Uniform representative sampling weights quotient states by orbit size and is physical only if representative counting is the declared base measure. A representative Markov kernel must also be quotient-lumpable: the transition probability to an orbit cannot depend on which representative of the source orbit is used.

**Sampler correctness.** A production sampler is a separate interface from a deterministic settler:

$$\texttt{settle}(q) \to n_r(q), \qquad \texttt{sample}(\mu_r) \to q.$$

For $w(q) = m(q)e^{-S(q)}$ and a proposal kernel $R(q, q')$ with reciprocal support, the Metropolis–Hastings acceptance rule

$$a(q, q') = \min\left\{1, \frac{w(q')R(q', q)}{w(q)R(q, q')}\right\}$$

gives detailed balance

$$\mu(q)R(q, q')a(q, q') = \mu(q')R(q', q)a(q', q),$$

and hence $\mu P = \mu$. Irreducibility and aperiodicity give uniqueness and convergence on the finite state space. If a repair-informed proposal is $q' = R_{\mathrm{repair}}(q) + \eta$ with density $g$, the Hastings ratio must include

$$\frac{g(q - R_{\mathrm{repair}}(q'))}{g(q' - R_{\mathrm{repair}}(q))};$$

dropping this term generally changes the stationary law. For a lazy reversible finite chain with spectral gap $\gamma$,

$$\|P^n(x, \cdot) - \mu\|_{\mathrm{TV}} \le \frac{1}{2}\sqrt{\mu(x)^{-1} - 1}\,(1 - \gamma)^n,$$

and the usual autocorrelation bound follows from the nontrivial spectral radius. Exact small regulators should therefore enumerate $Q_r$, $w_r$, $P_r$, detailed-balance error, stationarity error, refinement residuals, lumpability/orbit-size checks, and the exact spectral gap.

**Gaussian screen continuation.** For a real reduced screen field space $V_r$ after background and dipole modes are removed, let $K_r$ be positive definite and impose fixed expected quadratic release energy,

$$\mathbb{E}[q] = 0, \qquad \mathbb{E}[q^{\mathsf{T}} K_r q] = d_r A_r, \qquad d_r = \dim V_r.$$

The unique maximum-entropy density relative to the declared Euclidean volume is

$$p_r(q) = \frac{\sqrt{\det K_r}}{(2\pi A_r)^{d_r/2}} \exp\left[-\frac{1}{2A_r} q^{\mathsf{T}} K_r q\right], \qquad \mathrm{Cov}(q) = A_r K_r^{-1}.$$

This is the relative-entropy proof against the displayed Gaussian. If instead every sample obeys $q^{\mathsf{T}} K_r q = d_r A_r$, the MaxEnt law is uniform on the ellipsoid, not Gaussian. Under a linear coarse field map $C_{sr} : V_s \to V_r$, exact Gaussian refinement is the covariance criterion

$$A_r K_r^{-1} = C_{sr}(A_s K_s^{-1})C_{sr}^{\mathsf{T}}.$$

For harmonic truncation, choose real modes $Y_{\ell m}$, $2 \le \ell \le L$, set

$$K_L Y_{\ell m} = \kappa_\ell Y_{\ell m}, \qquad \kappa_\ell = [\ell(\ell + 1)]^{1+\theta/2},$$

and sample independent coefficients

$$a_{\ell m} \sim \mathcal{N}(0, A/\kappa_\ell).$$

Projection from $L'$ to $L$ then discards high modes and is exactly refinement-compatible. For irregular meshes, covariance should be primary under coarse-graining:

$$\Sigma_r = C_{sr}\Sigma_s C_{sr}^{\mathsf{T}}, \qquad K_r = A_r \Sigma_r^{-1}.$$

Equivalently, when eliminating fine variables from a precision matrix

$$K_s = \begin{pmatrix} K_{cc} & K_{cf} \\ K_{fc} & K_{ff} \end{pmatrix},$$

the coarse precision is the Schur complement

$$K_r = K_{cc} - K_{cf} K_{ff}^{-1} K_{fc},$$

not the principal block $K_{cc}$.

**Vacuum and primordial promotion gates.** A stationary sampler is not a physical vacuum: the identity transition kernel leaves every measure stationary. A finite vacuum theorem needs a physical transfer operator. If $T_r$ is real symmetric, positive definite, and strictly positive, Perron–Frobenius gives a simple largest eigenvalue $\lambda_{0,r}$ with positive eigenvector $\Omega_r$. Then

$$H_r = -a_t^{-1} \log(T_r/\lambda_{0,r})$$

is positive semidefinite, $\Omega_r$ is the unique zero-energy state, and the Doob kernel

$$P_r(x, y) = \frac{T_r(x, y)\Omega_r(y)}{\lambda_{0,r}\Omega_r(x)}$$

is stochastic and detailed-balanced with

$$\mu_r(x) = \frac{\Omega_r(x)^2}{\sum_z \Omega_r(z)^2}.$$

Continuum promotion further requires reflection positivity or an equivalent reconstruction certificate plus refinement compatibility of the transfer family. A screen covariance alone does not determine an unrestricted primordial curvature spectrum. Even the complete noiseless one-shell sequence for the thin-shell map

$$C_\ell = 4\pi \int_0^\infty \frac{dk}{k} \Delta_\zeta^2(k) j_\ell^2(k\chi_\star)$$

determines the three-dimensional correlation only on $0 \le s \le 2\chi_\star$ and has an infinite-dimensional kernel that persists under positivity. Source-derived uniqueness requires a physical dilation-intertwiner theorem or complete radial cross-covariance tomography. A finite prior gives a conditional continuation. Conventional free-scalar and lattice-gauge baselines may therefore be recorded, but those baselines carry closed OPH-native-vacuum and primordial-promotion receipts unless these gates are supplied.

# 5 Finite Screen Spectrum Theorem Package

This fragment owns the screen-level spectrum theorem used by the staged cosmology branch. It does not own TT, TE, EE, lensing, likelihoods, or physical CMB promotion. Those remain downstream Boltzmann and data-contract gates.

**Definition 5.1** (finite screen regulator)**.** *For regulator $r$, a finite screen regulator is*

$$\mathfrak{S}_r = (\mathcal{T}_r, M_r, \Gamma_r, \mathcal{Q}_r, C_{sr}, J_{rs}),$$

*where $\mathcal{T}_r$ is a finite cellulation of the screen, $M_r \succ 0$ is the area/quadrature mass matrix, $\Gamma_r$ is the hidden-presentation groupoid, $\mathcal{Q}_r = \Sigma_r/\Gamma_r$ is the physical quotient, and $C_{sr}$, $J_{rs}$ are the coarse-graining and interpolation maps. Shape regularity and quadrature convergence require*

$$f^\mathsf{T} M_r g \to \int_{S^2} f g \, d\Omega$$

*with an explicit band-limited error bound*

$$\left| f^{\mathsf{T}} M_r g - \int f g \, d\Omega \right| \le \varepsilon_{M,r}(L) \|f\|_{H^s} \|g\|_{H^s}.$$

*Patch count alone is therefore not an angular-resolution certificate.*

**Definition 5.2** (geometric screen scalar). *Let the quotient-visible collar-volume readout be*

$$J_{X,r}(x) = \lambda_r(x) \sqrt{\det \sigma_{AB,r}(x)} > 0,$$

*and let $\bar{J}_{X,r}$ be emitted by an independently defined homogeneous or frozen-background branch, not by a CMB fit. Define*

$$q_{0,r} = \frac{1}{3} \log \frac{J_{X,r}}{\bar{J}_{X,r}}.$$

*On a certified spherical chart, with $B_r = [1, n_x, n_y, n_z]$,*

$$\Pi_{<2,r} = B_r (B_r^{\mathsf{T}} M_r B_r)^{-1} B_r^{\mathsf{T}} M_r, \qquad q_r = (I - \Pi_{<2,r}) q_{0,r}.$$

*On an irregular screen, $B_r$ is replaced by the certified generalized eigenprojector onto the finite $\ell = 0, 1$ subspace. Hard-coded feature z-scores or observer-summary weights are diagnostic proxies, not $q_r$.*

**Proposition 5.3** (quotient invariance). *If a recharting $U$ preserves the screen inner product and geometric readout,*

$$U^{\mathsf{T}} M_r' U = M_r, \qquad q_{0,r}' = U q_{0,r}, \qquad B_r' = U B_r,$$

*then $\Pi_{\ge 2,r}' U = U \Pi_{\ge 2,r}$, and $q_r' = U q_r$.*

*Proof.* Substitution gives

$$\begin{aligned} \Pi_{<2,r}' U &= U B_r (B_r^{\mathsf{T}} U^{\mathsf{T}} M_r' U B_r)^{-1} B_r^{\mathsf{T}} U^{\mathsf{T}} M_r' U \\ &= U B_r (B_r^{\mathsf{T}} M_r B_r)^{-1} B_r^{\mathsf{T}} M_r = U \Pi_{<2,r}. \end{aligned}$$

Subtracting from $U$ gives the high-mode identity. □

**Definition 5.4** (physical scalar precision and action). *The scalar precision $K_r$ must have a physical origin on $V_r = \operatorname{im} \Pi_{\ge 2,r}$. Allowed branches are:*

$$K_r^{\mathrm{Hess}} = \nabla^2 \Phi_r \Big|_{q=0},$$

*for a quotient-visible mismatch or release free energy; a repair Dirichlet form*

$$K_r^{\mathrm{diss}} = H_r - T_r^{\mathsf{T}} H_r T_r, \qquad T_r^{\mathsf{T}} H_r T_r \preceq H_r;$$

*or the reversible generator form*

$$K_r^{\mathrm{Dir}} = -\frac{1}{2} (L_r + L_r^{\dagger_H}).$$

*A raw nonsymmetric repair matrix or caller-supplied $\kappa$ is not a precision operator. The absolute normalization must be fixed independently, for example by*

$$K_0 Y_{\ell m} = \ell(\ell+1) Y_{\ell m}$$

*on the $\theta = 0$ branch. Once normalized,*

$$S_{\mathrm{scr},r}[q] = \frac{1}{2A_{q,r}} \langle q, K_r q \rangle_r = \frac{1}{2A_{q,r}} q^{\mathsf{T}} M_r K_r q.$$

**Theorem 5.5** (finite MaxEnt screen covariance)**.** *Let $K_r e_i = \kappa_i e_i$ with an $M_r$-orthonormal basis of $V_r$, $\kappa_i > 0$, and $d_r = \dim V_r$. Among continuous densities in coefficients $q = \sum_i a_i e_i$ with $\mathbb{E}[a_i] = 0$ and*

$$\frac{1}{2}\mathbb{E}\langle q, K_r q\rangle_r = E^{\rm src}_{q,r},$$

*the unique entropy maximizer is*

$$d\mu_r(q) = Z_r^{-1} \exp\left[-\frac{1}{2A_{q,r}}\langle q, K_r q\rangle_r\right] dq, \qquad A_{q,r} = \frac{2E^{\rm src}_{q,r}}{d_r},$$

*and*

$$\mathbb{E}[a_i a_j] = \delta_{ij}\frac{A_{q,r}}{\kappa_i}.$$

*Proof.* The Euler–Lagrange equation for entropy with normalization and quadratic-energy constraints gives a Gaussian density with inverse temperature $\beta_r$. The expected energy is

$$E^{\rm src}_{q,r} = \frac{1}{2}\sum_i \kappa_i \frac{1}{\beta_r \kappa_i} = \frac{d_r}{2\beta_r}.$$

Thus $A_{q,r} = \beta_r^{-1} = 2E^{\rm src}_{q,r}/d_r$. Strict concavity of entropy gives uniqueness. □

**Remark 5.6** (discrete quotient branch)**.** *For a discrete finite quotient with intrinsic base weight $m_r(q)$, the exact Gibbs law is*

$$\mu_r(q) = Z_r^{-1} m_r(q) e^{-\beta_r E_r(q)}.$$

*A Gaussian statement on that branch requires a separate Laplace or central-limit theorem with controlled higher-order terms.*

**Definition 5.7** (source release energy)**.** *Let $m_r^0$ be the tracially pointed base weight on the finite physical collar quotient. Let $F_{r,a}$ be a finite ledger of primitive collar observables: scalar occupancy, protected-center presence, released volume, conserved source charges, and the release clock. The ledger excludes screen energy, sky spectra, CMB residuals, likelihoods, fitted amplitudes, and measurement-calibrated proxies. For source-emitted constraint values $c^{\rm col}_{r,a}$, the selected release law is*

$$\nu_r^{\rm rel}(x) = \frac{m_r^0(x)\exp[-\sum_a \lambda_{r,a} F_{r,a}(x)]}{Z_r}, \qquad \mathbb{E}_{\nu_r^{\rm rel}} F_{r,a} = c^{\rm col}_{r,a}.$$

*Strict concavity of relative entropy gives uniqueness on full support. A boundary optimum uses the same form on its certified support. The constraint ledger, base weight, source DAG, multipliers, support, and refinement maps belong to the receipt.*

*For this independently selected quotient ensemble on released collar normal forms, define*

$$E^{\rm src}_{q,r} = \frac{1}{2}\int \langle q_r(x), K_r q_r(x)\rangle_r \, d\nu_r^{\rm rel}(x).$$

*The receipt must expose $\nu_r^{\rm rel}$, the base measure, $K_r$, $d_r$, $E^{\rm src}_{q,r}$, $A_{q,r}$, no-observation ancestry, and the same operator normalization used in the action. MaxEnt alone does not determine amplitude.*

**Proposition 5.8** (microscopic-law necessity)**.** *For fixed positive $K_r$, the Gaussian family $Z(A)^{-1}\exp[-\langle q, K_r q\rangle/(2A)]$ exists for every $A > 0$. The quadratic MaxEnt form therefore leaves one positive scale free. The source release law in Definition 5.7, or an equivalent source-only vacuum law, supplies the energy that fixes $A_{q,r} = 2E^{\rm src}_{q,r}/d_r$.*

**Theorem 5.9** (rotational screen spectrum)**.** *If the continuum precision $K_\theta$ commutes with the $SO(3)$ action on scalar functions, then by Schur's lemma each $\mathcal{H}_\ell$ is an eigenspace. For the exact conformal-shell family,*

$$\Lambda_\ell(\theta) = \frac{\Gamma(\ell + 2 + \theta/2)}{\Gamma(\ell - \theta/2)}, \qquad \ell \geq 2.$$

*Then $\Lambda_\ell(0) = \ell(\ell+1)$, and the MaxEnt covariance gives*

$$C_\ell^q = A_q \frac{\Gamma(\ell - \theta/2)}{\Gamma(\ell + 2 + \theta/2)}.$$

*Consequently $\mathcal{D}_\ell^q = \ell(\ell+1)C_\ell^q/(2\pi) \sim (A_q/2\pi)\ell^{-\theta}$.*

**Definition 5.10** (source-derived conformal precision)**.** *Let $L_r \succeq 0$ be the normalized detailed-balanced scalar collar Dirichlet operator on $V_r$, with continuum limit $-\Delta_{S^2}$. Set*

$$B_r = (L_r + \tfrac{1}{4}I)^{1/2}, \qquad K_{\theta,r} = \frac{\Gamma(B_r + \frac{3}{2} + \frac{\theta}{2})}{\Gamma(B_r - \frac{1}{2} - \frac{\theta}{2})}.$$

*For $-2 < \theta < 4$ on the retained $\ell \geq 2$ branch, this operator is positive. The gamma recurrence gives $K_{0,r} = L_r$ exactly. The finite receipt carries detailed balance, positivity, the $\ell(\ell+1)$ normalization, anisotropy splitting, and refinement residuals.*

**Theorem 5.11** (finite refinement error)**.** *Suppose that for $2 \leq \ell \leq L$*

$$\left|\frac{\lambda_{\ell m,r}}{\Lambda_\ell(\theta)} - 1\right| \leq \varepsilon_{K,r}(L), \qquad \left|\frac{A_{q,r}}{A_q} - 1\right| \leq \varepsilon_{A,r}, \qquad \varepsilon_{K,r} < 1.$$

*Then $C_{\ell m,r}^q = A_{q,r}/\lambda_{\ell m,r}$ satisfies*

$$\left|\frac{C_{\ell m,r}^q}{A_q/\Lambda_\ell(\theta)} - 1\right| \leq \frac{\varepsilon_{A,r} + \varepsilon_{K,r}}{1 - \varepsilon_{K,r}}.$$

*Proof.* Write $A_{q,r} = A_q(1 + a_r)$ and $\lambda_{\ell m,r} = \Lambda_\ell(1 + b_{\ell m,r})$. Then

$$\frac{C_{\ell m,r}^q}{A_q/\Lambda_\ell} = \frac{1 + a_r}{1 + b_{\ell m,r}},$$

and the displayed bound follows from $|a_r| \leq \varepsilon_A$, $|b_{\ell m,r}| \leq \varepsilon_K$. □

**Theorem 5.12** (refinement-semigroup tilt)**.** *Let $R_b$ be the scalar refinement map for scale ratio $b > 1$. If one isolated scalar covariance mode has positive eigenvalue $\lambda(b)$, $\lambda(b_1 b_2) = \lambda(b_1)\lambda(b_2)$, and $\lambda$ is continuous, then there is a unique real $\theta$ with*

$$\lambda(b) = b^{-\theta}, \qquad \theta = -\frac{\log \lambda(b)}{\log b}.$$

*Proof.* Set $g(t) = -\log \lambda(e^t)$. The semigroup law gives $g(t + s) = g(t) + g(s)$. Continuity makes $g(t) = \theta t$, hence the result. □

**Definition 5.13** (edge-center reserve generator receipt)**.** *Write* $s = \log b$ *for logarithmic refinement thickness. Let* $u_{\text{full}}(s)$ *be the scalar-conditioned covariance-survival cocycle across a full oriented collar, and let* $u_q(s)$ *be its source-facing half-collar restriction. The receipt requires*

$$u(s+t) = u(s)u(t), \qquad u(0) = 1,$$

*strong continuity at zero, the source-derived full-collar density*

$$-u'_{\text{full}}(0) = \frac{P_\star}{24},$$

*and the orientation-reversal coarea identity*

$$-u'_q(0) = \frac{1}{2}[-u'_{\text{full}}(0)] = \frac{P_\star}{48}.$$

*These are infinitesimal generator statements. A one-step survival probability has exponent* $-\log u_q(\log b)/\log b$.

**Theorem 5.14** (edge-center tilt and repair-clock reconciliation)**.** *Under Definition 5.13,*

$$u_q(s) = e^{-\theta s}, \qquad \theta = \frac{P_\star}{48}, \qquad n_s = 1 - \frac{P_\star}{48}.$$

*In the coordinate* $\theta = \kappa_{\text{rep}}(P_\star - \varphi)$*, the same branch has*

$$\kappa_{\text{rep}}^{\text{edge}} = \frac{P_\star}{48(P_\star - \varphi)}.$$

*The value* $e$ *is a separate diagnostic hypothesis unless an additional identity equates it with this source-derived coordinate.*

*Proof.* For $g(s) = -\log u_q(s)$, the cocycle law gives the continuous Cauchy equation. Hence $g(s) = \theta s$. Differentiation at zero and the half-collar identity give $\theta = P_\star/48$. The formula for $\kappa_{\text{rep}}^{\text{edge}}$ is algebraic. ☐

**Theorem 5.15** (homogeneous radial source family)**.** *Assume source-stress closure, a single clock, freezeout, multicenter consistency, and translation/rotation covariance. Let* $C_\zeta = M_{\Delta^2_\zeta}$ *on the common physical* $d\ln k$ *mode basis. If a scale-natural source embedding transports refinement to physical dilation and its finite operator residual converges to*

$$D_s^{-1} C_\zeta D_s = e^{-\theta s} C_\zeta, \qquad (D_s f)(k, \hat{k}) = f(e^{-s}k, \hat{k}),$$

*then every positive measurable representative satisfies*

$$\Delta^2_\zeta(k) = A_\zeta (k/k_\star)^{-\theta}$$

*almost everywhere. Continuity gives the pointwise identity. A single shell has an infinite-dimensional radial kernel, including positive ambiguities; complete radial cross-covariances give the independent spherical-Hankel tomography route.*

*Proof.* Conjugation of the multiplication operator gives $\Delta^2_\zeta(e^s k) = e^{-\theta s}\Delta^2_\zeta(k)$ almost everywhere. In logarithmic coordinates, rational-translation invariance on a common full-measure set forces $e^{\theta t}\Delta^2_\zeta(e^t)$ to be constant almost everywhere. The one-shell and tomography statements follow from the correlation-restriction and spherical-Hankel theorems in the primordial bridge packet. ☐

**Theorem 5.16** (thin-shell power-law lift). *Assume* $q(\hat{n}) = Z_\star \Pi_{\ell\geq 2}\zeta_\star(R_\star \hat{n})$ *and*

$$\Delta_\zeta^2(k) = A_\zeta (k/k_\star)^{-\theta}.$$

*Then*

$$C_\ell^q = 4\pi Z_\star^2 \int d\ln k\, \Delta_\zeta^2(k) j_\ell^2(kR_\star),$$

*and, for* $-2 < \theta < 4$,

$$A_q = \pi^{3/2} Z_\star^2 A_\zeta (k_\star R_\star)^\theta \frac{\Gamma(1+\theta/2)}{\Gamma(3/2+\theta/2)}.$$

*Thus*

$$A_\zeta = \frac{A_q}{\pi^{3/2} Z_\star^2 (k_\star R_\star)^\theta} \frac{\Gamma(3/2+\theta/2)}{\Gamma(1+\theta/2)}.$$

**Proposition 5.17** (finite-window bound). *Let* $\Psi_\ell(k) = \int dr\, W(r) j_\ell(kr)$, $W \geq 0$, $\int W dr = 1$, *with mean* $R_\star$ *and variance* $\sigma_R^2$. *If*

$$\delta_\ell(k) = \frac{k^2\sigma_R^2}{2} \sup_{r\in \operatorname{supp} W} |j_\ell''(kr)|,$$

*then*

$$|C_{\ell,W}^q - C_{\ell,\mathrm{shell}}^q| \leq 4\pi Z_\star^2 \int d\ln k\, \Delta_\zeta^2(k) \left[2\delta_\ell(k) + \delta_\ell(k)^2\right].$$

**Proposition 5.18** (radial non-identifiability). *For a finite radial basis,* $C = Ap$. *If* $A \in \mathbb{R}^{N_\ell \times N_k}$ *has rank* $r$*, then* $\dim\ker A = N_k - r$*, and every* $p+v$ *with* $v \in \ker A$ *gives the same screen spectrum. Radial promotion therefore requires either a source theorem restricting* $p$*, or a declared prior with a published null basis, resolution kernels, positivity checks, and prior-sensitivity report.*

**Theorem 5.19** (conditional source-spectrum promotion). *One hash-locked source DAG may emit*

$$(q_r, S_{\mathrm{scr},r}, \theta, A_{q,r}, C_{\ell,r}^q, A_{\zeta,r}, \Delta_{\zeta,r}^2)$$

*as a conditional primordial packet only when the geometric-scalar, collar-law, release-energy, reserve-generator, conformal-precision, refinement, source-stress, single-clock, freezeout, adiabaticity, isocurvature, phase-coherence, thin-shell, radial-null, finite-window, and forward- residual receipts pass on that DAG. Any measurement, likelihood, fitted parameter, or measurement-calibrated ancestor fails the source packet. Physical temperature and polarization spectra require the independent Boltzmann, recombination, nuisance, covariance, and likelihood gates.*

**Target 5.20** (screen-spectrum receipt set). *A concrete source-derived spectrum requires the following receipt families before primordial promotion:*

1. *geometry, scalar quotient, low-mode projector, scalar precision, and operator normalization;*
2. *quotient ensemble selection, scalar release energy, and MaxEnt screen covariance;*
3. *the infinitesimal reserve generator, scalar refinement tilt, and angular screen spectrum with a finite error budget;*
4. *source-stress, clock, freezeout, radial-window, radial-null, forward-residual, and transfer firewall receipts.*

*The theorem and algorithm packet is complete. Construction of one finite source DAG that passes this receipt set is work in progress.*

### 5.0.1 Source-only primordial bridge theorem

The finite screen covariance theorem is a screen theorem. To read it as a source-only primordial curvature spectrum one must add the following bridge objects and receipts. The dependency chain is

$$\boxed{\begin{gathered}\mathrm{nf}(\mathcal{Q}_r) \to (\chi_r, u_r, h_r) \to \mathsf{BackgroundCurvatureStatus}_r \to (g, T_I) \to q_\zeta = \zeta_\rho \\ \to \text{single clock} \to \text{freeze-out} \to \text{coherent growing mode} \to \Delta^2_\zeta(k).\end{gathered}}$$

No successful fit of a screen $C_\ell$ or a TT spectrum may replace one of these hypotheses.

**Definition (background curvature branch label).** The primordial bridge carries a finite record

$$\mathsf{BackgroundCurvatureStatus}_r = (\mathsf{BranchType}, \kappa, I_K, I_{\Omega_K}, \mathsf{Basis}, \mathsf{Top})$$

where

$$\mathsf{BranchType} \in \{\textsc{FlatExact}, \textsc{FlatAssumed}, \textsc{OpenCurved}, \textsc{ClosedCurved}, \textsc{Unresolved}\}.$$

FlatExact is allowed only after a direct theorem or conditional CMH receipt. FlatAssumed may run flat kernels but its outputs remain assumption-conditioned. Unresolved blocks physical promotion. The $H^3$ observer-facing atlas is not this record; this record comes from the clocked FLRW slice and its spatial Levi–Civita holonomy or declared branch input.

**Definition (source stress readout).** On a reconstructed branch with relational metric $g_{ab}$, let

$$S_{\mathrm{src}} = \sum_I S_I[g, \Psi_I], \qquad T^I_{ab} = -\frac{2}{\sqrt{-g}}\frac{\delta S_I}{\delta g^{ab}}, \qquad T^{\mathrm{tot}}_{ab} = \sum_I T^I_{ab},$$

where $I$ ranges over every active sector, including ordinary matter, radiation, neutrinos, OPH anomaly or repair sectors, and any auxiliary source kept in the run. At finite regulator this is a quotient-visible map

$$\mathsf{StressRead}_r : \mathrm{nf}(\mathcal{Q}_r) \longrightarrow \{T^I_{r,ab}\}_I.$$

All source stress components must be read from the same global normal form as the collar scalar.

**Source-stress closure theorem.** Assume each $S_I$ is relationally diffeomorphism invariant, the source equations hold, sector exchanges satisfy

$$\nabla_a T^{ab}_I = Q^b_I, \qquad \sum_I Q^b_I = 0,$$

and $\mathsf{StressRead}_r$ descends to the physical quotient and converges. Then

$$\nabla_a T^{ab}_{\mathrm{tot}} = 0.$$

On the recovered Einstein branch,

$$G_{ab} + \Lambda g_{ab} = 8\pi G\, T^{\mathrm{tot}}_{ab}.$$

The evidence bundle must also report the non-circular comparison

$$E^{ab}_T = T^{ab}_{\mathrm{src}} - \frac{G^{ab} + \Lambda g^{ab}}{8\pi G};$$

using the geometric right-hand side alone is a geometry diagnostic, not a source-only primordial receipt.

**Total-energy frame and curvature covector.** When $T^a{}_b$ has a unique future timelike unit eigenvector $u^a$, define

$$T^a{}_b u^b = -\rho u^a, \qquad h_{ab} = g_{ab} + u_a u_b,$$

and decompose

$$T^{ab} = \rho u^a u^b + p h^{ab} + \pi^{ab}, \qquad u_a \pi^{ab} = 0, \qquad \pi^a{}_a = 0.$$

Let $\Theta = \nabla_a u^a$ and let $\sigma_{ab}$ be the shear. Total energy conservation gives

$$\dot\rho + \Theta(\rho + p) + \sigma_{ab}\pi^{ab} = 0, \qquad \dot f := u^a \nabla_a f.$$

On an expanding branch, define

$$p_{\text{eff}} = p + \frac{\sigma_{ab}\pi^{ab}}{\Theta};$$

then $\dot\rho + \Theta(\rho + p_{\text{eff}}) = 0$.

**Theorem (exact total-curvature evolution).** Let

$$\dot\alpha = \Theta/3, \qquad \zeta_a = \nabla_a \alpha - \frac{\dot\alpha}{\dot\rho}\nabla_a \rho,$$

and

$$\Gamma_a^{\text{eff}} = \nabla_a p_{\text{eff}} - \frac{\dot p_{\text{eff}}}{\dot\rho}\nabla_a \rho.$$

Then, exactly and without a long-wavelength approximation,

$$\mathcal{L}_u \zeta_a = -\frac{\Theta}{3(\rho + p_{\text{eff}})}\Gamma_a^{\text{eff}}.$$

Hence a barotropic effective source, $p_{\text{eff}} = p_{\text{eff}}(\rho)$, conserves $\zeta_a$.

**Collar scalar corollary.** For relational rods Lie-dragged by the total-energy flow, the collar-volume Jacobian satisfies

$$J_X = \lambda\sqrt{\det \sigma_{AB}}, \qquad \frac{d}{d\tau}\frac{1}{3}\log J_X = \frac{\Theta}{3}.$$

Relative to a frozen background,

$$q_\zeta = \frac{1}{3}\log\frac{J_X}{\bar J_X}.$$

On the uniform-total-density cut,

$$q_\zeta = \zeta_\rho + \text{constant}.$$

After removing the monopole and dipole,

$$q = \Pi_{\ell\geq 2} q_\zeta = \Pi_{\ell\geq 2}\zeta_\rho, \qquad Z_q = 1.$$

This is the source-only quotient scalar on the OPH screen; by itself it has type ScreenCurvature, not PrimordialCurvature.

**Definition (rank-one single-clock branch).** Let

$$\mathcal{F}(x) = (\rho, p_{\text{eff}}, \{\rho_I, p_I, Q_I\}_{I=1}^N)$$

be the scalar source map on the source moduli space. A branch is rank-one single-clock when $\operatorname{rank} d\mathcal{F} = 1$, $d\rho \neq 0$, every level set of $\rho$ is connected, all sector velocities agree with $u^a$, and orthogonal momentum transfer vanishes or is interval-bounded. At finite cutoff this requires a quotient-visible clock $\chi_r$ and a factorization

$$\mathcal{F}_r = \widehat{\mathcal{F}}_r \circ \chi_r.$$

**Theorem (single-clock adiabaticity).** On a rank-one single-clock branch,

$$p_{\mathrm{eff}} = p_{\mathrm{eff}}(\rho), \qquad S_a^{IJ} = 3(\zeta_a^I - \zeta_a^J) = 0,$$

and the intrinsic nonadiabatic pressure and transfer covectors obey

$$\Gamma_a^I = 0, \qquad \Xi_a^I = 0.$$

Therefore $\mathcal{L}_u \zeta_a = 0$. The proof is the chain rule through the single clock: every source scalar is a function of the same $\chi$, and connected $\rho$-fibers remove residual branch labels.

**Theorem (entropy repair gap).** Let the linearized scalar repair/evolution block be

$$\delta Y_{n+1} = L_r \delta Y_n + \eta_n,$$

and let $P_\perp$ project away gauge directions, constrained zero modes, and the adiabatic growing direction. If there is a positive metric $H_r$ such that

$$\|P_\perp L_r P_\perp\|_{H_r} \le e^{-\gamma_r} < 1,$$

then

$$\|P_\perp \delta Y_n\|_{H_r} \le e^{-n\gamma_r} \|P_\perp \delta Y_0\|_{H_r} + \sum_{j=0}^{n-1} e^{-(n-1-j)\gamma_r} \|P_\perp \eta_j\|_{H_r}.$$

In the unforced case the orthogonal entropy, decaying, and repair scalar modes vanish. Eigenvalues alone do not certify this theorem for nonnormal repair matrices; a norm bound or a discrete Lyapunov inequality

$$L_\perp^\top H L_\perp \preceq e^{-2\gamma} H$$

is required.

**Theorem (growing-mode coherence).** Let $X(k)$ collect the scalar initial-condition variables for curvature, radiation, baryons, CDM/anomaly, neutrinos, and repair-sector scalars, and let $v_+(k)$ be the normalized adiabatic growing vector. If the single-clock and repair-gap theorems hold through readout and no independent stochastic source acts afterward, then

$$X(k) = v_+(k)\zeta(k), \qquad \mathcal{P}_X(k) = \mathcal{P}_\zeta(k) v_+(k) v_+(k)^\dagger.$$

Thus rank $\mathcal{P}_X(k) = 1$, source-side isocurvature fractions vanish, and all species share a deterministic growing-mode phase. Finite runs must report $\lambda_2/\lambda_1$, $\beta_{\mathrm{iso}}(k)$, decaying-mode fraction, and phase-convention defects as source diagnostics.

**Theorem (screen-to-primordial bridge).** Assume the geometric record and causal-area metric receipts, the collar identity $q_\zeta = \zeta_\rho$, source-stress closure, rank-one normal form, positive repair gap, freeze-out, adiabatic growing-mode, isocurvature, and phase-coherence receipts all pass. Also assume the physical scale-bridge theorem: a source embedding with `source_embedding_hash`, a physical mode basis with `physical_mode_basis_id`, a calibrated comoving coordinate $\chi$, mode-normalization and density-of-states receipts, and a no-post-hoc-calibration receipt, and a

$\mathsf{BackgroundCurvatureStatus}_r$ that is not UNRESOLVED. Then the low-mode-removed OPH screen scalar is the source-only primordial curvature field restricted to the declared shell or radial window:

$$q(\hat{n}) = \Pi_{\ell_{\rm src}\geq 2} \int d\chi\, W(\chi)\zeta_\star(\chi\hat{n}).$$

For a thin shell, $q(\hat{n}) = \Pi_{\ell_{\rm src}\geq 2}\zeta_\star(R_\star\hat{n})$. The forward spectrum is exactly

$$C_\ell^q = 4\pi \int_0^\infty d\ln k\, \Delta_\zeta^2(k)|\Psi_\ell^{\mathsf{BranchType}}(k)|^2,$$

$$\Psi_\ell^{\mathsf{BranchType}}(k) = \begin{cases} \int dr\, W(r) j_\ell(kr), & \mathsf{BranchType} = \textsc{FlatExact} \text{ or } \textsc{FlatAssumed}, \\ \int dr\, W(r)\Phi_{\ell k}^{(-)}(r), & \mathsf{BranchType} = \textsc{OpenCurved}, \\ \int dr\, W(r)\Phi_{\ell k}^{(+)}(r), & \mathsf{BranchType} = \textsc{ClosedCurved}, \end{cases}$$

and for a thin shell

$$C_\ell^q = 4\pi \int_0^\infty d\ln k\, \Delta_\zeta^2(k) \left|\Psi_\ell^{\mathsf{BranchType}}(k; R_\star)\right|^2.$$

At finite regulator the continuum integral is represented by the finite spectral measure

$$d\nu_r(k) = \sum_j w_{rj}\delta(k - k_{rj}),$$

so the actual finite evidence receipt is

$$C^q_{\ell_{\rm src},r} = 4\pi \sum_j w_{rj}^{\log k}\, \Delta^2_{\zeta,r}(k_{rj})|\Psi^{\mathsf{BranchType}}_{\ell_{\rm src},r}(k_{rj})|^2.$$

Every radial node, window, and mode bin is bound to the geometry hash, source-embedding hash, scale-certificate hash, boundary-condition hash, and mode-basis identifier, and the evidence bundle must emit `RADIAL_KERNEL_GEOMETRY_COMPATIBILITY_RECEIPT`. The observed multipole $L_{\rm CMB}$ is not $\ell_{\rm src}$; it is the output index of the line-of-sight transfer solver. Cap eigenmodes, inverse cap-opening angles, and relabeled unit strings remain diagnostic by the dimensional, angular, cap-mode, and unit-string no-go propositions.

**Physical scale dependency.** The $k_{rj}$ grid in the preceding equation is not a free producer field. It is consumed only after

`PHYSICAL_SPATIAL_K_RECEIPT`

recomputes the comoving $k$-intervals from the physical geometry, scale certificate, scale factor, finite eigenvalues, normalization, and refinement residuals. The radial kernel and source-screen projection additionally require

`SCREEN_TO_PHYSICAL_K_ASSOCIATION_RECEIPT`.

The source covariance is a finite positive operator on the common physical mode projectors:

$$C_{\zeta,r} \succeq 0.$$

Homogeneity and isotropy on the safe band require

$$P_I C_{\zeta,r} P_I = \mathcal{P}_\zeta(k_I)P_I + E_I$$

with a declared residual bound, together with an off-diagonal leakage report

$$\sum_{I\neq J} |P_I C_{\zeta,r} P_J|^2.$$

The same physical projector family is used by primordial covariance and anomaly response:

$$\texttt{COMMON\_PRIMORDIAL\_ANOMALY\_MODE\_BASIS\_RECEIPT}.$$

Mode freezeout and common initial surface are separate gates:

$$\texttt{PHYSICAL\_MODE\_FREEZEOUT\_MAP\_RECEIPT}, \qquad \texttt{PHYSICAL\_FREEZEOUT\_SURFACE\_RECEIPT}.$$

The latter contains initial data and normal derivatives on one common spacelike surface.

**Radial prior, null space, and residual receipt.** The map above is a forward projection, not a free inversion. With finite bins $p_j$ for $\Delta_\zeta^2$,

$$A_{\ell j} = 4\pi \int_{\text{bin } j} d\ln k\, b_j(k)|\Psi_\ell(k)|^2, \qquad C^q = Ap.$$

The evidence bundle must declare the basis, support, prior $p_0, Q$, positivity conditions, singular spectrum of $AQ^{-1/2}$, effective rank, null basis, resolution kernels, prior sensitivity, and every forward residual

$$r_\ell = C_\ell^q - \widehat{C}_\ell.$$

A finite prior, regularizer, positivity constraint, or square discretization selects one representative from a nonunique shell equivalence class. Its typed output is ConditionalRadialContinuation. A source-derived primordial packet requires the common source, scale, mode, null-space, positivity, ancestry, and non-fitting forward-residual receipts together with one exact uniqueness branch from the radial theorem packet below. If the geometry packet is imported, the strongest possible claim tier is `CONDITIONAL_PHYSICAL`; `OPH_NATIVE_PHYSICAL` requires the native $\mathsf{CosmoGeomRead}_r$ theorem.

**Complete radial-lift theorem packet.** The screen-to-radial map is a forward restriction with a nonunique unrestricted inverse. For a flat source window $W$ and geometric normalization $Z_q$,

$$C_{\ell,W}^q = 4\pi Z_q^2 \int_0^\infty d\ln k\, \Delta_\zeta^2(k)|\Psi_{\ell,W}(k)|^2, \qquad \Psi_{\ell,W}(k) = \int W(dr) j_\ell(kr).$$

The physical radial packet must select exactly one of the two uniqueness branches below; a finite prior continuation has the type ConditionalRadialContinuation.

**Theorem (one-shell correlation restriction and exact no-go).** For a thin shell of radius $R$, the complete angular covariance satisfies

$$K_R(\mu) = Z_q^2 \xi_\zeta\Big(R\sqrt{2-2\mu}\Big) = \sum_{\ell \ge 0} \frac{2\ell+1}{4\pi} C_{\ell,R}^q P_\ell(\mu).$$

Hence even the noiseless sequence $\{C_{\ell,R}^q\}_{\ell\ge 0}$ determines the three-dimensional correlation only on $0 \le s \le 2R$. The full one-shell operator has an infinite-dimensional kernel, including positivity-preserving ambiguities. Explicitly, for any nonzero $g \in C_c^\infty((2R,\infty))$,

$$h_g(k) = \frac{2k^3}{\pi} \int_0^\infty r^2 g(r) j_0(kr)\, dr$$

has correlation perturbation $\xi_{h_g} = g$ and therefore changes no shell multipole. Choosing $B = |h_g| + w$ with any strictly positive integrable $w$ makes $B$ and $B + h_g$ two distinct positive spectra with the same complete shell covariance.

*Proof.* The spherical-Bessel addition theorem gives the displayed restriction identity. The map $\mu \mapsto R\sqrt{2-2\mu}$ covers exactly $[0,2R]$. Radial Fourier inversion gives $\xi_{h_g} = g$ and is injective, so the kernel contains an infinite-dimensional copy of $C_c^\infty((2R,\infty))$. The positive pair differs by the same null element. $\square$

**Corollary (exact low-mode-removed kernel).** For a continuous signed correlation perturbation $\xi_h$, invisibility to every shell multipole $\ell \geq 0$ is equivalent to $\xi_h(s) = 0$ on $0 \leq s \leq 2R$. Invisibility to every retained OPH multipole $\ell \geq 2$ is equivalent to

$$\xi_h(s) = a + b\left(1 - \frac{s^2}{2R^2}\right), \qquad 0 \leq s \leq 2R,$$

for constants $a, b$. Legendre completeness gives the first statement. Removing $\ell = 0, 1$ leaves the span of $P_0$ and $P_1$ invisible, which gives the second.

**Producer theorem (source-refinement orbit to physical dilation).** Let $\mathcal{H}_r^{\rm src}$ be the scale-labelled scalar mode space generated by the cofinal refinement orbit of the released screen, including its scale label. A single angular cut is insufficient. Let $U_r$ map the orbit unitarily onto the physical rank-one adiabatic source subspace. If

$$D_{s,r}U_r = U_r R_{s,r}, \qquad C_{\zeta,r} = U_r C_{{\rm src},r} U_r^*,$$

then

$$D_{s,r}^{-1} C_{\zeta,r} D_{s,r} - u_r(s) C_{\zeta,r} = U_r\left(R_{s,r}^{-1} C_{{\rm src},r} R_{s,r} - u_r(s) C_{{\rm src},r}\right) U_r^*.$$

Thus the source and physical operator residuals have equal norm. If the finite covariances converge strongly with a uniform norm bound, the finite dilation maps and their inverses converge strongly, $u_r(s) \to u(s)$, and this residual tends to zero, then $D_s^{-1} C_\zeta D_s = u(s) C_\zeta$ in the continuum.

*Proof.* Conjugate the source covariance by $U_r$ and use the commuting square; unitary invariance gives the residual identity. For the limit, add and subtract the finite conjugated expression. Strong convergence, uniform boundedness, scalar convergence, and the norm-residual hypothesis send the three resulting terms to zero. $\square$

**Theorem (source-dilation uniqueness).** Let $C_\zeta = M_{\Delta_\zeta^2}$ on the common physical $L^2(d\ln k\, d\Omega_k)$ mode basis. Define

$$(D_s f)(k, \hat{k}) = f(e^{-s}k, \hat{k}).$$

If the source refinement packet proves

$$D_s^{-1} C_\zeta D_s = u_q(s) C_\zeta, \qquad u_q(s+t) = u_q(s)u_q(t), \qquad -\frac{d}{ds}\ln u_q(s)\bigg|_{s=0} = \theta,$$

then $u_q(s) = e^{-\theta s}$ and there is $A > 0$ such that

$$\Delta_\zeta^2(k) = A k^{-\theta} \quad \text{almost everywhere.}$$

For a continuous source representative and pivot $k_\star$,

$$\boxed{\Delta_\zeta^2(k) = A_\zeta (k/k_\star)^{-\theta}}, \qquad A_\zeta = \Delta_\zeta^2(k_\star).$$

*Proof.* Strong continuity solves the multiplicative Cauchy equation for $u_q$. Conjugation of the multiplication operator gives $\Delta_\zeta^2(e^s k) = e^{-\theta s}\Delta_\zeta^2(k)$ almost everywhere for each $s$. In logarithmic coordinates, $e^{\theta t}\Delta_\zeta^2(e^t)$ is invariant under all rational translations on one common full-measure set;

bounded-transform convolution makes it constant almost everywhere. Continuity upgrades the identity to every $k$. $\square$

**Theorem (exact flat thin-shell lift).** On the source-dilation branch, for $-2 < \theta < 4$ and $\ell \geq 2$,

$$\int_0^\infty \frac{dx}{x} x^{-\theta} j_\ell^2(x) = \frac{\sqrt{\pi}}{4} \frac{\Gamma(1+\theta/2)}{\Gamma(3/2+\theta/2)} \frac{\Gamma(\ell-\theta/2)}{\Gamma(\ell+2+\theta/2)},$$

so

$$C^q_{\ell,R_\star} = A_q \frac{\Gamma(\ell-\theta/2)}{\Gamma(\ell+2+\theta/2)},$$

with the complete amplitude conversion

$$\boxed{A_\zeta = \frac{A_q}{\pi^{3/2} Z_q^2 (k_\star R_\star)^\theta} \frac{\Gamma(3/2+\theta/2)}{\Gamma(1+\theta/2)}.}$$

*Proof.* Insert the source power law into the exact forward map, set $x = kR_\star$, and apply the Weber–Schafheitlin square integral. The convergence strip follows from $j_\ell(x) \sim x^\ell/(2\ell+1)!!$ at zero and $j_\ell(x) = O(x^{-1})$ at infinity. $\square$

**Corollary (source-family injectivity and multipole consistency).** Fix $\theta, Z_q, k_\star$, and a source window for which the forward coefficient is positive and finite. The forward map is injective in $A_\zeta$ on the family $\Delta_\zeta^2 = A_\zeta (k/k_\star)^{-\theta}$. For each retained multipole,

$$A_\zeta^{(\ell,W)} = \frac{C^q_{\ell,W}}{4\pi Z_q^2 k_\star^\theta \int d\ln k\, k^{-\theta} |\Psi_{\ell,W}(k)|^2}.$$

Every $A_\zeta^{(\ell,W)}$ must agree. A disagreement falsifies the declared window/dilation packet and cannot be repaired with an $\ell$-dependent amplitude.

**Theorem (finite-window bound).** For $0 < \theta < 2\ell$, define

$$I_\ell(\theta) = \int d\ln x\, x^{-\theta} j_\ell^2(x), \quad J_\ell(\theta) = I_\ell(\theta-2) - \left[\ell(\ell+1) - \frac{\theta(\theta+1)}{2}\right] I_\ell(\theta),$$

$$\eta_{\ell,W} = \frac{2\sqrt{J_\ell(\theta)}}{\theta} \int W(dr) |r^{\theta/2} - R_\star^{\theta/2}|.$$

Then

$$|C^q_{\ell,W} - C^q_{\ell,R_\star}| \leq 4\pi Z_q^2 A_\zeta k_\star^\theta \eta_{\ell,W} \left(2R_\star^{\theta/2}\sqrt{I_\ell(\theta)} + \eta_{\ell,W}\right).$$

*Proof.* In the Hilbert norm $\|f\|_\theta^2 = \int d\ln k\, k^{-\theta}|f(k)|^2$, $\|j_\ell(kr)\|_\theta = r^{\theta/2}\sqrt{I_\ell}$ and $\|\partial_r j_\ell(kr)\|_\theta = r^{\theta/2-1}\sqrt{J_\ell}$. The fundamental theorem of calculus, Minkowski, and $|\|f\|^2 - \|g\|^2| \leq \|f-g\|(\|f\|+\|g\|)$ give the bound. $\square$

**Theorem (finite singular-value and prior continuation).** For a declared finite radial basis, write $C = Ap$. If $A \in \mathbb{R}^{N_\ell \times N_k}$ has rank $r$, then $\dim\ker A = N_k - r$, and every exact solution is $p = p_{\rm part} + V_0 a$ for a right-null basis $V_0$. Given a prior center $p_0$ and $Q \succ 0$, the unique minimizer of $\frac{1}{2}(p-p_0)^\mathsf{T} Q (p-p_0)$ subject to $Ap = C$ is

$$\boxed{p_* = p_0 + Q^{-1}A^\mathsf{T}(AQ^{-1}A^\mathsf{T})^+(C - Ap_0).}$$

The run publishes the singular values, rank threshold, right-null basis, resolution operator, unresolved projector, positivity active set, and prior sensitivity. The output type is ConditionalRadialContinuation.

**Theorem (unrestricted tomographic uniqueness).** For

$$C_\ell(r,r') = \frac{2}{\pi}\int_0^\infty k^2 P_\zeta(k) j_\ell(kr) j_\ell(kr')\, dk,$$

the covariance operator on $L^2(r^2dr)$ obeys

$$Q_\ell = \mathcal{H}_\ell^{-1} M_{P_\zeta} \mathcal{H}_\ell.$$

Thus a complete radial cross-covariance operator determines $P_\zeta$ uniquely almost everywhere. Fixing one radius $r_0$, the family $C_\ell(r, r_0)$ known for all $r$ suffices: its spherical-Hankel transform is $\sqrt{2/\pi}\, P_\zeta(k) j_\ell(kr_0)$, and the Bessel zeros are discrete.

*Proof.* Apply spherical-Hankel inversion in the first radial variable. Unitary conjugation exposes the multiplication operator; equal multiplication operators have equal multipliers almost everywhere. □

**Theorem (branch-typed radial boundary).** The ordinary spherical-Bessel and gamma-function formulas apply to FlatExact and FlatAssumed branches. Open and closed spatial branches use their declared hyperspherical eigenfunctions and spectral measures. An unresolved curvature branch blocks physical radial promotion, and the flat thin-shell amplitude formula cannot certify a curved branch.

**Promotion rule.** A source-derived primordial packet requires all source-stress, single-clock, freezeout, growing-mode, phase, physical-scale, source-amplitude, null-space, positivity, and non-fitting forward-residual receipts, together with either a radial-dilation intertwiner certificate or a radial-tomography certificate. A finite inverse selected before evaluation is a conditional radial continuation. It does not promote temperature or polarization spectra to physical observables.

**Corollary (transfer firewall).** Radial promotion changes no temperature, polarization, lensing, recombination, foreground, nuisance, covariance, or likelihood status. Those outputs require the independent transfer and likelihood system.

| | |
|---|---|
| `TOTAL_STRESS_CLOSURE_RECEIPT` | $T_I^{ab}, Q_I^a, E_T, \nabla_a T^{ab}$ |
| `SINGLE_CLOCK_NORMAL_FORM_RECEIPT` | $\chi_r$, rank $d\mathcal{F}$, connected fibers |
| `ENTROPY_REPAIR_GAP_RECEIPT` | $L_\perp, H, \gamma$, forcing/refinement bounds |
| `CURVATURE_EVOLUTION_RECEIPT` | $\zeta_a, \Gamma_a^{\text{eff}}$, evolution residual, freeze bound |
| `ADIABATIC_MODE_RECEIPT` | $S_{IJ}, \Gamma_I, \Xi_I$, growing-mode projection |
| `ISOCURVATURE_BOUND_REPORT` | $\beta_{\text{iso}}(k)$ and source covariance |
| `PRIMORDIAL_PHASE_COHERENCE_RECEIPT` | $\lambda_2/\lambda_1$, phase and decaying-mode defects |
| `SCREEN_TO_RADIAL_LIFT_RECEIPT` | $W(r), R_\star, \Psi_\ell(k)$, source hashes |
| `RADIAL_NULL_SPACE_REPORT_RECEIPT` | singular values, null basis, resolution kernels |
| `FORWARD_PROJECTION_RESIDUAL_RECEIPT` | $r_\ell$, aggregate norms, quadrature and solver errors. |

**Source-provenance and pooled-reducer firewall.** Let the source artifact DAG have parent-to-child edges and let $M$ be the set of measurement, residual, likelihood, posterior, fitted-parameter, or measurement-calibrated nodes. Define

$$\tau(v) = \mathbf{1}_{v\in M} \vee \bigvee_{p\in \text{Parents}(v)} \tau(p).$$

Then $\tau(v) = 1$ exactly when $v$ has a forbidden measurement ancestor. The proof is induction over a topological ordering of the DAG. Every promoted source node for $\eta_R, \Gamma_{\text{rec}}, A_\zeta, q_{\text{IR}}, \ell_{\text{IR}}$, $B_A(k,a)$, $\rho_A(a)$, and $N_{\text{CRC}}$ must have $\tau = 0$. Unknown source metadata fails closed.

For nonlinear estimates, shards may not average nonlinear quantities. Each row must emit additive sufficient statistics in a commutative monoid $(\mathcal{M}, \oplus)$, for example

$$\sum w_i, \qquad \sum w_i x_i, \qquad \sum w_i x_i x_i^{\mathsf{T}}$$

or, for weighted response regression,

$$X^{\mathsf{T}} W X, \qquad X^{\mathsf{T}} W y, \qquad y^{\mathsf{T}} W y.$$

Associativity, commutativity, complete coverage, and duplicate removal make the pooled statistic partition-invariant; the deterministic estimator is then evaluated only after pooling.

**Three-dimensional homogeneous lift.** A rotationally invariant screen covariance about one observer is not a statistically homogeneous three-dimensional primordial field by itself. A frozen lift

$$\mathcal{L}_r : V_r^{\text{screen}} \to \mathcal{H}_r^{(3)}$$

must define

$$\Sigma_r^{(3)} = \mathcal{L}_r \Sigma_r^{\text{screen}} \mathcal{L}_r^*$$

and certify positivity, refinement convergence, rotation covariance, translation covariance or an equivalent multicenter consistency theorem, and explicit null-space treatment. If the limiting covariance commutes with translations and rotations, Fourier diagonalization gives

$$\langle \zeta_a(\mathbf{k}) \zeta_b^*(\mathbf{k}') \rangle = (2\pi)^3 \delta^{(3)}(\mathbf{k} - \mathbf{k}') P_{ab}(|\mathbf{k}|),$$

with $P_{ab}(k)$ positive semidefinite.

**Transfer-ready initial data.** The source-to-transfer map must emit a complete initial state

$$Y(k, \eta_i) = \mathcal{M}_{\text{init}}(k; \mathfrak{B}_{\text{src}}) \xi(k),$$

including metric, photon, polarization, baryon, neutrino, anomaly, recipient, and auxiliary response variables. Near a singular early-time point, regular Frobenius modes solve

$$Y' = \left( \frac{M_{-1}}{\eta} + M_0 + O(\eta) \right) Y, \qquad Y = \eta^s (v_0 + v_1 \eta + \cdots), \quad M_{-1} v_0 = s v_0.$$

The branch must classify regular adiabatic growing modes, allowed isocurvature modes, decaying modes, gauge modes, and singular interaction modes. On the adiabatic branch,

$$S_{IJ} = 3(\zeta_I - \zeta_J) = 0$$

for all active sectors. The finite evidence receipt must bound the initial Hamiltonian and momentum constraints and the propagated residual

$$\epsilon_{\text{constraint}} = \sup_{k,\eta} \| \mathcal{C}(k, \eta) Y(k, \eta) \|.$$

**Receipt contract.** Every ensemble run must record its ensemble id, claim tier, regulator, representative schema, gauge action, quotient canonicalizer, base measure definition, action coefficients, coarse maps, zero-mode projector, amplitude convention, sampler kernel, partition-invariant random-event schema, smoothing policy, source provenance, and explicit nonclaims. The seed belongs to the run receipt rather than the ensemble definition. Claim tiers are:

$$\begin{aligned} E0:&\quad \text{seed noise, proposal noise, repair jitter,}\\ E1:&\quad \text{conventional reference ensemble,}\\ E2:&\quad \text{OPH-native quotient ensemble,}\\ E3:&\quad \text{OPH vacuum,}\\ E4:&\quad \text{OPH primordial field,}\\ E5:&\quad \text{observable cosmological prediction.}\end{aligned}$$

Smoothing must preserve raw coefficients, raw spectra, smoothing kernels, smoothed coefficients, smoothed spectra, and hashes of each stage; it is not part of $S_r$ unless explicitly declared. For bounded coarse observables $O$, if the implemented law satisfies $\|\widehat{\mu}_s - \mu_s\|_{\mathrm{TV}} \le \epsilon_{\mathrm{samp}}$ and the refinement defects sum to $\epsilon_{\mathrm{ref}}$, then

$$\left|\mathbb{E}_{\widehat{\mu}_s}[O \circ c_{sr}] - \mathbb{E}_{\mu_r}[O]\right| \le 2\|O\|_\infty(\epsilon_{\mathrm{samp}} + \epsilon_{\mathrm{ref}}),$$

which is the finite evidence accuracy statement attached to this theorem surface.

**Theorem 2.3 (EC from regulated overlap gluing).** For a collar $B_\delta$ around a cap boundary $\Sigma$, there is a canonical decomposition

$$H_{B_\delta} = (\tilde{\mathcal{H}}_{B_L} \otimes \tilde{\mathcal{H}}_{B_R})^{G_\Sigma} = \bigoplus_\alpha (H_{b_L^\alpha} \otimes H_{b_R^\alpha}),$$

and the sector-preserving collar algebra

$$A_{\mathrm{EC}}(B_\delta) := \bigoplus_\alpha (B(H_{b_L^\alpha}) \otimes B(H_{b_R^\alpha}))$$

with

$$Z(A_{\mathrm{EC}}(B_\delta)) = \bigoplus_\alpha \mathbb{C}\,\mathbf{1}_\alpha,$$

such that $\mathcal{A}(A_\delta B_\delta)$ acts only on $H_{b_L^\alpha}$ and $\mathcal{A}(B_\delta D_\delta)$ acts only on $H_{b_R^\alpha}$ within each block.

**Proof.** Split the collar into half-collars $B_L$ and $B_R$ meeting on $\Sigma = \partial C$. By the realized regulator presentation above, the physical collar Hilbert space is the diagonal invariant subspace $(\tilde{\mathcal{H}}_{B_L} \otimes \tilde{\mathcal{H}}_{B_R})^{G_\Sigma}$. Decompose each side into irreps:

$$\tilde{\mathcal{H}}_{B_L} = \bigoplus_\alpha (V_\alpha \otimes H_{b_L^\alpha}), \qquad \tilde{\mathcal{H}}_{B_R} = \bigoplus_\beta (V_\beta^* \otimes H_{b_R^\beta}).$$

Then

$$\tilde{\mathcal{H}}_{B_L} \otimes \tilde{\mathcal{H}}_{B_R} = \bigoplus_{\alpha,\beta} (V_\alpha \otimes V_\beta^*) \otimes (H_{b_L^\alpha} \otimes H_{b_R^\beta}).$$

By Schur's lemma,

$$(V_\alpha \otimes V_\beta^*)^{G_\Sigma} \cong \begin{cases} \mathbb{C}, & \alpha = \beta, \\ 0, & \alpha \ne \beta. \end{cases}$$

Therefore the invariant subspace is

$$H_{B_\delta} = \bigoplus_\alpha (H_{b_L^\alpha} \otimes H_{b_R^\alpha}),$$

as claimed. The induced sector-preserving collar algebra is

$$A_{\mathrm{EC}}(B_\delta) = \bigoplus_\alpha (B(H_{b_L^\alpha}) \otimes B(H_{b_R^\alpha})),$$

so the center is generated by the block projectors. Adjacent region algebras act on the left or right factor only because the gluing action is supported on $\Sigma$. QED.

**Remark.** On the central-defect subbranch, replace $G_\Sigma$ by its central extension. The sector label $\alpha$ then ranges over irreps of the extension; the decomposition is unchanged.

We refer to the decomposition in Theorem 2.3 as **edge-center completion (EC)**.

**Exact Markov route.** If $I(A_\delta : D_\delta \mid B_\delta) = 0$, the Hayden–Jozsa–Petz–Winter (HJPW) structure theorem yields a blockwise factorization of the state over *some* direct-sum/tensor decomposition of $\mathcal{H}_{B_\delta}$; that decomposition is state-dependent, and HJPW does not by itself identify it with the preselected EC factors $b_L^\alpha, b_R^\alpha$ of Theorem 2.3. The exact identity used for literal Markov-modular equalities below is the **EC-aligned** normal form

$$\rho_{A_\delta B_\delta D_\delta} = \bigoplus_\alpha p_\alpha (\rho_{A_\delta b_L^\alpha} \otimes \rho_{b_R^\alpha D_\delta}), \qquad I_\omega(A_\delta : D_\delta \mid B_\delta) = 0,$$

and the identification of the HJPW factors with the EC factors is a genuine additional state condition, the **Markov-split alignment hypothesis (MSA)**. It is not implied by exact Markovity: with four qubits $A, B_L, B_R, D$ and $\rho = \Phi_{AB_R} \otimes \Phi_{B_L D}$ (Bell pairs across the cut, one center block), one has $I(A : D \mid B) = 0$ but $I(A : B_R) = 2\ln 2$, so the state is exactly Markov yet not of the displayed EC-aligned form, which would force $I(A : B_R) = 0$. All literal Markov-modular equalities below therefore assume exact Markovity *plus* MSA, or an explicitly stated idealized limit that supplies both; the compact paper states MSA formally and records this counterexample. MSA is checkable state by state: the compact paper proves it equivalent to a splitting of $\log \rho$ into commuting one-sided terms supported on $Ab_L^\alpha$ and $b_R^\alpha D$, to the existence of a $\rho$-preserving conditional expectation onto either one-sided edge algebra (a Takesaki commuting square over the collar center [32]), and to blockwise vanishing one-sided mutual information $I(Ab_L^\alpha : b_R^\alpha D) = 0$. On the MaxEnt branch this reduces the derivation of MSA to a sharp condition on the effective Hamiltonian: every cross-cut coupling must act through the central sector label, as it does in lattice-gauge-type regulators where the interface energy is a function of the conserved flux. On that declared **central-interface branch** the compact paper derives both MSA and exact collar Markovianity as theorems (descent of one-sided invariants and boundary charges through the edge-center quotient), and it adopts the central-interface normal form as an explicit axiom-level branch clause stated with its MaxEnt axiom, so the literal Markov-modular identities hold on the declared branch without a separate state hypothesis. The clause is independent of the repair/consensus axioms: the compact paper exhibits a finite regulator package that satisfies overlap consistency, schedule-independent transactional repair, and an exactly closed refinement-channel family while its retained constraint family carries a boundary-invariant noncentral cross-cut coupling, so overlap-consistent repair does not force the clause and it stays a permanent declared branch input. Off the declared branch MSA is carried as an explicit hypothesis.

Interpreting collar refinement along the rate-controlled family stated in Axiom 3, Theorem 2.3 supplies the kinematic edge-center decomposition. Exact Markovity is one idealized route. The following lemma and theorem provide the quantitative decay route used when the manuscript keeps the approximation explicit.

**Lemma 2.6 (MaxEnt with local constraints implies local Gibbs form).** Under the local MaxEnt branch of Axiom 3, with its homogeneous global-sum constraints $\langle\sum_x O_a(x)\rangle = C_a$, and the finite-dimensional regulator realization above, the MaxEnt state has the Gibbs form

$$\omega = \frac{e^{-H_{\mathrm{eff}}}}{\operatorname{Tr} e^{-H_{\mathrm{eff}}}}, \qquad H_{\mathrm{eff}} = \sum_x \sum_a \lambda_a O_a(x) + H_{\mathrm{sec}},$$

where the sum runs over UV cells $x$ and density labels $a$, with one cell-independent multiplier $\lambda_a$ per label. For the finite-range collar theorem, $H_{\mathrm{sec}}$ is either a sum of uniformly bounded finite-range charge densities or is central and fixed on the chosen superselection sector. A genuinely nonlocal noncentral term falls outside the theorem. Under this condition the effective Hamiltonian is finite-range in UV graph units, and the multiplier count matches the constraint count $N_{\mathrm{con}}$ plus the number of retained sector terms on every lattice.

**Proof.** On a finite-dimensional algebra, the unique state maximizing $S(\rho) = -\operatorname{Tr}(\rho\log\rho)$ subject to linear constraints $\operatorname{Tr}(\rho O_i) = c_i$ is given by Lagrange multipliers:

$$\rho = \frac{e^{-\sum_i \lambda_i O_i}}{\operatorname{Tr} e^{-\sum_i \lambda_i O_i}}.$$

Strict concavity of von Neumann entropy ensures uniqueness. Here the constrained operators are the $N_{\mathrm{con}}$ global sums $O_a := \sum_x O_a(x)$ plus the retained sector terms, so the exponent has one shared multiplier per density label. MaxEnt gives the Gibbs form; the displayed locality restriction on $H_{\mathrm{sec}}$ is separate and load-bearing. Had the constraints instead been imposed per cell, $\langle O_a(x)\rangle = c_a(x)$, the multipliers would be cell-dependent fields $\lambda_a(x)$ and the family would grow with the lattice; that enlarged family belongs to the non-equilibrium approximation regime of the clarification above and is not the branch used in the refinement argument. QED.

**Definition 2.6a (closure defect and induced refinement map).** Fix successive regulator scales $\ell$ (finer) and $L$ (coarser), let $\{\omega_L(\lambda')\}$ be the homogeneous global-sum exponential family at scale $L$, and let $\Phi_{\ell\to L}$ be an admissible coarse-graining channel. For fine-scale multipliers $\lambda$ set $\sigma := \Phi_{\ell\to L}(\omega_\ell(\lambda))$ and define the **closure defect**

$$\varepsilon_{\ell\to L}(\lambda) := \inf_{\lambda'} D\big(\sigma \,\|\, \omega_L(\lambda')\big),$$

with $D$ the quantum relative entropy. When the infimum is attained at a unique $\lambda^*$, set $R_{\ell\to L}(\lambda) := \lambda^*$.

**Lemma 2.6b (I-projection residual bound).** If the constrained operators at scale $L$ together with the identity are linearly independent, then $\lambda' \mapsto D(\sigma\|\omega_L(\lambda'))$ is smooth and strictly convex; for faithful $\sigma$ the minimizer $\lambda^* = R_{\ell\to L}(\lambda)$ exists, is unique, and is characterized by moment matching $\langle S_c\rangle_{\omega_L(\lambda^*)} = \langle S_c\rangle_\sigma$ for every constrained operator $S_c$; and

$$\big\|\sigma - \omega_L(R_{\ell\to L}(\lambda))\big\|_1 \le \sqrt{2\,\varepsilon_{\ell\to L}(\lambda)},$$

so the family reproduces every bounded expectation up to $\|B\|\sqrt{2\varepsilon_{\ell\to L}(\lambda)}$. The refinement-closure clause of Axiom 3 is exactly the statement $\varepsilon_{\ell\to L}(\lambda) = 0$ along the realized branch, in which case $\sigma = \omega_L(R_{\ell\to L}(\lambda))$ and the induced refinement map is this moment-matching I-projection.

**Proof.** Writing $\log\omega_L(\lambda') = -\sum_c \lambda'_c S_c - \log Z_L(\lambda')$, the objective is $-S(\sigma) + \sum_c \lambda'_c \langle S_c\rangle_\sigma + \log Z_L(\lambda')$, and the Hessian of $\log Z_L$ is the Duhamel (Kubo–Mori) covariance matrix of the constrained operators, positive definite under the stated independence; this gives strict convexity, uniqueness, and the moment-matching first-order condition, while attainment for faithful $\sigma$ is standard finite-dimensional exponential-family duality [30]. The displayed residual bound is the quantum Pinsker inequality $D(\rho\|\tau) \geq \frac{1}{2}\|\rho-\tau\|_1^2$ [31] evaluated at $\lambda^*$, and $\varepsilon_{\ell\to L}(\lambda) = 0$ forces $\sigma = \omega_L(\lambda^*)$ by strict positivity of relative entropy off the diagonal. QED.

**Acceptance check.** A finite two-lattice test implementing this package (the constraint/multiplier count against the displayed $N_{\mathrm{con}}$-dimensional family on successive lattices, the moment-matching projection, the residual bound, a generic decimation channel with strictly positive closure defect, and the exactly closed transverse-field subfamily where $R_{\ell\to L}$ acts as the identity) is included with the repository sources.

**Strong conditional mixing premise.** Let $G_\ell$ be the UV cell graph. Assume local dimension at most $q$, interaction degree at most $\Delta_0$, interaction diameter at most $r_0$ graph layers, and

$$\sup_x \sum_{X\ni x} \|\beta\Phi_\ell(X)\| \leq J_0$$

with constants independent of the cut and refinement stage. Let $m_\ell$ be the resolved collar width in graph layers and set $\delta = m_\ell \ell$. For the faithful Gibbs state, define

$$\mathbf{J}_\rho(A : D \mid B) := \log\rho_{ABD} + \log\rho_B - \log\rho_{AB} - \log\rho_{BD}.$$

The strong mixing premise is a boundary-anchor expansion

$$\mathbf{J}_\rho(A_\delta : D_\delta \mid B_\delta) = \sum_{z\in\partial^{\mathrm{UV}}_{r_0} C} E_{z,\ell,\delta}, \qquad \|E_{z,\ell,\delta}\|_\infty \leq \kappa e^{-(m_\ell - r_0)/\zeta},$$

with $\kappa, \zeta$ uniform in the cut, boundary condition, and stage. This conditional or matrix mixing condition is stronger than ordinary two-point exponential clustering. Local Gibbs form, a Lieb–Robinson bound, or a Hamiltonian spectral gap does not imply it for a general noncommuting Gibbs state.

**Theorem 2.5 (finite-range conditional Gibbs mixing gives collar recovery).** Under Lemma 2.6 and the strong conditional mixing premise,

$$\boxed{I_\omega(A_\delta : D_\delta \mid B_\delta) \leq \kappa|\partial C|_{\mathrm{UV}} e^{-(m_\ell - r_0)/\zeta} \leq c|\partial C|_{\mathrm{UV}} e^{-\delta/\xi_\ell}}$$

with

$$\xi_\ell = \zeta\ell, \qquad c = \kappa e^{r_0/\zeta}.$$

**Proof.** Embedding the marginal logarithms in the tripartite algebra gives

$$I(A : D \mid B)_\rho = \mathrm{Tr}[\rho_{ABD}\mathbf{J}_\rho(A : D \mid B)]\,.$$

The boundary expansion and trace-norm duality bound this expectation by the sum of the boundary term norms. The second form follows from $\delta = m_\ell \ell$. QED.

For one uniform family, the sharp sufficient limit condition is

$$\frac{\delta}{\xi_\ell} - \log|\partial C|_{\mathrm{UV}} \longrightarrow +\infty.$$

If

$$|\partial C|_{\mathrm{UV}} \le a(\ell_0/\ell)^p, \qquad \delta(\ell) \ge \zeta(p+\eta)\ell\log(\ell_0/\ell), \quad \eta > 0,$$

then $\delta \to 0$, $\delta/\ell \to \infty$, and

$$I(A_\delta : D_\delta \mid B_\delta) \le ca(\ell/\ell_0)^\eta \longrightarrow 0.$$

The ratio $\delta/\ell \to \infty$ alone is insufficient. For $\ell_n = \ell_0 e^{-n^2}$, $\delta_n = \zeta n \ell_n$, and $|\partial C_n|_{\mathrm{UV}} = e^{n^2}$, that ratio diverges while the upper envelope grows as $e^{n^2-n}$.

A finite receipt records the interaction range and norm bound, treatment of global sector terms, boundary-cell count, graph separation, regional CMI in nats, matrix-defect norm, predeclared $\kappa, \zeta$, log-envelope slack, Fawzi–Renner error, held-out cuts, and the rate margin $\delta/(\zeta\ell) - \log|\partial C|_{\mathrm{UV}}$. Fitted constants and local random-triplet CMI remain finite proxies. They do not certify a cofinal limit.

This bound is the quantitative hinge for constructive gluing.

### 5.0.2 Concrete UV realization: quantum link models

Section 2.3 internalizes the regulator package: the fixed-cutoff type-I algebra and boundary fixed-point structure are the realized form of the screen net plus overlap gluing. Quantum link models are included here only as an explicit microscopic example of that structure, not as a separate axiom layer.

**UV regulator.** Triangulate $S^2$ at scale $\ell_{\mathrm{UV}}$, giving vertices $v$, oriented links $\ell$, and plaquettes $p$. Refinement corresponds to $\ell_{\mathrm{UV}} \to 0$ with increasing lattice size.

**Degrees of freedom.** Attach to every oriented link $\ell$ a **finite-dimensional** Hilbert space $\mathcal{H}_\ell$. In ordinary Wilson lattice gauge theory, the continuum/refinement-limit edge description is modeled by $L^2(G)$ (infinite-dimensional for continuous $G$), but the microscopic OPH regulator is instead a **quantum link model** with finite-dimensional link Hilbert spaces that preserve gauge symmetry in operator form [49]. Optionally attach matter Hilbert spaces $\mathcal{H}_v$ at vertices. Then:

$$\tilde{\mathcal{H}}_{\mathrm{total}} = \bigotimes_\ell \mathcal{H}_\ell \otimes \bigotimes_v \mathcal{H}_v,$$

finite-dimensional on any finite lattice. This is a concrete realization of the extended type-I presentation used in Section 2.3.

**Boundary gluing as Gauss-law invariants.** Define a local gauge transformation group $G_v$ at each vertex $v$ acting on incident links (and matter at $v$). Physical states satisfy:

$$|\psi\rangle \in \mathcal{H}_{\mathrm{phys}} \quad \Longleftrightarrow \quad U(g_v)|\psi\rangle = |\psi\rangle \;\; \forall\, v,\, g_v \in G_v.$$

Equivalently: $\mathcal{H}_{\mathrm{phys}} = \tilde{\mathcal{H}}_{\mathrm{total}}^{\prod_v G_v}$.

**Region algebras.** For any region $R \subset S^2$, define an extended Hilbert space $\tilde{\mathcal{H}}_R$ from the links/vertices in $R$. The **boundary gauge group** $G_{\partial R}$ acts on the cut degrees of freedom (the "half-links" ending on $\partial R$). Define:

$$\mathcal{A}_{\mathrm{inv}}(R) = \mathcal{B}(\tilde{\mathcal{H}}_R)^{G_{\partial R}}.$$

**This is the concrete quantum-link instance of the lifted fixed-point presentation used in Section 2.3.** The same gauge constraints then induce the sector-preserving EC algebra on collars.

**Why EC follows immediately.** Take a cap $C$ and a collar $B_\delta$ around $\partial C$. Because the *only* coupling between inside and outside is through the boundary gauge constraint, the collar Hilbert space decomposes into superselection blocks labeled by boundary irreps:

$$\mathcal{H}_{B_\delta} \cong \bigoplus_\alpha (H_{b_L^\alpha} \otimes H_{b_R^\alpha}),$$

with center generated by the projectors $P_\alpha$. This is precisely the Schur-lemma mechanism of Theorem 2.3. The labels $\alpha$ are the familiar edge-mode / electric-flux labels appearing whenever one factorizes gauge theories across an entangling cut [50]. Exact Markovity depends on the state and is not forced by the decomposition alone; nor does exact Markovity align the HJPW factors with these edge-mode labels, which is the separate Markov-split alignment hypothesis of Section 2.3.

**Dynamics and MaxEnt.** The natural Hamiltonian is a 2+1D lattice gauge Hamiltonian on the screen worldvolume: plaquette ("magnetic") terms, electric terms on links, vertex Gauss terms as constraints, plus local matter couplings. In quantum link form this is finite-dimensional per link while behaving like gauge theory in the continuum limit. Then the MaxEnt assumption becomes concrete: the MaxEnt state is a Gibbs state $\rho \propto e^{-\sum_i \lambda_i O_i}$ with quasi-local $O_i$, precisely the local-Gibbs regime.

**Geometry and $G$.** This microphysics naturally supplies the emergent geometric objects:

- **Edge entropy / area operator:** $L_C = \sum_\alpha (\log d_\alpha) P_\alpha$ becomes "log of boundary irrep dimension" in the gauge link model.
- **Newton constant $G$:** the conversion factor between edge entropy density per boundary UV cell and macroscopic geometric area.

Thus area is an operator living in the center of the boundary algebra, because in gauge systems the center is where the cut labels live.

**Scope of this example.** The quantum-link realization makes the fixed-cutoff matrix/fixed-point bookkeeping concrete. Modular flow on caps becomes geometric conformal dilation with the $2\pi$ KMS normalization through the support-visible BW scaling theorem on the geometric subnet. Viable architectures for this include holographic quantum error-correcting codes [51] and quantum double / string-net Hamiltonians [52], but any use of QECC distance, min-cut resilience, or Knill–Laflamme correction requires the corresponding code subspace, logical-operator, error-family, and recovery certificate.

### 5.0.3 Conformal-modular fixed-point microphysics

On the local finite-constraint MaxEnt branch of the third axiom, the logarithm of the selected state is a quasi-local UV generator, and the refinement-stable branch lies inside one common finite-dimensional multiplier family. At each regulator stage the patch and cap algebras are type-I matrix algebras. The finite regulator class is not refinement-closed, so the scaling-limit observer algebra may leave that class. The fixed-cutoff analysis proves a local-Gibbs/refinement-stable branch with propagation and endpoint-Lipschitz control. The geometric cap pair with geometric modular flow is obtained in the support-visible sense: the target algebra is the extracted geometric subnet, the support-visible modular matrix elements converge under regularization, and weak-$*$/GNS extraction plus support-readable modular covariance and ordered cut-pair rigidity gives the geometric cap automorphism.

**Local finite-constraint MaxEnt branch.** The constraint family $\mathcal{C}$ is generated by finitely many gauge-invariant local densities $\{O_a(x)\}$ of UV range $O(\ell_{\mathrm{UV}})$, constrained through their homogeneous global sums $\langle \sum_x O_a(x) \rangle = C_a$, with the same finite label set retained under refinement by the refinement-closure clause of the third OPH axiom. This clause restates that axiom at regulator level and adds no postulate beyond it; the closure clause itself is the substantive assumption.

**Theorem 2.6 (Local constraints imply a local-Gibbs form).** If the MaxEnt constraints are expectations of finitely many quasi-local operators $\{O_a\}$ with bounded support size at scale $\ell_{\mathrm{UV}}$, then the entropy maximizer is

$$\omega \propto \exp\left( - \sum_a \lambda_a O_a \right),$$

so the MaxEnt generator $H_{\mathrm{MaxEnt}} = -\log \omega$ is a UV-range quasi-local sum. This is exactly the local-Gibbs form used later.

**Proof.** Standard exponential-family result: maximum entropy subject to linear constraints $\langle O_a \rangle = c_a$ yields the Gibbs state with Lagrange multipliers $\lambda_a$. QED.

**Derived propagation control on the same branch.** Because $H_{\mathrm{MaxEnt}}$ is a finite-range or quasi-local sum on a finite type-I regulator net, standard Lieb–Robinson estimates [13] apply to the automorphism group it generates, or to any branch generator lying in the same bounded-support algebraic closure. Thus there is a finite propagation velocity $v_{\mathrm{LR}}$ and constants $C, \xi$ such that for local observables $A_X, B_Y$,

$$\big\| [\tau_t(A_X), B_Y] \big\| \le C \, \|A_X\| \, \|B_Y\| \, \min(|X|, |Y|) \, e^{-(d(X,Y) - v_{\mathrm{LR}}|t|)/\xi}.$$

Because changing a bounded interval endpoint only adds or removes an $O(|\Delta v|)$ collar of local terms once the central endpoint piece is separated off, the same local branch also gives bounded-interval endpoint-Lipschitz matrix elements,

$$\big| \langle \psi, (K[I'] - K[I]) \phi \rangle \big| \le C_{\psi,\phi,I_{\max}} \, |I' \Delta I|,$$

used later in the null-modular bridge. So quasi-local propagation and endpoint-Lipschitz control are not external regularity selectors; they are the local-constraint MaxEnt branch written in dynamical form.

**Refinement-stable multiplier branch.** By the refinement-closure clause of the third OPH axiom (a substantive renormalization condition, since coarse-graining a finite-range Gibbs family generically generates interactions outside any fixed finite density list), the same finite constraint family is preserved under coarse-graining, and the regulator states lie in one common finite-dimensional multiplier family rather than in unrelated state spaces at different cutoffs. The "refinement-stable" language used later therefore means persistence along the stable or fixed branch of this multiplier family. This state-side notion is enough to compare realized states across cutoffs. FixedCutoffBosonicSectorCategory constructs the fixed-stage tensor-generated category and its forgetful fiber. RefinementFunctorAndFiberDescent constructs the refinement pullbacks and compatible fibers only on a cofinal tail carrying the compact-gauge refinement receipt. The cofinal compact-gauge witness uses the same receipt, while the Standard Model selection step is the later MAR application to sector packages. The third OPH axiom supplies the realized state branch; it does not supply the matrix-algebra embeddings, compact-group surjections, or tensor-realization data needed for the sector colimit.

**Scaling limit and algebraic type.** The regulator presentation gives a family of finite type-I algebras

$$\mathcal{A}_\ell(C) \cong \mathcal{B}(\mathcal{H}_{C,\ell}).$$

A scaling limit of this family need not be type I. In the continuum-QFT case of interest one expects the local limit algebra $\mathcal{A}_\infty(C)$ to be non-type-I, typically type III. Accordingly the fundamental modular datum in the limit is the automorphism group of the pair $(\mathcal{A}_\infty(C), \omega_\infty^C)$, not a density matrix inside $\mathcal{A}_\infty(C)$.

**Corollary 2.6 (Geometric modular action on certified caps).** For any support-visible extracted scaling-limit geometric cap pair $(\mathcal{A}_\infty^{\mathrm{geo,sv}}(C), \omega_\infty^{\mathrm{geo},C})$ emitted by the finite cap-normal BW certificate theorem, let $\alpha_{\lambda_{\widehat{C}}(s)}$ be the automorphism induced by the independently normalized BW-framed cap flow. Then

$$\sigma_t^{\omega_\infty^{\mathrm{geo},C}} = \alpha_{\lambda_{\widehat{C}}(2\pi t)}.$$

If the limit cap algebra happens to be type I, this may be written as

$$K_C = 2\pi B_C + Z_C, \qquad Z_C \in Z(\mathcal{A}(C))_{\mathrm{sa}}.$$

In the generic continuum case, the automorphism identity is the complete statement.

**Proof.** This is the compact paper's finite cap-net BW theorem applied to the extracted geometric cap pair. The theorem's hypotheses include BW framing, support-order faithfulness, cap-family-uniform mixed-GNS convergence, held-out oriented cross-ratio rigidity, independently normalized geometric $2\pi$-KMS convergence, and wrong-normalization controls. QED.

**Alternative derivation via net regularity.** The same modular-covariance property can also be read off from a scaling-limit support map when the net satisfies the outer-regularity / minimal-support condition used later. This clarifies how the geometric labeling of the support-visible extracted limit net is read.

**(NR) Outer regularity / minimal support.** For any operator $O$, the intersection of all connected regions $P$ with $O \in \mathcal{A}(P)$ is again a connected region, denoted $\mathrm{supp}(O)$.

**Proposition 2.7 (Modular covariance from net regularity).** Under (NR), define for any region $R \subset C$

$$f_t^C(R) := \bigcup_{O \in \mathcal{A}(R)} \mathrm{supp}\Big(\sigma_t^{\omega,C}(O)\Big).$$

Then $\sigma_t^{\omega,C}(\mathcal{A}(R)) = \mathcal{A}(f_t^C(R))$, which is exactly the desired modular-covariance property.

**Proof.** Since $\sigma_t^{\omega,C}$ is an automorphism of $\mathcal{A}(C)$, and (NR) allows one to read support from the net labeling, the map $R \mapsto f_t^C(R)$ is well-defined and consistent. QED.

**Null-surface modular structure.** On the same support-visible scaling branch and extracted geometric-subnet branch, the null-surface modular machinery narrows as follows:

- **Fixed-cutoff null-strip bridge.** The null-strip package proves transferred cut-center data, the theorem-local inherited left/right strip-split condition needed for the spatial-collar-type tensor decomposition, exact-or-controlled four-term strip additivity on one inherited strip model, renormalized endpoint-Lipschitz control up to the weak tail generator, and the derived half-sided modular pair whose Borchers–Wiesbrock consequence is an explicit positive null-translation generator on its Stone domain with the affine half-line modular relation; the same half-line family then fixes the generator/charge identification internally.

- **Derived half-sided modular inclusion.** Nested null half-line algebras satisfy half-sided modular inclusion on the declared geometric scaling branch by Corollary 5.2e; Borchers–Wiesbrock then identifies the positive null-translation generator on its Stone domain together with the affine half-line modular relation [17].

- **Weak continuity and finite variation.** The bounded-interval and half-line endpoint-Lipschitz control follow from the local MaxEnt branch and define the weak tail generator at fixed cutoff. The support-visible scaling theorem together with the extracted geometric cap pair supplies the continuum null-generator setting in which that weak-tail data can be matched to the geometric null modular action of the relativistic phase.

**Constraint set specification.** On the local finite-constraint branch, the "correct fixed-cap constraint set" becomes explicit: constraints are the local conserved charges of the symmetries used in the derivation:

1. **Edge/cap label constraints:** fix the distribution of collar-sector labels, equivalently $\langle L_C \rangle$ for each cap size, giving the area term.

2. **Gauge charges:** fix boundary flux or charge operators.

3. **Geometric (conformal) charges:** fix the expectation of the conformal Killing charges that preserve the cap, i.e. the generator $B_C$ or its microscopic lattice approximation.

MaxEnt therefore selects the unique finite-stage invariant state compatible with those conserved charges. The support-visible BW scaling theorem is the continuum statement: if the limit algebra is non-type-I, the geometric modular action on that branch is outer rather than an inner density-matrix identity.

**Focusing status.** QNEC has rigorous QFT proofs in broad settings [26, 28], with hypotheses (null-surface locality and analyticity, modular-flow/causality/defect-OPE structure) that have not been verified on the reconstructed net, and QFC is strictly stronger than QNEC [27]. The Recoverable Generalized Entropy axiom therefore carries the focusing content of this branch itself (§5.10):

- EC + MaxEnt derive $S_{\text{gen}} = S_{\text{bulk}} + \langle L_C \rangle$ (Section 5.4), the object the axiom constrains.

- The null-stress and Einstein theorem branch supplies the kinematic inputs a future internal QNEC/QFC proof must consume; the derivation itself is open.

**Summary.** The CMFP package therefore separates the dependency structure into two layers:

- **Internal branch consequences:** the local-Gibbs form, quasi-local propagation, bounded-interval endpoint-Lipschitz control, and the realized state-side refinement branch along which later transport questions are asked, all from the local finite-constraint MaxEnt branch.

- **Support-visible scaling statement:** the scaling limit emits the support-visible geometric cap pair and ordered cut-pair rigidity holds on it. On that branch the limit algebra may be non-type-I, the modular action may be outer, and the $2\pi$ normalization and later null half-sided-inclusion bridge follow.

The theorem route works on the support-visible quotient. The realized transported cap-local system packages the geometric cap-local test family, the projectively compatible transported marginal family, and the asymptotic transport-equivalence certificate. The regularized transport theorem below replaces the unavailable full-algebra lower spectral floor. Local weak-$*$ extraction and GNS gluing then emit the support-visible scaling-limit cap pair, support-readable modular covariance reads the modular group as a cap-local support map, and ordered cut-pair rigidity collapses the residual cap-preserving conformal freedom to the unique hyperbolic subgroup.

**Proposition 2.6a (Recoverability is not modular geometry: common-floor collapse countermodel).** Exact or asymptotically exact Markov recovery at finite cutoff does not by itself imply a full-algebra lower spectral floor along refinement. In a two-dimensional matrix algebra, let

$$\rho_n = \begin{pmatrix} e^{-n} & 0 \\ 0 & 1-e^{-n} \end{pmatrix}.$$

Each $\rho_n$ is faithful at finite $n$, and tensoring it with any fixed finite exact-Markov collar factor gives a full-rank exact-Markov collar family. Nevertheless $\lambda_{\min}(\rho_n) = e^{-n} \to 0$, so there is no positive lower bound along the refinement family. On the off-diagonal matrix unit $E_{12}$, the modular generator carries the logarithmic gap

$$[-\log \rho_n, E_{12}] = n\, E_{12} + O(1)E_{12},$$

and the unregularized modular transport has no finite common-floor limit on that direction. Thus collar Markovity and finite-stage faithfulness are not enough for the false stronger full-algebra BW lift. The observer-facing theorem uses the support-visible regularized replacement, with the exact-Markov comparison family checked only on fixed collars and made replacement-independent after weak-$*$/GNS extraction.

**Theorem 2.6b (regularized support-visible modular transport).** Fix a local collar model of finite dimension $d_{m,\delta}$. Let $\rho_n$ be the transported physical collar marginal, $\widehat{\rho}_n$ the exact-Markov comparison marginal on the same collar model, and $\Delta_n = \|\rho_n - \widehat{\rho}_n\|_1$. For $a > 0$, set $K_a(\rho) = -\log(\rho + a\mathbf{1})$. For every bounded collar observable $O$ in the support-visible algebra,

$$\left|\operatorname{Tr} \rho_n O\big(K_a(\rho_n) - K_a(\widehat{\rho}_n)\big)\right| \le \|O\| \left( \frac{4\Delta_n}{a} + d_{m,\delta} a + 4\Delta_n |\log a| \right).$$

Consequently, if $a_n \downarrow 0$ is chosen with

$$\Delta_n/a_n \to 0, \qquad d_{m,\delta} a_n \to 0, \qquad \Delta_n |\log a_n| \to 0,$$

then the regularized support-visible modular matrix elements converge on that fixed collar model.

**Proof sketch.** On the spectral interval $[a,\infty)$, the logarithm is operator-Lipschitz at the scale used above by its integral representation. Splitting the comparison into the support above $a$, the $a$-tail, and the trace-distance error gives the displayed bound. The three displayed conditions make the three error terms vanish. QED.

**Definition 2.6c (explicit multiresolution reference branch).** The finite-to-continuum bridge is evaluated on a declared reference branch, not on an unstructured sequence of finite plots. A regulator is

$$r = (m, L, b),$$

where $m$ is the refinement depth, $L$ is the finite support or volume scale, and $b$ is a boundary/phase label. The screen branch uses nested icosahedral subdivisions of $S^2$; Euclidean branches use dyadic mesh $a_m = a_0 2^{-m}$, with an additional temporal spacing $a_t$ when a transfer matrix is present. Each added cell, collar, edge-center, or support-visible channel $j$ carries

$$D_j = \bigoplus_{\alpha \in S_j} M_{d_{j,\alpha}}(\mathbb{C})$$

and a faithful detail state $\tau_j$. At regulator $r$,

$$\widetilde{M}_r = \bigotimes_{j \in \mathcal{J}_r} D_j, \qquad \widetilde{\omega}_r = \bigotimes_{j \in \mathcal{J}_r} \tau_j.$$

A finite-depth local presentation circuit $\Gamma_r : \widetilde{M}_r \to M_r$ changes from multiresolution coordinates to the OPH patch, port, collar, and gauge coordinates. For $r \preceq s$, write

$$\widetilde{M}_s = \widetilde{M}_r \otimes D_{s\setminus r}, \qquad \tau_{s\setminus r} = \bigotimes_{j\in\mathcal{J}_s\setminus\mathcal{J}_r} \tau_j.$$

The canonical refinement and coarse-graining maps are

$$\iota_{rs}(A) = \Gamma_s\Big[\Gamma_r^{-1}(A)\otimes \mathbf{1}\Big],$$

and

$$Q_{sr}(X) = \Gamma_r\Big[(\mathrm{id}\otimes\tau_{s\setminus r})(\Gamma_s^{-1}(X))\Big].$$

The embedded conditional expectation is $E_{sr} = \iota_{rs}Q_{sr}$. The reference state is $\widehat{\omega}_r = \widetilde{\omega}_r \circ \Gamma_r^{-1}$.

**Theorem 2.6d (exact modular-compatible reference tower).** The data

$$(M_r, \widehat{\omega}_r, \iota_{rs}, E_{sr})_{r\preceq s}$$

form a directed tower. The maps $\iota_{rs}$ are unital injective $*$-homomorphisms with $\iota_{st}\iota_{rs} = \iota_{rt}$. The $Q_{sr}$ are faithful state-preserving completely positive coarse-grainings with $Q_{tr} = Q_{sr}Q_{ts}$. The maps $E_{sr}$ are faithful conditional expectations onto $\iota_{rs}(M_r)$, and

$$\widehat{\omega}_s \circ \iota_{rs} = \widehat{\omega}_r, \qquad \widehat{\omega}_r \circ Q_{sr} = \widehat{\omega}_s, \qquad \widehat{\omega}_s \circ E_{sr} = \widehat{\omega}_s.$$

The finite reference modular groups are also exactly compatible:

$$\sigma_t^{\widehat{\omega}_s}(\iota_{rs}A) = \iota_{rs}(\sigma_t^{\widehat{\omega}_r}(A)).$$

This theorem uses no full-algebra spectral floor uniform in $r$; in bare coordinates it is only the identity $A \mapsto A\otimes\mathbf{1}$, partial expectation against faithful detail states, and conjugation by $\Gamma_r, \Gamma_s$.

**Theorem 2.6e (canonical renormalization and correlation Cauchy bound).** Let $\mathfrak{A}$ be the $C^*$-inductive limit with embeddings $j_r : M_r \to \mathfrak{A}$. There is a unique state $\widehat{\omega}$ with $\widehat{\omega}\circ j_r = \widehat{\omega}_r$, and the maps $Q_{sr}$ induce state-preserving conditional expectations

$$\mathbb{E}_r : \mathfrak{A} \to j_r(M_r)$$

such that $\|\mathbb{E}_r(A) - A\| \to 0$ for every $A \in \mathfrak{A}$. The finite-regulator realization of a continuum test observable is

$$\mathcal{R}_r(A) = j_r^{-1}(\mathbb{E}_r A).$$

For $\|A_k\| \le M$,

$$\left|\widehat{\omega}_r\left(\prod_{k=1}^n \mathcal{R}_r(A_k)\right) - \widehat{\omega}\left(\prod_{k=1}^n A_k\right)\right| \le M^{n-1}\sum_{k=1}^n \|A_k - \mathbb{E}_r A_k\|.$$

Thus the operator mixing matrix and additive subtraction at finite cutoff are outputs of $\mathcal{R}_r$, not free fits to the target continuum answer. Independent lattice-spacing, volume, and boundary sweeps must publish the corresponding martingale-tail or physical-error envelopes before finite regulator data are read as continuum evidence.

**Theorem 2.6f (compact-time modular convergence on fixed local algebras).** Let $\rho_{s\downarrow r}^{\mathrm{phys}}$ be the density matrix of the physical state at fine stage $s$, restricted to the embedded fixed local algebra $M_r$, and let

$$\epsilon_{s,r} := \|\rho_{s\downarrow r}^{\mathrm{phys}} - \widehat{\rho}_r\|_1.$$

For a regularizer $\eta_s > 0$, define

$$K_{s\downarrow r}^{(\eta_s)} = -\log(\rho_{s\downarrow r}^{\mathrm{phys}} + \eta_s \mathbf{1}).$$

If $\lambda_r = \lambda_{\min}(\widehat{\rho}_r) > 0$, then for every $A \in M_r$ and every $T < \infty$,

$$\sup_{|t|\le T} \left\| e^{-itK_{s\downarrow r}^{(\eta_s)}} A e^{itK_{s\downarrow r}^{(\eta_s)}} - \sigma_t^{\widehat{\omega}_r}(A) \right\| \le 2T\|A\| \left[ \frac{\epsilon_{s,r}}{\eta_s} + \log\left(1 + \frac{\eta_s}{\lambda_r}\right) \right].$$

Hence $\eta_s \to 0$ and $\epsilon_{s,r}/\eta_s \to 0$ give uniform-on-compact-time convergence on each fixed support-visible local algebra. For a countable cofinal local test tower, pointwise $\epsilon_{s,r} \to 0$ admits a diagonal subsequence and one cutoff schedule. The phrase "finite-stage modular action" below means this transported, regularized local action; it does not mean the unregularized full finite-cap modular group.

**Cap-family uniformity clause.** For application to the finite cap-net BW theorem, the compact-time estimate is required uniformly over the declared finite nondegenerate cap family and the fixed separating support-visible test tower. The operator-norm estimate then implies the quadratic mixed-GNS Cauchy residual because $\|XD\|_{2,\omega} \le \|X\|\|D\|_{2,\omega}$. The reference tower supplies the analytic engine; the finite cap-normal certificate adds the support-order, BW-frame, oriented cross-ratio, geometric $2\pi$-KMS, and wrong-scale controls.

**Theorem 2.6g (finite positive transfer and Lorentzian handoff gate).** On Euclidean repair branches, let $P_{C,r}$ be weighted resampling inside the fibers of the complete repaired visible datum for active collar $C$. Ref. [1] proves that this finite operator is the orthogonal conditional expectation and gives the exact support, equal-fiber-row, and weighted detailed-balance matrix receipt that recognizes it. A concrete active collar repair kernel is identified with $P_{C,r}$ only when its independently derived transition matrix passes that receipt; constructing the matrix from the target formula is not a verification. Choose $c_{C,r} \ge 0$, and set

$$L_r^{\mathrm{rep}} = \sum_C c_{C,r}(I - P_{C,r}), \qquad T_r(a_t) = e^{-a_t L_r^{\mathrm{rep}}}.$$

Then $L_r^{\mathrm{rep}} = L_r^{\mathrm{rep}\,*} \ge 0$, $0 < T_r(a_t) \le I$, $T_r(a_t)\mathbf{1} = \mathbf{1}$, and the stationary Euclidean path measure is reflection positive:

$$\mathbb{E}_r[(\Theta F)F] \ge 0$$

for positive-time cylinder functions $F$. A Lorentzian unitary statement requires this finite positivity plus either exact shell-additive direct-limit semigroup identities or a Mosco / strong-resolvent substitute, including vacuum-projection and intertwiner convergence. Weak-$*$ state convergence alone is not a transfer theorem.

**BW-side closure boundary.** Proposition 2.6a blocks the stronger unregularized full-algebra conclusion from the proved finite-cutoff package. Theorems 2.6c–2.6g supply the explicit reference-tower certificate needed for the finite-to-continuum reading: continuum correlations come from the conditional-expectation renormalization map, compact-time modular group convergence is only for transported regularized physical marginals on fixed support-visible local algebras, and Lorentzian transfer needs reflection positivity or an equivalent positive-transfer certificate. Theorem 4.2 then uses exactly the observer-facing support-visible content to close the BW/geometric cap-pair automorphism statement. Without the multiresolution regulator certificate, finite regulator data remain regulator evidence. A black-box AQFT/BW reconstruction route would require separately verifying conformal-net properties such as locality, additivity, duality, standardness, positive energy, and suitable modular inclusions.

## 5.1 Overlap Consistency and Gluing

The constructive part of overlap consistency is the tree-gluing theorem below. The structural part is the origin of the gluing redundancy itself. On the ordinary or central-defect branch, gauge-as-gluing is the finite-regulator overlap redundancy of local chart presentations: once the overlap algebras are realized in finite-dimensional charts, overlap-consistent rechartings of a cut form a compact unitary transition system. The collar theorem of Section 2.3 should therefore be read with its boundary group $G_\Sigma$ understood as shorthand for this derived compact boundary action, not as a model-specific add-on. On the genuinely noncentral branch, the same weak gluing data are encoded by a compact crossed-module change system, so the fixed-cutoff collar theorem upgrades to a higher-gauge statement rather than failing. The fixed-cutoff topological package is therefore closed on all three branches. A second structural point is UV underdetermination. If each patch Hilbert space is tensored with an inert finite ancillary factor and the observable patch algebras are embedded as $\mathcal{A}(P) \otimes \mathbf{1}$, then the physical observables, collar conditional mutual information, carried Markov errors, and quotient normal forms are unchanged. So the physical UV branch is fixed only modulo gauge or implementation hiding together with such ancillary stabilization, not a unique microscopic presentation; the unique theorem-grade object is the induced family of terminal expectation functionals on the declared physical observable algebras, not a preferred microscopic representative.

### 5.1.1 Constructive gluing on tree covers

**Theorem 3.1 (tree gluing).** Let a rooted tree of patches satisfy a tree-ordered overlap structure and a tripartite factorization $(A_k, B_k, C_k)$ at step $k$. If a target state $\rho^*$ obeys $I(A_k : C_k \mid B_k) \le \varepsilon_k$, then there exist recovery maps $\mathcal{R}_k$ such that

$$\|\rho^*_{A_k B_k C_k} - (\mathrm{id}_{A_k} \otimes \mathcal{R}_k)(\rho^*_{A_k B_k})\|_1 \le \delta_k,$$

with

$$\delta_k = 2\sqrt{1 - e^{-\varepsilon_k}} \le 2\sqrt{\varepsilon_k},$$

The iteratively glued state $\hat{\rho}$ satisfies

$$\|\hat{\rho} - \rho^*\|_1 \le \min\left(2, \sum_{k=2}^{n} \delta_k\right).$$

**Proof.** Induct on $k$. The recovery error contracts under CPTP maps, so the errors add. QED.

### 5.1.2 Gauge-as-gluing and loops

At finite regulator scale, the fixed-cutoff gauge-as-gluing package identifies the overlap-consistency redundancy of local chart presentations. Choose finite-dimensional local presentations of the overlap algebras on a connected cut $\Sigma$. Because the overlap algebras are matrix algebras, any overlap-consistent change of chart is inner and is implemented by a unitary on the cut Hilbert space. Fixing a reference chart, the compact closure of the subgroup generated by all such recharting unitaries is the boundary gluing group $K_\Sigma$; before fixing the reference chart, the same data form a compact unitary groupoid.

**Proposition 3.2a (Derived gauge-as-gluing at finite regulator).** Let a finite regulator chart be chosen for the patches meeting along a connected interface $\Sigma$. Then overlap consistency determines a compact unitary transition system on the cut data. On the ordinary or central-defect branch, this transition system reduces to a compact boundary group $K_\Sigma$, and when the triple-overlap defect is central its projective composition law lifts to a genuine action of a compact central extension $\widehat{K}_\Sigma$. Gauge is therefore the overlap redundancy itself, not an extra primitive.

**Proof.** On a finite-dimensional matrix algebra every $^*$-automorphism is inner, so each overlap-consistent recharting is conjugation by a unitary on the cut Hilbert space. The subgroup generated by those unitaries has compact closure inside a finite-dimensional unitary group. If triple-overlap defects are central, the resulting projective composition law lifts to a central extension. QED.

**Lemma 3.2b (trees vs loops).** If the patch adjacency graph is a tree, one can choose local charts $h_i$ so that the overlap labels satisfy $g_{ij} = h_i^{-1} h_j$ on all edges. If loops exist, the loop holonomy

$$H(\gamma) = g_{i_1 i_2} g_{i_2 i_3} \cdots g_{i_n i_1}$$

is invariant under local frame changes. Nontrivial holonomy is the obstruction to global trivialization. QED.

**Corollary 3.2c (Collar consequence on the ordinary or central-defect branch).** Let $B_\delta = B_L \cup B_R$ be a collar around a cap boundary $\Sigma$, and set $\widehat{K}_\Sigma = K_\Sigma$ on the ordinary branch. Then the EC theorem of Section 2.3 has the derived interpretation

$$H_{B_\delta} = (\tilde{\mathcal{H}}_{B_L} \otimes \tilde{\mathcal{H}}_{B_R})^{\widehat{K}_\Sigma} \cong \bigoplus_\alpha (H_{b_L^\alpha} \otimes H_{b_R^\alpha}),$$

with

$$Z(A_{\mathrm{EC}}(B_\delta)) = \bigoplus_\alpha \mathbb{C}\,\mathbf{1}_\alpha.$$

The right half-collar carries the contragredient action because it sees the inverse transport across the same cut. Exact Markov normal forms used later require the additional state hypothesis $I_\omega(A_\delta : D_\delta \mid B_\delta) = 0$ *together with* the Markov-split alignment hypothesis of Section 2.3, or the explicitly stated idealized recoverability limit supplying both; the block decomposition itself is kinematic and follows from the derived boundary action.

**Proof sketch.** Decompose the left boundary data into irreps $(V_\alpha \otimes H_{b_L^\alpha})$ of $\widehat{K}_\Sigma$ and the right boundary data into the dual modules $(V_\beta^* \otimes H_{b_R^\beta})$. Then Schur's lemma leaves a singlet only when $\alpha = \beta$, producing the displayed direct sum. QED.

### 5.1.3 Loop obstruction class (central defect)

On the central-defect subbranch, identify the overlap centers with one fixed abelian unitary coefficient group $Z_\Sigma$ and assume overlap transport acts trivially on it. Choose unitary overlap implementers normalized by $U_{ji} = U_{ij}^{-1}$ and define central defects $z_{ijk}$ at the implementer level by

$$\Omega_{ijk} := U_{ij} U_{jk} U_{ki} = z_{ijk} \in Z_\Sigma,$$

or equivalently $U_{ij} U_{jk} = z_{ijk} U_{ik}$. This implementer-level relation records the projective multiplier; the automorphism-level expression $\mathrm{Ad}(z_{ijk})$ alone would be the identity and would not determine $z_{ijk}$.

Then $\{z_{ijk}\}$ is an untwisted Čech 2-cocycle with coefficients in $Z_\Sigma$, and its cohomology class $[z]$ is gauge invariant. Central defects do not obstruct the collar block decomposition: they only replace $K_\Sigma$ by its central extension $\widehat{K}_\Sigma$. The condition $[z] = 0$ is exactly the strictifiability of the projective triangle defect. It is not by itself global path independence: at least one resulting strict edge 1-cocycle must also have trivial represented holonomy around every closed overlap loop. With a nontrivial action on varying centers, the corresponding statement uses the twisted/local-system Čech differential instead. (A full proof for the fixed-coefficient subbranch appears in Section 6.4 below, in the algebra-net language.)

#### 5.1.4 Non-central obstruction (2-group cocycle)

When defects are not central, the natural coefficient data is a crossed module $(H \to G)$ with an action of $G$ on $H$ by conjugation. Here $G$ is the reconstructed gauge group, and $H$ is the unitary group acting on edge multiplicity spaces, with boundary map $\partial : H \to G$.

A crossed module is a homomorphism $\partial : H \to G$ together with an action of $G$ on $H$ such that

$$\partial(g \triangleright h) = g\,\partial(h)\,g^{-1}, \qquad \partial(h) \triangleright h' = hh'h^{-1}.$$

On a good cover $\{P_i\}$, a weakly coherent gluing is encoded by:

$$g_{ij} : P_{ij} \to G, \qquad h_{ijk} : P_{ijk} \to H,$$

obeying the 2-cocycle conditions

$$g_{ij}g_{jk} = \partial(h_{ijk})g_{ik},$$

and on quadruple overlaps,

$$h_{jkl}h_{ijl} = (g_{ij} \triangleright h_{ikl})h_{ijk}.$$

Gauge changes act by 1- and 2-cochains in the standard way for crossed-module cohomology.

Write

$$q_\Sigma = [(g, h)] \in \check{H}^2(N_\Sigma, H_\Sigma \to G_\Sigma)$$

for the full crossed-module gluing orbit. It classifies the equivalence orbit of the pair $(g, h)$ but does not in general determine a unique ordinary $G_\Sigma$-valued 1-cocycle class after strictification, because the map below need not be injective. Let

$$\begin{aligned} \iota_\Sigma : \check{H}^1(N_\Sigma, G_\Sigma) &\longrightarrow \check{H}^2(N_\Sigma, H_\Sigma \to G_\Sigma), \\ [g] &\longmapsto [(g, 1)] \end{aligned}$$

be the natural strict-locus map. Define the separate $\{0, 1\}$-valued strictification obstruction by

$$\begin{aligned} o^{(2)}_\Sigma(g, h) = 0 \quad &\Longleftrightarrow \quad q_\Sigma \in \mathrm{im}(\iota_\Sigma) \\ &\Longleftrightarrow \quad [(g, h)] \text{ has a representative } (g^{\mathrm{str}}, 1) \\ &\qquad\qquad \text{with } g^{\mathrm{str}}_{ij} g^{\mathrm{str}}_{jk} = g^{\mathrm{str}}_{ik}. \end{aligned}$$

and set $o^{(2)}_\Sigma(g, h) = 1$ otherwise.

**Theorem 3.4 (non-central higher-defect strictification).** The higher associator defect of $(g_{ij}, h_{ijk})$ is removable iff $o^{(2)}_\Sigma(g, h) = 0$. In that case a higher-gauge change gives $h_{ijk} = 1$ and a genuine $G$-valued edge 1-cocycle. Strict endpoint-only transport additionally requires that at least one allowed strict representative have trivial represented holonomy. The full orbit $q_\Sigma$ need not determine a unique ordinary $H^1$ class.

**Proof sketch.** Removing the higher associator corresponds to $h_{ijk} = 1$ and $g_{ij}g_{jk} = g_{ik}$. Crossed-module coboundaries preserve the full orbit, so this strictification exists exactly when that orbit meets the image of the ordinary $G$-valued 1-cocycle locus, which is the definition of $o^{(2)}_\Sigma = 0$. The map $\iota_\Sigma$ need not be injective: $H$-valued edge changes can relate strict representatives and alter their ordinary holonomy through $\partial$. Thus the invariant residual test is whether at least one allowed strict representative has trivial represented holonomy, as in Theorem 3.4b. QED.

The central-defect case is the abelian truncation with $H$ central and trivial action, which reduces to Section 3.3. A genuinely noncentral class is the point where the fixed-cutoff collar theorem upgrades from the ordinary-group package to the higher-gauge one below.

**Corollary 3.4a (Fixed-cutoff higher-gauge EC and transportability).** For a finite regulator chart on a connected cut $\Sigma$, the genuinely noncentral branch admits a compact crossed-module change system

$$\mathcal{T}_\Sigma = C^1(N_\Sigma, H_\Sigma) \rtimes C^0(N_\Sigma, G_\Sigma).$$

The physical higher-gauge collar is

$$\mathcal{H}^{2g}_B = (\widetilde{\mathcal{H}}_{B_L} \otimes \widetilde{\mathcal{H}}_{B_R})^{\mathcal{T}_\Sigma} \cong \bigoplus_\lambda (\mathcal{H}_{b^\lambda_L} \otimes \mathcal{H}_{b^\lambda_R}),$$

with

$$Z(\mathcal{A}_{2g}(B)) = \bigoplus_\lambda \mathbb{C}\, \mathbf{1}_\lambda,$$

and the full orbit $q_\Sigma$ is invariant under local rechartings and classifies the fixed-cutoff genuinely noncentral gluing datum. Its separate invariant property $o^{(2)}_\Sigma = 0$ holds iff the higher associator is removable. The resulting strict representatives can have different ordinary $G_\Sigma$-valued 1-holonomies, so transportability is the separate existential test of Theorem 3.4b rather than a uniquely defined holonomy attached to $q_\Sigma$.

**Proof sketch.** Pair-overlap rechartings are inner, while triple-overlap associators strictify to compact crossed-module data. Finite-dimensional unitary $\mathcal{T}_\Sigma$-modules decompose semisimply, and Schur matching leaves only diagonal left/right blocks. Crossed-module cohomology classifies the higher defect; Theorem 3.4b separately tests the ordinary loop holonomy that remains after strictification. QED.

**Theorem 3.4b (TransportabilityFromOverlapGluing).** Fix a connected cut $\Sigma$ and a finite regulator charting nerve $N_\Sigma$. Transport of a collar charge is defined by the path composite in the overlap recharting groupoid: along $p = (i_0 \to \cdots \to i_m)$, the charge block is moved by $U_p = U_{i_{m-1}i_m} \cdots U_{i_0 i_1}$. This construction uses the overlap unitary transition system itself, not an added DHR transportability assumption.

After a central or crossed-module strictification, let $U^{\mathrm{str}}$ denote a resulting genuine edge 1-cocycle. Write $\mathcal{C}_\alpha$ for the full transported collar-charge block, including its boundary carrier and multiplicity/intertwiner data, and define

$$\mathrm{Hol}_{\alpha,U^{\mathrm{str}}} : \pi_1(|N_\Sigma|, i_*) \longrightarrow U(\mathcal{C}_\alpha), \qquad [\gamma] \longmapsto U^{\mathrm{str}}_\gamma\big|_{\mathcal{C}_\alpha}.$$

On the ordinary branch, transport is strictly path-independent iff every closed overlap loop has trivial holonomy on the collar-sector block. On the central branch, elementary triangle moves accumulate the central cocycle $z_{ijk}$. Strict path-independent transport exists iff $[z]_\Sigma = 0$ and there is a central 1-cochain strictification for which $\mathrm{Hol}_{\alpha,U^{\mathrm{str}}} = 1$. The first condition removes the projective triangle defect; the second removes the residual ordinary loop action. If $[z]_\Sigma = 0$ but the second condition fails, the lifts form a genuine 1-cocycle and transport is path dependent.

On the genuinely noncentral branch, path comparisons are $H_\Sigma$-valued 2-morphisms in the crossed module $H_\Sigma \to G_\Sigma$. Strict ordinary transport exists iff $o^{(2)}_\Sigma = 0$ and at least one crossed-module strictification gives a genuine $G_\Sigma$-valued 1-cocycle with trivial represented holonomy on $\mathcal{C}_\alpha$.

If $o_\Sigma^{(2)} \neq 0$, the sector has higher transport rather than ordinary compact-group DR transport. If $o_\Sigma^{(2)} = 0$ but every allowed strict representative has nontrivial represented $G_\Sigma$-holonomy, the higher associator has strictified but ordinary path dependence remains. The full orbit $q_\Sigma = [(g,h)]$ need not determine a unique ordinary $H^1$ class and is not itself the strictification test. Thus "zero obstruction" below means the combined associator-strictification and trivial-holonomy criterion, not vanishing of triangle or higher-associator data alone. QED.

**Triangle-free cycle check.** Let $N_\Sigma = C_n$, $n \geq 4$, have no filled 2-simplices. On $\mathcal{C}_\alpha = \mathbb{C}^2$, put identity transport on every oriented cycle edge except one, where the sector action is the non-scalar unitary $V = \mathrm{diag}(-1, 1)$, and use inverse transports on reversed edges. All central 2-data and higher associator data are vacuously trivial, so $[z]_\Sigma = 0$ and $o_\Sigma^{(2)} = 0$. Its ordered cycle holonomy is $V \neq 1$, which no central rephasing removes. For a genuinely noncentral instance, take the crossed module

$$\mathrm{SU}(2) \hookrightarrow \mathrm{U}(2)$$

with conjugation action and the standard action on $\mathcal{C}_\alpha$. The band $\mathrm{U}(2)/\mathrm{SU}(2) \cong \mathrm{U}(1)$ is detected by the determinant, and the cycle has band holonomy $\det V = -1$. An SU(2)-valued edge change has determinant one and a vertex-frame change only conjugates the based loop, so no allowed strict representative has trivial holonomy. The two routes between the endpoints of the distinguished edge differ, and the combined criterion rejects both the central and noncentral assignments.

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## 5.2 Modular Flow and Lorentz Kinematics

### 5.2.1 Modular additivity in the Markov collar limit

Consider a collar tripartition $A : B : D$ around a cap boundary, with the EC decomposition of Section 2.3 understood. Define, for a faithful reference state $\omega$,

$$\Delta K(\omega) := K_{ABD}(\omega) - K_{AB}(\omega) - K_{BD}(\omega) + K_B(\omega).$$

Exact modular additivity is not a consequence of EC alone. It is available only on the exact Markov set, or along a controlled fixed-cutoff family that approaches that set on one fixed collar model. Three quantities must therefore be kept separate:

1. the raw collar conditional mutual information $I(A : D \mid B)$;
2. the constructive Fawzi–Renner comparison error

$$r_{\mathrm{FR}}(\varepsilon) := 2\sqrt{1 - e^{-\varepsilon}} \leq 2\sqrt{\varepsilon};$$

3. the fixed-collar exact-Markov replacement modulus

$$\delta^{\mathrm{M}}_{A:B:D}(\varepsilon) := \sup\left\{ \inf_{\sigma \in \mathfrak{M}_{A:B:D}} \|\rho - \sigma\|_1 : I(A : D \mid B)_\rho \leq \varepsilon \right\},$$

where

$$\mathfrak{M}_{A:B:D} := \{\sigma_{ABD} : I(A : D \mid B)_\sigma = 0\}.$$

**Lemma 4.1a (EC-aligned exact Markov implies exact additivity up to a central term).** If $I(A : D \mid B)_\omega = 0$ and $\omega$ satisfies the Markov-split alignment hypothesis of Section 2.3, then the state takes the EC-aligned HJPW block form

$$\omega_{ABD} = \bigoplus_\alpha p_\alpha \, \omega^{(\alpha)}_{Ab^\alpha_L} \otimes \omega^{(\alpha)}_{b^\alpha_R D},$$

and $\Delta K(\omega)$ is central. On the canonical HJPW block model one may take

$$\Delta K(\omega) = 0.$$

Equivalently, there exists a central operator $K_{\partial,ABD}(\omega) \in Z(\mathcal{A}(B))$ such that

$$K_{ABD}(\omega) = K_{AB}(\omega) + K_{BD}(\omega) - K_B(\omega) + K_{\partial,ABD}(\omega).$$

**Proof.** By alignment, the HJPW blocks are the EC blocks. On each block the modular Hamiltonians of $ABD$, $AB$, $BD$, and $B$ are the logarithms of tensor-product states with the same classical block weight $p_\alpha$. The blockwise logarithms therefore cancel exactly in the combination $K_{ABD} - K_{AB} - K_{BD} + K_B$. If one keeps the blockwise endpoint-label bookkeeping explicit rather than absorbing it into the canonical block identification, the remainder is a direct sum of block constants, hence central in $Z(\mathcal{A}(B))$. Without alignment, exact Markovity cancels the four logarithms on the state's own HJPW blocks, but the resulting block constants need not lie in $Z(\mathcal{A}(B))$, so the central form used here is unavailable. QED.

**Proposition 4.1b (Controlled exact-Markov replacement on a fixed collar).** Because the state space is compact and conditional mutual information is continuous in finite dimension,

$$\delta^{\mathrm{M}}_{A:B:D}(\varepsilon) \longrightarrow 0 \qquad (\varepsilon \downarrow 0).$$

Thus small collar CMI converges to the exact HJPW normal form only in this controlled fixed-cutoff sense; the manuscript does not claim a dimension-free one-shot trace-norm bound from small $I(A : D \mid B)$ directly to an exact Markov state. The limit points are exact Markov states in the CMI-zero sense; identifying their factorization with the preselected EC factors is the Markov-split alignment hypothesis of Section 2.3, which is carried separately and is not produced by small CMI (the Bell-pair counterexample of Section 2.3 has zero CMI at fixed distance from every EC-aligned state).

**Theorem 4.1b' (Finite-collar Markov replacement stability).** On one fixed faithful finite-dimensional collar model, the qualitative modulus above can be made collar-local and quantitative. If all relevant marginals stay above a chosen floor $\lambda_* > 0$, then there are constants $C_{A:B:D,\lambda_*} > 0$ and $\theta_{A:B:D,\lambda_*} > 0$, depending only on that collar model and floor, such that

$$d_{\mathrm{M}}(\rho) \leq C_{A:B:D,\lambda_*} \, I(A : D \mid B)_\rho^{\theta_{A:B:D,\lambda_*}}.$$

This is a fixed-collar Lojasiewicz-type rate for the analytic function $I(A : D \mid B)$ on the faithful state manifold. It is not a dimension-free stability theorem for arbitrary tripartite quantum systems.

**Corollary 4.1c (Carried collar defect operator).** Let $\omega_\varepsilon$ satisfy $I(A : D \mid B)_{\omega_\varepsilon} \leq \varepsilon$, and choose $\sigma_\varepsilon \in \mathfrak{M}_{A:B:D}$ with

$$\|\omega_\varepsilon - \sigma_\varepsilon\|_1 \leq \delta^{\mathrm{M}}_{A:B:D}(\varepsilon).$$

Define the carried defect operator

$$\mathfrak{D}_{A:B:D}(\omega_\varepsilon, \sigma_\varepsilon) := \Delta K(\omega_\varepsilon) - \Delta K(\sigma_\varepsilon).$$

Then

$$K_{ABD}(\omega_\varepsilon) = K_{AB}(\omega_\varepsilon) + K_{BD}(\omega_\varepsilon) - K_B(\omega_\varepsilon) + K_{\partial,ABD}(\sigma_\varepsilon) + \mathfrak{D}_{A:B:D}(\omega_\varepsilon, \sigma_\varepsilon),$$

where $K_{\partial,ABD}(\sigma_\varepsilon) = \Delta K(\sigma_\varepsilon)$ is central. For every bounded observable $X$ supported on $A \cup B$,

$$|\mathrm{Tr}[X(\omega_\varepsilon - \sigma_\varepsilon)]| \leq \|X\|_\infty \, \delta^{\mathrm{M}}_{A:B:D}(\varepsilon).$$

If the relevant marginals of $\omega_\varepsilon$ and $\sigma_\varepsilon$ are uniformly faithful with lower spectral bound $\lambda_* > 0$, then

$$\|\mathfrak{D}_{A:B:D}(\omega_\varepsilon, \sigma_\varepsilon)\|_\infty \leq 4\lambda_*^{-1} \, \delta^{\mathrm{M}}_{A:B:D}(\varepsilon).$$

Independently, Fawzi–Renner recovery supplies a constructive comparison state $\omega_\varepsilon^{\mathrm{rec}}$ with

$$\|\omega_\varepsilon - \omega_\varepsilon^{\mathrm{rec}}\|_1 \leq r_{\mathrm{FR}}(\varepsilon), \qquad r_{\mathrm{FR}}(\varepsilon) := 2\sqrt{1 - e^{-\varepsilon}} \leq 2\sqrt{\varepsilon}.$$

This $O(\varepsilon^{1/2})$ term is the finite-stage observable error. The modulus $\delta^{\mathrm{M}}_{A:B:D}(\varepsilon)$ is instead the fixed-collar quantity that justifies replacing the physical state by an exact Markov normal form in the later geometric modular arguments.

Accordingly, the manuscript's later exact collar formulas take one of two forms:

1. literal exactness when the reference state is exact Markov on the relevant collar; or
2. a controlled collar family for which $\delta^{\mathrm{M}}_{A:B:D}(\varepsilon_\delta) \to 0$, with the constructive Fawzi–Renner remainder and the carried operator defect $\mathfrak{D}_{A:B:D}$ kept explicit at finite stage.

**Theorem 4.1d (Finite-stage modular-defect propagation and dimension quarantine).** For any fixed downstream branch calculation that uses finitely many collar or strip modular-additivity identities, replacing each exact identity by its controlled finite-stage form changes every bounded downstream modular observable $O$ by at most

$$\mathcal{P}_O\Big(\{r_{\mathrm{FR}}(\varepsilon_j)\}, \{\delta^{\mathrm{M}}_j(\varepsilon_j)\}, \{\eta^{\mathrm{reg}}_j\}\Big),$$

where $\mathcal{P}_O \to 0$ as all listed errors vanish. The polynomial-continuity modulus depends only on the declared fixed collar models, support-visible dimensions, bounded observable class, faithful floors or regularization schedules, and bounded modular-time intervals. No BW/Lorentz, null-modular, or Einstein scaling step uses a dimension-free trace-norm stability theorem from small CMI directly to exact Markov normal form.

### 5.2.2 Theorem: $\mathrm{BW}_{S^2}$ on the extracted geometric subnet

The collar analysis proves a fixed-cutoff statement on the finite type-I regulator net: the reduced cap state has a literal density matrix, its modular Hamiltonian exists, and its nonadditive part is confined to a shrinking collar up to carried errors. The Lorentz claim is therefore not a fixed-cutoff matrix-algebra identity and not the slogan "finite cells imply Lorentz invariance." Its target is the support-visible refinement-limit geometric cap pair $(\mathcal{A}^{\mathrm{geo,sv}}_\infty(C), \omega^{\mathrm{geo},C}_\infty)$. The branch theorem is the conditional implication

$$\mathsf{FiniteCapBWCertificate} \Longrightarrow \sigma_t^{\omega_\infty^{\mathrm{geo},C}} = \alpha_{\lambda_{\widehat{C}}(2\pi t)}.$$

The finite cap-normal certificate consists of cap-normal density and nondegeneracy, BW framing, support-order faithfulness, cap-family-uniform mixed-GNS modular convergence, support covariance, held-out oriented cross-ratio convergence, independently normalized geometric $2\pi$-KMS convergence, and wrong-normalization separation. The support-visible theorem below uses regularized modular transport, projectively compatible exact-Markov replacement on fixed collars, weak-$*$/GNS extraction, support-readable modular covariance, and that finite cap-normal certificate; it does not require a type-I continuum algebra or a full-algebra unregularized common spectral floor. Bare finite consensus does not imply the certificate without the separate Einstein branch-entry production theorem.

Fix a cap $C \subset S^2$ and a shrinking collar family $(A_\delta, B_\delta, D_\delta)$ around $\partial C$. Write

$$\varepsilon_\delta := I(A_\delta : D_\delta \mid B_\delta)_\omega, \qquad r_{\mathrm{FR}}(\varepsilon_\delta) := 2\sqrt{1 - e^{-\varepsilon_\delta}} \le 2\sqrt{\varepsilon_\delta},$$

and, on each fixed faithful collar model,

$$\eta^{\mathrm{M}}_\delta := 4\lambda_*^{-1}\, \delta^{\mathrm{M}}_{A_\delta:B_\delta:D_\delta}(\varepsilon_\delta),$$

where $\lambda_* > 0$ is the lower spectral bound for comparing modular Hamiltonians. All exact collar formulas in this subsection are therefore to be read either literally at exact Markovity or along a controlled collar family satisfying

$$\delta^{\mathrm{M}}_{A_\delta:B_\delta:D_\delta}(\varepsilon_\delta) \to 0 \qquad (\delta \downarrow 0),$$

with $r_{\mathrm{FR}}(\varepsilon_\delta)$ and $\eta^{\mathrm{M}}_\delta$ carried explicitly.

For each cap $C$, let $\lambda_C(s) \subset \mathrm{Conf}^+(S^2)$ denote the standard cap-preserving conformal one-parameter subgroup, normalized so that the null blow-up near a smooth cut acts by $v \mapsto e^{-s}v$, and let $\alpha_{\lambda_C(s)}$ denote the induced automorphism of the scaling-limit cap net.

**Theorem 4.2 (Finite cap-net support-visible $\mathrm{BW}_{S^2}$ scaling theorem).** Let $\widehat{C} = (C, n_C, p_C^-, p_C^+)$ be a nondegenerate BW-framed cap. For every OPH-realized observer-supporting refinement branch satisfying the five OPH axioms, the derived fixed-cutoff regulator/collar/consensus package, and the finite cap-normal BW certificate above, the support-visible geometric cap net admits a weak-$*$/GNS scaling-limit cap pair $(\mathcal{A}^{\mathrm{geo,sv}}_\infty(C), \omega^{\mathrm{geo},C}_\infty)$. Then the scaling-limit modular automorphism group is

$$\sigma_t^{\omega_\infty^{\mathrm{geo},C}} = \alpha_{\lambda_{\widehat{C}}(2\pi t)}.$$

At finite regulator stage the nonadditive part of

$$K_C^{(\delta)} := -\log \rho_C^{(\delta)}$$

is confined to the shrinking collar up to the carried errors $r_{\mathrm{FR}}(\varepsilon_\delta)$, the fixed-collar replacement modulus $\delta^{\mathrm{M}}$, and the regularized support-visible modular transport bound. The support-visible modular limit is independent of the exact-Markov replacement sequence after projective compatibility and GNS quotienting, and support-readable modular covariance turns the limiting modular group into a cap-local support map before the certificate's framed-cap and oriented cross-ratio rigidity clauses are applied. No separate cap-isotropy/SO(2)-equivariance selector, finite-cell Lorentz-invariance premise, or full-algebra unregularized common floor is used. Record/pointer and interface-inert auxiliary registers are outside the extracted geometric subnet. If the scaling-limit cap algebra is type I, the same automorphism identity may be represented by

$$K_C = 2\pi B_C + Z_C, \qquad Z_C \in Z(\mathcal{A}(C))_{\mathrm{sa}};$$

otherwise the automorphism identity is the full statement and the action is outer. Consequently, on a branch carrying the multiresolution reference-tower certificate of Theorems 2.6c–2.6f, the transported regularized modular action on every fixed support-visible geometric local algebra converges uniformly on compact modular-time intervals to the geometric cap-dilation action. Without that certificate, this theorem identifies only the scaling-limit cap-pair modular automorphism and does not assert compact-time convergence of finite-regulator modular groups.

**Proof.** Markov locality localizes the modular defect to the shrinking collar, with carried finite-stage errors $r_{\mathrm{FR}}(\varepsilon_\delta)$ and $\delta^{\mathrm{M}}$. Proposition 2.6a shows why a full-algebra unregularized floor is unavailable in general. Theorem 2.6b supplies the matrix-element replacement on fixed collar models, and Theorem 2.6f supplies the stronger compact-time group estimate only for transported regularized local actions with the stated reference-tower certificate. Projective replacement compatibility and the support-visible convergence audit ensure that the emitted modular limit is independent of the chosen exact-Markov comparison family. Support-visible cap-pair extraction on the local GNS support quotient uses weak-$*$ compactness of finite-stage state spaces and the consensus refinement mechanism to supply local limit states, then applies GNS construction to obtain the scaling-limit cap pair. Support-readable modular covariance reads the modular group as a cap-local support map. The finite cap-normal certificate supplies support-order faithfulness, BW framing, held-out oriented cross-ratio rigidity, and compact-time equicontinuity, so the support flow is the framed cap-preserving conformal flow up to normalization:

$$\sigma_t^{\omega_\infty^{\mathrm{geo},C}} = \alpha_{\lambda_C(\kappa_C t)}$$

for some $\kappa_C > 0$, with the certificate-selected frame understood in $\lambda_C$. The certificate then passes the independently normalized geometric $2\pi$-KMS condition to $s \mapsto \alpha_{\lambda_{\widehat{C}}(s)}$. KMS uniqueness, together with wrong-$\beta$ separation and nontriviality, fixes $\kappa_C = 2\pi$. Thus

$$\sigma_t^{\omega_\infty^{\mathrm{geo},C}} = \alpha_{\lambda_{\widehat{C}}(2\pi t)}.$$

If the limit algebra is type I, this automorphism identity is represented by $K_C = 2\pi B_C + Z_C$ with $Z_C$ central; otherwise the automorphism identity is the full statement and the action is outer. QED.

**Definition 4.2a (BW-branch observer-relative time reading).** On the branch satisfying the hypotheses of Theorem 4.2, OPH uses the modular automorphism parameter $t$ of the extracted cap pair $(\mathcal{A}_\infty^{\mathrm{geo,sv}}(C), \omega_\infty^{\mathrm{geo},C})$ as that cap observer's relative time parameter. This is a declared BW-branch reading of the geometric modular flow derived in Theorem 4.2; it is not an additional proof that arbitrary operational clocks, global time, or the full problem of time follow from the axioms.

**Scope note.** The theorem fixes the $2\pi$ normalization and the carried refinement limit on the support-visible extracted geometric cap pair under the finite cap-normal certificate. The UV-side scaffold is the realized transported geometric cap-local system, regularized support-visible modular transport on fixed local collars, support-visible cap-pair extraction on the local GNS support quotient, support-readable modular covariance, BW framing, oriented cross-ratio rigidity, and geometric KMS normalization. No separate cap-isotropy input is used, and the conformal support-map clause is the explicit scaling-limit branch condition rather than a finite-regulator Lorentz premise. Compact-time convergence of finite-regulator modular groups is available only through the transported-local certificate of Theorem 2.6f; a black-box conformal-net BW reconstruction would be an additional validation route, not a hidden premise.

### 5.2.3 Theorem: cap-normal $H^3$ reconstruction

**Theorem 4.3 (Cap-normal $H^3$ reconstruction, imported from the compact proof).** On the support-visible round-cap BW branch, write the celestial null section as

$$q(\Omega) = (1, \Omega), \qquad \Omega \in S^2.$$

For a nondegenerate oriented round cap

$$C(\mathbf{c},\alpha)=\{\Omega:\mathbf{c}\cdot\Omega\geq\cos\alpha\},\qquad 0<\alpha<\pi,$$

define

$$n_C=(\cot\alpha,\csc\alpha\,\mathbf{c}).$$

Then, for the time-first Lorentz form $\eta=-dt^2+d\mathbf{x}^2$,

$$\eta(n_C,n_C)=1,\qquad \eta(n_C,q(\Omega))=\frac{\mathbf{c}\cdot\Omega-\cos\alpha}{\sin\alpha}.$$

Thus $\eta(n_C,q(\Omega))=0$ exactly on $\partial C$, with positive sign in the cap interior and negative sign outside. For $g\in\mathrm{Conf}^+(S^2)$ with Lorentz representative $\Lambda_g$,

$$\Lambda_g q(\Omega)=\omega_g(\Omega)q(g\Omega),\qquad n_{gC}=\Lambda_g n_C.$$

Consequently

$$\mathrm{Conf}^+(S^2)\cong\mathrm{PSL}(2,\mathbb{C})\cong\mathrm{SO}^+(3,1),$$

oriented round caps form the de Sitter normal space

$$\mathrm{Cap}^{\mathrm{or}}_{\mathrm{round}}(S^2)\cong dS_3^{\mathrm{cap}}\cong\mathrm{SO}^+(3,1)/\mathrm{SO}^+(2,1),$$

and future unit timelike observer frames form the distinct hyperbolic space

$$H^3\simeq\mathrm{SO}^+(3,1)/\mathrm{SO}(3),\qquad \dim H^3=3.$$

Each cap normal determines the geodesic half-space

$$\mathcal{H}_C^+=\{u\in H^3:\eta(n_C,u)\geq 0\},$$

not a preferred observer point.

**Proof.** This is the cap-normal reconstruction theorem in the compact paper, which supplies the projective-null-cone model, the cap-normal bijection, Lorentz equivariance, the $H^3$ observer-frame hyperboloid, the ideal-boundary identification, and the cap/half-space duality. The BW theorem supplies the cap-preserving conformal subgroup and its $2\pi$ modular normalization; the cap-normal theorem supplies the explicit projective Lorentz representation and does not rederive BW from bare finite consensus. QED.

**Corollary 4.3a (Frame-space boundary).** The theorem proves the exact canonical Lorentz/$H^3$ observer-frame hyperboloid on the stated branch. Its points are future unit timelike frames, not event positions. It does not populate an event base, construct a chart-blind neutral bulk, derive a physical curvature radius $R_H$, produce stress-energy, or close the Einstein branch-entry umbrella. Record-conditioned cap responses can estimate frame data under the certificate imported next; event location, object families, neutral bulk, scale, stress, and Einstein branch entry are separate receipts. Theorems 4.3c–4.3h below import the compact paper's geometry-producer, null-net-standardness, event-manifold, stress, entropy-bridge, and composed branch-entry packets. Their composition is conditional on one source-derived common-domain tower with certified cofinal tails, universal coupling, a vacuum reference, and independent physical scale readouts. Construction and certification of such a tower are work in progress.

**Theorem 4.3b (Conditional record-conditioned $H^3$ frame estimate).** Fix a quotient-visible record token $i$ and clock slice $t$. On the frame space of Corollary 4.3, suppose its record-conditioned modular cap responses

$$R_i(C,t,O) = \omega_{i,O}\Big(\sigma_t^{C,O}(M_{C,0,O})\Big), \qquad \omega_{i,O}(A) = \frac{\omega_O(P_{i,O}AP_{i,O})}{\omega_O(P_{i,O})},$$

admit the calibrated frame-local factorization

$$y_{i,j} = F_j(X_i(t)) + e_{i,j}, \qquad F_j(X) = \psi_j\left(\frac{\eta(X,n_j)}{R_H}\right),$$

on a compact frame region $\Omega \subset H^3_{R_H}$. Assume the finite response map has quantitative observability

$$\alpha \bar{d}(X,Y) \le |F(X) - F(Y)|_W \le L\bar{d}(X,Y), \qquad \alpha > 0,$$

and the total error obeys $|e_i|_W \le \sigma_i$. If $\mathcal{N}_\varepsilon$ is an $\varepsilon$-net of $\Omega$ and $\widehat{X}_i(t)$ is a residual minimizer up to tolerance $\tau_i$, then the conditioned frame support is

$$S_i(t) = B_H\left(\widehat{X}_i(t), R_H\left[\frac{L}{\alpha}\varepsilon + \frac{2}{\alpha}\sigma_i + \frac{1}{\alpha}\tau_i\right]\right).$$

Exact continuous data have a unique minimizer. Finite noisy data produce the displayed frame ball; a unique finite frame value is licensed only when the certified residual gap $\Delta_{\mathrm{loc},i}(t) > 0$. If that gap fails, the output is AMBIGUOUS. This theorem does not locate an event, populate the event base, resolve unlabeled mixtures, identify particle species, produce stress-energy, or establish chart-blind neutral bulk.

**Theorem 4.3c (Quotient-intrinsic geometry producer and dimension-selection boundary).** On a refinement tower of repaired quotient normal forms, the support-visible incidence complex: vertices the quotient-visible patch classes, simplices the jointly supported patch sets: is computable from the normal form alone and is invariant under repair schedule, gauge, and refinement. If the tower carries the computable receipts (i) spherical incidence (connected closed orientable combinatorial surface with Euler characteristic 2), (ii) disk/mesh nondegeneracy, (iii) modular cross-ratio Cauchy convergence, and (iv) independently normalized geometric $2\pi$-KMS comparison with wrong-normalization separation, then the refinement limit produces the round $S^2$, its conformal structure, the oriented round-cap family with cap normals, and every clause of the finite cap-normal BW certificate, with quantitative error envelopes. Conversely, there are finite repair systems with isomorphic rewrite structure whose incidence complexes realize $S^2$, $T^2$, the 2-skeleton of $\partial\Delta^4$, a wedge of two spheres, and any modular temperature $\beta$: repair confluence selects no topology, dimension, orientation framing, or normalization. The screen $S^2$ is therefore *derived on the receipt branch and underdetermined off it*; the receipts are the irreducible geometric branch content of Einstein branch entry. (Compact paper, geometry-producer packet; machine receipts ship with the released geometry code.)

**Theorem 4.3d (Null-net standardness, derived translations, and the four-translation assembly).** On the producer branch, the null half-line/interval blow-up net at a produced cap boundary point lives on one common GNS space, and: (i) the reference vector is cyclic and separating for every half-line and interval algebra used, given the finite-stage faithfulness of the MaxEnt states, the mixed-GNS Cauchy clause, and one half-line cyclicity receipt, with interval cyclicity then derived by the Reeh–Schlieder/Borchers argument rather than assumed; (ii) the half-sided modular inclusion is derived from the produced geometric cap flow, without assuming any translation or dilation unitaries, and Borchers–Wiesbrock yields the positive null-translation generator on its Stone domain; (iii) on the exact-Markov/central-interface branch the reduced interval modular Hamiltonians split into local terms plus uniformly bounded interface blocks (endpoint-Lipschitz), while an explicit four-qubit nearest-neighbour Gibbs witness shows finite-range locality alone does *not* imply this (machine-verified); (iv) the two derived translation groups and the modular dilations generate Möbius covariance and the projective bounded-interval action; and (v) on the modular-intersection receipt branch the per-direction positive generators commute, transform with the null-vector weight under the produced Lorentz action, satisfy the dependent-family linearity relation, and assemble into one four-parameter translation representation with joint spectrum in the closed future cone: a positive-energy Poincaré representation. (Compact paper, null-net standardness packet.)

**Theorem 4.3e (Conditional Lorentzian event manifold).** Events are equivalence classes of record germs whose common-refinement distance tends to zero; cofinal box intersection is insufficient because it need not be transitive. Unconditionally the event space is a second-countable nonseparated uniform space. The manifold branch requires population plus germ realization (E1), separation (E2), a locally bi-Lipschitz response chart with an interior-ball receipt (E3), an affine overlap cocycle (E4), $C^{1,1}$ tetrads and inverse tetrads (E4′), a held-out quadratic-cone fit with inertia $(1, 3)$ and wrong-signature controls (E5), and local semantic causal reachability (E6). The conformal celestial $S^2$ checks the inferred cone boundary; it does not force a quadratic cone. On these receipts the event base is a Lorentzian four-manifold, while $H^3$ is the fiber of future unit timelike frames over each event. Stable causality and global hyperbolicity remain named assumptions. Explicit countermodels show that $\dim H^3 = 3$ by itself determines neither event dimension nor manifold structure. This packet supplies, conditionally, the locally Lorentzian $d = 4$ regime consumed by the Einstein section. (Compact paper, event-manifold packet; machine receipts ship with the released geometry code.)

**Theorem 4.3f (Local conserved stress tensor from modular charges).** A symmetric bilinear form on $(\mathbb{R}^4, \eta)$ is determined by its null-null values up to a multiple of $\eta$, and a directional charge family arises from a tensor iff it satisfies the finite dependent-family linearity relations; nine generic null directions reconstruct the trace-free part with Lipschitz stability. On the event region of Theorem 4.3e with the assembly branch of Theorem 4.3d, the smeared endpoint-derivative charges of the bounded-interval modular kernels satisfy these relations, so they define a semiclassical expectation tensor $\langle T_{ab} \rangle$: symmetric, local, Poincaré-covariant with one common normalization, equal on null projections to the positive Borchers generators on a common core, and weakly conserved ($\nabla^a \langle T_{ab} \rangle = 0$ a.e.) by translation invariance: with ambiguity exactly $\phi g_{ab}$ plus improvement terms. Countermodel: per-direction positive generators violating one linearity relation admit no rank-two source (irreducible tomography residual); scalar collar CMI can never be promoted to a rank-two source. Universal coupling of this tensor to the geometric branch is the named receipt UC. (Compact paper, stress packet; receipts ship with the released geometry code.)

**Theorem 4.3g (Repaired entropy bridge and absolute Einstein equation).** On the central-interface branch, write the state as a direct sum of bulk states tensored with maximally mixed edge factors. Its entropy splits exactly into $S_{\text{bulk}} = H(p) + \sum_\alpha p_\alpha S(\rho_\alpha^{\text{bulk}})$ and $S_{\text{edge}} = \sum_\alpha p_\alpha \log d_\alpha$. The finite first law is $\delta S = 2\pi\delta\langle B_C \rangle + \delta\langle Z_C \rangle$, with $Z_C = \sum_\alpha (\log d_\alpha) P_\alpha$. Hence $\delta S_{\text{bulk}} = 2\pi\delta\langle B_C \rangle$ and $\delta S_{\text{edge}} = \delta\langle Z_C \rangle$ throughout the declared central-interface class, including variations of the sector weights. The changing stress channel is supplied by the exact MaxEnt envelope identity $dS/dt = \lambda = 2\pi$, giving the Clausius balance $\delta S_{\text{edge}} = \lambda\delta t - 2\pi\delta\langle B_C \rangle$ on the coupled variation class. One common scaling family must carry the conditional-mixing, boundary-count, fixed-collar, sharp-rate, and $o(\ell^4)$ remainder bounds. Coverage of all local timelike directions follows from population plus the produced Lorentz orbit. The vacuum-reference premise fixes the first-variation integration constant. Independent scale readouts with trivial positive-rescaling stabilizer identify the structural coupling. These premises yield $G_{ab} + \Lambda g_{ab} = 8\pi G\langle T_{ab} \rangle$ per connected component. Without the vacuum reference, the result fixes the residue tensor only up to one covariantly constant metric term. (Compact paper, entropy-bridge and small-diamond packets.)

**Theorem 4.3h (Typed Einstein branch composition and realized status).** Bare finite consensus is not Einstein-complete: the same finite-consensus reduct admits extensions in which the Einstein equation holds and extensions in which it fails. The valid gravity result is conditional. Let one source-derived repaired quotient tower carry deterministic geometry, modular, event, stress, entropy, and scale readouts on a common domain, with commuting refinement diagrams. Assume the finite geometry and null-net receipts, cap-interior modular data, the event-manifold receipts, one certified cofinal family on which every relevant remainder is $o(\ell^4)$, the universal-coupling and vacuum-reference receipts, and two independent scale readouts with trivial positive-rescaling stabilizer. The stress-tomography, entropy, small-ball, timelike-polarization, Ward, and Bianchi results then compose to

$$G_{ab} + \Lambda g_{ab} = 8\pi G \langle T_{ab} \rangle$$

on each connected event component. The exact finite layer includes the nine-direction tomography frame with determinant 8192/27 and the entropy and MaxEnt identities. Conditional on the continuum diamond-kernel formula and the smooth fixed-volume small-ball area identity, exact arithmetic gives the coefficients $8\pi^2\ell^4/15$ and $-4\pi\ell^4/15$; the continuum formulas themselves remain explicit analytic premises. The finite receipt programs verify algebra, evaluator behavior, manifest and deletion logic, and synthetic controls. They do not prove that the typed branch is inhabited. Realized-branch nonemptiness requires one antecedent-only source tower, certified asymptotic tails, universal coupling, a source-derived vacuum reference, an identifiable scale, and isolated full-tower countermodels for semantic receipt minimality. This construction is work in progress. (Compact paper, composed branch-entry theorem.)

## 5.3 Gravity from Fixed-Cap Generalized-Entropy Stationarity

### 5.3.1 Cap first law

Section 4.2 fixes the cap modular statement first at the automorphism level. On the extracted geometric subnet, the scaling-limit cap modular flow is geometric with the standard $2\pi$ normalization,

$$\sigma_t^{\omega_\infty^{\mathrm{geo},C}} = \alpha_{\lambda_C(2\pi t)}.$$

If the realized scaling-limit cap algebra happens to be type I, one may choose a density matrix $\rho_C^{\omega_\infty^{\mathrm{geo},C}}$ and modular Hamiltonian

$$K_C := -\log \rho_C^{\omega_\infty^{\mathrm{geo},C}},$$

and then for a perturbation $\rho(\varepsilon)$ with $\rho(0) = \omega_\infty^{\mathrm{geo},C}$ the first-law identity reads

$$\delta S_C = \delta\langle K_C \rangle = 2\pi\, \delta\langle B_C \rangle.$$

This displayed variation is the special type-I/fixed-central-sector form of $K_C = 2\pi B_C + Z_C$, where the central term has no first-order contribution on the chosen sector. In the generic continuum/non-type-I case, Section 4.2 does not supply an inner operator identity; the theorem surface there is the geometric modular automorphism statement, so the cap first-law formula is restricted to this special realization.

### 5.3.2 Null-surface modular bridge

This subsection extends the fixed-cutoff weak-tail-generator boundary to the derived positive null-translation stage and the exact half-line generator/charge identification. At fixed cutoff one first transfers cut-center data to regulated null strips, then imposes the extra inherited split condition needed for the spatial-collar-type tensor decomposition, obtains exact or controlled strip additivity on one inherited strip model, and derives a weak tail generator for the renormalized half-line family. On the scaling-limit geometric-cap branch of Theorem 4.2, the null half-line blow-up net then inherits geometric dilation and therefore half-sided modular inclusion, so Borchers–Wiesbrock supplies an explicit positive null-translation generator on its Stone domain. Bounded-interval formulas are discharged by the E0.5 affine/projective kernel below, and the tensor upgrade carries the null-invisible metric ambiguity. Accordingly the null modular bridge consists of:

- null-cut center transfer yielding the central cut-label decomposition, together with the extra inherited split condition needed for the spatial-collar-type tensor decomposition;
- exact or controlled four-term strip additivity on one fixed inherited strip model;
- endpoint-Lipschitz control for the renormalized half-line family and the resulting weak tail generator;
- derived half-sided modular inclusion on the null half-line blow-up net, and the resulting Borchers positive translation generator.

The later density-upgrade template is kept below as an explicit downstream template rather than as a fixed-cutoff theorem proved in this subsection. Lemma 5.2f makes the positive null-translation generator itself explicit, and Theorem 5.2g closes the half-line generator/charge identification on that same family: they record the Borchers unitary group, positivity and self-adjointness on the Stone domain, the corrected affine commutator relation, the half-line modular-Hamiltonian identity $K_a = K_0 - 2\pi a P_\Omega$, and the exact identification of $P_\Omega$ with the local null-stress charge. Bounded-interval formulas are downstream.

**Proposition 5.2a (Null-cut center transfer and inherited split).** Fix a regulated null tripartition

$$I_- = (v_1, v_2), \qquad J = (v_2, v_3), \qquad I_+ = (v_3, v_4),$$

with cuts $\Gamma_- := \{v = v_2\}$ and $\Gamma_+ := \{v = v_3\}$. Under the fixed-cutoff realized presentation, assume that the two null cuts inherit the ordinary or central-defect boundary-redundancy data used in the spatial collar branch, so that in a compatible type-I regulator presentation one has

$$\tilde{\mathcal{H}}_{I_-} \cong \bigoplus_{\alpha_-} W_{\alpha_-} \otimes \mathcal{H}_{i_-^{\alpha_-}},$$

$$\tilde{\mathcal{H}}_J \cong \bigoplus_{\alpha_-,\alpha_+} W_{\alpha_-}^* \otimes \mathcal{H}_{j^{\alpha_-,\alpha_+}} \otimes W_{\alpha_+},$$

$$\tilde{\mathcal{H}}_{I_+} \cong \bigoplus_{\alpha_+} W_{\alpha_+}^* \otimes \mathcal{H}_{i_+^{\alpha_+}},$$

where opposite sides of each cut carry inverse transport. If the strip algebra is the commutant of the transported cut actions, then

$$\mathcal{A}(J) \cong \bigoplus_{\alpha_-,\alpha_+} \mathcal{B}(\mathcal{H}_{j^{\alpha_-,\alpha_+}}), \qquad Z(\mathcal{A}(J)) = \bigoplus_{\alpha_-,\alpha_+} \mathbb{C}\, P_{\alpha_-,\alpha_+}.$$

If, in addition, each multiplicity space factors as

$$\mathcal{H}_{j^{\alpha_-,\alpha_+}} \cong \mathcal{H}_{j_L^{\alpha_-,\alpha_+}} \otimes \mathcal{H}_{j_R^{\alpha_-,\alpha_+}},$$

with $\mathcal{A}(I_- \cup J)$ acting blockwise only on $\mathcal{H}_{i_-^{\alpha_-}} \otimes \mathcal{H}_{j_L^{\alpha_-,\alpha_+}}$ and $\mathcal{A}(J \cup I_+)$ acting blockwise only on $\mathcal{H}_{j_R^{\alpha_-,\alpha_+}} \otimes \mathcal{H}_{i_+^{\alpha_+}}$, then

$$\mathcal{A}(J) \cong \bigoplus_{\alpha_-,\alpha_+} \mathcal{B}(\mathcal{H}_{j_L^{\alpha_-,\alpha_+}}) \otimes \mathcal{B}(\mathcal{H}_{j_R^{\alpha_-,\alpha_+}}),$$

and

$$\mathcal{H}_{I_- \cup J \cup I_+} \cong \bigoplus_{\alpha_-,\alpha_+} \mathcal{H}_{i_-^{\alpha_-}} \otimes \mathcal{H}_{j_L^{\alpha_-,\alpha_+}} \otimes \mathcal{H}_{j_R^{\alpha_-,\alpha_+}} \otimes \mathcal{H}_{i_+^{\alpha_+}}.$$

**Proof.** Complete reducibility gives the displayed decompositions. Commuting with the two transported cut actions forces strip operators to preserve the sector pair $(\alpha_-, \alpha_+)$ and act only on the multiplicity space; Schur's lemma kills the off-diagonal intertwiners and leaves the direct-sum algebra above. Taking invariants across both cuts in the glued tripartition again uses Schur's lemma and leaves exactly the matching sector pairs. The left/right split of the multiplicity spaces is additional structure; it is not forced by the center transfer alone. QED.

**Corollary 5.2b (Exact or controlled four-term null modular relation on an inherited strip model).** On one fixed finite-dimensional strip model satisfying Proposition 5.2a, define

$$\mathfrak{M}_{I_-:J:I_+} := \{\sigma : \ I(I_- : I_+ \mid J)_\sigma = 0\},$$

and

$$\delta^{\mathrm{M}}_{I_-:J:I_+}(\varepsilon) := \sup \left\{ \inf_{\sigma \in \mathfrak{M}_{I_-:J:I_+}} \|\rho - \sigma\|_1 : I(I_- : I_+ \mid J)_\rho \le \varepsilon \right\}.$$

For any strip state $\eta$, let

$$\Delta K_J(\eta) := K_{I_- \cup J \cup I_+}(\eta) - K_{I_- \cup J}(\eta) - K_{J \cup I_+}(\eta) + K_J(\eta).$$

If the strip reference state $\omega$ is exact Markov and EC-aligned on the inherited split (the strip form of the Markov-split alignment hypothesis of Section 2.3), then $\Delta K_J(\omega)$ is central, and on the canonical inherited HJPW model one may take

$$\Delta K_J(\omega) = 0.$$

Equivalently, there exists a central operator $K_{\partial,J}(\omega) \in Z(\mathcal{A}(J))$ such that

$$K_{I_- \cup J \cup I_+}(\omega) = K_{I_- \cup J}(\omega) + K_{J \cup I_+}(\omega) - K_J(\omega) + K_{\partial,J}(\omega).$$

If instead $I(I_- : I_+ \mid J)_\omega \le \varepsilon$, choose $\widetilde{\omega}_J \in \mathfrak{M}_{I_-:J:I_+}$, assumed EC-aligned on the inherited split wherever the central identification below is used, with

$$\|\omega - \widetilde{\omega}_J\|_1 \le \delta^{\mathrm{M}}_{I_-:J:I_+}(\varepsilon).$$

Define

$$\mathfrak{D}_J(\omega, \widetilde{\omega}_J) := \Delta K_J(\omega) - \Delta K_J(\widetilde{\omega}_J).$$

Then

$$K_{I_- \cup J \cup I_+}(\omega) = K_{I_- \cup J}(\omega) + K_{J \cup I_+}(\omega) - K_J(\omega) + K_{\partial,J}(\widetilde{\omega}_J) + \mathfrak{D}_J(\omega, \widetilde{\omega}_J),$$

with $K_{\partial,J}(\widetilde{\omega}_J) = \Delta K_J(\widetilde{\omega}_J) \in Z(\mathcal{A}(J))$. Every bounded observable on $I_- \cup J$ or $J \cup I_+$ then differs from the exact-Markov reference by at most

$$\|O\|_\infty \, \delta^{\mathrm{M}}_{I_-:J:I_+}(\varepsilon).$$

If the relevant strip marginals are uniformly faithful with lower spectral bound $\lambda_* > 0$, then

$$\|\mathfrak{D}_J(\omega, \widetilde{\omega}_J)\|_\infty \le 4\lambda_*^{-1}\, \delta^{\mathrm{M}}_{I_-:J:I_+}(\varepsilon).$$

Independently, Fawzi–Renner gives a recovered comparison state with trace-norm error

$$r_{\mathrm{FR}}(\varepsilon) = 2\sqrt{1 - e^{-\varepsilon}} \le 2\sqrt{\varepsilon}.$$

Thus the exact four-term strip relation is available only at EC-aligned exact Markovity or in a controlled strip family on one fixed inherited strip model admitting EC-aligned replacements with $\delta^{\mathrm{M}}_{I_-:J:I_+}(\varepsilon_J) \to 0$, while $\mathfrak{D}_J$ is carried explicitly at finite stage.

**Proof.** On the inherited split of Proposition 5.2a, the EC-aligned exact-Markov case is the same HJPW block calculation as in the spatial collar theorem: alignment identifies the HJPW blocks with the inherited blocks, the four logarithms cancel blockwise on the canonical model, and any residual bookkeeping term is central. Without alignment, HJPW factorizes the state only over its own state-dependent split, and the centrality of $\Delta K_J$ in $Z(\mathcal{A}(J))$ is not available. The controlled replacement is the same fixed-model compactness argument used for ordinary collars, after relabeling $A, B, D \mapsto I_-, J, I_+$; the operator-norm bound on $\mathfrak{D}_J$ is the corresponding relabeling of the modular-transport estimate. QED.

**Definition (Renormalized null modular functional).** For a null interval $I$ on generator $\Omega$, let $K_\partial(I, \Omega)$ denote the central endpoint-label term singled out by Corollary 5.2b on the inherited strip model, or by its controlled exact-Markov replacement when that model is used as reference. Define

$$\widetilde{K}[I, \Omega] := K[I, \Omega] - K_\partial(I, \Omega), \qquad \widetilde{K}_a(\Omega) := \widetilde{K}[(a, \infty), \Omega].$$

**Proposition 5.2c (Endpoint-Lipschitz null modular families and weak tail generator).** For renormalized modular Hamiltonians on one null generator,

$$\widetilde{K}[I, \Omega] := K[I, \Omega] - K_\partial(I, \Omega), \qquad \widetilde{K}_a(\Omega) := \widetilde{K}[(a, \infty), \Omega],$$

the branch-internal endpoint-control coming from the local finite-constraint MaxEnt branch gives:

$$\left|\langle \psi, (\widetilde{K}[(a', b'), \Omega] - \widetilde{K}[(a, b), \Omega])\phi\rangle\right| \le C_{\psi,\phi,\Omega}(|a' - a| + |b' - b|)$$

for bounded intervals in a compact endpoint window, and

$$\left|\langle \psi, (\widetilde{K}_{a'}(\Omega) - \widetilde{K}_a(\Omega))\phi\rangle\right| \le C_{\psi,\phi,\Omega}|a' - a|$$

for half-lines. Hence

$$f_{\psi,\phi}(a) := \langle \psi, \widetilde{K}_a(\Omega)\phi\rangle$$

is locally Lipschitz and therefore absolutely continuous, and its distributional derivative defines the weak tail generator

$$\langle \psi, q(a, \Omega)\phi\rangle := -\frac{1}{2\pi}\, \partial_a f_{\psi,\phi}(a).$$

For $a < b$,

$$\langle \psi, (\widetilde{K}_b(\Omega) - \widetilde{K}_a(\Omega))\phi\rangle = -2\pi \int_a^b \langle \psi, q(v, \Omega)\phi\rangle\, dv.$$

$q(a, \Omega)$ is a weak tail generator rather than a local operator-valued density; its identification with the positive self-adjoint Borchers generator occurs only after Corollary 5.2e and Lemma 5.2f.

**Proof.** The derived endpoint-control estimate applies to the renormalized family after removal of the central endpoint term. For bounded intervals, the symmetric-difference length is bounded by $|a'-a|+|b'-b|$; for half-lines it is exactly $|a'-a|$. Therefore the matrix-element functions are Lipschitz in the endpoints. A Lipschitz function on a compact interval is absolutely continuous, so $f_{\psi,\phi}(a)$ has an $L^\infty_{\rm loc}$ derivative and obeys the fundamental theorem of calculus. Defining $q$ as the rescaled negative derivative gives the displayed integral relation. QED.

**Downstream c.12–c.13 boundary.** The fixed-cutoff bridge established above is the end of this subsection's proved content.

**Lemma 5.2d (Downstream density-upgrade template).** If, in addition, the weak tail generator $q(a,\Omega)$ is weakly differentiable in $a$ and both $f_{\psi,\phi}(a)$ and $q_{\psi,\phi}(a)$ vanish at $+\infty$, then one may define a density

$$\langle\psi, p(a,\Omega)\phi\rangle := -\partial_a\langle\psi, q(a,\Omega)\phi\rangle = \frac{1}{2\pi}\,\partial_a^2\langle\psi, \widetilde{K}_a(\Omega)\phi\rangle$$

and obtain the half-line formula

$$\langle\psi, \widetilde{K}_a(\Omega)\phi\rangle = 2\pi\int_a^\infty (v-a)\,\langle\psi, p(v,\Omega)\phi\rangle\,dv.$$

This is a downstream density-upgrade step and is not proved from the fixed-cutoff strip arguments above.

**Null half-line blow-up net.** Fix a smooth cut point and choose affine coordinate $v$ on generator $\Omega$ so that the cut sits at $v=0$. For $a\geq 0$, write

$$H_a := (a,\infty), \qquad \mathcal{M}_a(\Omega) := \overline{\bigvee_{a<c<d<\infty}\mathcal{A}((c,d),\Omega)}.$$

By isotony of the null interval net,

$$a\leq b \quad\Longrightarrow\quad \mathcal{M}_b(\Omega)\subseteq\mathcal{M}_a(\Omega).$$

**Corollary 5.2e (Derived half-sided modular inclusion on null half-lines).** On the scaling-limit geometric-cap branch of Theorem 4.2, the blow-up modular action near a smooth entangling cut acts on the null coordinate by

$$v\mapsto e^{-2\pi t}v.$$

Therefore, for every $a\geq 0$,

$$\sigma_t^\omega(\mathcal{M}_a(\Omega)) = \mathcal{M}_{e^{-2\pi t}a}(\Omega).$$

Hence, for every $a>0$,

$$\sigma_t^\omega(\mathcal{M}_a(\Omega))\subseteq\mathcal{M}_a(\Omega) \qquad (t\leq 0),$$

so the inclusion $\mathcal{M}_a(\Omega)\subset\mathcal{M}_0(\Omega)$ is half-sided modular. After the harmless convention change $t\mapsto -t$, this is the standard positive-time half-sided-inclusion form. The half-sided modular inclusion is therefore derived from the null-interval structure, isotony, and the scaling-limit geometric action rather than imported separately.

**Proof.** Blow up the cap modular flow of Theorem 4.2 near a smooth cut and restrict to the chosen null generator. In the tangent limit the cap-preserving flow becomes the null dilation $v\mapsto e^{-2\pi t}v$. For every bounded interval $(c,d)\subset H_a$, this sends $\mathcal{A}((c,d),\Omega)$ to $\mathcal{A}((e^{-2\pi t}c, e^{-2\pi t}d),\Omega)$, and taking the von Neumann closure of the interval net yields the displayed identity for $\mathcal{M}_a(\Omega)$. If $t\leq 0$, then $e^{-2\pi t}a\geq a$, so $H_{e^{-2\pi t}a}\subseteq H_a$, and isotony gives $\mathcal{M}_{e^{-2\pi t}a}(\Omega)\subseteq\mathcal{M}_a(\Omega)$. QED.

**Lemma 5.2f (Positive null-translation generator).** For the derived half-sided modular inclusion

$$\mathcal{M}_a(\Omega)\subset\mathcal{M}_0(\Omega) \qquad (a>0),$$

let $\Delta_0(\Omega)$ be the modular operator of the standard pair $(\mathcal{M}_0(\Omega), \omega)$ and define $K_0(\Omega) := -\log \Delta_0(\Omega)$. Borchers–Wiesbrock then yields a unique strongly continuous one-parameter unitary group

$$U_\Omega(a) = e^{iaP_\Omega}, \qquad a \in \mathbb{R},$$

such that

$$U_\Omega(a)\omega = \omega, \qquad U_\Omega(a)\mathcal{M}_b(\Omega)U_\Omega(a)^* = \mathcal{M}_{a+b}(\Omega) \quad (a, b \geq 0),$$

and whose generator $P_\Omega$ is positive and self-adjoint on the Stone domain

$$D(P_\Omega) := \left\{ \psi \in \mathcal{H} : \lim_{a \to 0} \frac{U_\Omega(a)\psi - \psi}{ia} \text{ exists} \right\}.$$

In addition,

$$\Delta_0(\Omega)^{it} U_\Omega(a) \Delta_0(\Omega)^{-it} = U_\Omega(e^{-2\pi t} a), \qquad \Delta_0(\Omega)^{it} P_\Omega \Delta_0(\Omega)^{-it} = e^{-2\pi t} P_\Omega,$$

and on the common invariant analytic core of $K_0(\Omega)$ and $P_\Omega$,

$$[K_0(\Omega), P_\Omega] = -\, i\, 2\pi P_\Omega.$$

If $\Delta_a(\Omega)$ denotes the modular operator of $(\mathcal{M}_a(\Omega), \omega)$ and $K_a(\Omega) := -\log \Delta_a(\Omega)$, then

$$\Delta_a(\Omega) = U_\Omega(a)\Delta_0(\Omega)U_\Omega(a)^*, \qquad K_a(\Omega) = U_\Omega(a)K_0(\Omega)U_\Omega(a)^*,$$

so

$$K_a(\Omega) = K_0(\Omega) - 2\pi a\, P_\Omega$$

as a quadratic-form identity on $D(K_0(\Omega)) \cap D(P_\Omega)$. Equivalently,

$$\langle \psi, (K_b(\Omega) - K_a(\Omega))\phi \rangle = -2\pi(b - a)\langle \psi, P_\Omega \phi \rangle$$

for all $\psi, \phi \in D(K_0(\Omega)) \cap D(P_\Omega)$. In the canonical normalization of Corollary 5.2b, the weak endpoint derivative from Proposition 5.2c is therefore exactly the Borchers generator. Bounded-interval modular-Hamiltonian formulas are downstream and require the separate interval-preserving branch recorded in Theorem 5.2g.

**Proof.** Corollary 5.2e gives the required half-sided modular inclusion. Borchers–Wiesbrock then yields the unique strongly continuous unitary group $U_\Omega(a)$, its positive self-adjoint generator $P_\Omega$, and the affine covariance relation with the modular group; Stone's theorem gives the displayed domain formula. Differentiating

$$\Delta_0(\Omega)^{it} U_\Omega(a) \Delta_0(\Omega)^{-it} = U_\Omega(e^{-2\pi t} a)$$

first at $a = 0$ and then at $t = 0$ on the common analytic core gives

$$[K_0(\Omega), P_\Omega] = -\, i\, 2\pi P_\Omega.$$

Because $U_\Omega(a)\omega = \omega$ and $U_\Omega(a)\mathcal{M}_0(\Omega)U_\Omega(a)^* = \mathcal{M}_a(\Omega)$, uniqueness of modular data for the translated standard pair gives the displayed formulas for $\Delta_a(\Omega)$ and $K_a(\Omega)$. Differentiating $K_a(\Omega) = U_\Omega(a)K_0(\Omega)U_\Omega(a)^*$ in $a$ yields $dK_a/da = -2\pi P_\Omega$ as a quadratic-form identity, and integrating from 0 to $a$ gives the stated half-line modular-Hamiltonian relation. QED.

**Theorem 5.2g (The half-line generator is the local null-stress charge).** In the canonical normalization of the preceding half-line construction,

$$\langle \psi, P_\Omega \phi \rangle = -\frac{1}{2\pi} \frac{d}{da}\Big|_{a=0^+} \langle \psi, \widetilde{K}_a(\Omega)\phi \rangle$$

for all $\psi, \phi \in D(K_0(\Omega)) \cap D(P_\Omega)$. In the effective spacetime description of that same renormalized half-line family, the local null-stress charge is defined by the identical endpoint derivative,

$$Q_{kk}(\Omega) := -\frac{1}{2\pi} \frac{d}{da}\Big|_{a=0^+} \widetilde{K}_a^{\mathrm{eff}}(\Omega),$$

so

$$P_\Omega = Q_{kk}(\Omega)$$

as a quadratic-form identity on the common domain. The only continuum input used here is the standard local modular-Hamiltonian form on the effective description of that same half-line family; it is used only to name, in continuum language, the operator fixed by the OPH half-line derivative. Bounded-interval transport through the affine-covariant kernel $g_I(v)$ is discharged by E0.5 below, and reconstruction of a full tensor from the directional charges is subject to the null-invisible metric ambiguity.

**Boundary of the null bridge.** The null bridge is an explicit theorem branch rather than an automatic fixed-cutoff identity. Its required data are:

- transferred cut-center data alone give the central sector decomposition of the null strip rather than the left/right HJPW split;
- the extra decomposition-inheritance condition of Proposition 5.2a is exactly what upgrades those sectors to the same collar-type block structure used in the spatial branch;
- the fixed-cutoff null-strip bridge is exact for exact Markov strip states on that inherited decomposition, and otherwise uses a controlled limit with the carried defect operator $\mathfrak{D}_J$;
- the local finite-constraint MaxEnt branch gives endpoint-Lipschitz control strong enough to define a weak tail generator for renormalized half-lines, and the scaling-limit geometric-cap branch of Theorem 4.2 then derives half-sided modular inclusion on the half-line blow-up net;
- Borchers–Wiesbrock therefore supplies a genuine positive null-translation generator $P_\Omega$ inside the bridge itself, together with its Stone domain and the half-line modular-Hamiltonian relation $K_a = K_0 - 2\pi a P_\Omega$;
- the half-line generator/charge identification is fixed inside the bridge itself, while bounded-interval transport and the later tensor upgrade are downstream.

This is the route from null-strip factorization to the later D5 Einstein branch, with the half-line generator/charge identification proved inside the null modular bridge itself.

### 5.3.3 Modular energy as stress-tensor charge on null half-lines

The effective-theory input here is the standard stress-tensor representation that names the operator fixed by the null bridge. In the canonical normalization of the preceding null-half-line construction, the positive null generator is the endpoint derivative of the renormalized half-line modular family:

$$\langle \psi, P_\Omega \phi \rangle = -\frac{1}{2\pi} \frac{d}{da}\Big|_{a=0^+} \langle \psi, \widetilde{K}_a(\Omega)\phi \rangle$$

for all $\psi, \phi \in D(K_0(\Omega)) \cap D(P_\Omega)$.

In the effective spacetime description of that same half-line family, the local null-stress charge is defined by the identical endpoint derivative,

$$Q_{kk}(\Omega) := -\frac{1}{2\pi}\frac{d}{da}\Big|_{a=0^+}\widetilde{K}_a^{\text{eff}}(\Omega),$$

so on the common quadratic-form domain

$$P_\Omega = Q_{kk}(\Omega).$$

Thus the null generator is exactly the local null-stress charge in the effective spacetime description; this step is not a separate scaling-limit input. When the effective description admits a local stress tensor, for example in a UV CFT regime or any local Lorentzian regime with local modular Hamiltonians, the same operator is the standard null-stress charge associated with the chosen null generator.

For comparison, in a CFT vacuum on a ball one has the familiar local formula

$$H_\zeta = \int_\Sigma T_{ab}\zeta^b \, d\Sigma^a,$$

so the present identification is the null-half-line version of that same modular-energy locality rather than an additional EFT postulate.

The downstream boundary is narrower. The E0.5 bounded-interval kernel transports this half-line statement to bounded null intervals with affine-covariant kernel $g_I(v)$, and the null-to-tensor upgrade determines $T_{ab}$ only up to the familiar ambiguity

$$T_{ab} \mapsto T_{ab} + \phi g_{ab}.$$

But the generator/charge identification itself is fixed inside the null bridge.

#### 5.3.4 Localized generalized entropy from Markov + MaxEnt

Using the collar decomposition, exact Markovity, and MSA, the declared central-interface branch has the following edge-aligned Markov normal form. On the nonexact branch, Theorem 2.5 supplies a recovered comparison state and a vanishing CMI envelope only under strong conditional mixing; convergence to an exact Markov state is a fixed-collar statement, and alignment with the displayed edge factors remains a separate MSA condition. Under those conditions MaxEnt selection within each edge sector gives

$$\rho_C = \bigoplus_\alpha p_\alpha \left(\rho^\alpha_{\text{bulk},C} \otimes \frac{\mathbf{1}^\alpha_{\text{edge}}}{d_\alpha}\right).$$

The entropy splits as

$$S(\rho_C) = H(p_\alpha) + \sum_\alpha p_\alpha S(\rho^\alpha_{\text{bulk},C}) + \sum_\alpha p_\alpha \log d_\alpha.$$

*Convention:* Throughout this paper, "log" denotes the natural logarithm (ln), so entropies are measured in **nats** (1 nat = $1/\ln 2 \approx 1.443$ bits). This is standard in thermodynamics and QFT; the Bekenstein-Hawking formula S = A/4G uses nats. When clarity requires it, we write $\log_2$ explicitly for bits.

Define

$$S_{\text{bulk}}(C) := H(p_\alpha) + \sum_\alpha p_\alpha S(\rho^\alpha_{\text{bulk},C}),$$

and the central area operator

$$L_C := \sum_\alpha (\log d_\alpha) P_\alpha.$$

Then

$$S_{\text{gen}}(C) := \operatorname{Tr}(\rho L_C) + S_{\text{bulk}}(C).$$

**Newton coupling as an edge-entropy area-law readout.**

In the collar double-scaling limit, the edge contribution becomes extensive along the entangling surface $\Sigma = \partial C$:

$$\operatorname{Tr}(\rho L_C) \approx N_\Sigma \cdot \bar{\ell}(t), \qquad \bar{\ell}(t) := \sum_\alpha p_\alpha \log d_\alpha,$$

where $N_\Sigma$ is the number of UV cut elements covering $\Sigma$ and $\bar{\ell}(t)$ is the **single-cell edge entropy** from the heat-kernel distribution (Theorem 6.20). Similarly, the geometric area is extensive:

$$A(C) \approx N_\Sigma \cdot a_{\text{cell}},$$

where $a_{\text{cell}}$ is the area per UV cut element in the emergent metric.

Matching these expressions gives the geometric area-law coupling:

$$G_{\text{geom}} = \frac{a_{\text{cell}}}{4\,\bar{\ell}(t)}$$

where:

- $a_{\text{cell}}$ is fixed operationally from the UV correlation/mixing length $\xi$ via $a_{\text{cell}} \sim \xi^2$ (from the exponential mixing hypothesis),
- $\bar{\ell}(t) = \sum_R p_R(t) \log d_R$ is computed from the heat-kernel edge distribution with $p_R \propto d_R e^{-t\lambda_R}$.

Explicitly:

$$\bar{\ell}(t) = \frac{\sum_R d_R e^{-t\lambda_R} \log d_R}{\sum_R d_R e^{-t\lambda_R}}.$$

On the OPH gravity branch this formula does not back-solve the scale. The selected no-$G$ scale certificate supplies $\gamma_\star = \ell_\star \nu_{\text{Cs}}/c$, equivalently $B_\star = 3\pi/\ell_\star^2$. The local cell readings are

$$a_{\text{cell}} = P\ell_\star^2, \qquad \bar{\ell}_{\text{shared}} = P/4,$$

so the Newton area-law readout gives

$$G_{\text{geom}} = \frac{P\ell_\star^2}{4(P/4)} = \ell_\star^2.$$

The SI display is $G_{\text{SI}} = c^3\ell_\star^2/\hbar$.

#### 5.3.5 Fixed-cap generalized-entropy stationarity from MaxEnt

MaxEnt selection implies that for variations preserving cap labels (fixed size and charges),

$$\delta S_{\mathrm{gen}}(C) = 0.$$

Using the split above and the first law for the bulk term,

$$\delta S_{\mathrm{gen}}(C) = \delta\langle L_C\rangle + \delta\langle K_{\mathrm{bulk}}\rangle.$$

#### 5.3.6 No-smuggling dependency discharge for the Einstein bridge

**Theorem E0 (Einstein bridge dependency discharge).** Write OPH5 for the five-axiom basis: the $S^2$ screen net, overlap consistency, local MaxEnt/refinement, recoverable generalized entropy, and MAR. Let one source-derived cofinal OPH regulator tower carry the typed common-domain geometry, modular, event, stress, entropy, and scale readouts of Theorem 4.3h, together with its finite, asymptotic, and physical-identification premises. On that branch, OPH5 and the additional receipts supply the inputs used by the Einstein bridge:

- geometry readout through quotient normal forms via the screen fold $\chi_{S,r} \circ n_r \circ \pi_r$;
- Lorentz/H3 readout from round cap pairs, since $\mathrm{Conf}^+(S^2) \cong \mathrm{PSL}(2,\mathbb{C}) \cong \mathrm{SO}^+(3,1)$ and $H^3 \simeq \mathrm{SO}^+(3,1)/\mathrm{SO}(3)$;
- $2\pi$-normalized support-visible BW flow on the extracted geometric subnet;
- the null generator/local null-stress charge identity on the same branch;
- the bounded-interval kernel of Lemma E0.5;
- fixed-cap generalized-entropy stationarity from the MaxEnt/refinement theorem;
- the standard fixed-volume small-ball area identity after geometry readout;
- $o(\ell^4)$ remainder control by Lemma E0.6; and
- all local timelike directions by the cap-pair coverage of Lemma E0.7.

Thus the finite consensus reduct supplies quotient normal forms. The typed branch receipts supply the geometric, modular, stress, entropy, coverage, tail, coupling, vacuum, and scale inputs needed by the Einstein bridge. Construction and certification of one source-derived tower carrying all of them are work in progress. Bare finite consensus does not imply the Einstein equation.

**Lemma E0.5 (Bounded-interval kernel from null projective covariance).** On the null blow-up of the support-visible geometric branch, the half-line generator/charge identity, endpoint-Lipschitz renormalized interval control, and affine/projective covariance give

$$K^{\mathrm{null}}_{(a,b)} = 2\pi \int_a^b \frac{(v-a)(b-v)}{b-a}\, T_{kk}(v)\, dv + R_{(a,b)}.$$

The remainder $R_{(a,b)}$ is the carried endpoint/collar term of the same refinement family. In the small-ball limit this is the ball modular kernel used below.

**Lemma E0.6 (Uniform remainder control).** Let $(r, \ell_r, \delta_r)$ be one declared cofinal family on which Theorem 2.5 holds uniformly over the event region and cap family, and set

$$\varepsilon_r = c|\partial C_r|_{\mathrm{UV}} e^{-\delta_r/\xi_r}.$$

Assume that the fixed-collar Markov replacements have one modulus $\delta_r^{\mathrm{M}} \le C_{\mathrm{M}} \varepsilon_r^{\theta}$, with $C_{\mathrm{M}} < \infty$ and $\theta > 0$, and that for some

$$s > \max\{8, 4/\theta\}, \qquad \omega_r \longrightarrow +\infty,$$

the same family satisfies

$$\frac{\delta_r}{\xi_r} - \log(c|\partial C_r|_{\mathrm{UV}}) \geq s \log(\ell_0/\ell_r) + \omega_r.$$

If the interval-kernel residual is also $o(\ell_r^4)$ on this family, then

$$2\sqrt{\varepsilon_r} = o(\ell_r^4), \qquad \delta_r^{\mathrm{M}} = o(\ell_r^4),$$

and the endpoint, collar, Markov, and long-wavelength derivative remainders do not alter the Einstein coefficient. Indeed, $\varepsilon_r \leq (\ell_r/\ell_0)^s e^{-\omega_r}$; the two conclusions follow from $s/2 > 4$ and $s\theta > 4$. Mere pointwise convergence of a collar error does not supply this common-family rate, and no per-radius diagonal choice is used.

**Lemma E0.7 (Cap-pair timelike coverage).** On the cap-normal $H^3$ branch, every future timelike vector $v$ has $r = \sqrt{-\eta(v,v)}$ and $u = v/r \in H^3$. For any unit $s \in u^\perp$, the vectors

$$q_\pm = u \pm s$$

are future null and

$$v = \frac{r}{2}(q_+ + q_-).$$

Thus normalized future null directions in observer skies cover every local timelike direction once the observer-frame chart is supplied. This does not say that two unnormalized projective rays alone select a unique observer frame; the observer normalization or equivalent causal-diamond scale data is also required.

**Proposition E2 (Recovered-core stress closure).** If the recovered stress tensor is reconstructed from the full compatible family of null and timelike modular charges on the recovered core, then $\nabla^a \langle T_{ab} \rangle = 0$. The overlap-compatible local charge family cancels on opposite faces of infinitesimal recovered-core diamonds; after the cap-pair tensor upgrade this cancellation is the covariant divergence-free condition.

### 5.3.7 Einstein equation from fixed-cap generalized-entropy stationarity

This subsection is a branch theorem. The finite consensus reduct supplies quotient normal forms; Theorem E0 discharges the OPH5 recovered-core inputs needed by the Einstein bridge: geometry readout, support-visible BW modular covariance, the null-stress bridge, bounded-interval transport, fixed-cap MaxEnt stationarity, the fixed-volume small-ball area variation, carried-remainder control, and the timelike scalar-to-tensor upgrade.

In the d=4 scaling regime, the small-ball bridge is internal to the derived gravity chain once the E0.5 bounded-interval kernel is included. The only extra standard geometric input is the fixed-volume area-variation identity for a small geodesic ball. The modular kernel itself is not imported from a separate EFT law: on the extracted geometric subnet, a sufficiently small cap is the tangent causal diamond $D_\ell$, whose preserving conformal Killing field is

$$\xi_{D_\ell} = \frac{1}{2\ell}\Big((\ell^2 - r^2 - t^2)\,\partial_t - 2t\,x^i\partial_i\Big).$$

On the t=0 slice its lapse is

$$\xi_{D_\ell} \cdot n = \frac{\ell^2 - r^2}{2\ell}.$$

The bounded-interval kernel comes from combining the D3 geometric cap generator with the E0.5 affine/projective transport of the D4 half-line stress bridge, so the D3 cap-modular theorem and the D4 stress bridge together yield

$$\delta S_{\rm bulk}(C) = 2\pi \int_{B_\ell} \frac{\ell^2 - r^2}{2\ell}\, \delta\langle T_{00}\rangle\, d^3x + \delta\langle E^{(\eta)}_{C,\ell}\rangle.$$

If $\delta\langle T_{00}\rangle$ is approximately constant across the ball and the carried remainder is $o(\ell^4)$, then

$$\delta S_{\rm bulk}(C) = \frac{8\pi^2\ell^4}{15}\, \delta\langle T_{00}\rangle + O(\ell^5\partial T) + o(\ell^4).$$

Stationarity of generalized entropy therefore gives the observer-covariant scalar relation

$$\delta\Big[(G_{ab} + \Lambda g_{ab})u^a u^b\Big] = 8\pi G\, u^a u^b\, \delta\langle T_{ab}\rangle$$

for the local diamond rest-frame four-velocity $u^a$. In the adapted rest frame $u^a = (1, 0, 0, 0)$, this is

$$\delta(G_{00} + \Lambda g_{00}) = 8\pi G\, \delta\langle T_{00}\rangle.$$

#### 5.3.8 Overlaps supply all timelike directions

This scalar-to-tensor upgrade is internal once the Lorentz branch is in place. Overlapping observers through the same bulk point realize every local timelike four-velocity u. Define

$$Y_{ab} := G_{ab} + \Lambda g_{ab} - 8\pi G\, \langle T_{ab}\rangle.$$

The local-diamond rest-frame relation says

$$u^a u^b\, \delta Y_{ab} = 0$$

for every such u. In local inertial coordinates,

$$u^a = \gamma(1, v^i), \qquad |\vec{v}| < 1,$$

so

$$0 = u^a u^b\, \delta Y_{ab} = \gamma^2\Big(\delta Y_{00} + 2v^i \delta Y_{0i} + v^i v^j \delta Y_{ij}\Big)$$

for all $|\vec{v}| < 1$. The polynomial therefore vanishes coefficientwise, hence

$$\delta Y_{00} = 0, \qquad \delta Y_{0i} = 0, \qquad \delta Y_{ij} = 0.$$

So the full tensor variation vanishes:

$$\delta Y_{ab} = 0.$$

Along any connected scaling branch of reference states, $Y_{ab}$ is therefore constant. The only purely local freedom left by the null reconstruction is the metric term, and fixing that term on one maximally symmetric reference state gives

$$G_{ab} + \Lambda g_{ab} = 8\pi G\, \langle T_{ab}\rangle.$$

Thus the tensor upgrade does not require a separate EFT handoff once the geometric-cap theorem, the null bridge, and the overlap-complete timelike family are in place.

#### 5.3.9 Non-tunable numerical constants

The gravity chain yields specific numerical constants as rigid outputs of the axiom chain.

**The $2\pi$ KMS normalization.** From Theorem 4.2, cap modular flow on the support-visible scaling-limit geometric cap pair is fixed by the KMS condition to the standard $2\pi$ normalization. This is the same rigidity that fixes Unruh/Hawking temperature normalization.

**The geometric coefficient $\Omega_{d-2}/(d^2-1)$.** This coefficient appears in both (a) the CFT-ball modular Hamiltonian weight integral and (b) the geometric area-variation identity. It is an exact integral identity:

$$\int_{B_\ell^{d-1}} \frac{\ell^2 - r^2}{2\ell}\, d^{d-1}x = \frac{\Omega_{d-2}\,\ell^d}{d^2-1}.$$

In $d = 4$:

$$\frac{\Omega_2}{4^2-1} = \frac{4\pi}{15} \approx 0.8377580409572781.$$

This is the reason prefactors cancel cleanly when going from $\delta S_{\rm gen} = 0$ to the Einstein equation (leaving $8\pi G$ with the $2\pi$ fixed by Theorem 4.2).

**What is predicted.** The framework cleanly separates:

- **Non-tunable constants**: $2\pi$ (KMS period), $\Omega_{d-2}/(d^2-1)$ (geometric coefficient), the existence of the Einstein form.
- **Global closure constants**: $G$ is the conversion between edge entropy and geometric area on the emitted local gravity branch. $\Lambda$ is not fixed by the local reference-state variation data alone; stable correctable-record closure plus identification of the record carrier with de Sitter horizon entropy fixes $\Lambda_\star \ell_\star^2 = 3\pi/N_\star$, while the SI curvature scale also uses the selected OPH scale certificate.

#### 5.3.10 Quantitative Markov error and controlled corrections

The role of this subsection is to keep three distinct quantities separate:

1. the raw collar conditional mutual information
$$\varepsilon_\delta := I(A_\delta : D_\delta \mid B_\delta)_\omega;$$
2. the constructive recovery error
$$r_{\rm FR}(\varepsilon_\delta) := 2\sqrt{1 - e^{-\varepsilon_\delta}} \le 2\sqrt{\varepsilon_\delta};$$
3. the exact-Markov replacement error on one fixed faithful collar model,
$$\eta_\delta^{\rm M} := 4\lambda_*^{-1}\, \delta^{\rm M}_{A_\delta:B_\delta:D_\delta}(\varepsilon_\delta),$$
where $\lambda_* > 0$ is the lower spectral bound needed to compare modular Hamiltonians.

The exact identity for the modular defect expectation is unchanged:

$$\langle \Delta K_\delta \rangle_\omega = -I(A_\delta : D_\delta \mid B_\delta)_\omega = -\varepsilon_\delta,$$

with

$$\Delta K_\delta := K_{A_\delta B_\delta D_\delta} - K_{A_\delta B_\delta} - K_{B_\delta D_\delta} + K_{B_\delta}.$$

What changes is the interpretation of later "exact" collar formulas. For each fixed finite-dimensional collar one may choose an exact Markov reference state $\sigma_\delta$ with

$$\|\omega_\delta - \sigma_\delta\|_1 \le \delta^{\mathrm{M}}_{A_\delta:B_\delta:D_\delta}(\varepsilon_\delta),$$

and a constructive recovered comparison state $\omega_\delta^{\mathrm{rec}}$ with

$$\|\omega_\delta - \omega_\delta^{\mathrm{rec}}\|_1 \le r_{\mathrm{FR}}(\varepsilon_\delta).$$

These two comparisons do different jobs:

**Bounded-observable control.** For every bounded observable $X$ supported on $A_\delta \cup B_\delta$,

$$|\mathrm{Tr}[X(\omega_\delta - \omega_\delta^{\mathrm{rec}})]| \le \|X\|_\infty\, r_{\mathrm{FR}}(\varepsilon_\delta),$$

and

$$|\mathrm{Tr}[X(\omega_\delta - \sigma_\delta)]| \le \|X\|_\infty\, \delta^{\mathrm{M}}_{A_\delta:B_\delta:D_\delta}(\varepsilon_\delta).$$

The first bound is constructive and dimension-free; the second is the fixed-collar route to the exact Markov set.

**Modular-additivity control.** If the relevant marginals are uniformly faithful with lower spectral bound $\lambda_* > 0$, then

$$\|\Delta K_\delta(\omega) - \Delta K_\delta(\sigma_\delta)\|_\infty \le \eta_\delta^{\mathrm{M}}.$$

Since $\Delta K_\delta(\sigma_\delta)$ is central by exact Markovity, the finite-stage modular defect is within $\eta_\delta^{\mathrm{M}}$ of the exact splice or additivity value.

Therefore the manuscript's exact collar identities arise as limits with a carried remainder

$$\eta_\delta := r_{\mathrm{FR}}(\varepsilon_\delta) + \eta_\delta^{\mathrm{M}},$$

together with the separate long-wavelength derivative remainders present in the small-ball expansion. Exact identities at finite collar width require exact Markovity; otherwise one works with $\eta_\delta$-controlled approximations and then lets $\eta_\delta \to 0$ in the controlled collar limit.

**Finite-cutoff Einstein remainder bound.** In the small-ball rest-frame calculation, write

$$\delta S_C = \frac{8\pi^2\ell^4}{15}\,\delta\langle T_{00}\rangle + \delta\langle E_C^{(\delta)}\rangle + O(\ell^5\partial T),$$

where $E_C^{(\delta)}$ packages the carried collar remainder. Then the derived fixed-cap generalized-entropy stationarity theorem gives

$$\delta(G_{00} + \Lambda g_{00}) = 8\pi G\,\delta\langle T_{00}\rangle + \mathcal{E}_\delta,$$

with

$$\mathcal{E}_\delta := \frac{15G}{\pi\ell^4}\,\delta\langle E_C^{(\delta)}\rangle + O(\ell\,\partial T),$$

and, by the bounds above,

$$|\delta\langle E_C^{(\delta)}\rangle| \le C_C\,\eta_\delta$$

for some collar-dependent constant $C_C$ on the fixed faithful model. Equivalently, before the controlled collar and small-ball limits, one may write

$$|\mathcal{E}_{\ell,\delta}| \le C_1 r_{\mathrm{FR}}(\varepsilon_\delta) + C_2\delta^{\mathrm{M}}_{A_\delta:B_\delta:D_\delta}(\varepsilon_\delta) + C_3\eta_\delta^{\mathrm{reg}} + C_4\ell\,\|\partial T\| + o_\delta(1) + o_\ell(1),$$

with constants depending only on the fixed collar model, the bounded observable class, and the small-ball chart. Thus the exact Einstein relation is the controlled limit, not a finite-cutoff identity inherited from approximate Markovity.

If one wishes to package this carried remainder as an effective anomalous energy density, the natural definition is

$$\delta\langle T_{00}^{\mathrm{anom}}\rangle := \frac{15}{8\pi^2\ell^4}\,\delta\langle E_C^{(\delta)}\rangle.$$

Then

$$\delta(G_{00}+\Lambda g_{00}) = 8\pi G\,\delta(\langle T_{00}\rangle + \langle T_{00}^{\mathrm{anom}}\rangle) + O(\ell\,\partial T).$$

This is a bookkeeping definition of the finite-stage remainder, not an upgrade of the recovered-core theorem. Its significance is exactly that it is controlled:

$$|\delta\langle T_{00}^{\mathrm{anom}}\rangle| \le \frac{15\,C_C}{8\pi^2\ell^4}\,\eta_\delta.$$

On a separate dark/anomaly continuation branch, this carried scalar remainder may be promoted to a repair stress only after a quotient-invariant source-localization receipt identifies the same collar operator with the small-ball Einstein remainder, and after a positive causal packet lift supplies the full tensor. In SI units that continuation has rest-energy normalization

$$T_A^{ab}u_a u_b = \frac{15\hbar c}{8\pi^2\ell^4}R_{r,\ell} + O(\epsilon_{\mathrm{loc}}),$$

where $R_{r,\ell}$ is measured in nats and $\ell$ is the proper geodesic small-ball radius. The scalar quantity $I(A:D\mid B)$ alone is not a stress tensor: packet factorization, causal directions, an allocation rule, a rest frame, source-localization map, and transport law are additional continuation data.

The framework carries the Markov error through three distinct controls:

1. $r_{\mathrm{FR}}(\varepsilon_\delta)$ controls bounded observables at one stage;
2. $\delta^{\mathrm{M}}_{A_\delta:B_\delta:D_\delta}(\varepsilon_\delta)$ controls convergence to the exact Markov normal form on a fixed collar;
3. $\eta_\delta^{\mathrm{M}}$ controls modular-Hamiltonian replacements once faithfulness is assumed;
4. the Einstein-branch remainder is a carried $O(\eta_\delta)$ term, not a silent exact identity at finite cutoff.

This is the concrete bridge from "axioms about screens" to "precision GR predictions plus a disciplined carried remainder".

#### 5.3.11 Focusing/QNEC status on the null-modular branch

The focusing content of the recovered core enters through the Recoverable Generalized Entropy axiom and stays an axiom. This subsection records the status of the two focusing statements on the null-modular branch and the exact reason no internal derivation is claimed.

**What the branch supplies.** The fixed-cutoff null bridge of §5.2 supplies transferred cut-center data, exact-or-controlled strip additivity on the inherited strip model, the derived half-sided modular pair with its Borchers–Wiesbrock translation generator, and the half-line generator/charge identification $[K,P] = -i2\pi P$. These are kinematic inputs a QNEC statement consumes; they are not a proof of it.

**Why relative-entropy monotonicity does not yield QNEC.** Monotonicity under restriction orders the relative entropies $S(\rho_{R(\lambda)}\|\omega_{R(\lambda)})$ along nested null cuts; it fixes the sign of the first $\lambda$-derivative and says nothing about the second. A monotone function of the cut need not be convex, so the step from monotonicity to $d^2S_{\mathrm{rel}}/d\lambda^2 \geq 0$ is unavailable. The entanglement first law $\delta S = \delta\langle K\rangle$ is a first-order state variation at fixed algebra and reference state; the QNEC second null shape variation is a different object, and expanding the first law to second order in the cut does not produce it. The published QNEC proofs supply exactly the missing machinery: null-surface locality and analyticity hypotheses in the free and super-renormalizable settings [26], and modular-flow, causality, and defect-OPE inputs in the general QFT setting [28]. None of those hypotheses is verified on the reconstructed patch net.

**Why QFC is not derived from QNEC.** The Quantum Focusing Conjecture is strictly stronger than QNEC: the foundational QFC statement obtains QNEC as its locally parallel limit [27], and no implication runs the other way. QNEC together with the Einstein branch (Theorem 5.1) and the Raychaudhuri identity controls the diagonal, stationary-cut part of the generalized-expansion variation and leaves the off-diagonal, shear, counterterm, and general nonstationary-cut content of QFC untouched. Focusing for $S_{\mathrm{gen}}$ therefore stands where the Recoverable Generalized Entropy axiom declares it and carries no independent internal support from this branch.

**Claim boundary.** The null-modular branch narrows what a future internal QNEC proof must construct: the half-sided modular structure is in place, and the open items are the QFT-side hypotheses named above, verified on the reconstructed net, plus the off-diagonal QFC content. Until such a proof lands as a dated artifact, every downstream use of focusing on this branch cites the axiom.

**Theorem 5.1 (OPH Einstein-branch closure in the scaling regime).** Under OPH5 on a recovered-core, support-visible scaling branch satisfying the E0 dependency discharge above, the fixed-cap generalized-entropy stationarity condition implies the rest-frame first-variation relation

$$\delta(G_{00} + \Lambda g_{00}) = 8\pi G\, \delta\langle T_{00}\rangle.$$

If this relation holds for all local directions and reference states in the locally Lorentzian scaling regime, the timelike scalar-to-tensor upgrade promotes it to the semiclassical Einstein equation modulo the expected $\Lambda g_{ab}$ ambiguity. The D6 capacity branch closes that metric term globally under stable whole-fiber correctable-record closure, the unique finite-size slack zero, and the horizon–record identification.

**Proof sketch.** The E0 discharge turns the OPH5 recovered-core branch into the exact hypothesis package required by the Jacobson-type bridge: quotient-safe geometry readout, $2\pi$-normalized BW cap flow, local null stress charge, bounded-interval ball kernel with $o(\ell^4)$ remainder, fixed-cap generalized-entropy stationarity, the fixed-volume small-ball area identity, and all timelike directions. The cap first law identifies $\delta S_C$ with $\delta\langle K_C\rangle$, while the null bridge together with the E0.5 bounded-interval kernel relates the bulk modular variation to the required null stress-tensor charge on the diamond. Since the reference state is maximally symmetric, the fixed-volume small-ball area identity enters at first order in the perturbation, and the derived fixed-cap generalized-entropy stationarity theorem for admissible fixed-cap MaxEnt variations on the realized cap-label-preserving MaxEnt family yields the displayed rest-frame relation. Lemma E0.7 supplies all local timelike directions, the scalar-to-tensor theorem upgrades the relation to the tensor equation modulo $\Lambda g_{ab}$, and the D6 capacity branch closes the remaining metric term under its separately typed stable-capacity and horizon-record premises. QED.

### 5.3.12 Black-hole structural statements and continuation-level spectroscopy

This subsection mixes two claim tiers and is organized accordingly. The retained structural black-hole package is a four-step chain: fixed-cutoff edge-center collar decomposition on the exact-Markov or idealized exact-recoverability horizon carrier, the small-CMI recoverability reading of interior encoding on that same carrier, Hawking/KMS normalization on the geometric modular branch, and the discrete area spectrum together with the Schwarzschild transition identity $\Delta E = k_B T_H \ln(d'/d)$ for an actual sector change $d \to d'$. Finite same-boundary records and finite reconstruction thresholds do not by themselves supply physical radiation entropy, exterior time, evaporation flux, a geometric island, a radiative quotient, or an asymptotic waveform readout. Any clean comb structure, QNM selector, Page-type linewidth estimate, PBH burst template, Kerr/LIGO horizon spectroscopy template, or Page-curve/island closure is continuation-level and uses continuation inputs stated below.

**Derived structural statements.**

**Area eigenvalues from edge sectors.** The central area operator (Section 5.4) is

$$L_C = \sum_\alpha (\log d_\alpha) P_\alpha,$$

where $d_\alpha \in \mathbb{N}$ is the dimension of the edge Hilbert space in sector $\alpha$. With the normalization $\mathrm{Tr}(\rho L_C) = \langle A \rangle / 4G_{\mathrm{geom}}$, the area eigenvalues are

$$A_\alpha = 4G_{\mathrm{geom}} \log d_\alpha = 4\ell_\star^2 \ln d_\alpha,$$

where $\ell_\star^2 = \hbar G_{\mathrm{SI}}/c^3$ is the scale-readout area displayed as the Planck area. Since $d_\alpha$ is a positive integer, **areas are discretely spaced** with logarithmic gaps.

**Hawking emission energy quantization.** For a Schwarzschild black hole with $A(M) = 16\pi G^2 M^2/c^4$, a transition between sectors $d \to d'$ changes the area by

$$\Delta A = 4\ell_\star^2 \ln(d'/d).$$

The corresponding ADM energy change of the hole is $\Delta(Mc^2) = c^2 \Delta M$, with $\Delta M = \Delta A/(dA/dM)$. This gives

$$\Delta(Mc^2) = \frac{\hbar c^3}{8\pi G M} \ln(d'/d).$$

Using the Hawking temperature $T_H = \hbar c^3/(8\pi G k_B M)$, whose $2\pi$ normalization is fixed by the $\mathrm{BW}_{S^2}$ tangent-limit normalization of Theorem 4.2:

$$\Delta(Mc^2) = k_B T_H \ln(d'/d).$$

**Continuation-level spectroscopy templates.** The remainder of this subsection is not part of the recovered core. It records what follows only if one adds extra discrete-horizon selection rules and, where stated, standard semiclassical inputs such as evaporation-power models, QNM/transition identifications, or greybody matching.

**Integer transitions.** If dominant emission steps reduce the edge dimension by an integer factor $k$ (i.e., $d' = d/k$), the emitted spectrum becomes a discrete comb:

$$\Delta E_k = k_B T_H \ln k, \qquad \Delta f_k = \frac{c^3}{16\pi^2 G M} \ln k.$$

**Structural condition: comb vs. generic discreteness.** The log-integer *comb* structure requires the additional dynamical assumption that integer-ratio emission steps ($d \to d/k$) dominate. If generic transitions between arbitrary integers dominate instead, the set of $|\ln(d'/d)|$ values becomes a dense log-rational set that may appear quasi-continuous after folding in linewidths and astrophysical effects. What follows directly from the axioms is the *discrete area spectrum*; the clean comb pattern follows only with that selection rule.

**Continuation-level template (Discrete Hawking spectrum).** If the additional integer-transition selection rule is realized, the Hawking emission spectrum consists of discrete lines with spacing $\Delta E_k = k_B T_H \ln k$, where $k$ is an integer characterizing the dominant sector transitions, instead of a continuous thermal profile.

**Mass-independent fractional linewidth (continuation-level estimate).** Using Page's semiclassical calculation for emission power $P(M) = p_0 \hbar c^6/(G^2 M^2)$ with $p_0 \approx 2 \times 10^{-4}$, the emission rate is $\dot{N} \approx P/\langle E \rangle$ where $\langle E \rangle = a\, k_B T_H$ with $a \sim \mathcal{O}(1-10)$. The natural linewidth $\Gamma \sim \hbar \dot{N}$ divided by the level spacing gives:

$$\frac{\Gamma}{\Delta E_k} \approx \frac{64\pi^2 p_0}{a \ln k},$$

a function of $(a, k)$. At $k = 2$ the declared range $a \in [1, 10]$ spans 1.8% to 18%; at $k = 3$, 1.1% to 11%. A few-percent value therefore requires the particle content and greybody model that pin $a$ inside a narrower interval (for instance $a \approx 4$–$6$ at $k = 2$); no such model is selected here. The fraction is mass-independent at every $(a, k)$.

**Connection to quasinormal modes (interpretive continuation).** The highly-damped Schwarzschild quasinormal modes have asymptotic real part (Motl, 2002):

$$\mathrm{Re}\,\omega \to \frac{c^3}{8\pi G M} \ln 3.$$

This matches exactly the $k = 3$ transition frequency $\Delta E_3/\hbar$. If one adopts a Bohr-type identification between quantum transition frequencies and asymptotic QNM frequencies, this selects

$$\Delta A = 4\ell_\star^2 \ln 3 \approx 4.39\, \ell_\star^2$$

as the fundamental area quantum.

**Scope statement.** The area quantization follows from the edge-sector structure (derived). The $k = 3$ selection requires the additional interpretive identification with QNM frequencies (not derived from axioms). The linewidth prediction uses standard semiclassical inputs.

**Numerical examples.** For Δf_k = ($c^3/16\pi^2$GM) ln k:

- **M = 30 M☉**: k=2 at 29.7 Hz, k=3 at 47.1 Hz
- **M = 1 M☉**: k=2 at 891 Hz, k=3 at 1412 Hz
- **M = $10^{12}$ kg (primordial)**: k=2 at 7.3 MeV, k=3 at 11.6 MeV

Within the integer-transition continuation, these frequencies track $k_B T_H \ln k$ exactly and are in principle distinguishable from a continuous thermal spectrum.

**Continuation-level PBH burst search template.**

Within the discrete-horizon continuation, the Hawking comb would provide a distinctive gamma-ray template. A template-level discriminant is **log-integer energy ratios**: if two emission lines are observed at energies $E_2$ and $E_3$, their ratio must satisfy

$$\frac{E_3}{E_2} = \frac{\ln 3}{\ln 2} \approx 1.585$$

exactly, independent of black hole mass. Within that continuation template, the ratio is fixed once the log-integer rule is assumed.

**Available instruments and energy coverage.** The $k = 2$ line energy $E_2 = k_B T_H \ln 2$ determines which instruments can see a given BH mass:

| Instrument | Energy band | BH mass range (k=2 in band) |
|---|---|---|
| Fermi GBM (BGO) | 0.15–40 MeV | $2\times10^{11}$–$5\times10^{13}$ kg |
| Fermi LAT | 0.1–300 GeV | $2\times10^{7}$–$7\times10^{10}$ kg |
| H.E.S.S. | 0.1–100 TeV | $7\times10^{4}$–$7\times10^{7}$ kg |
| LHAASO-WCDA | 1–15 TeV | $5\times10^{5}$–$7\times10^{6}$ kg |

**Detector resolution vs. intrinsic linewidth.** The continuation-level linewidth estimate is $64\pi^2 p_0/(a \ln k)$, spanning 1.8%–18% at $k = 2$ over the declared $a \in [1, 10]$ (mass-independent at every $(a, k)$). Published detector energy resolutions:

- Fermi GBM: $< 10\%$ (0.1–1 MeV), $\sim 4\%$ at 10 MeV (BGO)
- Fermi LAT: $< 10\%$ (1–100 GeV)
- H.E.S.S.: $\sim 15\%$ (TeV)
- LHAASO-WCDA: $\sim 33\%$ (TeV)

Within this template, the comb could be resolvable with GBM/LAT; at TeV energies it would appear as moderately broad bumps rather than sharp lines.

**Search protocol.** A dedicated OPH-comb search would:

1. Select burst-like candidates (10–120 s time windows, matching existing PBH burst search protocols).
2. Fit each candidate with null model (smooth continuum) vs. OPH comb model (peaks at $E_k = E_0 \ln k$ convolved with detector response).
3. Scan over the single scale parameter $E_0 = k_B T_H$ (equivalently, BH mass).
4. Require at least two lines satisfying log-integer ratio to claim detection.
5. Correct significance for trials (time windows $\times$ sky positions $\times$ $E_0$ scan).

**Observational context.** Dedicated PBH burst searches (H.E.S.S., LHAASO) report **no significant bursts**. An OPH-specific comb-template analysis of archival data would:

- Set upper limits on OPH-comb PBH burst rates
- Demonstrate direct testability of the discrete spectrum prediction
- Provide constraints comparable to or stronger than generic PBH burst limits

**Data availability.** Fermi GBM provides public Time-Tagged Event (TTE) burst data; Fermi LAT provides public photon event lists with documented analysis workflows. H.E.S.S. has a small public test data release.

**Continuation-level GW horizon spectroscopy template for Kerr remnants.**

The same continuation-level discrete-horizon branch extends to gravitational wave observables. For Kerr black holes, the thermodynamic first law is $\delta M = T_H \delta S + \Omega_H \delta J$, so the entropy change for absorbing a quantum with frequency $\omega$ and azimuthal number $m$ is:

$$\delta S = \frac{\hbar(\omega - m\Omega_H)}{k_B T_H}.$$

In the edge-sector framework, $\delta S = \ln(d'/d)$, so the discreteness condition becomes:

$$\hbar(\omega - m\Omega_H) = k_B T_H \ln k, \qquad k \in \{2, 3, 4, \ldots\}$$

Under those additional discrete-horizon assumptions, this gives the **GW horizon spectroscopy comb**: a continuation-level set of discrete resonant frequencies where the horizon can efficiently absorb or emit energy. A physical QNM or ringdown claim needs a separate exterior-background bridge, a radiative quotient, a finite operator converging to the continuum perturbation generator, future-horizon and future-null-infinity boundary conditions, and an asymptotic readout map. A finite repair spectrum or fitted clock scale is not that bridge.

**Kerr line frequencies.** For a remnant with mass $M$ and dimensionless spin $\chi = a_*/M$, define the spin correction factor:

$$g(\chi) = \frac{2\sqrt{1-\chi^2}}{1+\sqrt{1-\chi^2}}, \qquad \Omega_H(M,\chi) = \frac{c^3}{2GM} \cdot \frac{\chi}{1+\sqrt{1-\chi^2}}.$$

The line frequencies are:

$$f_{k,m}(M,\chi) = \frac{m\,\Omega_H(M,\chi)}{2\pi} + \frac{c^3}{16\pi^2 GM}\, g(\chi)\, \ln k$$

Within this continuation template, once LIGO/Virgo infers $(M,\chi)$ for a remnant, the line pattern is fixed by the inferred remnant parameters and the assumed discrete-horizon rule. The line pattern belongs to a continuation template outside recovered core.

**Line weights from GR envelope + discretization.** In this continuation template, the line strengths are modeled by matching to the known GR greybody absorption spectrum in the semiclassical limit. The discretization rule gives bin width $\Delta\omega_k \approx \omega_T \ln(1+1/k)$ where $\omega_T = k_B T_H/\hbar$. The net line weight (absorption minus stimulated emission) is:

$$W^{\text{net}}_{k,\ell m} = \Gamma^{\text{GR}}_{\ell m}(\omega_{k,m}) \cdot \Delta\omega_k \cdot \frac{k-1}{k}$$

where $\Gamma^{\text{GR}}_{\ell m}$ is the standard GR greybody factor and the $(k-1)/k$ factor arises from KMS detailed balance with $e^{(\omega - m\Omega_H)/T_H} = k$.

**Universal stacking coordinate.** Define the dimensionless rescaled frequency:

$$x := \frac{GM}{c^3 g(\chi)}(\omega - m\Omega_H).$$

Then the predicted line locations collapse to universal constants:

$$x_k = \frac{\ln k}{8\pi} \qquad (k = 2, 3, 4, \ldots)$$

Numerically: $x_2 = 0.02758$, $x_3 = 0.04371$, $x_4 = 0.05516$, $x_5 = 0.06404$.

**Stacking test.** Multiple BBH events can be mapped to this universal $x$ coordinate and stacked. If the comb is real, peaks align across events with different $(M,\chi)$; detector noise does not stack coherently.

**Comparison to existing work.** Prior area-quantization searches [53] used parameterized models with one free spacing constant. The OPH prediction is more constrained: multiple lines with exact $\ln k$ ratios, plus the $(k-1)/k$ weight hierarchy from detailed balance.

**Numerical example (GW170608).** Remnant parameters: $M_f \approx 18.0 M_\odot$, $\chi_f \approx 0.69$. For $m = 2$, the horizon rotation frequency is $m\Omega_H/(2\pi) \approx 719$ Hz. The **thermal comb spacing** (the part that encodes the area quantization) is:

| k | $\Delta f_k := \frac{c^3 g(\chi)}{16\pi^2 GM} \ln k$ (Hz) | Relative weight $(k-1)/k$ |
|---|---|---|
| 2 | 41.6 | 0.500 |
| 3 | 65.9 | 0.667 |
| 4 | 83.2 | 0.750 |
| 5 | 96.5 | 0.800 |
| 6 | 107.5 | 0.833 |

The full physical frequencies are $f_{k,2} = 719 + \Delta f_k$ Hz (i.e., 760–827 Hz), outside LIGO's most sensitive band for this remnant. However, the **stacking analysis** uses the rescaled coordinate $x = GM(\omega - m\Omega_H)/(c^3 g(\chi))$, which maps the thermal spacing to universal constants $x_k = \ln k/8\pi$ regardless of the rotation offset.

**Template-matching criterion.** After rescaling by $(M, \chi)$, spectral features in this continuation template must satisfy the offset-subtracted ratio

$$\frac{f_{k,m} - m\,\Omega_H/(2\pi)}{f_{2,m} - m\,\Omega_H/(2\pi)} = \frac{\ln k}{\ln 2}$$

exactly, equivalently $x_k/x_2 = \ln k/\ln 2$ in the universal coordinate defined above, independent of remnant parameters. The full frequencies $f_{k,m}$ do not satisfy $f_k/f_2 = \ln k/\ln 2$ at nonzero $\Omega_H$: the additive rotation offset breaks the ratio, and only the thermal comb pieces carry it. Absence of coherent stacking at the predicted $x_k$ values would challenge the discrete-horizon continuation template, not by itself the derived area-spectrum statement.

### 5.3.13 Classical mechanics from emergent GR

Once the Einstein equation is established, the framework inherits standard GR consequences. This section makes explicit how classical mechanics emerges.

**Stress-energy conservation is automatic.** The contracted Bianchi identity is geometric:

$$\nabla^a G_{ab} = 0.$$

Combined with the Einstein equation, this implies:

$$\nabla^a \langle T_{ab} \rangle = 0.$$

**Geodesic motion from dust limit.** For pressureless classical matter ("dust"), $T^{ab} = \rho\, u^a u^b$. Conservation yields:

$$\nabla_a(\rho u^a u^b) = 0 \quad \Rightarrow \quad u^b \nabla_a(\rho u^a) + \rho\, u^a \nabla_a u^b = 0.$$

Projecting orthogonally to $u^b$ using $h^b{}_c = \delta^b{}_c + u^b u_c$ kills the first term, giving:

$$\rho\, u^a \nabla_a u^b = 0 \quad \Rightarrow \quad u^a \nabla_a u^b = 0.$$

This is the geodesic equation: free classical bodies follow spacetime geodesics.

**Newtonian limit from weak-field GR.** Take the weak-field, slow-motion limit with metric:

$$g_{00} \approx -(1 + 2\Phi/c^2), \qquad g_{0i} \approx 0, \qquad g_{ij} \approx \delta_{ij}(1 - 2\Phi/c^2),$$

and velocities $|\mathbf{v}| \ll c$. Then $G_{00} \approx 2\nabla^2\Phi/c^2$ (leading order), and $T_{00} \approx \rho c^2$. The Einstein equation reduces to:

$$\nabla^2\Phi = 4\pi G\rho.$$

Geodesic motion reduces to:

$$\ddot{\mathbf{x}} = -\nabla\Phi.$$

These are Newton's gravitational law and Newton's second law. Classical mechanics is recovered as a controlled limit of the emergent GR dynamics.

**Precision classical predictions.** Once the field equation is fixed to Einstein form, the framework inherits the standard GR precision toolbox (post-Newtonian expansion, lensing, time delay, etc.), with no free "shape" parameters beyond $G$ and $\Lambda$.

Selected precision predictions (in the regime where the GR derivation applies):

*Light bending by mass $M$:* For impact parameter $b$,

$$\Delta\theta = \frac{4GM}{c^2 b}.$$

For the Sun with $b \approx R_\odot$: $\Delta\theta \approx 1.751$ arcsec.

*Mercury perihelion advance:* Per orbit,

$$\Delta\varpi = \frac{6\pi GM}{a(1-e^2)c^2}.$$

Using Mercury's orbital parameters: $\Delta\varpi \approx 42.98$ arcsec/century.

*Gravitational redshift:* Between two radii in a static potential,

$$\frac{\Delta\nu}{\nu} \approx \frac{\Delta\Phi}{c^2}.$$

For the Sun (surface to infinity): $z \approx 2.12 \times 10^{-6}$.

These predictions are fixed functions of $G$ and known source parameters, and are confirmed observationally to high precision. The framework contains them automatically on the derived Einstein branch.

### 5.3.14 Precision gravity predictions and experimental bounds

The pure Einstein linearization makes classical dispersion and polarization statements that can be confronted with the tightest available experimental bounds. This section translates those branch-scoped statements into the observables that experiments constrain; it does not infer a quantum graviton pole from the classical equation.

**Speed of gravitational waves.** On a suitable background, the two transverse-traceless classical modes of the pure Einstein linearization propagate on the same invariant null cone as the Maxwell modes:

$$\frac{c_{\mathrm{GW}} - c_\star}{c_\star} = 0 \text{ exactly on this action/background branch.}$$

*Published bound (GW170817 + GRB 170817A multi-messenger):*

$$-3 \times 10^{-15} < \frac{c_{\mathrm{GW}} - c}{c} < +7 \times 10^{-16} \quad \text{(90\% credibility).}$$

For a source at $\sim$ 40 Mpc, this fractional difference corresponds to only a few seconds of propagation-time mismatch across $\sim 10^8$ years of travel.

**Einstein quadratic mass parameter.** On the additional pure two-derivative Einstein–Hilbert action branch, the transverse-traceless quadratic operator has no Fierz–Pauli hard-mass parameter:

$$\mu^2_{\mathrm{FP,hard}} = 0.$$

This is an action-content statement, not a graviton-particle mass prediction from diffeomorphism redundancy alone. A quantum mass or pole claim requires Definition 6.18, while diffeomorphism- invariant Stueckelberg/bimetric completions and extra-field theories lie outside the pure-Einstein field-content branch.

*Published bound (GW dispersion analysis, PDG 2025):*

$$m_{\mathrm{grav,disp}} \leq 1.76 \times 10^{-23}\ \mathrm{eV}/c^2 \quad \text{(90\% credibility; model-dependent dispersion fit)}.$$

This corresponds to a reduced Compton wavelength $\bar{\lambda}_C \gtrsim 1.6 \times 10^{16}$ m, i.e., order $\sim 1.6$ light-years.

**No dipole radiation.** Many modified gravity theories predict extra channels (scalar/vector) producing dipolar radiation at $(-1)$PN order. The derived GR limit predicts no such channel.

*Published bound (GW170817 inspiral phasing, PDG 2025):*

$$-4 \times 10^{-6} < \delta\hat{p}_{-2} < 2 \times 10^{-5} \quad \text{(90\% credibility)}.$$

**Tensor polarizations on the selected field-content branch.** The pure Einstein linearization contains two tensor polarizations. It does not prove that every broader diffeomorphism-invariant EFT lacks additional scalar, vector, or massive modes. Pure non-tensor hypotheses are disfavored by observational constraints, and mixed tensor-scalar/vector models are tightly constrained.

**Equivalence principle tests.** Additional null checks from the derived GR structure:

- Universality of free fall (space tests): precision ~$10^{-15}$
- Nordtvedt parameter ($\eta = 4\beta$ - $\gamma$): (0.47 ± 0.55) × $10^{-4}$
- Binary pulsar radiative damping (PSR J0737-3039): 0.999963 ± 0.000063

#### 5.3.15 State–observable error propagation and the dynamical bridge

On the additional pure-Einstein action/background branch, the framework states the classical equalities recorded in Section 5.3.14. Finite Markov/recovery control supplies a separate and more limited quantitative statement. This subsection therefore keeps two error ledgers separate. The *state–observable ledger* tracks expectation values of specified bounded local observables in a recovered state; the Markov/recovery machinery controls this ledger. The *dynamical-parameter ledger* tracks spectral and dispersion properties of an evolution generator: propagation speeds, particle-pole masses, and correlator poles. The trace-distance chain alone does *not* control that ledger.

**The key quantitative hook.** From Theorem 3.1, if the target state satisfies

$$I(A_k : C_k \mid B_k) \leq \varepsilon_k,$$

then, writing $\rho$ for the target state and $\sigma$ for its recovered comparison state, the recovery map gives the trace-norm bound

$$\|\rho - \sigma\|_1 \le \delta_k := 2\sqrt{1 - e^{-\varepsilon_k}} \le 2\sqrt{\varepsilon_k}.$$

**State–observable ledger.** Trace distance gives immediate bounds on expectation-value errors for the declared operator class: bounded operators supported in the recovered region, compared at the time of recovery. Using the standard dual norm inequality:

$$|\langle O\rangle_\rho - \langle O\rangle_\sigma| \le \|O\|_\infty \|\rho - \sigma\|_1 = 2\|O\|_\infty D(\rho, \sigma),$$

where $D(\rho, \sigma) = \frac{1}{2}\|\rho - \sigma\|_1$ is the trace distance. This inequality is exactly as strong as it looks and no stronger: the norm $\|O\|_\infty$ is the Lipschitz constant only for bounded $O$, and the comparison is an equal-time statement about states.

**Exponential decay from the mixing hypothesis.** Writing $w$ for collar width, so it is not confused with the recovery error $\delta_k$, the mixing assumption (Section 2.3) provides

$$I_\omega(A_w : D_w \mid B_w) \le C_{\mathrm{mix}} |\partial C|_{\mathrm{UV}} e^{-w/\xi}.$$

Combining these gives an explicit precision dial for bounded local observables:

$$\delta_{\mathrm{step}} \lesssim 2\sqrt{C_{\mathrm{mix}} |\partial C|_{\mathrm{UV}}} \cdot e^{-w/(2\xi)}.$$

For an operator normalized by $\|O\|_\infty \le 1$, an independently chosen absolute expectation-error target $\tau_O$ is guaranteed at one step if $\delta_{\mathrm{step}} \le \tau_O$; the theorem therefore gives the sufficient CMI condition $\varepsilon \le (\tau_O/2)^2$. The target $\tau_O$ belongs only to the state–observable ledger and cannot be set by importing a gravitational-wave speed tolerance.

**What the dial does not certify: dynamical parameters.** The gravitational-wave propagation speed and any graviton pole or dispersion-mass parameter are not bounded local observables. They are spectral/dispersion properties of the dynamical generator (poles of retarded correlators, or long-time wave-packet phase and group velocities), so the trace-distance inequality supplies neither their norm nor their Lipschitz constant. Two states can be close in trace distance at one time while being vacua of generators with different dispersion relations, and a small equal-time expectation error does not control phase accumulation over the $\sim 10^8$-year propagation baseline of GW170817. Consequently, $\delta \lesssim 10^{-15}$ is *not* a theory-side certificate for the multimessenger bound $|c_{\mathrm{GW}}/c - 1| \lesssim 10^{-15}$ of Section 5.3.14, and the recovery error must not be presented as the precision dial for that published bound. The ideal classical relation $c_{\mathrm{GW}} = c_\star$ and the vanishing pure-Einstein hard parameter $\mu^2_{\mathrm{FP,hard}} = 0$ belong to the additional action-level branch recorded there; neither stability away from that branch nor a quantum graviton pole follows from the trace-distance chain.

**Required bridge (dynamical-parameter ledger; open).** A sufficient acceptance gate can be stated precisely, but its hypotheses have not been derived from the Markov remainder in this paper. Let $\mathcal{B} = [k_{\min}, k_{\max}]$, with $0 < k_{\min} < k_{\max}$, be the declared experimental wave-number band. Suppose a future construction supplies, for each tensor helicity, a $C^1$ self-adjoint finite-regulator generator $H_\nu^{\mathrm{TT}}(k)$ and the GR reference generator $H_0^{\mathrm{TT}}(k)$, expressed in angular-frequency units ($\hbar = 1$) on one specified common finite-dimensional helicity fiber or after a declared $C^1$ unitary identification of the two fibers. Assume that the reference TT tensor-mode eigenvalue $\omega_0(k) = c_\star k$ is simple in the helicity block and is separated from the rest of the spectrum by a uniform gap $g_{\mathcal{B}} > 0$. Define

$$M_{\mathcal{B}} := \frac{1}{c_\star} \sup_{k \in \mathcal{B}} \left\| \partial_k H_0^{\mathrm{TT}}(k) \right\|_\infty$$

and the dimensionless dynamical error

$$\eta_{\mathrm{dyn},\nu} := \max\left\{ \frac{1}{g_{\mathcal{B}}} \sup_{k\in\mathcal{B}} \left\| H_\nu^{\mathrm{TT}}(k) - H_0^{\mathrm{TT}}(k) \right\|_\infty , \frac{1}{c_\star} \sup_{k\in\mathcal{B}} \left\| \partial_k H_\nu^{\mathrm{TT}}(k) - \partial_k H_0^{\mathrm{TT}}(k) \right\|_\infty \right\}.$$

For $\eta_{\mathrm{dyn},\nu} < 1/4$, let $\omega_\nu(k)$ be the continued isolated eigenvalue and define its group velocity by $c_{\mathrm{GW},\nu}(k) := \partial_k \omega_\nu(k)$. The standard isolated-eigenvalue projector estimate and the Hellmann–Feynman formula then give, uniformly over $\mathcal{B}$,

$$\sup_{k\in\mathcal{B}} \left| \frac{c_{\mathrm{GW},\nu}(k)}{c_\star} - 1 \right| \le L_{\mathcal{B}}\, \eta_{\mathrm{dyn},\nu}, \qquad L_{\mathcal{B}} := 1 + 4M_{\mathcal{B}}.$$

Indeed, if $P_\nu(k)$ and $P_0(k)$ are the rank-one spectral projectors, then $\|P_\nu - P_0\|_\infty \le 2\eta_{\mathrm{dyn},\nu}$, hence $\|P_\nu - P_0\|_1 \le 4\eta_{\mathrm{dyn},\nu}$. Differentiating the isolated eigenvalue therefore bounds the group-velocity change by the derivative perturbation plus $4M_{\mathcal{B}} c\, \eta_{\mathrm{dyn},\nu}$. This displays a dimensionless constant derived from a specified generator and uniform over the declared band. If $q := L_{\mathcal{B}} \eta_{\mathrm{dyn},\nu} < 1$, then stationary propagation at fixed group velocity over path length $D$ obeys the group-delay bound

$$|\Delta t| \le \frac{D}{c_\star} \frac{q}{1-q}.$$

The same bound applies to varying propagation only if it holds pointwise along the entire path; on a cosmological path, the redshifted wave number must remain inside $\mathcal{B}$. A wave-packet phase bound likewise requires uniform control of the inverse dispersion $k_\nu(\omega)$ on the corresponding frequency band. A model-dependent particle dispersion-mass bound is a separate spectral estimate at $k = 0$; it is not a corollary of the band-limited speed estimate.

The OPH-specific bridge requires three additional constructions:

1. construct $H_\nu^{\mathrm{TT}}$, or an equivalent retarded two-point kernel, from the finite and scaling OPH dynamics;
2. prove a norm-resolvent, quadratic-form, or correlator estimate of the form $\eta_{\mathrm{dyn},\nu} \le C_{\mathcal{B}} r_N$, where $r_N := \min(2, \sum_{k=2}^N \delta_k)$ is the accumulated trace-norm recovery bound of Theorem 3.1 for the associated $N$-step construction and $C_{\mathcal{B}}$ is derived rather than assumed; and
3. apply the resulting band-uniform dispersion estimate to the detector band and propagation baseline, including phase accumulation where used by the likelihood.

Only after the second item is proved may one combine the estimates to obtain

$$\sup_{k\in\mathcal{B}} \left| \frac{c_{\mathrm{GW},\nu}(k)}{c} - 1 \right| \le L_{\mathcal{B}} C_{\mathcal{B}} r_N.$$

No estimate $\eta_{\mathrm{dyn},\nu} \le C_{\mathcal{B}} r_N$ is proved here. Substituting $\delta_k$, $\delta_{\mathrm{step}}$, or any equal-time trace-distance error for $\eta_{\mathrm{dyn},\nu}$ without that bridge is invalid.

**Precision summary.** The framework provides:

1. The ideal pure-Einstein linearization statements $\mu^2_{\mathrm{FP,hard}} = 0$ and $c_{\mathrm{GW}} = c_\star$, without promotion to a quantum graviton pole.
2. Translation of those ideal equalities into the specific observables experiments constrain (Section 5.3.14).
3. Explicit bounds on how far expectation values of bounded local observables in the recovered state can drift, using the conditional mutual information → trace distance → observable error chain.

It therefore provides quantitative error control on recovered states and bounded observables, but no present theory-side certificate for the published GW-speed or model-dependent particle dispersion-mass bounds. Dynamical-parameter certification remains in the separate open ledger above.

### 5.3.16 Dark-sector response from the modular anomaly

The D12 dark-sector continuation promotes scalar repair occupation to a canonical pair $(n, \theta)$. Its lattice action is a proposed dynamical completion rather than a recovered-core theorem.

The candidate anomaly-tensor slot $T_{ab}^{\text{anom}}$ is available only after the finite covariant source packet of Section 5.9 supplies source localization, CMI-to-modular-source matching when that route is used, rank-two reconstruction, conservation, normalization, and universal coupling. On that packet it supplies one structural ingredient for a possible dark-sector continuation without introducing additional particle species. Raw scalar collar CMI supplies only a recovery diagnostic.

**Conditional source identification.** Once the source packet has identified an anomalous local modular energy, its rest-frame normalization is

$$\langle T_{00}^{\text{anom}} \rangle = \frac{15}{8\pi^2} \cdot \frac{\delta \langle K_C^{(\text{anom})} \rangle}{\ell^4}$$

The quantity can be called dark only on a branch whose packet also establishes gravitational coupling and absence of an electromagnetic coupling. These properties motivate the dark-sector interpretation; neither they nor the observational identification follow from the collar-CMI theorem.

**Conditional repair-charge action.** The phase equation supplies a conserved repair current with explicit boundary flux. Its dilute homogeneous phase has $\rho_R \propto a^{-3}$, while the cubic condensed link energy gives $a_R = \sqrt{a_b a_0}$ and $v^4 = GM_b a_0$ on the spherical deep branch. The same action couples coherent matter through $q_\star \chi_\nu^{\text{can}} S_{\text{coh}}^{\text{can}}$ and carries the field stress required for momentum closure.

**Scope.** The canonical pair, compact-phase completion, baryonic source map, dimensional couplings, full constitutive law, relativistic refinement limit, abundance, and physical likelihoods are work in progress. A compact neutral coherent source has no monopole and cannot support a closed device.

**Cosmology/Boltzmann contract.** The conditional background scaling above is not an FLRW perturbation kernel. A cosmological dark/anomaly claim must instead expose the variables needed by an Einstein–Boltzmann implementation:

$$\bar{\rho}_A(a), \quad \bar{\rho}_{A,\text{eq}}(a), \quad w_A(a), \quad c_{s,A}^2(k,a), \quad \sigma_A(k,a), \quad Q_A^\mu, \quad B_A(k,a), \quad \Gamma_{\text{rec}}(k,a).$$

The first, source-side no-go is that $\bar{\rho}_A(a)$, $\bar{\rho}_{A,\text{eq}}(a)$, and $B_A(k,a)$ alone do not define a physical Boltzmann source. A physical source must be emitted by a finite covariant collar-packet parent

$$\mathcal{P}_r[X,g] = (C_r, Z_r, A_r, R_r, G_r, \pi_r[X], L_r[X], Q_r, D_r),$$

whose finite packet states split the anomaly and any repair recipient, whose repair generator and reaction channels close the total stress tensor, and whose causal response data set the pressure, sound speed, anisotropic stress, exchange current, and repair rate. The parent must also declare exactly one local source route: fixed-reference anomalous modular energy, or nonlinear CMI stress with a CMI-to-modular-source matching theorem/hypothesis and residual ledger. Raw collar CMI is otherwise a recoverability diagnostic, not the $15/(8\pi^2)$ BW source. $B_A$ is gauge-invariant only when built from the anomaly-frame baryon density $n_b^{(A)} = -u_{A\mu} J_b^\mu$. If the exchange branch is nonzero, explicit recipient stress and an equal-and-opposite exchange current are mandatory; otherwise the repair term is not a closed stress-energy source. A transition spectral number is only $\gamma_{\text{repair step}}$ without active-fiber, physical-clock, and common-parent response-pole receipts for a physical $\Gamma_{\text{rec}}$. A physical likelihood comparison also requires declared source, solver, dataset, covariance, and nuisance provenance.

**Canonical finite covariant collar-packet parent.** For a regulator $r$ and background $(X, g)$, the source object is

$$\mathcal{P}_r[X,g] = (\mathcal{C}_r, g_r, e_r, U_r, Z_r, \omega_r, p_r, f_r, \Phi_r, \mathcal{R}_r, \mathcal{G}_r, \pi_r, \mathsf{Read}_r).$$

Here $\mathcal{C}_r$ is a finite causal cell complex with oriented faces, cell four-volumes, causal adjacency, and finite parallel transports $U_{F\to c}$; $e_r$ is the tetrad/local-frame data; $Z_r = \bigsqcup_s Z_{s,r}$ is the packet state space for anomaly, recipient, radiation, standard matter, and any interaction sector; $\omega_r, p_r, f_r$ are invariant weights, local momenta, and occupations; $\Phi_r$ is the face flux; $\mathcal{R}_r$ is the reaction channel list; $\mathcal{G}_r$ is the finite gauge/quotient data; $\pi_r$ is the restriction/refinement map; and $\mathsf{Read}_r$ is the local observable readout. With signature $(-+++)$, every packet used in a kinetic stress branch must satisfy $g_{ab}p_z^a p_z^b = -m_z^2$, $p_z^0 > 0$, and $\omega_z f_{c,z} \geq 0$, with the invariant measure convention declared explicitly.

**Scalar-to-stress nonuniqueness.** A scalar collar row, even a finite row for $\rho_A(a)$, $\rho_{A,\text{eq}}(a)$, or $B_A(k,a)$, does not determine a stress tensor. Many inequivalent packet parents can have the same scalar density while differing in pressure, sound speed, anisotropic stress, exchange current, gauge-invariant perturbation variables, and causal response. Therefore no evidence bundle may promote scalar rows to physical CMB, lensing, growth, or $S_8$ inputs unless the parent supplies the missing stress and response variables.

**Finite MaxEnt lift.** When only finitely many one-cell source readings $e_x(u_i) = T^A_{ab}(x)u_i^a u_i^b$ are available, a stress lift is not unique until the branch supplies a finite selection rule. The admissible MaxEnt lift minimizes no hidden observational residual: it maximizes the declared packet entropy subject to the source readings, mass shells, conserved charges, local frame covariance, and refinement restrictions. The lift is physical only when held-out timelike probes pass a quadraticity/tomography test and the variational stress agrees with the packet-moment stress in the predeclared norm.

**Packet stress and exchange closure.** For a kinetic parent,

$$T^{ab}_{I,r}(c) = \frac{1}{V_c}\sum_{z\in Z_{I,r}(c)} \omega_z f_{c,z} p_z^a p_z^b, \qquad J^a_{I,r}(c) = \frac{1}{V_c}\sum_{z\in Z_{I,r}(c)} \omega_z f_{c,z} p_z^a.$$

More general packet parents may add an internal stress $\Sigma_z^{ab}$, but then $\Sigma_z^{ab}$ is part of the primitive parent data and must obey the same local-frame and refinement checks. The finite transport/reaction equation must imply

$$\mathrm{Div}_r\, T^{ab}_{I,r} = Q^b_{I,r}, \qquad \sum_I Q^b_{I,r} = 0.$$

If the repair exchange is nonzero, a row labelled "recipient" is not enough: the recipient stress $T_R^{ab}$, exchange current $Q_R^b$, and closure residual $Q_A^b + Q_R^b$ must be emitted and certified. If exchange is off, the parent instead emits a repair-exchange-off receipt.

**Response, rate, and gauge receipts.** The finite causal response must provide a finite domain of dependence, subluminal characteristic bound, retarded support residual, and response stability residual. The transition-matrix diagnostic may be called $\gamma_{\text{repair step}}$ until the active fiber, conserved-sector decomposition, physical clock $\Delta\tau_{\text{physical}}$, and common parent response pole are certified. Only then may one write

$$\Gamma_{\text{rec}} = -\operatorname{Re}\lambda_{\text{active}}/\Delta\tau_{\text{physical}}.$$

Gauge checks must be stated in gauge-invariant or rest-frame variables; raw Newtonian/synchronous agreement is a diagnostic, not a gauge-independence receipt.

**Certificate hierarchy.** The finite parent certificate is the conjunction of primitive evidence, not a producer-declared Boolean. The source side must certify the route and entropy unit, modular nonadditivity, localization and stress tomography, finite-packet kinematics and mass shell, covariant transport and four-momentum, stress readout and variational agreement, local-frame and quotient invariance, total stress and exchange-current closure, cosmological gauge invariance, finite domain of dependence, subluminal and retarded response, stability under refinement, and the cold-dark-matter limit. Frozen source, solver, likelihood, and data hashes are later promotion receipts. They do not belong to the parent theorem and cannot rescue a failed parent.

The same firewall applies to the promoted CMB-source inputs

$$\eta_R, \quad \gamma_{\text{repair step}} \text{ or certified } \Gamma_{\text{rec}}, \quad A_\zeta, \quad q_{\text{IR}}, \quad \ell_{\text{IR}}, \quad B_A(k,a), \quad \rho_A(a), \quad N_{\text{CRC}}.$$

Each one must have a source-provenance DAG node recording the source report, parents, no-CMB-data flag, and measurement-use flags. Contradictory provenance, such as declaring no CMB data use while also fitting to Planck, fails closed. Reducers must pool additive sufficient statistics globally before any nonlinear estimate of tilt, amplitude, inverse precision, rank, condition number, isocurvature leakage, $B_A$, $\rho_A$, or $\Gamma_{\text{rec}}$; raw transition diagnostics are $\gamma_{\text{repair step}}$. Shard-local nonlinear averages and shard-local `any()` likelihood rollups are diagnostic only. $N_{\text{CRC}}$ is the D6 consensus invariant unless a separate additive capacity schema proves disjoint coverage. The cold transported limit is the check case: when exchange, pressure, sound-speed, and anisotropic stress corrections are turned off, the anomaly slot must reduce to a CDM-like component before recombination. Any nonzero-field response, late-time growth suppression, or $S_8$-relief branch has to be emitted by the finite-collar parent evaluator through $B_A(k,a)$ and $\Gamma_{\text{rec}}(k,a)$ after the physical-clock and response receipts pass, not fitted as a free environmental kernel to CMB, weak-lensing, SPARC, or cluster data. The compressed $H_0$, $\Omega_m$, $\sigma_8$, and $S_8$ rows are plumbing diagnostics for a low-$H_0$, Planck-like branch, not theorem-grade cosmology.

**Finite cosmological source and prediction eligibility.** The continuation-level physical cosmology gate is the conjunction

$$\mathsf{CMB\_READY} = \mathsf{SOURCE} \wedge \mathsf{SCALE} \wedge \mathsf{PARENT} \wedge \mathsf{KERNEL} \wedge \mathsf{INIT} \wedge \mathsf{TRANSFER} \wedge \mathsf{FREEZE} \wedge \mathsf{LIKE}.$$

$\mathsf{SOURCE}$ is the source-only DAG with transitive measurement-taint rejection and globally pooled sufficient statistics; $\mathsf{SCALE}$ is the physical mode/clock/lift bridge; $\mathsf{PARENT}$ is the finite covariant dark/anomaly parent with packet kinematics, mass shell, packet stress readout, reaction-channel four-momentum, recipient stress and exchange-current closure for nonzero exchange, total stress closure, local-frame/quotient invariance, cosmological gauge invariance, finite domain of dependence, subluminal retarded response, response stability, refinement, and CDM-limit recovery receipts; $\mathsf{KERNEL}$ emits physical $\rho_A$, $\rho_{A,\text{eq}}$, $B_A$, and certified $\Gamma_{\text{rec}}$; the source-only $\rho_A$ row additionally requires the anomaly abundance selector

$$\mathcal{E}_{A,r} = \mathbb{E}_{\mu^{\text{rel}}_{A,r}}[\mathsf{L}_{A,r}]$$

and its no-data-use and refinement receipts; INIT supplies regular initial modes and Einstein constraints; TRANSFER supplies Boltzmann well-posedness and numerical convergence; FREEZE fixes the model generation; and LIKE is the official likelihood execution. While this conjunction is false, physical CMB, BAO, lensing, growth, RSD, and $S_8$ outputs are diagnostic or conditional, not recovered-core predictions.

This branch is a D12 completion contract, not a physical dark-matter theory.

**Falsifiability.** The repair-charge condensate requires a source-derived action, relativistic limit, and joint galaxy, Solar-System, lensing, cluster, and cosmological likelihood. These constructions are work in progress, so the falsification program contains no cosmological target or verdict.

#### 5.3.17 De Sitter holography: static patch vs boundary-at-infinity

A natural question arises: how does this framework relate to the "unsolved problem" of de Sitter holography?

**What the usual dS holography problem is.** When people say "dS holography is unsolved," they typically mean that we do not have anything as sharp as AdS/CFT, where the bulk has a timelike asymptotic boundary supporting a well-defined dual CFT with a precise dictionary. For de Sitter, there is no asymptotic timelike boundary in the static patch where one can simply place the dual theory. The classic dS/CFT proposal at future infinity has familiar difficulties, including non-unitarity worries and complex conformal weights.

**Static-patch/horizon-screen setup.** The framework begins with an observer's static patch and its horizon screen $S^2$, building a net of subregion algebras on that screen. At finite cutoff those algebras are type-I regulators. The Lorentz statement, when invoked, is a support-visible scaling-limit statement about the refinement-limit observer net, and that limit may leave the regulator class. By Theorem 5.2.2, the realized scaling-limit cap modular action on the extracted geometric cap pair is geometric and, in the non-type-I case of interest, generally outer.

This is therefore a fundamental fork away from AdS/CFT-style holography:

| AdS/CFT | This framework |
|---|---|
| Codimension-1 boundary at infinity | Codimension-2 horizon screen ($S^2$) |
| Single global boundary theory | Observer-dependent patches that overlap |
| Dual CFT required | Only algebras + consistency conditions |
| Negative $\Lambda$ | Positive $\Lambda$ natural |

This aligns with the static-patch/complementarity intuition in the dS literature, where the fundamental description is patch-based and different static patches are related by consistency rules, not by a single global boundary theory.

**The mechanism: $\Lambda$ as global capacity, not local physics.** A key structural result is that null modular data reconstruct the stress tensor only up to an additive metric term. This is the statement that vacuum-energy or cosmological-constant shifts are invisible to the local null-data route. The Einstein equation derived from the fixed-cap generalized-entropy stationarity theorem is therefore fixed only up to $\Lambda g_{ab}$.

**Conditional theorem boundary: correctable public-record closure.** Freeze a capacity carrier $\mathcal{H}_{\mathrm{cap},r,D}$ with $D = \dim \mathcal{H}_{\mathrm{cap},r,D}$ and $N = \log D$. For every terminal world $q$, compatible record atoms, endogenous reachability, a frozen publicness policy, and the complete source-supplied global checkpoint family define a compound confusability graph $G_q$. The exact public capacity and whole-fiber readback are

$$M_0(q) = \alpha(G_q), \qquad \mathfrak{F}_{r,0}(D) = \{M_0(q) : q \in \widetilde{\Omega}_{r,D}\}.$$

When the entire nonempty terminal fiber has one common readback, define $M_0(\mathfrak{U}_N) := \widehat{F}_{r,0}(e^N)$. The universe-level closure is then

$$\boxed{N = \log M_0(\mathfrak{U}_N)}.$$

The notation is typed: $M_0$ is always a multiplicative record count, while $N$ is its logarithm. Stable direct closure and its exact finite-size selector are

$$\mathfrak{F}_{r,0}(D_\star) = \{D_\star\}, \qquad s(D) := \log D - \log M_0(D), \qquad s(D_\star) = 0 < s(D) \quad (D \neq D_\star).$$

On the exact reversible branch every checkpoint generator is injective, so

$$M_0(q) = |X_{\text{reach}}(q)|,$$

reducing the finite computation to exact CSP or model counting. This is the preferred first implementation of the source-only finite public checkpoint packet $\mathcal{C}^{\text{pub}}_{r,D}$.

The local null-data route is blind to metric-term shifts, so the Einstein branch is fixed locally only modulo $\Lambda g_{ab}$. Conditional on stable direct closure and the independent horizon–record identification,

$$N_\star = \log D_\star = \frac{A_{\text{dS}}}{4\ell_\star^2}, \qquad \Lambda_\star \ell_\star^2 = \frac{3\pi}{N_\star}.$$

With the selected scale certificate the same branch has the display

$$G_{ab} + \frac{3\pi}{GN_\star}\, g_{ab} = 8\pi G\, \langle T_{ab} \rangle.$$

The independent common screen/electroweak load-carrier identification tests the same closed $N$ against the weak/Higgs load through

$$R_{\text{EW}}(P, N) = \alpha_U(P) \log\left(\frac{N}{\pi}\right) - \frac{6\pi}{P} = 0, \qquad N_{\text{bridge}} = \pi \exp\left[\frac{6\pi}{P\alpha_U(P)}\right].$$

Neither physical bridge constructs the direct capacity map.

**Capacity readout.** The branch uses $N_{\text{scr}}$ as the entropy capacity. The bare radius-squared ratio is $N_{\text{patch}} = (r_{\text{dS}}/\ell_P)^2$, and

$$N_{\text{scr}} = \pi N_{\text{patch}} = \frac{3\pi}{\Lambda \ell_P^2}.$$

The observed late-time scale gives $N_{\text{patch}} \simeq 1.05 \times 10^{122}$, with a Planck-$\Lambda$ central entropy capacity $N_\Lambda \simeq 3.313 \times 10^{122}$. The conditional electroweak bridge value $N_{\text{EW}} \simeq 3.532 \times 10^{122}$ is about 6.6 percent higher.

**What this solves vs. what it assumes.** The model does **not** solve the classic "give me a unitary CFT at future infinity" problem. It does not aim there. It also does **not** prove that every refinement-stable MaxEnt branch lands in the static-patch geometric modular branch with emitted cap pair and standard modular action. What it does provide is a coherent route to **patch holography** in which de Sitter static patches are natural:

1. the fundamental object is a horizon screen in a static-patch description;

2. $\Lambda$ is a capacity parameter tied to finite Hilbert-space dimension, not a locally reconstructible vacuum-energy term;

3. Einstein-like dynamics emerge up to $\Lambda g_{ab}$;

4. on the support-visible BW scaling branch, the realized scaling-limit cap modular action is geometric and may be outer on a non-type-I observer algebra.

**BW-side boundary.** The Lorentz side is closed at the support-visible scaling level by Theorem 4.2. The theorem intentionally avoids the false stronger route through a full-algebra unregularized common floor; the automorphism statement on the observer-facing geometric cap pair is the required static-patch content.

**Many observers, one $\Lambda$.** In this framework, each timelike observer is associated with a horizon patch rather than a single global description. On the conditional D6 branch, the dimensionless de Sitter capacity relation is the shared global constraint across overlap-consistent descriptions; its SI $\Lambda$ display uses the selected scale certificate.

**Summary.** The model moves the holographic screen from "infinity" to an observer's horizon. Direct N closure is a correctable public-record theorem candidate, not a checkpoint-fixed or marginal-information count. Horizon saturation and the weak/Higgs common-load equation are independent downstream commuting squares. The weak/Higgs bridge and the late-time de Sitter coordinate differ by about 6.6 percent; that comparison becomes physical only after the direct packet and both bridges share a certified carrier.

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## 5.4 Gauge Reconstruction and Standard Model Structure

### 5.4.1 Edge sector category and gauge group reconstruction

At any fixed UV cutoff, edge-center completion provides finitely many visible seed charges and finite-dimensional intertwiners. Their rigid tensor closure is defined below and may have infinitely many simple classes. The local MaxEnt / collar-mixing package established above controls only fixed-cutoff recoverability, modular-support localization, and carried error terms on the realized branch. It is logically separate from the question whether zero-obstruction edge sectors survive refinement. The fixed-stage category is theorem-produced; the refinement/fiber ladder and cofinal compact-gauge witness require the explicit compact-gauge refinement receipt defined below.

On the ordinary or central-defect branch, path-independent movement of collar charges is supplied by overlap gluing. TransportabilityFromOverlapGluing constructs transport from overlap paths and proves the exact combined criterion: the central triangle class $[z]_\Sigma$ must vanish and at least one allowed strictification must have trivial represented sector holonomy. On the genuinely noncentral branch, $o^{(2)}_\Sigma = 0$ strictifies the higher associator, after which at least one allowed strict $G_\Sigma$-valued representative must have trivial represented holonomy. A nonzero $o^{(2)}_\Sigma$ remains a higher-gauge sector; an orbit for which every strict representative has residual loop action is also outside the ordinary path-independent DR sector. The full orbit $q_\Sigma$ need not determine a unique ordinary $H^1$ class and is not itself the strictification test. Thus the overlap obstruction calculus classifies and routes sectors; it does not by itself select the Standard Model.

For the tensor construction at cutoff $r$, choose one common stagewise strict edge cocycle $U^{\mathrm{str}}_r$ from the allowed representatives (the ordinary branch uses its given strict system). Retain only seeds on which this same representative has trivial action around every closed overlap loop. Sectors requiring incompatible strictifications define alternative fixed-stage categories; they are not unioned before tensor closure.

**Classification is not realization.**

**Proposition (obstruction neutrality of the Standard Model selection step).** The ordinary trivial-holonomy condition, the combined central condition $[z]_\Sigma = 0$ plus a trivial-holonomy strictification, and the combined noncentral condition $o^{(2)}_\Sigma = 0$ plus an allowed strict representative with trivial represented $G_\Sigma$-holonomy are transportability conditions. After one common stagewise strict representative is chosen, they permit an ordinary transportable bosonic sector category generated only by sectors trivial under that choice; they do not select

$$\frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}.$$

On a cofinal tail carrying the compact-gauge refinement receipt, DR/Tannaka reconstruction returns $G = \mathrm{Aut}_\otimes(\mathcal{F})$ for the constructed sector category and fiber functor. If that category is trivial, $G$ is trivial; in general $G$ is whatever compact group the tensor-fiber data reconstructs. The Standard Model quotient enters only after MAR is applied to a nonempty realized one-Higgs chiral sector package. QED.

**Definition (fixed-cutoff tensor-generated sector category).** Let $S_r = \{W_{\alpha,r} : P_{\alpha,r}$ is visible and has trivial holonomy under the common $U^{\mathrm{str}}_r\}$. Define

$$\mathsf{Sect}^{\mathrm{bos}}_r = \langle \mathbf{1}, S_r, S^*_r \rangle_{\oplus,\otimes,\text{subobjects},\text{isomorphism}} \subset \mathrm{Rep}_{\mathrm{fd}}(\widehat{K}_{\Sigma,r}).$$

This is the replete full additive Karoubi rigid tensor subcategory generated by the finitely many visible carriers. The original projectors $P_{\alpha,r}$ identify only one-collar seeds. The simple objects are all irreducible summands of finite tensor words in seeds and conjugates; they need not have projectors in the original one-collar algebra, and there may be infinitely many of them even though every object and Hom space is finite-dimensional. A subobject $X \subset T$ is supported by its invariant idempotent in $\mathrm{End}_{\widehat{K}_{\Sigma,r}}(T)$, or by the corresponding block in a supplied finite concatenated-collar realization of that tensor word.

**Theorem (FixedCutoffBosonicSectorCategory).** At every fixed regulator cutoff, on the ordinary or central-defect zero-obstruction branch and in the bosonic internal-gauge sector of the $3+1$-dimensional EFT regime, the preceding construction is a semisimple rigid symmetric $C^*$-tensor category. Complete reducibility of finite-dimensional representations of the compact group gives semisimplicity and subobjects; the operator adjoint and norm give the $C^*$-structure; conjugate representations give duals; and the standard flip gives the bosonic symmetry. For a closed loop, $U^{\mathrm{str}}_r$ acts as the identity on every retained seed, hence as the identity on direct sums, tensor products, conjugates, and invariant subobjects. Thus the whole generated category has one monoidally coherent path-independent transport, rather than an objectwise choice of strictification. Collar concatenation realizes tensor words whenever the corresponding finite tensor-realization receipt is supplied. The canonical carrier-forgetful functor

$$F_r : \mathsf{Sect}^{\mathrm{bos}}_r \longrightarrow \mathsf{Hilb}_{\mathrm{fd}}$$

is faithful, $^*$-preserving, and symmetric strong monoidal. Fermionic signs, spinorial matter, and chirality are not part of this bosonic internal-gauge category; they belong to the later super-Tannakian or matter-sector lift.

**Definition (compact-gauge refinement receipt).** On a cofinal tail, first require a surjectivity/finite-extendability certificate for each finite-state restriction $\rho_{sr} : Q_s \to Q_r$. Only then is the commutative pullback $C(Q_r) \to C(Q_s)$, $f \mapsto f \circ \rho_{sr}$, injective. This does not produce a map of noncommutative collar algebras. For

$$\mathcal{A}_{\mathrm{EC},r} = \bigoplus_{\alpha} M_{d_{\alpha,r}}(\mathbb{C}), \qquad \mathcal{A}_{\mathrm{EC},s} = \bigoplus_{\beta} M_{d_{\beta,s}}(\mathbb{C}),$$

the receipt separately supplies multiplicities $m^{rs}_{\beta\alpha}$ and unitaries $V^{rs}_{\beta}$, with $d_{\beta,s} = \sum_\alpha m^{rs}_{\beta\alpha} d_{\alpha,r}$, and defines

$$\big(\jmath_{rs}(a)\big)_\beta = V^{rs}_\beta \left[\bigoplus_\alpha (a_\alpha \otimes \mathbf{1}_{m^{rs}_{\beta\alpha}})\right] (V^{rs}_\beta)^*.$$

Injectivity requires every coarse $\alpha$ to occur. The whole center maps into the fine center exactly when each $\beta$ has one coarse shadow; on that branch $\jmath_{rs}(P_{\alpha,r}) = \sum_{\beta:\mathrm{sh}_{sr}(\beta)=\alpha} P_{\beta,s}$. Composition requires $m^{rt}_{\gamma\alpha} = \sum_\beta m^{st}_{\gamma\beta} m^{rs}_{\beta\alpha}$ and coherent $V$'s. The same receipt supplies continuous surjections

$$\pi^\Sigma_{sr} : \widehat{K}_{\Sigma,s} \twoheadrightarrow \widehat{K}_{\Sigma,r}, \qquad \pi^\Sigma_{tr} = \pi^\Sigma_{sr} \circ \pi^\Sigma_{ts},$$

and one common strict representative $U^{\mathrm{str}}_r$ at each stage. The pullbacks carry coarse generators into $\mathsf{Sect}^{\mathrm{bos}}_s$, intertwine $U^{\mathrm{str}}_r$ with $U^{\mathrm{str}}_s$, preserve trivial action of every fine overlap loop without changing representative object by object, and agree with the displayed collar supports. The receipt also supplies compatible finite concatenated-collar realizations for every finite tensor word and invariant subobject projector. The finite-state consensus maps alone imply none of these matrix, center, compact-group, or tensor-realization data.

**Theorem (RefinementFunctorAndFiberDescent).** On a cofinal tail carrying that receipt, define

$$U_{rs} := (\pi^\Sigma_{sr})^* : \mathsf{Sect}^{\mathrm{bos}}_r \longrightarrow \mathsf{Sect}^{\mathrm{bos}}_s.$$

Then $U_{rs}$ is fully faithful, symmetric strong monoidal, and $^*$-preserving; it preserves the tensor unit, duals, irreducibles, and the full zero-obstruction transport criterion, with $U_{st}U_{rs} = U_{rt}$. For the canonical forgetful fibers, $F_s U_{rs} = F_r$ literally on carrier Hilbert spaces and linear maps.

**Proof.** For $X = (H_X, \varrho_X)$, set $U_{rs}X = (H_X, \varrho_X \circ \pi^\Sigma_{sr})$ and $U_{rs}(f) = f$. Surjectivity gives the exact identity

$$\mathrm{Hom}_{\widehat{K}_{\Sigma,s}}(U_{rs}X, U_{rs}Y) = \mathrm{Hom}_{\widehat{K}_{\Sigma,r}}(X, Y),$$

so pullback is fully faithful and a coarse simple cannot split. Pullback commutes with tensor products, unit, conjugates, duality maps, and the bosonic flip, while composition of the $\pi$'s gives composition of the $U$'s. The receipt intertwines the common stagewise strict representatives, so zero holonomy persists for the whole tensor-generated category without a sector-specific gauge change. Since pullback changes only the action, the comparison $F_s U_{rs} X = H_X = F_r X$ is the identity; naturality, monoidality, and refinement coherence are identity diagrams. The displayed block formula, not finite-set duality, separately proves the noncommutative algebra embedding and its center behavior. QED.

**The** $\mathrm{U}(1)$ **charge-one test.** If $W_1(z) = z$ is visible and retained by the common stagewise representative $U^{\mathrm{str}}_r$, the chosen category contains $W_n = W_1^{\otimes n}$ for $n \geq 0$ and $W_{-n} = (W_1^*)^{\otimes n}$ for $n > 0$. Charge two is $W_2 = W_1 \otimes W_1$; its subobject idempotent, equivalently the unique nonzero central projection in its intertwiner algebra, is $\mathbf{1}_{W_2} \in \mathrm{End}_{\mathrm{U}(1)}(W_1^{\otimes 2}) \cong \mathbb{C}$, represented physically in the certified two-collar algebra, not necessarily in the original one-collar algebra. At the next certified refinement, $U_{rs}W_2 = ((\pi^\Sigma_{sr})^* W_1)^{\otimes 2}$ remains simple and $F_s(U_{rs}W_2) = \mathbb{C} = F_r(W_2)$ with identity comparison. Thus the fixed collar remains finite while the tensor-generated simple skeleton may be infinite. A finite fusion/truncation alternative would require a separately stated associative fusion quotient.

Consequently, on a cofinal tail carrying the compact-gauge refinement receipt, the EFT branch has a directed ladder $(\mathsf{Sect}_r^{\mathrm{bos}}, U_{rs}, F_r)$ of tensor-generated bosonic edge-sector categories satisfying that zero-obstruction criterion, with fully faithful monoidal pullback functors and compatible forgetful fibers. Write

$$\mathsf{Sect}_\infty := \varinjlim_r \mathsf{Sect}_r^{\mathrm{bos}},$$

for the directed colimit retaining the sectors and intertwiners that persist in that certified system. Without the receipt, only the fixed-stage categories and forgetful functors are constructed; $\mathsf{Sect}_\infty$ and $\mathrm{Aut}_\otimes(\mathcal{F})$ are conditional objects. The theorem below is neutral about whether $\mathsf{Sect}_\infty$ is trivial or nontrivial: if the realized branch furnishes only the tensor unit, the reconstructed compact group is the trivial group.

**Persistence lemma.** Write $U_{rs} : \mathsf{Sect}_r^{\mathrm{bos}} \to \mathsf{Sect}_s^{\mathrm{bos}}$ for the pullback functor on a receipt-certified refinement tail. If a realized zero-obstruction edge-sector class $\alpha_{r_0}$ appears on a sufficiently fine collar, has overlap-visible support, and has representatives $\alpha_s \in \mathsf{Sect}_s^{\mathrm{bos}}$ with $U_{st}\alpha_s \simeq \alpha_t$ for all later $t \succeq s$, then it determines a unique object $[\alpha] \in \mathsf{Sect}_\infty$. Fully faithful pullback along a surjective compact-group map cannot erase or split a coarse simple, and the receipt preserves its full zero-obstruction data. This is a persistence result conditional on the certified ladder, not a consequence of the finite-state refinement maps and not a proof that a nontrivial colimit exists.

This gives a refinement-limit bosonic tensor category $\mathsf{Sect}_\infty$ of edge charges:

- objects: zero-obstruction sector labels that persist in the assumed directed system,
- morphisms: intertwiners between sectors,
- tensor product: fusion by collar concatenation, $\alpha \otimes \beta = \bigoplus_\gamma N_{\alpha\beta}{}^{\gamma}\, \gamma$,
- duals: orientation reversal / charge conjugation $\alpha \leftrightarrow \bar{\alpha}$,
- symmetric braiding in the EFT regime (no anyonic statistics in 3+1D).

The monoidal refinement maps descend the tensor unit, associators, duals, and bosonic symmetry to the colimit, while the compatible stagewise forgetful carriers descend to a faithful bosonic fiber functor $\mathcal{F} : \mathsf{Sect}_\infty \to \mathsf{Hilb}_{\mathrm{fd}}$. The fibers are finite-dimensional objectwise even though $\mathsf{Sect}_\infty$ can have infinitely many simple objects.

**Theorem 6.1 (Receipt-conditional bosonic sector category and Tannaka/DR reconstruction).** On the ordinary or central zero-obstruction bosonic EFT branch, suppose a cofinal tail carries the compact-gauge refinement receipt used by RefinementFunctorAndFiberDescent. Then the colimit $\mathsf{Sect}_\infty$ is a rigid symmetric $C^*$ tensor category with faithful bosonic fiber functor $\mathcal{F}$. Let $\mathrm{Aut}_\otimes(\mathcal{F})$ denote the group of unitary symmetric monoidal natural automorphisms of $\mathcal{F}$. Then

$$G := \mathrm{Aut}_\otimes(\mathcal{F})$$

is a compact group and $\mathsf{Sect}_\infty \simeq \mathrm{Rep}(G)$. In particular $G$ is unique up to isomorphism, and this group is a compact subgroup of a product of unitary groups.

**Proof.** The MaxEnt/local-Gibbs/collar-mixing package is not used here to prove nontriviality or to supply the refinement receipt. The fixed-cutoff categories are constructed by FixedCutoff-BosonicSectorCategory, and on the certified tail the monoidal pullback functors and compatible forgetful fibers are constructed by RefinementFunctorAndFiberDescent. Since those functors preserve tensor products, unit, duals, and $^*$-structure, the directed colimit inherits a rigid symmetric $C^*$-tensor structure, and the compatible stagewise carrier spaces descend to a faithful bosonic fiber functor $\mathcal{F}$. Fix a small skeleton of $\mathsf{Sect}_\infty$. For every object $X$, a unitary symmetric monoidal natural automorphism $\eta \in \mathrm{Aut}_\otimes(\mathcal{F})$ has a unitary component $\eta_X \in U(\mathcal{F}(X))$, so

$$\mathrm{Aut}_{\otimes}(\mathcal{F}) \hookrightarrow \prod_X U(\mathcal{F}(X)), \qquad \eta \mapsto (\eta_X)_X.$$

For each intertwiner $f : X \to Y$, naturality gives the closed relation $\mathcal{F}(f)\eta_X = \eta_Y \mathcal{F}(f)$. For each pair $X, Y$, the monoidal structure isomorphism $J_{X,Y}$ of $\mathcal{F}$ gives the closed relation

$$\eta_{X\otimes Y} = J_{X,Y}\,(\eta_X \otimes \eta_Y)\,J_{X,Y}^{-1}, \qquad \eta_{\mathbf{1}} = \mathrm{id}_{\mathbb{C}}.$$

Hence $G = \mathrm{Aut}_{\otimes}(\mathcal{F})$ is a closed subgroup of the compact product $\prod_X U(\mathcal{F}(X))$, so $G$ is compact. The DR/Tannaka hypotheses are therefore satisfied on the constructed pair $(\mathsf{Sect}_\infty, \mathcal{F})$, and the Doplicher–Roberts/Tannaka reconstruction theorem gives a symmetric $C^*$-tensor equivalence $\mathsf{Sect}_\infty \simeq \mathrm{Rep}(G)$ [37, 38]. If another compact group $G'$ gave the same category through a faithful bosonic fiber functor, the induced symmetric tensor equivalence $\mathrm{Rep}(G) \simeq \mathrm{Rep}(G')$ compatible with the forgetful functors would identify both groups as the automorphism group of the same fiber functor. Thus $G \cong G'$, so the reconstruction is unique up to isomorphism. QED.

**Definition 6.1r (DHR/DR realization hypotheses).** The field-algebra step below uses the following net-realization package, stated explicitly because it is logically additional to the Tannakian data of Theorem 6.1:

1. **(R1) Observable net.** An isotonous net $O \mapsto \mathcal{A}(O)$ of $C^*$-algebras over the directed set of small-region collar localizations of the EFT regime, with quasi-local algebra $\mathcal{A}$ and a faithful vacuum representation.
2. **(R2) Locality, duality, Property B.** Spacelike commutativity; Haag duality $\mathcal{A}(O')' = \mathcal{A}(O)''$ in the vacuum representation, or the explicitly weaker essential-duality variant if that is the version invoked; and Property B.
3. **(R3) Localized endomorphisms.** Each persistent class $[\alpha] \in \mathsf{Sect}_\infty$ is realized by an endomorphism $\rho^\alpha_O$ of $\mathcal{A}$ acting trivially on $\mathcal{A}(O_1)$ for every localization region $O_1$ disjoint from $O$.
4. **(R4) Transportability.** For every admissible pair of regions $O, \widetilde{O}$ there exists a unitary $u \in \mathcal{A}$ with $u\,\rho^\alpha_O(\cdot)\,u^* = \rho^\alpha_{\widetilde{O}}(\cdot)$. This is a net-level statement about unitary intertwiners. Vanishing collar/represented holonomy supplies path-independent collar transport at the categorical level and is a necessary input to this hypothesis, but it is not by itself equivalent to DHR transportability.
5. **(R5) Statistics and conjugates.** Each $\rho^\alpha$ has finite statistics with permutation-symmetric (bosonic) statistics operator matching the symmetry of $\mathsf{Sect}_\infty$, and a conjugate endomorphism realizing the categorical dual $\bar{\alpha}$.
6. **(R6) Completeness.** The assignment $[\alpha] \mapsto [\rho^\alpha]$ is an equivalence of symmetric tensor $C^*$-categories between $\mathsf{Sect}_\infty$ and the category of localized transportable finite-statistics endomorphisms of $\mathcal{A}$.

This paper constructs the categorical side: the fixed-cutoff categories, the refinement functors and fibers, and the colimit pair $(\mathsf{Sect}_\infty, \mathcal{F})$. It does not construct the net realization (R1)–(R6); in particular no observable net is specified here on which the persistent sectors act as localized transportable endomorphisms. (R1)–(R6) therefore enter as explicit hypotheses, and every fixed-point identity $\mathcal{A} = \mathcal{F}_{\mathrm{net}}^G$ in this paper is conditional on them.

**Corollary 6.1 (two-level gauge reconstruction on zero-obstruction sectors).** The reconstruction conclusion splits into two levels with strictly different hypotheses.

*(i) Tannakian level (proved).* If in the small-region limit the edge sectors satisfy the zero-obstruction transport criterion of Theorem 3.4b and lie on a cofinal tail carrying the compact-gauge

refinement receipt, then Theorem 6.1 yields a compact group $G = \mathrm{Aut}_\otimes(\mathcal{F})$ and a symmetric tensor equivalence $\mathsf{Sect}_\infty \simeq \mathrm{Rep}(G)$. This level uses only the constructed category and fiber functor and involves no observable net.

*(ii) DHR/DR field-net level (conditional).* If in addition the realization hypotheses (R1)–(R6) of Definition 6.1r hold, then the Doplicher–Roberts reconstruction theorem [37, 38] supplies a field algebra $\mathcal{F}_{\mathrm{net}}$ with normal (here bosonic) commutation relations, carrying an action of $G$ with field operators implementing every persistent sector, such that

$$\mathcal{A} = \mathcal{F}_{\mathrm{net}}^G.$$

**Proof.** Level (i) is Theorem 6.1. For level (ii): under (R1)–(R6) the localized transportable finite-statistics endomorphisms of $\mathcal{A}$ form a rigid symmetric $C^*$-tensor category equivalent to $\mathsf{Sect}_\infty$ by (R6), hence to $\mathrm{Rep}(G)$ by level (i). The Doplicher–Roberts theorem [37, 38], whose locality, duality, Property B, transportability, statistics, conjugate, and completeness inputs are exactly (R2)–(R6) on the net (R1), then produces the crossed-product field net $\mathcal{F}_{\mathrm{net}}$, its field operators, and the identification $\mathcal{A} = \mathcal{F}_{\mathrm{net}}^G$ with $G$ as gauge group. Without (R1)–(R6) the displayed fixed-point identity is not asserted; the unconditional content of this corollary is level (i). QED.

**Corollary 6.1a (The compact-gauge setup and realized witness).** On the central branch, the categorical collar-transport input to Corollary 6.1 is the combined theorem-level condition $[z]_\Sigma = 0$ plus trivial represented holonomy for an allowed strictification; this condition feeds the sector-category construction and hypothesis (R4) of Definition 6.1r, but it is not itself the net-level DHR-transportability statement, which additionally requires localized endomorphisms and unitary transporters on an observable net. On the genuinely noncentral branch, strict ordinary transport requires both $o_\Sigma^{(2)} = 0$ and at least one allowed strict representative with trivial represented $G_\Sigma$-holonomy; a nonzero $o_\Sigma^{(2)}$ remains a higher-gauge fixed-cutoff sector, while an orbit for which every strict representative has residual loop action remains nontransportable for the ordinary DR corollary. The full orbit $q_\Sigma$ classifies the crossed-module representatives but need not determine a unique ordinary $H^1$ class. FixedCutoffBosonicSectorCategory constructs the fixed-stage category. RefinementFunctorAndFiberDescent constructs the fully faithful pullback functors and compatible forgetful fibers only from the explicit compact-gauge refinement receipt. Thus DR/Tannaka reconstruction yields a compact group on the receipt-certified branch at the Tannakian level; the field-algebra identity $\mathcal{A} = \mathcal{F}_{\mathrm{net}}^G$ additionally requires the DHR/DR realization hypotheses (R1)–(R6); MAR, not the cocycle calculus alone, selects the realized Standard Model branch.

**Theorem (gauge-sector classification-selection factorization).** On the OPH compact-gauge lane, conditional on a cofinal tail carrying the compact-gauge refinement receipt, the realized Standard Model claim factors as

$$\begin{aligned}&\text{overlap/gluing data} \longrightarrow \text{associator strictification test} \longrightarrow \text{strict-representative holonomy test}\\&\qquad \longrightarrow \mathsf{Sect}_\infty \longrightarrow G = \mathrm{Aut}_\otimes(\mathcal{F}) \longrightarrow \mathfrak{S}_{\mathrm{MAR}}.\end{aligned}$$

The first arrow computes and, where possible, removes the central or crossed-module associator defect. The second computes represented holonomy across the allowed strict representatives. The third keeps only sectors for which at least one such representative has trivial loop action. The fourth reconstructs the compact group from the persistent tensor category and fiber functor. These are classification and reconstruction steps. The final arrow is the realization/selection step: MAR acts on realized admissible sector packages, not on obstruction classes alone. QED.

### 5.4.2 Selecting the SM factors (derived from MAR)

Theorem 6.1 yields *some* compact $G$. Axiom 5 (MAR, Minimal Admissible Realization) acts next on admissible sector packages; on the explicit one-Higgs chiral matter branch below, it selects the realized Standard Model quotient.

---

**Axiom 5 (MAR): Minimal Admissible Realization.** Among all OPH-realizable sector packages $\mathfrak{S}$ consisting of the connected Lie gauge-sector image relevant in the low-energy EFT, its admissible light chiral matter content, and one Higgs doublet, and which are (i) loop-coherent / transportable (the central triangle class vanishes, or the noncentral higher associator is strictifiable, and at least one allowed strict 1-cocycle representative has trivial represented sector holonomy), (ii) anomaly-free, (iii) refinement-stable with light chiral matter, (iv) single-Higgs Yukawa-completable with one connected abelian charge factor acting nontrivially on the coupled carrier, (v) intrinsically quark-sector CP-capable, (vi) weak-sector UV-completable on that same one-Higgs branch, the realized low-energy package is the lexicographically minimal one under

$$C(\mathfrak{S}) = (\chi_{\mathrm{cpl}},\ N_{\mathrm{nonab}},\ N_c,\ N_g).$$

The complete formal statement, admissibility definitions, and proof route are given in the rest of Section 6.2.

Here $\chi_{\mathrm{cpl}}$ is the **coupled edge capacity**: the dimension of the minimal unitary carrier containing a common irreducible block on which the admissible pseudoreal and complex nonabelian charge types both act nontrivially. This is intentionally stronger than the abstract minimal faithful representation dimension. For $S(U(3) \times U(2))$, the block-diagonal action on $\mathbb{C}^3 \oplus \mathbb{C}^2$ is faithful of dimension 5, but it is not coupled and therefore does not enter MAR. The object minimized by MAR is the sector package $\mathfrak{S}$, not the bare tensor category by itself. MAR is thus an explicit structural-economy selector on the admissible class, not a theorem derived from the preceding axioms.

**Definition 6.1b (MAR realization space and physical equivalence).** Let $\mathfrak{A}_{\mathrm{MAR}}$ be the set of isomorphism classes of finite low-energy sector packages

$$\mathfrak{S} = (G^0, \mathcal{R}_{\mathrm{light}}, H, \mathcal{Y}, \mathcal{F})$$

on the ordinary or central zero-obstruction bosonic branch of Theorem 6.1, together with the explicit realized one-Higgs chiral matter package used below. Here $G^0$ is the connected Lie gauge-sector image, $\mathcal{R}_{\mathrm{light}}$ is the light chiral matter representation family, $H$ is one Higgs doublet, $\mathcal{Y}$ denotes the Yukawa-completable charge data, and $\mathcal{F}$ is the descended finite bosonic fiber functor on the retained transportable sector package. A package is MAR-admissible exactly when it satisfies the six clauses in Axiom 5: zero obstruction, anomaly cancellation, refinement stability with light chiral matter, one-Higgs Yukawa completability with one connected abelian charge factor acting nontrivially on the coupled carrier, intrinsic quark-sector CP capability, and weak-sector UV completability on the same one-Higgs branch.

Two packages are physically equivalent when they are related by a compact-group isomorphism and a fiber-compatible symmetric monoidal equivalence that preserve the observer-visible representations, Yukawa invariants, anomaly polynomial, hypercharge lattice up to the allowed U(1) normalization, and the one-Higgs branch, modulo generation relabeling, charge-conjugation convention, gauge-center quotienting, implementation hiding, and inert ancillary stabilization.

**Definition 6.1c (MAR order).** For $\mathfrak{S} \in \mathfrak{A}_{\mathrm{MAR}}$, define

$$C(\mathfrak{S}) = (\chi_{\mathrm{cpl}}, N_{\mathrm{nonab}}, N_c, N_g) \in \mathbb{N}^4,$$

where $N_{\mathrm{nonab}}$ is the number of connected nonabelian simple factors acting nontrivially on the coupled carrier, $N_c$ is the dimension of the minimal intrinsically complex color-type role, and $N_g$ is the generation count on the realized one-Higgs chiral branch. MAR orders packages by the lexicographic order on $C(\mathfrak{S})$, then quotients ties by the physical equivalence relation above.

**Proposition 6.1d (well-founded MAR minima).** Every nonempty MAR-admissible class has a nonempty set of MAR-minimal packages. More precisely, if $\mathcal{A} \subseteq \mathfrak{A}_{\mathrm{MAR}}$ is nonempty, then the set $C(\mathcal{A}) \subseteq \mathbb{N}^4$ has a lexicographically least element. The MAR-minimal packages in $\mathcal{A}$ are exactly the packages whose complexity vector is that least element.

**Proof.** The lexicographic order on $\mathbb{N}^4$ is well-founded: first minimize $\chi_{\mathrm{cpl}}$, then $N_{\mathrm{nonab}}$ on that fiber, then $N_c$, then $N_g$. Each step minimizes a nonempty subset of $\mathbb{N}$. QED.

**Proposition 6.1e (meaning of MAR uniqueness).** MAR uniqueness is uniqueness of the observer-visible low-energy package modulo the physical equivalence relation in Definition 6.1b. It is not uniqueness of a microscopic regulator representative. The subsequent lemmas prove that, within the connected positive-dimensional Lie admissible class with one connected abelian factor and faithful action on the minimal coupled carrier, all MAR-minimal representatives realize the same connected SM gauge image, the same exact hypercharge lattice after normalization, the same structural electroweak force content, $N_c = 3$, and $N_g = 3$. Any remaining differences are gauge-center quotient choices, relabelings, or inert implementation data and therefore do not change the physical predictions recorded in D8–D9.

**Note on transport obstruction.** The gluing obstruction is tracked explicitly so the gauge lane states its branch conditions. TransportabilityFromOverlapGluing identifies the central criterion as $[z]_\Sigma = 0$ together with an allowed trivial-holonomy strictification, and the genuinely noncentral criterion as $o^{(2)}_\Sigma = 0$ together with at least one allowed strict representative having trivial represented $G_\Sigma$-holonomy. When $o^{(2)}_\Sigma \neq 0$, the sector is handled by crossed-module data rather than by the ordinary DR field-algebra corollary; when $o^{(2)}_\Sigma = 0$ but every strict representative has residual holonomy, it also does not enter that corollary. The full orbit $q_\Sigma$ classifies the crossed-module representatives but need not determine one ordinary $H^1$ class. FixedCutoffBosonicSectorCategory constructs the fixed-cutoff category. On a cofinal tail carrying the compact-gauge refinement receipt, RefinementFunctorAndFiberDescent constructs the refinement/fiber descent and the realized witness theorem supplies nonempty ordinary/central compact-gauge witness data. The realized Standard Model branch enters only after MAR is applied to the admissible one-Higgs sector packages.

**What MAR derives.** Product structure, minimal sector content, and coupled edge-capacity minimality are consequences of MAR applied to the admissible class:

- **Product structure**: follows from the minimal coupled carrier $\mathbb{C}^3 \otimes \mathbb{C}^2$, which enforces commuting color and weak actions.
- **Minimal sector content**: the pseudoreal doublet and complex triplet are the minimal nonabelian representations satisfying the admissibility conditions, while the connected abelian charge factor is an explicit admissibility input.
- **Coupled edge-capacity minimality**: this is the first component of MAR's complexity vector.

With MAR stated, the SM derivation proceeds via standard lemmas:

**Lemma 6.2 (Product factorization implies product group).** If $\mathsf{Sect} \simeq \mathrm{Rep}(G)$ and $\mathsf{Sect} \simeq \mathsf{Sect}_1 \boxtimes \mathsf{Sect}_2$, then

$$G \cong G_1 \times G_2, \qquad \mathsf{Sect}_i \simeq \mathrm{Rep}(G_i).$$

QED.

**Lemma 6.3 (SU(2) from a pseudoreal doublet).** Let $H$ be a positive-dimensional compact connected Lie group with a faithful *irreducible* 2D pseudoreal unitary representation $V$. Then the connected derived subgroup of the image is $SU(2)$ acting as the fundamental doublet. Irreducibility is load-bearing: $U(1)$ acting by $\mathrm{diag}(z, z^{-1})$ is faithful and pseudoreal but reducible, and contains no $SU(2)$. Proof: a compact connected abelian group has only one-dimensional irreducibles, so the image is nonabelian; by Schur the center acts by scalars, so the derived subgroup acts irreducibly on $\mathbb{C}^2$; the only connected compact groups acting irreducibly on $\mathbb{C}^2$ have derived subgroup $SU(2)$. Finite or disconnected counterexamples are not part of the theorem package, which only applies the lemma to the identity component on the relevant nonabelian image. QED.

**Lemma 6.4 (SU(3) from an intrinsically complex triplet).** Let $H$ be a positive-dimensional compact connected Lie group with a faithful irreducible 3D unitary representation $W$ that is *intrinsically complex on its nonabelian factor*: the connected derived subgroup of the image acts by an irreducible representation not equivalent to its conjugate. Abelian twisting does not qualify: a character twist can make the full representation inequivalent to its conjugate while the nonabelian factor stays real or pseudoreal, and such twists are excluded by the hypothesis. Then the connected derived subgroup of the image is conjugate to $SU(3)$ acting as the fundamental triplet, up to finite central kernel. The hypothesis rejects the defining representation of $SO(3)$ on $\mathbb{C}^3$ (real on the nonabelian factor) and its twists $(z, R) \mapsto zR$ under $U(1) \times SO(3)$ (real on the nonabelian factor), and retains the fundamental of $SU(3)$. Proof: by Schur the center acts by scalars, so the derived subgroup acts irreducibly; two nontrivial simple factors would force a tensor decomposition of dimension at least 4, so exactly one simple factor acts; among compact simple Lie algebras only $\mathfrak{su}(2)$ and $\mathfrak{su}(3)$ carry nontrivial irreducibles of dimension at most 3; the 3D representation of $\mathfrak{su}(2)$ is real, so the intrinsically complex hypothesis leaves the fundamental of $\mathfrak{su}(3)$. This restates Lemma `lem:su3class` of the compact paper. The intrinsically-complex-nonabelian hypothesis is load-bearing: without it, $SO(3)$ on $\mathbb{C}^3$ satisfies every remaining hypothesis and refutes the conclusion. Finite or disconnected counterexamples are not part of the theorem package, which only applies the lemma to the identity component on the relevant nonabelian image. QED.

**Lemma 6.5 (Connected abelian factor criterion).** If the admissible sector package contains a connected abelian charge factor acting nontrivially on the coupled carrier, then the identity component of the abelianized reconstructed group contains a one-torus. Under the single connected abelian-factor admissibility condition, this factor is $U(1)$. QED.

**Proposition 6.6 (physical group quotient).** If the realized matter spectrum has hypercharges quantized in sixths, then the kernel acting trivially on all realized sectors is $\mathrm{Z}_6$, so

$$G_{\mathrm{phys}} = \frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}.$$

QED.

**Proposition 6.6a (SM from MAR).** Under Axiom 5 (MAR):

- By the definition of $\chi_{\mathrm{cpl}}$, the realized MAR package contains a light chiral pseudoreal weak-type role and a light chiral complex color-type role on a common coupled carrier; clause (iii) imposes refinement stability on that branch, and clause (iv) adds one connected abelian charge factor acting nontrivially on the same carrier together with one-Higgs Yukawa completeness.

- The minimal faithful pseudoreal representation is the doublet ($\chi = 2$), giving $SU(2)$. No intrinsically complex 2D irrep qualifies in the connected compact Lie class, because the connected derived image of any irreducible 2D unitary representation lies in $SU(2)$, so the nonabelian action is pseudoreal up to abelian twist. The first intrinsically complex case is therefore the triplet ($\chi = 3$), giving $SU(3)$.
- The minimal coupled carrier for both is $\mathbb{C}^3 \otimes \mathbb{C}^2$, giving coupled edge capacity $\chi_{\mathrm{cpl}} = 6$.
- The block-diagonal faithful representation $\mathbb{C}^3 \oplus \mathbb{C}^2$ of $S(U(3) \times U(2))$ has dimension 5, but it is not coupled and therefore is not the MAR minimizer.
- The maximal compact subgroup of $U(6)$ acting on $\mathbb{C}^3 \otimes \mathbb{C}^2$ with commuting actions is $(SU(3) \times SU(2) \times U(1))/(\text{finite center})$.
- The commutant of $SU(3) \times SU(2)$ inside $U(6)$ is exactly $U(1)$, so any connected abelian factor acting nontrivially on the coupled carrier is necessarily a single $U(1)$, and no additional continuous factors appear without increasing $\chi_{\mathrm{cpl}}$.
- The tensor-product carrier follows from the coupled-carrier admissibility clause, which itself demands a pseudoreal weak-type role and an intrinsically complex color-type role coexisting on one block. That clause is a declared MAR input with its menu counted in the closure ledger, and no competition against non-product simple groups (for instance $SU(6)$) is run.

Combined with Proposition 6.6 (hypercharges quantized in sixths from the realized spectrum), this yields:

$$G_{\mathrm{phys}} = \frac{SU(3) \times SU(2) \times U(1)}{Z_6}.$$

The full proof route is given in the remainder of Section 6.2.

**Theorem 6.6aa (compact-gauge witness landing under the realized matter-package hypothesis).** On the ordinary or central zero-obstruction bosonic branch, suppose a cofinal tail carries the compact-gauge refinement receipt, and suppose, as a named input hypothesis (the realized matter-package hypothesis), that a realized one-Higgs chiral matter package with the witness labels below exists on the branch. Then the OPH compact reference architecture admits an OPH-realizable cofinal heat-kernel edge-sector witness

$$\mathcal{W}_{\mathrm{SM}} = \{Q_i, u_i^c, d_i^c, L_i, e_i^c, H\}_{i=1}^3$$

with

$$Q_i = (3, 2)_{1/6}, \quad u_i^c = (\bar{3}, 1)_{-2/3}, \quad d_i^c = (\bar{3}, 1)_{1/3}, \quad L_i = (1, 2)_{-1/2}, \quad e_i^c = (1, 1)_1, \quad H = (1, 2)_{1/2}.$$

The fixed-cutoff edge heat-kernel law gives $p_R(t) \propto d_R e^{-tC_2(R)}$, hence positive support for every finite-dimensional witness sector at finite $t$. This positivity concerns the bosonic internal-gauge sector labels only; positive heat-kernel weight on a boundary label does not supply a light chiral fermion multiplet carrying that label, and existence of the displayed matter plus one-Higgs package is the realized matter-package hypothesis. The pullback functors constructed from the receipt carry those zero-obstruction labels cofinally. The witness is anomaly-free and one-Higgs Yukawa-complete by the hypercharge theorem, contains the weak-type and color-type roles used by MAR, and satisfies the CP-capability and weak-sector UV clauses that force $N_g = 3$ once $N_c = 3$. Therefore, under the realized matter-package hypothesis, the MAR-admissible class is nonempty; realized-branch nonemptiness for the matter package itself is open, parallel to the treatment of the geometric branch in Theorem 4.3h. Under the same hypothesis, every OPH-admissible microscopic UV completion of the realized branch projects, modulo physical equivalence, to

$$\frac{SU(3)\times SU(2)\times U(1)}{Z_6}, \qquad N_c=3, \qquad N_g=3,$$

with the Standard Model hypercharge lattice. Microscopic regulator representatives remain underdetermined up to gauge presentation, implementation hiding, and inert ancillary stabilization. QED.

**Theorem (Realized Standard Model branch).** Assume the ordinary or central zero-obstruction bosonic branch; a cofinal tail carrying the compact-gauge refinement receipt; the fixed-cutoff bosonic sector category, refinement/fiber descent, and compact-group reconstruction above; a nonempty realized one-Higgs chiral MAR-admissible class witnessed by $\mathcal{W}_{\mathrm{SM}}$; weak-type and intrinsically complex color-type roles on a common coupled carrier; one connected abelian charge factor acting nontrivially on that carrier; and the anomaly-free, one-Higgs Yukawa-complete, CP-capable, and weak-sector UV clauses of Axiom 5. Then MAR minima exist, and every MAR-minimal observer-visible package in that class is physically equivalent to

$$\left(\frac{\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)}{\mathbb{Z}_6}, \mathcal{R}_{\mathrm{SM}}, H, \mathcal{Y}_{\mathrm{SM}}\right), \qquad N_c=3, \qquad N_g=3,$$

with

$$\mathcal{W}_{\mathrm{SM}}=\{Q_i,u_i^c,d_i^c,L_i,e_i^c,H\}_{i=1}^3,$$

$$Q_i=(3,2)_{1/6}, \quad u_i^c=(\bar{3},1)_{-2/3}, \quad d_i^c=(\bar{3},1)_{1/3}, \quad L_i=(1,2)_{-1/2}, \quad e_i^c=(1,1)_1, \quad H=(1,2)_{1/2}.$$

The proof is the well-founded MAR-minima proposition above plus the weak/color classification lemmas, the minimal coupled-carrier result $\mathbb{C}^3\otimes\mathbb{C}^2$, the $U(1)$ commutant, hypercharge theorem, three-color corollary, MAR-branch generation theorem, and the $\mathbb{Z}_6$ kernel proof. QED.

**Corollary 6.6b (Three colors on the realized branch).** On the realized one-generation matter package carried by the minimal coupled block of Proposition 6.6a, the color multiplicity is fixed by the same MAR derivation: the color factor is the fundamental triplet of SU(3), so

$$N_c=3.$$

The selector is inside the stated theorem stack. QED.

**Corollary 6.6c (structural electroweak force and charges).** On the realized one-Higgs branch, the electroweak gauge factor has Lie algebra

$$\mathfrak{su}(2)_L\oplus\mathfrak{u}(1)_Y.$$

The Higgs doublet $H=(1,2)_{1/2}$ selects a nonzero neutral vacuum vector

$$\phi_0=\frac{v}{\sqrt{2}}\begin{pmatrix}0\\1\end{pmatrix}, \qquad v\neq 0,$$

with

$$Q\phi_0=(T_3+Y)\phi_0=0,$$

so its vector stabilizer has unbroken generator

$$Q=T_3+Y,$$

and the unbroken gauge factor is $\mathrm{U}(1)_Q$. The charged weak generators give

$$W^{\pm} = \frac{1}{\sqrt{2}}(W^1 \mp iW^2),$$

while the neutral $\mathrm{SU}(2)_L$ and $\mathrm{U}(1)_Y$ gauge fields span the $Z/A$ basis on the D10 quantitative branch. Explicitly,

$$e^{i\alpha T_3} e^{i\beta Y} \phi_0 = e^{i(\beta-\alpha)/2} \phi_0,$$

so exact vector invariance ties the phases by $\beta = \alpha$, locally, and leaves $\mathrm{U}(1)_Q$. The projective ray $[\phi_0]$, however, is unchanged for independent $\alpha$ and $\beta$: its stabilizer is the two-torus $\mathrm{U}(1)_{T_3} \times \mathrm{U}(1)_Y$, modulo the inherited finite center. Projectivization therefore classifies the carrier ray but cannot by itself select the electromagnetic diagonal. Thus the recovered structural package contains the weak charged-current carriers $W^{\pm}$, the neutral weak carrier $Z$, the electromagnetic carrier $A$, and the exact charge operator $Q = T_3 + Y$. The mixing angle, $v$, and numerical $W/Z$ masses belong to the D10 running/matching surface rather than to this recovered-core structural corollary. QED.

**Corollary 6.6d (conditional Maxwell sector on the realized electromagnetic branch).** On the ordinary one-Higgs branch, let $A_Q$ be the support-visible connection component on the unbroken electromagnetic factor $\mathrm{U}(1)_Q$. The abelian restriction of the compact-gauge curvature gives

$$F_Q = dA_Q.$$

Assume in addition the Lorentzian low-energy Maxwell action

$$S_{\mathrm{EM}}[A_Q, J_Q] = -\frac{1}{2g_Q^2} \int F_Q \wedge *F_Q + \int A_Q \wedge *J_Q,$$

the Euler-Lagrange equations are

$$dF_Q = 0, \qquad d * F_Q = g_Q^2 * J_Q.$$

After canonical electromagnetic normalization, these are Maxwell's equations. The numerical value of $g_Q$, equivalently $\alpha_{\mathrm{em}}$, belongs to the Ward-projected electromagnetic readout. The compact-group reconstruction and connection label do not themselves supply this action or its nonzero kinetic coefficient. QED.

### 5.4.3 Refinement stability and unprotected relevant operators

The state-side refinement stability used later in the gauge branch is contained in the third OPH axiom through its refinement-closure clause. By that clause the same finite local constraint family is preserved across cutoffs, so the realized UV states lie in one common finite-dimensional MaxEnt family rather than in a new coupling space at every refinement step. The selected stable branch of this family is therefore not an extra state axiom beyond the MaxEnt branch itself. The closure clause it relies on is a substantive renormalization condition: coarse-graining a finite-range Gibbs family generically generates interactions outside any fixed finite density list. The closure defect, the moment-matching I-projection realizing $R_{\ell \to L}$, and its trace-norm residual bound are quantified in Definition 2.6a and Lemma 2.6b; the clause is exactly the statement that the defect vanishes along the realized branch, and that vanishing remains assumed rather than proved. This state statement does not supply the compact-gauge refinement receipt.

Concretely, if

$$\omega_\ell(\lambda) = Z_\ell(\lambda)^{-1} \exp\left( -\sum_x \sum_a \lambda_a O_a(x) - \sum_b \mu_b Q_b \right),$$

then any refinement channel $\Phi_{\ell\to L}$ compatible with Axiom 3, including its refinement-closure clause, acts on the realized branch by an induced finite-dimensional map

$$\Phi_{\ell\to L}(\omega_\ell(\lambda)) = \omega_L(R_{\ell\to L}(\lambda)).$$

The resulting refinement-stable state branch is simply a trajectory or invariant subset of this finite-dimensional multiplier map. Accordingly, when sections speak of a refinement-stable directed colimit of sectors, persistence along this MaxEnt branch is only the state-side input. The additional compact-gauge receipt must supply finite extendability, center-compatible block embeddings, coherent surjective compact-group maps, preservation of the full transport obstruction, and finite tensor realizations. The MaxEnt/mixing package supplies none of those data and does not decide whether the resulting certified colimit is trivial or nontrivial.

The consensus paper makes the state-side coarse-graining connection explicit. Given quotient normal-form maps $n_r$, obstruction maps $h_r$, and coarse-graining maps $(\rho_{sr}, \chi_{sr})$, reconciliation commutes with coarse-graining at the macroscopic readout scale whenever the two square defects

$$d_r^{Q}(\rho_{sr} n_s(x), n_r \rho_{sr}(x)), \qquad d_r^{\mathcal{H}}(\chi_{sr} h_s(x), h_r \rho_{sr}(x))$$

are controlled. Exact refinement naturality gives zero defect; approximate RG matching is theorem-grade only to the extent that these defects are bounded on the selected branch. This keeps the MaxEnt refinement map $R_{\ell\to L}$ connected to the reconciliation law without promoting arbitrary coarse-graining channels to OPH laws.

Relevant operators that are neither symmetry-forbidden nor retained in the selected constraint family are precisely the directions that try to push the flow off that stable branch.

**Lemma 6.7 (refinement-stable MaxEnt branch forbids unprotected relevant operators).** Assume the local finite-constraint MaxEnt/refinement branch of the third OPH axiom. Let $\mathcal{O}$ be a gauge-invariant Lorentz-scalar relevant deformation in the emergent EFT sense ($\Delta < 4$ in $3+1$D), allowed by symmetry and absent from the retained constraint family. Then the selected branch can keep the coupling of $\mathcal{O}$ at zero only if symmetry forbids that direction or the constraint family explicitly retains it. Otherwise generic refinement induces a nonzero component along that direction and the flow leaves the selected branch. This is a branch-persistence statement, not a universal entropy-ordering theorem for arbitrary off-branch phases.

**Proof sketch.** Linearize the induced refinement map $R_{\ell\to L}$ on the finite-dimensional multiplier space around the selected branch. A relevant operator gives an unstable direction with scaling exponent $y > 0$. If $\mathcal{O}$ is not fixed by symmetry and is not part of the retained constraint family, generic UV mismatch produces a nonzero component along that direction. Because $y > 0$, repeated coarse-graining amplifies the component and pushes the flow off the selected branch unless one fine-tunes it away at every scale or protects it by symmetry or by the declared constraint family. QED.

**Corollary 6.8 (chirality selector).** A gauge-invariant Dirac mass term is a relevant scalar. If both chiralities exist in conjugate representations, the mass term is allowed and will be generated under refinement unless symmetry-forbidden or explicitly retained as a protected constraint. Therefore keeping light fermions on the selected refinement-stable branch requires chiral matter content or an explicit protecting mechanism. QED.

### 5.4.4 Generation number from CKM CP capability and weak-sector completability (derived from MAR)

Anomaly cancellation is generation-by-generation, so it does not fix the number of generations. On the realized one-Higgs branch, the lower and upper bounds come from the CP-capability and weak-sector UV clauses declared in MAR, and MAR then selects the minimum.

**Proposition 6.9 (The number of generations is N_g = 3).** On the realized one-Higgs quark branch of Proposition 6.6a, with the derived $N_c = 3$ from Corollary 6.6b and with the CP-capability and weak-sector UV clauses contained in Axiom 5, the generation number is

$$N_g = 3.$$

**Inputs.**

1. **Intrinsic CKM CP capability** is part of the realized quark branch through clause (v) of Axiom 5.
2. **Weak-sector UV completability** is part of the same realized one-Higgs branch through clause (vi) of Axiom 5.
3. **MAR minimality** acts on the same realized branch once the first three complexity entries are fixed.
4. Use the derived $N_c = 3$ from Corollary 6.6b.

**Step 1: CKM CP-capability lower bound.** The number of physical CP-violating phases in an N_g × N_g CKM matrix is:

$$\#(\text{CP phases}) = \frac{(N_g - 1)(N_g - 2)}{2}.$$

- For N_g = 1, 2: this is 0 → **no intrinsic CKM CP capability**.
- For N_g = 3: this is 1 → **intrinsic CKM CP capability is available**.

So intrinsic CKM CP capability requires:

$$N_g \geq 3.$$

**Step 2: SU(2) asymptotic freedom upper bound.** The one-loop coefficient is:

$$b_{\mathrm{SU}(2)} = \frac{22}{3} - \frac{1}{3} N_g (N_c + 1) - \frac{1}{6},$$

where the final $-1/6$ is the contribution of one complex Higgs doublet. Asymptotic freedom means $b_{\mathrm{SU}(2)} > 0$, i.e.,

$$N_g (N_c + 1) < \frac{43}{2}.$$

With N_c = 3, we have N_c + 1 = 4, so:

$$4N_g < \frac{43}{2} \quad \Rightarrow \quad N_g \leq 5.$$

Combining: 3 ≤ N_g ≤ 5.

**Step 3: MAR selection.** These are not extra post-MAR selectors: they are the numerical content of clauses (v) and (vi) of Axiom 5 on the realized one-Higgs branch. With MAR lexicographic minimality selecting the window minimum, the upper bound $N_g \leq 5$ from the SU(2) clause is a consistency check and does no selection work. Given the allowed window {3, 4, 5}, MAR (fourth component of the complexity vector $C(\mathfrak{S})$) selects the smallest viable choice:

$$N_g = 3.$$

QED.

**Why this is convincing.**

- The clause set selects a **single integer** inside the admissible window $\{3, 4, 5\}$; clause (v) (CKM CP capability) is a declared input that sets the lower edge, counted in the closure ledger.
- It uses **two explicit MAR clauses** (intrinsic CKM CP capability and weak-sector UV completability) plus MAR's lexicographic minimality, rather than a separate post hoc admissibility menu.
- It is not a fit to a continuous number.
- Under the stated gauge-selection hypotheses, this is a derived result.

**Corollary 6.9a (MAR-minimal SM package is unique up to physical equivalence).** On the ordinary or central zero-obstruction bosonic branch, assume the MAR-admissible class is nonempty and contains the explicit realized one-Higgs chiral matter package used in Theorem 6.1 through Proposition 6.9. Then every MAR-minimal representative in that branch has the same observer-visible Standard Model package:

$$\frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}, \qquad N_c = 3, \qquad N_g = 3,$$

with the hypercharge lattice of Proposition 6.6 and the finite quotient fixed by Proposition 6.6. The only residual freedom is physical equivalence in the sense of Definition 6.1b.

**Proof.** Proposition 6.1d supplies the MAR minima. Lemmas 6.3–6.5 and Proposition 6.6a fix the minimal weak, color, coupled-carrier, and connected abelian roles at the least first three complexity entries. Corollary 6.6b fixes $N_c = 3$, Proposition 6.9 fixes $N_g = 3$, and the hypercharge and quotient propositions fix the visible charge lattice and $\mathbb{Z}_6$ kernel. The equivalence relation removes only relabelings, gauge-center conventions, and inert implementation data. QED.

### 5.4.5 Hilbert-space formulation of gluing data

Let {P_i} be a good cover of the screen. For each patch, fix a representation

$$\pi_i : \mathcal{A}_i \to \mathcal{B}(\mathcal{H}_i).$$

For each overlap, choose a unitary intertwiner

$$U_{ij} : \mathcal{H}_j \to \mathcal{H}_i$$

such that for all $O \in \mathcal{A}_{ij}$,

$$\pi_i(O) = U_{ij} \pi_j(O) U_{ij}^{\dagger}.$$

Normalize U_ii = 1 and U_ji = U_ij†.

**Lemma 6.10 (centrality on triple overlaps).** On a triple overlap define

$$\Omega_{ijk} := U_{ij} U_{jk} U_{ki}.$$

For all $O \in \mathcal{A}_{ijk}$,

$$\Omega_{ijk} \pi_i(O) = \pi_i(O) \Omega_{ijk}.$$

**Proof.** Conjugation by U_ki sends π_i(O) to π_k(O), by U_jk to π_j(O), by U_ij back to π_i(O). Thus conjugation by Ω_ijk fixes π_i(O), so Ω_ijk commutes with π_i(O). QED.

**Lemma 6.11 (gauge behavior).** If Ũ_ij = V_i U_ij V_j† with V_i acting trivially on overlap observables, then

$$\tilde{\Omega}_{ijk} = V_i \Omega_{ijk} V_i^{\dagger}.$$

In particular, if Ω_ijk is central, its class is gauge invariant. QED.

### 5.4.6 Loop obstruction class (central defect)

Assume the defect is central, the overlap centers are identified with one fixed abelian unitary coefficient group $Z_\Sigma$, and overlap transport acts trivially on $Z_\Sigma$. With $U_{ji} = U_{ij}^{-1}$, define the implementer-level multiplier by

$$\Omega_{ijk} = U_{ij} U_{jk} U_{ki} = z_{ijk} \in Z_\Sigma,$$

equivalently $U_{ij} U_{jk} = z_{ijk} U_{ik}$. This is the abelian truncation of the full 2-group obstruction in Section 3.4. The multiplier must be defined at the implementer level: $\mathrm{Ad}(z_{ijk})$ is the identity for central $z_{ijk}$ and does not by itself record the defect.

**Theorem 6.12 (central defect strictification and residual holonomy).** The family $\{z_{ijk}\}$ is an untwisted Čech 2-cocycle with coefficients in $Z_\Sigma$, and its class $[z] \in \check{H}^2(N_\Sigma, Z_\Sigma)$ is gauge invariant. On any quadruple overlap $P_{ijkl}$,

$$z_{jkl} z_{ikl}^{-1} z_{ijl} z_{ijk}^{-1} = 1.$$

A central edge rephasing by a cochain in $C^1(N_\Sigma, Z_\Sigma)$ removes every triangle defect iff $[z] = 0$. After such a rephasing, write $U_{ij}^{\mathrm{str}}$ for the resulting genuine edge 1-cocycle. Let $\mathcal{C}_\alpha$ be the full transported charge block, including its carrier and multiplicity/intertwiner data. Loop-coherent, endpoint-only transport exists on that block iff, in addition,

$$\mathrm{Hol}_{\alpha, U^{\mathrm{str}}}([\gamma]) = U_\gamma^{\mathrm{str}}\big|_{\mathcal{C}_\alpha} = \mathbf{1}_{\mathcal{C}_\alpha} \qquad \text{for every } [\gamma] \in \pi_1(|N_\Sigma|, i_*).$$

Thus strict path independence is equivalent to the combined conditions $[z] = 0$ and trivial represented holonomy for one allowed strictification.

**Proof.** The fixed-coefficient and trivial-action hypotheses place every multiplier in the same group $Z_\Sigma$. Associativity compares $(U_{ij} U_{jk}) U_{kl}$ with $U_{ij}(U_{jk} U_{kl})$ on a quadruple overlap. Substituting $U_{ij} U_{jk} = z_{ijk} U_{ik}$ gives the displayed Čech cocycle identity. A vertex-frame change $U_{ij} \mapsto V_i U_{ij} V_j^{\dagger}$ conjugates $z_{ijk}$, as in Lemma 6.11, and therefore leaves a central multiplier unchanged. Separately, a central edge rephasing $U_{ij} \mapsto a_{ij}^{-1} U_{ij}$, with $a \in C^1(N_\Sigma, Z_\Sigma)$, replaces $z$ by the corresponding Čech coboundary. Hence $[z] = 0$ is equivalent to removing all triangle multipliers by a central 1-cochain. This yields a strict 1-cocycle, not automatically endpoint-only transport. For two paths $p, p'$ with the same endpoints, the remaining discrepancy is the holonomy of the closed loop $p'^{-1} p$; it vanishes for every pair exactly when the displayed holonomy representation on $\mathcal{C}_\alpha$ is trivial. Conversely, strict path independence makes every triangle comparison and every represented closed-loop action trivial. QED.

If overlap transport acts nontrivially on a local system of centers, the same comparison contains the transported factor $\varphi_{ij}(z_{jkl})$ and must be written with the corresponding twisted Čech differential; that branch is not the untwisted theorem stated here.

### 5.4.7 EFT reduction to anomaly cancellation

Assume ExtEFT: a low-energy 3+1D chiral gauge theory exists with group $G$. Then the obstruction class $[z]$ is structurally analogous to the EFT 't Hooft anomaly class and is expected to map to it after a separate anomaly-descent construction. That map lies outside this manuscript, so $[z] = 0$ is used here as the internal triangle-defect strictifiability condition rather than a proved equivalence to EFT anomaly cancellation; the separate trivial-holonomy condition of Theorem 6.12 is required for transportability.

### 5.4.8 Hypercharge from anomaly freedom and Yukawas

**Theorem 6.13 (Hypercharge from anomaly freedom and Yukawas).** Assume gauge group SU(N_c) × SU(2) × U(1)_Y and one generation of left-handed Weyl fermions (Q, $u^c$, $d^c$, L, $e^c$), with a Higgs doublet H and Yukawa terms

$$QHu^c, \qquad QH^\dagger d^c, \qquad LH^\dagger e^c.$$

Then anomaly freedom and Yukawa invariance fix the hypercharges up to an overall normalization, yielding the Standard Model pattern for N_c = 3.

**Proof.** Yukawa invariance gives

$$Y_u = -(Y_Q + Y_H), \quad Y_d = -Y_Q + Y_H, \quad Y_e = -Y_L + Y_H.$$

Anomaly cancellation yields

$$\begin{aligned} SU(2)^2U(1): &\quad N_cY_Q + Y_L = 0, \\ \text{grav}^2U(1): &\quad 2N_cY_Q + N_cY_u + N_cY_d + 2Y_L + Y_e = 0. \end{aligned}$$

Solving gives

$$Y_L = -N_cY_Q, \quad Y_H = N_cY_Q, \quad Y_u = -(N_c+1)Y_Q, \quad Y_d = (N_c-1)Y_Q, \quad Y_e = 2N_cY_Q.$$

With these relations, SU(N_c)$^2$U(1) and U(1)$^3$ anomalies vanish automatically. Fixing the normalization by Q = $T_3$ + Y and Q($\nu$_L)=0 gives

$$Y_Q = \frac{1}{2N_c}.$$

For N_c = 3,

$$Y_Q = \frac{1}{6}, \quad Y_L = -\frac{1}{2}, \quad Y_e = 1, \quad Y_u = -\frac{2}{3}, \quad Y_d = \frac{1}{3}, \quad Y_H = \frac{1}{2}.$$

Without Yukawas, the cubic anomaly leaves two discrete branches (Y_u, Y_d exchange). Yukawa invariance selects the branch with a single Higgs doublet. QED.

**Corollary 6.13a (Exact rational hypercharges).** With the derived N_c = 3, the hypercharge assignments are uniquely fixed to exact rational values:

$$Y_Q = \tfrac{1}{6}, \quad Y_L = -\tfrac{1}{2}, \quad Y_u = -\tfrac{2}{3}, \quad Y_d = \tfrac{1}{3}, \quad Y_e = 1, \quad Y_H = \tfrac{1}{2}.$$

**Why this is convincing.**

- These are **exact rationals**, not approximate numbers.

- Their ratios are fixed by anomaly freedom + Yukawa invariance, and the absolute lattice is fixed by the standard normalization, with no continuous parameters to adjust.
- This high-precision set of numbers strongly constrains the realized matter package and matches observation exactly.

### 5.4.9 Witten anomaly on the realized color branch

**Theorem 6.14 (Witten anomaly consistency on the realized color branch).** On the realized one-generation package emitted by Proposition 6.6a and Corollary 6.6b, the global SU(2) anomaly is absent generation by generation.

**Inputs.**

1. The realized color factor is the SU(3) triplet from Corollary 6.6b.
2. The matter content per generation includes:
    - one left-handed quark doublet Q which is an SU(2) doublet and carries color,
    - one left-handed lepton doublet L which is an SU(2) doublet and color singlet.
3. **Witten's global SU(2) anomaly constraint** (Witten, 1982): the number of left-handed SU(2) doublets must be even.

**Proof.** Count SU(2) doublets per generation:

- Quark doublets: N_c copies (one per color),
- Lepton doublets: 1 copy.

Total doublets per generation:

$$N_c + 1.$$

Substituting the realized value $N_c = 3$ gives

$$N_c + 1 = 4.$$

Witten anomaly cancellation therefore requires an even number of doublets, and the realized branch satisfies it exactly:

$$4 \equiv 0 \pmod 2.$$

So the realized triplet-doublet package is globally consistent generation by generation. QED.

**Theorem-stack role.** The D8 minimal-coupled-carrier derivation emits the specific value $N_c = 3$ and carries it into D9. Witten's anomaly is retained as a nontrivial consistency check on that emitted triplet-doublet package, not as a residual selector that upgrades oddness to 3.

**Why this is convincing.**

- It checks a **global anomaly constraint** on the emitted branch without introducing a new selector.
- The parity constraint is independent of continuous parameters, RG running, masses, or Yukawa values.
- It confirms that the realized SU(3) triplet plus lepton doublet package is globally consistent generation by generation.

### 5.4.10 Bond-dimension gatekeeping

In tensor-network or code realizations, gauge actions act on edge factors of size $\chi$, so emergent compact gauge groups embed in $U(\chi)$. This shows a capacity constraint: accommodating SU(3) color and SU(2) weak factors shows $\chi \geq 6$ in the minimal case, consistent with the MAR-derived gauge group.

### 5.4.11 Classical carrier modes and the quantum-particle gate

An abstract compact group identifies a symmetry type, and a classical field equation identifies a classical dynamical branch. Neither datum alone constructs a propagating quantum particle. The following receipt separates those levels.

The upstream structural results remain separate receipts. Conditional on the compact-gauge refinement receipt, Theorem 6.1 constructs the refinement-limit bosonic edge-sector category and its compact group; a field-algebra realization with that group as a local gauge symmetry requires the DHR/DR hypotheses (R1)–(R6) of Definition 6.1r. On the realized electroweak branch, a chosen nonzero neutral Higgs vacuum *vector* has stabilizer $U(1)_Q$, whereas its projective ray alone has the larger two-torus stabilizer recorded above. On the gravitational side, Theorems 4.3c–4.3e and 5.1 conditionally supply the observer-facing geometry, Lorentzian event-manifold, and classical Einstein branch. None of these structural outputs supplies the action Hessian or quantum data required below.

**Definition 6.18 (carrier-mode and quantum-particle receipts).** Fix a structural invariant speed $c_\star$. This is the common invariant null-cone conversion speed. Its representation as $299\,792\,458\ \mathrm{m\,s^{-1}}$ is exact by the SI definition of the metre and is not a prediction of that decimal magnitude. A *classical massless carrier-mode receipt* for a field $q_X$ consists of: (C1) a stated background and phase; (C2) an explicit quadratic action with positive nonzero physical kinetic coefficient; (C3) a gauge fixing or constraint reduction with a finite physical projector $\Pi_X$; and (C4) a positive reduced Hamiltonian. Separately, the action Hessian restricted to the physical subspace must have the form

$$K_X^{\mathrm{phys}}(\omega, \mathbf{k}) = Z_X(\omega^2 - c_\star^2|\mathbf{k}|^2)\Pi_X, \qquad Z_X > 0.$$

Equivalently, the free reduced Green function exhibits the action-level pole

$$D_X^{\mathrm{phys}}(\omega, \mathbf{k}) = \frac{i\,\Pi_X/Z_X}{\omega^2 - c_\star^2|\mathbf{k}|^2 + i0}.$$

This receipt proves a classical massless carrier mode and the absence of an algebraic mass term in that declared quadratic action. It does not prove a particle.

A *quantum-particle receipt* additionally requires: (Q1) a positive-energy vacuum quantization and a physical Hilbert space obtained by constraint reduction or BRST cohomology; (Q2) a physical two-point function whose Källén–Lehmann measure contains a positive-residue massless contribution

$$d\rho_X(\mu^2) = Z_X^{\mathrm{pole}}\,\delta(\mu^2)d\mu^2 + d\rho_X^{\mathrm{cont}}(\mu^2), \qquad Z_X^{\mathrm{pole}} > 0;$$

or an equivalent joint energy–momentum spectrum containing a positive-residue $p^0 > 0,\ p^2 = 0$ mass shell (at fixed $\mathbf{k}$, $\omega = c_\star|\mathbf{k}|$); and (Q3) the stability/asymptotic-state or LSZ hypotheses appropriate to the phase, including a deconfined asymptotic sector for a colored carrier. Only a receipt passing (C1)–(C4) and (Q1)–(Q3) licenses a quantum-particle claim. A reconstructed group, a connection label, or a classical field equation by itself fails this gate.

**Theorem 6.18 (action-level modes on the Maxwell, pure-Yang–Mills, and Einstein branches).** The following statements hold on the stated additional action and phase branches.

i. **Electromagnetic branch.** Suppose the source-free ordinary electromagnetic vacuum $J_Q = 0$ has an unbroken $\mathrm{U}(1)_Q$ connection with the Lorentzian Maxwell action

$$S_Q = -\frac{1}{4g_Q^2}\int F_{Q,\mu\nu}F_Q^{\mu\nu}\, d^4x, \qquad 0 < g_Q^2 < \infty,$$

about the ordinary vacuum, with no Higgs, Stueckelberg, medium, or nonlocal quadratic mass operator. In radiation gauge the reduced variables are the two transverse components and

$$H_Q^{(2)} = \frac{1}{2g_Q^2}\int(|\mathbf{E}_T|^2 + |\mathbf{B}|^2)d^3x, \qquad K_{Q,T} = g_Q^{-2}(\omega^2 - c_\star^2|\mathbf{k}|^2)\Pi_T, \qquad \operatorname{rank}\Pi_T = 2.$$

Equivalently, covariant $\xi$-gauge gives

$$D_{\mu\nu}^Q(k) = \frac{-ig_Q^2}{k^2 + i0}\left(\eta_{\mu\nu} - (1-\xi)\frac{k_\mu k_\nu}{k^2 + i0}\right),$$

whose longitudinal part decouples from a conserved current. Thus this branch has two classical massless electromagnetic carrier modes. Calling them photons requires the quantum-particle receipt.

ii. **Pure Yang–Mills branch.** Suppose a compact group $G$ is supplied with the Lorentzian two-derivative pure Yang–Mills action with $0 < g^2 < \infty$, expanded about the topologically trivial background in a gauge $\bar{A}_\mu = 0$, in a perturbative or deconfined phase with no Higgs condensate. Writing the perturbation as $a_\mu^a$, its quadratic action is

$$S_{\mathrm{YM}}^{(2)} = -\frac{1}{4g^2}\sum_a\int(\partial_\mu a_\nu^a - \partial_\nu a_\mu^a)(\partial^\mu a^{a,\nu} - \partial^\nu a^{a,\mu})d^4x.$$

Lorenz gauge gives the standard covariant inverse, while Gauss' law and radiation gauge leave two transverse components per generator, with

$$H_{\mathrm{YM}}^{(2)} = \frac{1}{2g^2}\sum_a\int(|\mathbf{E}_T^a|^2 + |\mathbf{B}^a|^2)d^3x \geq 0,$$

with strict positivity on nonzero reduced finite-energy modes of $\mathbf{k} \neq 0$ under the stated boundary conditions. Their quadratic transverse kernel and free Green function are

$$K_{\mathrm{YM},T}^{ab} = \delta^{ab}g^{-2}(\omega^2 - c_\star^2|\mathbf{k}|^2)\Pi_T, \qquad D_{\mathrm{YM},T}^{ab} = \frac{ig^2\delta^{ab}\Pi_T}{\omega^2 - c_\star^2|\mathbf{k}|^2 + i0}, \qquad \operatorname{rank}\Pi_T = 2$$

so there are $2\dim G$ perturbative classical transverse modes and no hard mass in that quadratic expansion. It does not produce a gauge-invariant asymptotic gluon. In the confining QCD phase, and on a gauge-invariant Yang–Mills branch with a positive physical gap, the colored free pole need not occur in the physical spectrum.

iii. **Pure Einstein branch.** Suppose, in addition to the derived classical Einstein equation, that the field-content branch is the two-derivative Einstein–Hilbert action with $G > 0$ about a Minkowski vacuum with $\Lambda = 0$, without higher-curvature kinetic terms, bimetric mixing, or extra scalar/vector fields. After de Donder gauge and residual-gauge reduction, the two transverse-traceless components obey

$$S_{\mathrm{TT}}^{(2)} = \frac{1}{64\pi G}\int\left[(\partial_t h_{ij}^{\mathrm{TT}})^2 - c_\star^2(\nabla h_{ij}^{\mathrm{TT}})^2\right]d^4x,$$

$$H^{(2)}_{\mathrm{TT}} = \frac{1}{64\pi G}\int\Big[(\partial_t h^{\mathrm{TT}}_{ij})^2 + c_\star^2(\nabla h^{\mathrm{TT}}_{ij})^2\Big]\,d^3x, \qquad D^{\mathrm{TT}}_{ij,kl}(k) \propto \frac{i\Pi^{\mathrm{TT}}_{ij,kl}}{k^2+i0}, \qquad \operatorname{rank}\Pi^{\mathrm{TT}} = 2.$$

Thus the pure Einstein linearization has two classical massless spin-two wave modes on this background. On a curved Einstein background the relevant statement is instead about the Lichnerowicz operator and its boundary conditions; a Poincaré particle pole is not automatic. The present OPH derivation does not construct the metric-field quantization, physical graviton Hilbert space, or positive-residue two-point pole required by (Q1)–(Q3).

**Proof.** For (i), expand the Maxwell action to second order. Gauss' law removes the longitudinal canonical pair, leaving two transverse polarizations with the displayed positive Hamiltonian and dispersion $\omega^2 = c_\star^2|\mathbf{k}|^2$. Inverting the gauge-fixed Hessian gives the displayed covariant Green function. For (ii), write $F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + O(A^2)$; the quadratic Hessian is one Maxwell Hessian per Lie-algebra generator, while the nonabelian interactions begin at cubic order. For (iii), the second variation of the Einstein–Hilbert action is the Fierz–Pauli kinetic form. De Donder gauge and its residual gauge freedom reduce it to the transverse-traceless action shown, with the + and × polarizations. These Hessian calculations establish (C1)–(C4). None constructs the quantum data in (Q1)–(Q3), so no unconditional particle conclusion follows. □

**Corollary 6.18a (conditional particle interpretation).** If a branch in Theorem 6.18 is also supplied with its quantum-particle receipt, its positive-residue $\delta(\mu^2)$ term defines a massless particle pole with the displayed physical polarizations. Without that receipt, this paper claims only the corresponding classical or perturbative carrier mode. In particular, the Tannaka group alone does not imply a Maxwell kinetic term or deconfined phase, and the classical Einstein equation alone does not imply a graviton state.

**Hard-term versus physical-spectrum boundary.** The vanishing quadratic parameters $\mu^2_{Q,\mathrm{quad}}$, $\mu^2_{\mathrm{YM,quad}}$, and $\mu^2_{\mathrm{EH,TT}}$ in the displayed actions are action-level statements, not universal exact particle masses. Gauge or diffeomorphism redundancy can coexist with Higgs/Stueckelberg completions, plasma or medium self-energies, confinement, higher-derivative poles, bimetric sectors, or additional massive fields. Excluding those possibilities requires the declared field-content and phase hypotheses rather than the symmetry label alone.

#### 5.4.12 Quotient-protected charge quantization

**Theorem 6.19 (Charge quantization and no fractional color singlets).** If the global gauge group is

$$G_{\mathrm{phys}} = \frac{\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)}{Z_6},$$

as derived in Proposition 6.6, then

All color-singlet states have integer electric charge.

Equivalently: no stable isolated particles with charges like $\pm 1/3$ can exist as color singlets.

**Proof.** The $Z_6$ quotient identifies the center elements (e^{2πi/3}, -1, e^{iπ/3}) ∈ SU(3) × SU(2) × U(1) with the identity. For a color-singlet state ($\tau = 0$), the SU(3) factor acts trivially. The resulting identification requires the SU(2) × U(1) quantum numbers to satisfy

$$(-1)^{2j}\cdot e^{i\pi n/3} = 1,$$

where j is the SU(2) spin and n = 6Y is the integer hypercharge label. This gives n ≡ -6j (mod 6), i.e., n ≡ 0 (mod 6) for integer j and n ≡ 3 (mod 6) for half-integer j. Equivalently: Y is integer when j is integer, and Y is half-integer when j is half-integer.

After electroweak breaking, Q = $T_3$ + Y. For integer j, $T_3 \in \mathbb{Z}$ and Y ∈ ℤ, so Q ∈ ℤ. For half-integer j, $T_3 \in \mathbb{Z} + 1/2$ and Y ∈ ℤ + 1/2, so Q = (half-integer) + (half-integer) ∈ ℤ. In both cases, Q ∈ ℤ. QED.

**Experimental bound**: No fractionally charged color-singlet particles have been observed. Three independent high-precision bounds confirm this:

1. **Neutrality of matter** (PDG 2024): The proton-electron charge sum satisfies

$$|q_p + q_e|/e < 1 \times 10^{-21},$$

   confirming charge quantization to 21 decimal places.

2. **Fractional charge searches in bulk matter**: Silicone oil drop experiments limit fractionally charged particle abundance to

$$\text{(fractionally charged particles)/nucleon} \lesssim 10^{-22}.$$

3. **Collider searches** (CMS, PRL 134, 2025): Exclusions for stable particles with $q \in [e/3, 0.9e]$ up to masses ∼ 640 GeV (95% CL).

Existing searches are fully consistent with these structural consequences.

These are exact symmetry- or quotient-protected outputs: the empirical content is that violations are excluded up to available experimental sensitivity.

#### 5.4.13 Coupling extraction from edge-sector probabilities

The edge-center completion (Theorem 2.3) yields sector probabilities p_$\alpha$ on collar boundaries. These probabilities encode the renormalized gauge coupling through a heat-kernel/Laplacian weighting law.

**Abelian case (Z_n).** For a Z_n gauge theory, the edge sectors are labeled by charge q ∈ {0, 1, ..., n-1}. The correct "Casimir" eigenvalue is the Laplacian eigenvalue of the boundary random walk:

$$\lambda_q = 4\sin^2\left(\frac{\pi q}{n}\right).$$

Note: only in the limit n → ∞ and q ≪ n does λ_q ≈ (2πq/n)$^2$ ∝ q$^2$. For finite n, the exact form is essential.

The sector probabilities follow a heat-kernel law:

$$p_q \propto e^{-t(\mu)\lambda_q},$$

where t($\mu$) is the "modular time" parameter encoding the scale. The extraction formula is:

$$t(\mu) = -\frac{\log(p_q/p_0)}{\lambda_q}, \qquad g^2_{\text{ent}}(\mu) = \frac{t(\mu)}{2\pi}.$$

Consistency requires that t extracted from different charges q agrees; this is verified numerically (see Section 6.14).

**Electric-center measurement.** The edge sectors are measured using the *electric-center* prescription. For a region A and boundary vertex v ∈ ∂A, define the restricted star operator:

$$Q_v^{(A)} = \prod_{\ell \in \mathrm{star}(v) \cap A} X_\ell^{\pm 1},$$

where X_ℓ is the shift operator on link ℓ. The sector projectors are:

$$P_{v,q} = \frac{1}{n} \sum_{m=0}^{n-1} \omega^{-mq} \left(Q_v^{(A)}\right)^m, \qquad \omega = e^{2\pi i/n},$$

and the probabilities are p_{v,q} = ⟨P_{v,q}⟩. This electric-center operator, built from X's rather than Z's, correctly captures the boundary gauge charge/flux that labels entanglement edge sectors.

**Non-abelian generalization.** For SU(N) gauge theories, the edge sectors are labeled by irreducible representations with probabilities:

$$p_j \propto d_j \, e^{-t(\mu) C_2(j)},$$

where d_j is the dimension and $C_2$(j) the quadratic Casimir. Extraction:

$$t(\mu) = -\frac{\log(p_j/p_0)}{C_2(j)}, \qquad g^2_{\mathrm{ent}}(\mu) = \frac{t(\mu)}{2\pi}.$$

**Theoretical derivation.** The fixed-cutoff same-overlap edge law is carried by the regulated microphysics package once the declared overlap-sector projectors, reversible thermal branch, and quasi-local local-Gibbs generator of Theorem 2.6 are in hand. On this synthesis surface, the point is to summarize that fixed-cutoff source package and to state the refinement/Peter–Weyl lift boundary explicitly rather than to relocate the leaf-level source surface. The Laplacian spectrum selection and the one-sided sector rank do not follow from MaxEnt, gauge invariance, and locality alone; both are named hypotheses of the theorem.

**Theorem 6.20 (Fixed-cutoff edge-sector law and refinement lift boundary).** Under the OPH axioms, the fixed-cutoff overlap-gauge realization of Sections 2.3 and 3.2, the quasi-local local-Gibbs form supplied by Theorem 2.6, and hypotheses (EH-1) and (EH-2) below, the regulated same-overlap edge-sector probability distribution satisfies:

$$p_R = \frac{d_R \, e^{-t\lambda_R}}{\sum_{R'} d_{R'} \, e^{-t\lambda_{R'}}}$$

where λ_R is the Laplacian eigenvalue on the R-isotypic component and t is determined by the collar Gibbs parameter.

**Hypothesis (EH-1, edge Hamiltonian).** The MaxEnt edge generator is the factorwise group Laplacian, $H_{\mathrm{edge}} = t\,\Delta_G$ (for a product group $G = \prod_i G_i$, a positive combination $\sum_i c_i \Delta_{G_i}$); for finite groups, the Cayley-graph Laplacian with $\lambda_R = |S| - (1/d_R)\sum_{s\in S} \chi_R(s)$. This is a declared microphysics input, not a MaxEnt consequence: gauge invariance and the Gibbs form force only $H_{\mathrm{edge}} = \sum_R h_R P_R$, and any function of the sector label (for instance $h_R = C_2(R)^2$, or a finite projector energy) is gauge invariant and realizable by a spatially local lattice Hamiltonian at fixed cutoff. Differential order on group space is not spatial finite-range locality, so no locality argument excludes those alternatives. Within second-order bi-invariant differential operators the Laplacian is unique up to the stated scale freedom, and the appendix supplies a model-specific mechanism that derives (EH-1) for one declared Markov chain: a proposal kernel with weighted symmetry and a Casimir acceptance rule has the stated law as its stationary distribution. That derivation covers that chain, not every OPH/MaxEnt edge branch; universality of (EH-1) across branches is open and tested numerically in Section 6.14.

**Hypothesis (EH-2, one-sided edge algebra).** The probability space is the one-sided electric-center edge algebra $\mathcal{A}_{\text{edge}} = \bigoplus_R B(V_R)$, the Gauss-law constrained algebra visible from one side of the cut, on which the sector projector has rank $d_R$. The convention is fixed here once: on the full two-sided space $L^2(G) \cong \bigoplus_R V_R \otimes V_R^*$ the R-isotypic projector has rank $d_R^2$, and a Gibbs trace taken there gives sector weights $\propto d_R^2 e^{-t\lambda_R}$ (the closed-surface partition sum $\sum_R d_R^2 e^{-tC_2(R)}$ is that two-sided object). The $d_R$ law of the theorem is a statement about the one-sided algebra and is not obtained by tracing the two-sided Gibbs state, since the two-sided reduction keeps weight $d_R^2 e^{-t\lambda_R}$. (EH-2) declares the one-sided state assignment directly, consistent with the Markov normal form in which the edge factor on one side contributes $\log d_R$ to the entropy.

**Proof.**

*Step 1 (Edge Hilbert space).* From the fixed-cutoff overlap-gauge realization, the edge degrees of freedom at a boundary circle $\Sigma = \partial C$ live in a Hilbert space transforming under the gauge group G. Microscopically this is a finite-dimensional quantum-link edge space on the declared overlap interface; the effective representation-theoretic description used only for the refinement lift is modeled by $L^2(G)$ with its Peter–Weyl decomposition [47], and the physical one-sided carrier is the (EH-2) algebra.

*Step 2 (Gauge invariance).* The Gauss law constrains physical states. For an entanglement cut at $\Sigma$, the physical edge observables from one side decompose sectorwise as $\mathcal{A}_{\text{edge}} = \bigoplus_R B(V_R)$ by (EH-2).

*Step 3 (Edge Hamiltonian).* By (EH-1) the MaxEnt generator restricted to edge modes is $H_{\text{edge}} = \sum_R \lambda_R P_R$ with $P_R$ the (EH-2) sector projector of rank $d_R$.

*Step 4 (MaxEnt selection).* MaxEnt with the (EH-1) generator selects the Gibbs state on the (EH-2) algebra:

$$\rho_{\text{edge}} = \frac{1}{Z} e^{-tH_{\text{edge}}} = \frac{1}{Z} \sum_R e^{-t\lambda_R} P_R.$$

*Step 5 (Sector probabilities).* The probability of sector R is $p_R = \mathrm{Tr}(\rho_{\text{edge}} P_R)$, and on the (EH-2) algebra $\mathrm{Tr}\, P_R = d_R$, so

$$p_R = \frac{d_R\, e^{-t\lambda_R}}{Z}.$$

QED.

**Scope.** The fixed-cutoff same-overlap derivation is the theorem-bearing part of the edge-law package, conditional on (EH-1) and (EH-2). The quasi-local local-Gibbs generator used here is derived from Theorem 2.6: if MaxEnt constraints are expectations of finitely many quasi-local operators, the entropy maximizer is automatically a Gibbs state with a quasi-local generator. The additional representation-theoretic step summarized here is the compact-group / Peter–Weyl refinement lift. The Casimir-ratio predictions, coupling extraction, $\alpha_U$ closure, and the 2D-YM/worldsheet bridge downstream of this law are valid on models satisfying (EH-1)/(EH-2), not as consequences of the listed axioms alone; the numerical tests of Section 6.14 probe (EH-1) on explicit models. Model-family uniqueness and large-edge continuations have their own claim boundaries.

**Normalization anchor: 2D Yang-Mills.** The parameter t can be exactly matched to a conventional coupling in 2D Yang-Mills, where the physical Hamiltonian is literally the group Laplacian:

$$H = \frac{g^2}{2}\Delta_G, \qquad \Delta_G\chi_R = -C_2(R)\chi_R \quad \Rightarrow \quad E_R = \frac{g^2}{2}C_2(R).$$

Euclidean evolution for "time" A (the area of a cylinder in 2D YM) gives weight(R) $\propto$ exp(-A E_R) = exp(-g$^2$ A C$_2$(R)/2). Comparing with the heat-kernel expansion K_t(U) = Σ_R d_R $\chi$_R(U) e^{-t C$_2$(R)} yields the exact identification:

$$t_{\text{phys}} = \frac{g^2 A}{2} \quad \text{(in 2D YM, no ambiguity).}$$

This shows that the Laplacian + MaxEnt → heat-kernel structure is exact in a solvable two-dimensional Yang-Mills case. The coefficient in front of C$_2$ is fixed. This normalization anchor is a two-dimensional edge-theory check, separate from the compact paper's conditional support-visible compact-gauge repair-gap theorem. In any regime where the edge theory reduces to an effective 2D YM with known "Euclidean thickness" A_eff:

$$g^2(\mu) = \frac{2}{A_{\text{eff}}(\mu)} \cdot \frac{\Delta_R(\mu)}{C_2(R)},$$

and the RHS must be R-independent. This R-independence is an internal precision consistency test; the formula itself is the normalization map that connects t to the conventional gauge coupling.

#### 5.4.14 Numerical validation of the heat-kernel law

The heat-kernel/Laplacian weighting of edge sectors is validated in explicit 2D Z_n gauge models on closed geometries.

**Model.** A 2×2 periodic lattice gauge theory (8 links) with Z_n link Hilbert spaces and Hamiltonian:

$$H = -K\sum_p \text{Re}(B_p) - h\sum_\ell \text{Re}(X_\ell) - \Gamma\sum_v \text{Re}(A_v),$$

where X_$\ell$ is the Z_n shift on link $\ell$, B_p is the oriented plaquette operator (product of Z's around plaquette p), and A_v is the oriented star/Gauss operator (outgoing X, incoming X†). With K = 1 and Γ = 5, the ground state satisfies ⟨A_v⟩ = 1 at all vertices to numerical precision.

**Region and edge operator.** Region A consists of links whose tail has x = 0 ("half-lattice" cut). At each boundary vertex v, the electric-center edge charge is the restricted star Q_v^{(A)} = ∏_{$\ell$ ∈ star(v) ∩ A} X_$\ell$^{±1}.

**Results for $\mathbf{Z_2}$.** With $\lambda_1$ = 4sin$^2$($\pi$/2) = 4:

| h | $p_0$ | $p_1$ | t | g_ent |
|---|---|---|---|---|
| 0.5 | 0.8266 | 0.1734 | 0.391 | 0.249 |
| 1.0 | 0.9612 | 0.0388 | 0.803 | 0.357 |
| 2.0 | 0.9917 | 0.0083 | 1.194 | 0.436 |

Z$_2$ has a single nontrivial Laplacian eigenvalue, so a one-parameter fit of t can always be made exactly for any admissible nontrivial weight. The Z$_2$ table is therefore an implementation and convention check, not a test of the heat-kernel form.

**Results for $\mathbf{Z_3}$ (symmetry-forced consistency check).** With $\lambda_1 = \lambda_2$ = 4sin$^2$($\pi$/3) = 3:

| h | $p_0$ | $p_1$ | $p_2$ | t(q=1) | t(q=2) | g_ent | m_plaq |
|---|---|---|---|---|---|---|---|
| 0.2 | 0.4395 | 0.2803 | 0.2803 | 0.1500 | 0.1500 | 0.154 | 2.22 |
| 0.5 | 0.7509 | 0.1245 | 0.1245 | 0.5989 | 0.5989 | 0.309 | 1.75 |
| 1.0 | 0.9606 | 0.0197 | 0.0197 | 1.2956 | 1.2956 | 0.454 | 4.07 |
| 1.5 | 0.9851 | 0.0074 | 0.0074 | 1.6288 | 1.6288 | 0.509 | 7.06 |
| 2.0 | 0.9921 | 0.0039 | 0.0039 | 1.8440 | 1.8440 | 0.542 | 10.10 |

Both equalities in this table are forced by symmetry before any heat-kernel ansatz is imposed. The equality $p_1 = p_2$ is exact charge-conjugation symmetry in $Z_3$, and the two nontrivial Fourier modes share the degenerate eigenvalue $\lambda_1 = \lambda_2 = 3$, so extracting t from q = 1 and q = 2 must give the same value for *any* conjugation-invariant distribution. The observed agreement (t_{q=1} ≈ 1.2956389318579 versus t_{q=2} ≈ 1.2956389318521 at h = 1.0, i.e. ~$10^{-14}$) therefore validates the implementation and the symmetry conventions only; it cannot distinguish the heat-kernel form from an arbitrary conjugation-symmetric distribution on $Z_3$, and it is not counted as an independent confirmation of the heat-kernel law. The substantive overconstrained tests are the $Z_5$ and $S_3$ calculations below, which involve two distinct nonzero eigenvalues.

**Region-choice stability.** At h = 1, the extracted g_ent is nearly independent of region size:

- 2 links (one vertex's outgoing links): g_ent ≈ 0.453
- 4 links (half-lattice): g_ent ≈ 0.454
- 6 links (three vertices): g_ent ≈ 0.453

This locality confirms that the coupling is dominated by physics near the cut, not global bookkeeping, exactly what is expected if this behaves like a local QFT observable.

**Results for $Z_5$ (golden ratio test).** The $Z_5$ case provides a stringent test because the Laplacian eigenvalues have a distinctive ratio involving the golden ratio $\phi = (1+\sqrt{5})/2$:

$$\lambda_q = 4\sin^2\left(\frac{\pi q}{5}\right), \qquad \frac{\lambda_2}{\lambda_1} = \frac{\sin^2(72°)}{\sin^2(36°)} = \phi^2 \approx 2.618.$$

This ratio distinguishes the Laplacian law from naive alternatives: a linear model (λ_q ∝ q) would predict ratio 2, while a quadratic model (λ_q ∝ $q^2$) would predict ratio 4.

Direct diagonalization of the displayed Hamiltonian on the full 2×2-torus link basis ($5^8$ states, K = 1, Γ = 5), measuring the restricted-star edge charge at a boundary vertex of the half-lattice region, gives:

| h | Measured ratio $\ln(p_2/p_0)/\ln(p_1/p_0)$ | Deviation from $\phi^2$ |
|---|---|---|
| 0.50 | 2.876 | 9.9% |
| 0.20 | 2.750 | 5.1% |
| 0.10 | 2.671 | 2.0% |
| 0.05 | 2.643 | 1.0% |

In the weak-field limit h → 0 the measured ratio converges to the golden ratio squared, and the stated limiting direction agrees with the tabulated sequence: the deviation decreases monotonically as h decreases. (The table reports the direct link-basis run, reproducible with the held-out validation script in the released edge-sector code; a dual/flux-basis parameterization carries a tabulated parameter that runs opposite to the h of the displayed Hamiltonian and is not used. At large h the ratio instead drifts toward the perturbative order-counting value 2, which is why the weak-field direction is the declared convergence regime for this model.) The test is genuinely overconstrained: $Z_5$ has two distinct nonzero eigenvalues, t is fitted on the q = 1 sector alone, the q = 2 weight is a held-out prediction, and the deviation column reports the held-out residual. Convergence of that residual confirms that the vacuum entanglement spectrum encodes the precise geometric structure of the gauge group Laplacian.

**Significance.** This validates the mathematical law (sector probabilities weighted by Laplacian eigenvalues) in explicit 2D gauge-invariant models with non-flat sector distributions. Because the nontrivial $Z_3$ eigenvalues are degenerate, it is the $Z_5$ test (and the $S_3$ test below) that is structurally identical to SU(2)/SU(3): two or more distinct nonzero eigenvalues overconstrain the slope, and held-out agreement confirms the mechanism works before jumping to nonabelian groups.

**Results for $S_3$ (first nonabelian test).** The abelian tests above use charge-sector projectors that reduce to Fourier modes. For nonabelian groups, the edge-sector projector must be generalized to character projectors:

$$P_{v,R} = \frac{d_R}{|G|} \sum_{h \in G} \chi_R(h^{-1}) Q_v^{(A)}(h),$$

where d_R is the dimension of irrep R, $\chi$_R is its character, and Q_v^{(A)}(h) is the restricted gauge action at boundary vertex v acting only on links in region A.

For $S_3$ (the smallest nonabelian group, order 6), there are three irreps: trivial (d=1), sign (d=1), and standard (d=2). The Cayley-graph Laplacian eigenvalues for the transposition generating set are:

$$\lambda_{\text{triv}} = 0, \qquad \lambda_{\text{sign}} = 6, \qquad \lambda_{\text{std}} = 3.$$

**Exact reduction on one plaquette.** For the single-plaquette model (4 links), imposing Gauss's law at all vertices means the physical wavefunction depends only on the plaquette holonomy's conjugacy class. Since $S_3$ has exactly 3 conjugacy classes, the gauge-invariant Hilbert space is 3-dimensional, spanned by the character states {|$\chi$_R⟩}. In this basis, the edge-sector probabilities are exactly p_R = |c_R|$^2$ where $|\psi_0\rangle$ = Σ_R c_R |$\chi$_R⟩. This is an exact identity for the one-plaquette gauge-invariant sector.

The heat-kernel ansatz predicts p_R ∝ d_R exp(-t $\lambda$_R). Extracting t independently from the sign and standard irreps provides an overconstrained test: the ratio $\lambda$_sign/$\lambda$_std = 6/3 = 2 is a parameter-free prediction. Results from a single-plaquette $S_3$ lattice gauge model (K=1, Γ=5):

| h | p_triv | p_sign | p_std | t (sign) | t (std) | Δt/t | log-ratio |
|---:|---:|---:|---:|---:|---:|---:|---:|
| 0.5 | 0.909 | 0.0013 | 0.089 | 1.09 | 1.01 | 8.4% | 2.17 |
| 1.0 | 0.980 | $7.5\times10^{-5}$ | 0.020 | 1.58 | 1.54 | 2.8% | 2.06 |
| 2.0 | 0.996 | $4.3\times10^{-6}$ | 0.004 | 2.06 | 2.04 | 1.0% | 2.02 |
| 5.0 | 0.9993 | $1.0\times10^{-7}$ | 0.00066 | 2.68 | 2.67 | 0.3% | 2.006 |
| 12 | 0.9999 | $3.0\times10^{-9}$ | 0.00011 | 3.27 | 3.27 | 0.1% | 2.002 |
| 100 | 1.0000 | $6.1\times10^{-13}$ | $2.0\times10^{-6}$ | 4.69 | 4.69 | 0.009% | 2.0002 |

The "Δt/t" column shows the fractional difference (t_sign - t_std) / $\bar{t}$. The "log-ratio" column shows log(p_sign/$p_0$) / log(p_std/(2 $p_0$)), which should equal $\lambda$_sign/$\lambda$_std = 2 if the heat-kernel form holds exactly. Equivalently, t is fitted from the standard sector alone and the sign-sector weight is a held-out prediction; log-ratio minus 2 and $\Delta t/t$ are the held-out residuals. Unlike $Z_3$, the two nonzero eigenvalues are distinct (6 versus 3), so no symmetry forces this agreement.

As h increases, both diagnostics converge: $\Delta t/t$ drops below $10^{-4}$ and the log-ratio approaches 2.000. This is exactly the expected behavior: finite-size corrections are largest at strong coupling; the heat-kernel form becomes exact as the perturbative regime is approached.

This gives a nonabelian validation of the edge-sector extraction mechanism. The structure (character projectors, Laplacian eigenvalues from the group's Cayley graph, overconstrained t extraction) is identical to the SU(2) and SU(3) continuation setup.

**Parameter-free predictions for SU(2) and SU(3).** The heat-kernel law yields exact, parameter-free ratio predictions that require no scheme matching. Define the "Casimir log-gap":

$$\Delta_R \equiv \ln\left(\frac{p_0}{d_0}\right) - \ln\left(\frac{p_R}{d_R}\right) = t\, C_2(R).$$

Ratios of Δ_R cancel all unknowns (t, partition function):

$$\frac{\Delta_{R_1}}{\Delta_{R_2}} = \frac{C_2(R_1)}{C_2(R_2)} \quad \text{(exact, parameter-free)}.$$

*SU(2) predictions.* Irreps labeled by spin j have d_j = 2j+1 and $C_2(j) = j(j+1)$. The framework predicts:

- $\Delta_1/\Delta_1/_2 = 2/(3/4) =$ **8/3 ≈ 2.667**
- $\Delta_3/_2/\Delta_1/_2 = (15/4)/(3/4) =$ **5**
- $\Delta_3/_2/\Delta_1 = (15/4)/2 =$ **15/8 = 1.875**

*SU(3) predictions.* Irreps labeled by Dynkin indices (p,q) have $C_2(p,q) = (p^2 + q^2 + pq + 3p + 3q)/3$. Using the fundamental **3** = (1,0) with $C_2 = 4/3$ as the reference:

- $\Delta_8/\Delta_3 = 3/(4/3) =$ **9/4 = 2.25**
- $\Delta_6/\Delta_3 = (10/3)/(4/3) =$ **5/2 = 2.5**
- $\Delta_{10}/\Delta_3 = 6/(4/3) =$ **9/2 = 4.5**
- $\Delta_{15}/\Delta_3 = (16/3)/(4/3) =$ **4**
- $\Delta_{27}/\Delta_3 = 8/(4/3) =$ **6**

These are the SU(2)/SU(3) analogs of the $Z_5$ golden-ratio test: exact rational numbers fixed entirely by group theory, with no adjustable parameters.

**Preliminary SU(3) results.** A one-plaquette SU(3) "quantum link" model (finite truncated irrep basis, n_max = 12, $\kappa = 2$) extracts t from 14 different irreps simultaneously. The results show internal consistency at the 1-3% level:

| bare $g^2$ | extracted t (mean±std) | g_ent | gap |
|---|---|---|---|
| 0.3 | 0.314±0.0005 | 0.224 | 1.92 |
| 0.5 | 0.539±0.0025 | 0.293 | 1.83 |
| 0.8 | 0.896±0.012 | 0.378 | 1.72 |
| 1.0 | 1.144±0.025 | 0.427 | 1.64 |

The standard deviation across irreps provides a built-in error estimate. This is a QCD proton-physics surrogate rather than a full proton calculation; it lacks dynamical quarks and operates on a single plaquette. It nevertheless demonstrates that the nonabelian extraction machinery produces self-consistent outputs without tuning.

**Extracting the normalization factor A_eff.** The 2D YM anchor (Section 6.13) gives t = $g^2$ A / 2, so the "effective Euclidean thickness" is

$$A_{\text{eff}} = \frac{2t}{g^2}.$$

Computing this from the SU(3) table:

| bare $g^2$ | extracted t | A_eff |
|---:|---:|---:|
| 0.3 | 0.314 | 2.093 |
| 0.5 | 0.539 | 2.156 |
| 0.8 | 0.896 | 2.240 |
| 1.0 | 1.144 | 2.288 |

**Mean**: A_eff ≈ 2.19 with point-to-point scatter ~4%.

**Extrapolation to weak coupling.** The systematic drift in A_eff shows fitting A_eff($g^2$) = $A_0$ + a · $g^2$. A weighted linear fit gives:

$$A_0 = 2.004 \pm 0.012$$

with $\chi^2$/dof ≈ 0.09, indicating excellent consistency. This strongly shows that, in this toy UV completion, the "missing normalization" converges to A_eff → 2 as $g^2$ → 0.

The normalization factor behaves like a quasi-constant and extrapolates to a simple value (≈ 2) in the weak-coupling limit. This provides a concrete path to absolute coupling predictions: once A_eff is determined from microphysics, the conversion $g^2$ = 2t/A_eff fixes the gauge coupling without additional free parameters.

**Internal validation summary.** The finite-group calculations separate into symmetry-forced implementation checks and genuinely overconstrained tests:

- $\mathbf{Z_2}$**,** $\mathbf{Z_3}$: implementation and convention checks. $Z_2$ has one nontrivial eigenvalue (one parameter, one datum); in $Z_3$ the equalities $p_1 = p_2$ and t_{q=1} = t_{q=2} are forced by charge conjugation plus the degenerate eigenvalue $\lambda_1 = \lambda_2$, so the ~$10^{-14}$ agreement is expected from symmetry alone
- $\mathbf{Z_5}$: overconstrained golden-ratio test ($\lambda_2/\lambda_1 = \phi^2$, two distinct nonzero eigenvalues), held-out residual 1.0% at h = 0.05, decreasing monotonically as h → 0
- $\mathbf{S_3}$: overconstrained nonabelian Casimir log-ratio test (λ_sign/λ_std = 2), held-out residual 0.01%
- **SU(3)**: 14-irrep simultaneous extraction, internal scatter 1-3%

The $Z_3$ equality is forced by symmetry and is retained only as a regression check of the implementation; it is not an independent confirmation of the heat-kernel law. The substantive validation rests on the $Z_5$, $S_3$, and SU(3) tests, each of which fits t using fewer spectral sectors than it predicts, involves two or more distinct nonzero eigenvalues, and reports held-out residuals that converge in the declared regime of the corresponding model (h → 0 for the $Z_5$ torus model; h large for the single-plaquette $S_3$ model). This provides internal validation of the mechanism "MaxEnt + Laplacian => heat-kernel sector weights" before applying it to physical gauge groups. A self-contained held-out refit script for the finite-group models is provided with the public code release; it fits t on a declared subset of sectors and prints held-out residuals for the remaining sectors.

#### 5.4.15 Frozen IBM Quantum Cloud engineering archive

The former IBM Quantum Cloud program is frozen as an engineering and reproducibility archive. It checked whether OPH-motivated reduced-sector structures survived preparation and readout on real superconducting-qubit devices, and whether a programmed record-gated controller beat restricted controller nulls. Standard quantum mechanics predicts every prepared state and dynamic circuit used in that program. The runs therefore have no power to distinguish OPH from quantum mechanics, regardless of shot count or blinding quality. No new quantum-cloud campaign is promoted as OPH evidence until a source-closed observable with a different numerical QM prediction and an identifiable effect size is derived.

Detailed circuits, representative derived summaries, the complete blinded-run receipts, and archived rerun instructions are public in the frozen project write-up and the accompanying code and data bundle.

- **Stage 1 (Markov/recoverability benchmark).** On `ibm_marrakesh` and `ibm_fez`, the structured state reconstructs below both controls in conditional mutual information and above both controls in Petz fidelity. On `ibm_marrakesh`, the observed ordering is $0.2309 < 0.3890 < 0.9474$ for CMI and $0.9297 > 0.8649 > 0.6066$ for Petz fidelity; on `ibm_fez`, it is $0.1498 < 0.4992 < 0.9166$ and $0.9479 > 0.8117 > 0.6449$. This reproduces the programmed ordering on two real backends; the Petz calculation is offline and the result is not theory-discriminating.
- **$Z_3$ hardware sanity check.** The two-sector $Z_3$ extraction (whose two-sector agreement is expected a priori from the degenerate eigenvalues and charge-conjugation symmetry, so this is an implementation check, not an overconstrained test) passes cleanly on hardware: across prepared $t = 0.30, 0.60, 0.90$, the mean extracted $t$ is within about 0.02 of the target, the two independent extractions agree to within 0.0142, 0.0034, and 0.0043, and leakage stays below 0.1%. This shows that the reduced-sector preparation and readout path is internally coherent on-device before sharper ratio claims are interpreted.
- **$Z_5$ exact-ratio preparation.** The programmed target is $\Delta_2/\Delta_1 = \varphi^2 \approx 2.618$. The measured values span approximately 2.407 to 2.878, and the focused high-shot $t = 0.90$ interval lies below the exact target. Because the target amplitudes are directly prepared, QM predicts the same ideal ratio.
- **$S_3$ nonabelian layout diagnostic.** The first hardware run revealed a real layout-dependent bias. Reversing the qubit layout moved the mitigated ratio from 1.8724 to 2.0299, and a repeat returned 2.0657. Because the better mapping was selected after observing the bias, this is a post-hoc device diagnostic rather than a blinded confirmation.

These frozen runs document preparation, readout, layout, and feedback engineering on real devices. They add no evidence for OPH over standard quantum mechanics because the two descriptions make no different prediction for the implemented interventions. Their proper role is reproducibility and controller validation, not theory confirmation.

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### 5.5 Claim Boundaries, Tests, and Scope

#### 5.5.1 Classification of results and dependencies

The documentary node labels D1–D12 are used only to keep the dependency boundary compact. The phase split is:

$$\text{Phase I} = (D1\text{–}D5) \cup (D7\text{–}D9), \qquad \text{Phase II} = D6 \cup D10, \qquad \text{Phase III} = D12.$$

| Phase | Nodes | Meaning |
|---|---|---|
| I | D1–D5, D7–D9 | Recovered core: fixed-cutoff overlap repair and collar structure; the Lorentz/three-dimensional spatial-chart/null-modular/Einstein branch; receipt-conditional bosonic compact gauge reconstruction; the support-visible four-dimensional Euclidean Yang–Mills form and conditional repair-gap theorem on the compact-gauge continuum/transfer branch; realized-branch Standard Model quotient, exact hypercharge, structural electroweak force content, $N_c = 3$, and $N_g = 3$ under the explicit matter-package and admissibility inputs. |
| II | D6, D10 | Quantitative closures: screen-capacity closure of the same Einstein branch and the $P$-driven electroweak/gauge-coupling branch. These depend on declared external or branch-specific inputs. |
| III | D12 | Continuations: flavor details beyond the stated theorem surfaces, $H^3$ record-worldline stitch certificates and the event-manifold receipts `E1`–`E6` of Theorem 4.3e, dark-sector proposals, conditional screen-spectrum and CMB/inflation-replacement kernels, $H_0/S_8$ and growth branches, the finite-quotient baryogenesis anomaly/current theorem with its open anomalous-record-generator branch, spectroscopy, hadrons, and string/worldsheet effective-description branches. |

Phase I is the theorem-bearing core using the stated scaling, transport, and gauge theorem stack. Phase II supplies quantitative closures on top of that core. Phase III contains program branches whose extra ansätze are not part of the recovered-core theorem package. In particular, the conditional screen-spectrum branch reaches a MaxEnt Gaussian screen covariance only after the quotient-ensemble selector, quotient-normal-form scalar field, positive repair operator, operator-tilt, and fixed expected quadratic scalar-release-energy receipts pass. A primordial curvature spectrum also needs the common physical mode and radial receipts together with a source-dilation intertwiner or radial cross-covariance tomography. Physical TT/TE/EE spectra require finite covariant source, source-provenance, pooled-reducer, Boltzmann transfer, and frozen-likelihood gates. Low-$\ell$ CMB kernels, parity envelopes, inflation-replacement claims, and dark/anomaly $H_0/S_8$ growth modifications are continuation gates unless their finite-collar source functions, recipient-stress closures, refinement ladders, and immutable likelihood contracts are supplied.

**No semantic promotion by relabeling.** A finite OPH diagnostic does not become a physical observable merely because it is renamed. Capacity bookkeeping is not mass without an independent energy readout; a record archive is not radiation without a source, propagation, and detector channel; finite repair eigenvalues are not continuum normal modes without an operator and readout bridge; exact finite recovery is not a Page curve without physical radiation entropy and exterior time calibration; and a finite conditional-mutual-information score is not a geometric island, entanglement wedge, quantum extremal surface, or replica saddle. These are not rhetorical distinctions: deterministic relabeling preserves the same information ancestry, so physical promotion requires source-separated bridge objects, residual ledgers, negative controls, and frozen validation targets.

The effective low-energy summary is

$$\mathcal{L}_{\text{eff}}^{\text{OPH}} \approx \sqrt{-g}\left[\frac{1}{16\pi G}(R-2\Lambda) + \mathcal{L}_{\text{SM}}^{\text{realized branch}}\right] + \sum_i \frac{c_i}{M_*^{\Delta_i-4}}\mathcal{O}_i.$$

Here $\mathcal{L}_{\text{SM}}^{\text{realized branch}}$ denotes the realized $\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)/\mathbb{Z}_6$ sector. The higher-dimension terms absorb UV-sensitive and continuation-level corrections outside the recovered core.

### 5.5.2 Structural assessment

- **Dynamics**: the GR chain runs through geometry readout, geometric modular covariance, the null bridge, bounded-interval transport, fixed-cap generalized-entropy stationarity, fixed-volume small-ball area variation, carried-remainder control, and the tensor upgrade. The scaling branch uses the support-visible BW scaling theorem on the geometric subnet.
- **Gauge structure**: compact gauge reconstruction uses the transportable bosonic sector package on a cofinal tail carrying the compact-gauge refinement receipt. MAR plus anomaly algebra and the explicit CKM/weak-sector clauses on the same one-Higgs branch select the realized Standard Model quotient, $N_c = 3$, $N_g = 3$, and the structural electroweak force content $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y \to \mathrm{U}(1)_Q$.
- **Microscopic theory**: finite quantum-link style presentations give concrete fixed-cutoff examples. They do not select a unique microscopic representative.
- **Noncentral gluing**: the crossed-module higher-gauge package closes the fixed-cutoff topological branch. The realized zero-obstruction bosonic compact-gauge branch is carried by a separate theorem stack on the ordinary or central branch.

### 5.5.3 Known-force and charge coverage

The recovered stack does not leave any known long-range or Standard Model gauge force unassigned:

- **Gravity**: D3–D6 recover Lorentz kinematics, the three-dimensional observer-frame hyperboloid, the null-stress bridge, and the Jacobson-type Einstein branch, with $T_{ab}$ as the stress-energy source. On the additional pure Einstein–Hilbert/Minkowski action branch, Theorem 6.18 gives two classical transverse-traceless modes; a graviton quantum requires the separate particle receipt.
- **Strong interaction**: D8–D9 recover the $\mathrm{SU}(3)_c$ color factor, the color triplet $N_c = 3$, quark color triplet/antitriplet assignments, and the eight gluon generators $(8, 1, 0)$. Confinement and hadron spectra are separate infrared QCD questions, not missing gauge-charge assignments.
- **Weak interaction**: D8–D9 recover the $\mathrm{SU}(2)_L$ weak doublet structure, the one-Higgs branch, and the charged weak carriers $W^{\pm}$ from the broken $\mathrm{SU}(2)_L$ generators. D10 supplies quantitative running/matching chart coordinates for $v$ and the $W/Z$ prescription audits.
- **Hypercharge and electromagnetism**: The hypercharge theorem, the $\mathbb{Z}_6$ quotient, and Corollary 6.6c fix the $\mathrm{U}(1)_Y$ lattice, the unbroken generator $Q = T_3 + Y$, integer charge for color singlets, and the electromagnetic connection label $A_Q$. Corollary 6.6d gives $F_Q = dA_Q$, $dF_Q = 0$, and $d{*}F_Q = g_Q^2 {*}J_Q$ only after the low-energy Maxwell action is supplied. Theorem 6.18 then gives two classical massless transverse modes; a photon particle additionally requires its quantum-pole receipt. The Thomson-limit audit separates a source/root witness, a mixed-provenance diagnostic, an external-data empirical closure, and the measured endpoint. The missing source-only hadronic transport prevents promotion of that endpoint.

### 5.5.4 Testable Items and Phenomenological Branches

The detailed prediction surfaces live in the companion papers. The synthesis-level list is:

- **Structural and action-level tests**: exact color and electroweak charge assignments, exact hypercharge quantization, $N_c = 3$, $N_g = 3$, and no simple-GUT $X/Y$ gauge channel on the product-group branch; plus the classical Maxwell, perturbative pure-Yang–Mills, and pure-Einstein quadratic carrier-mode tests of Theorem 6.18. Quantum particle poles are separate receipt-gated claims.

- **Information-theoretic gravity bound**: modular-additivity defects give explicit upper bounds on GR deviations wherever the Markov/mixing hypotheses apply.
- **Edge-sector tests**: Casimir-ratio predictions, including $\Delta_8/\Delta_3 = 9/4$ for the SU(3) edge-sector benchmark, test the heat-kernel mechanism.
- **Quantitative particle checks**: Ref. [5] separates the source-audit $W/Z$ running/chart branch, the selected-carrier chart, the conditional value law, and the reference-fitted inverse adapter. The source-audit coordinates are (80.330, 91.119) GeV, and the value-law coordinates are (80.3770000154, 91.1879780779) GeV. The comparison values 80.3692(133) and 91.1880(20) GeV are PDG mass-dependent-width Breit–Wigner parameters; complex-pole masses are a different convention. The chart coordinates are not commensurate with either physical mass definition and carry no near-hit evidence. The two-loop audit returned $(79.115335, 89.802735)$ GeV under an inconsistent MSSM-1L+SM-2L hybrid prescription, formed by adding an SM two-loop increment to an MSSM one-loop baseline. The pole audit returned approximately $(79.53284, 89.71232)$ GeV under a partial PRTS/Feynman-gauge prescription. The definition of $v$, tadpole treatment, field-content and threshold matching, scale, and higher-order gates are work in progress. These audits establish neither a unique scheme conversion nor a unique 1–2% defect, and they do not exhaust the physical prescription family. The Higgs/top pair is read on the double-criticality branch ($\lambda = 0$, $\beta_\lambda = 0$ at one source scale), whose frozen boundary-scale candidate gives $(m_H, m_t) =$ (125.77, 172.63) GeV at two loops, against measured 125.13(11) and 172.60(30) GeV, with $m_H = 125.72$ GeV on the fit-free curve at the measured top; the target-anchored declared-surface fit (125.1995304097, 172.3523553288) GeV is kept separate. These values hold on their named charts and carry no promoted complex-pole receipt. The reciprocal-ray quark candidate fails on common-scale dimensionless Yukawas, and the exact generic interface has six scalar coordinates. The mixed-convention residual chart has no prediction status and is fit better by its lower-order ablation. The particle simulator has no Yukawa coupling or Yukawa information, and generation-blind flavor-singlet data do not select a physical flavor orbit. No nonzero source-only physical mass is emitted in the electroweak, charged-lepton, quark, or neutrino lanes. The symmetry-protected massless Maxwell, perturbative Yang–Mills, and Einstein kernels remain classical action statements until their quantum-pole gates pass.
- **Fine-structure endpoint**: the certified source/root witness is $136.994835177413\ldots$, the mixed-provenance no-hadron diagnostic is $137.035959513609\ldots$, and an external $e^+e^- \to$ hadrons compilation with $\Delta\alpha^{(5)}_{\mathrm{had}}(M_Z) = 0.0267591200 \pm 0.0007524955$ gives the empirical endpoint 136.2636589431 on $[136.1583832834, 136.3690975240]$. The source/root value is the unique root of an incomplete declared numerical map; no derivation identifies it with the physical fine-structure endpoint. The certified self-consistent gauge-width fixed point is $137.035660136946577\ldots$; the $137.035959513609\ldots$ diagnostic mixes the certified source root with $\alpha_U$ at the CODATA-derived comparison pixel and is not a single-map fixed point. The measured endpoint is 137.035999177(21); its same-scheme gap from the empirical interval is $[0.6485541111, 0.8547920666]$ inverse-alpha units. A source-only bridge requires the OPH hadron construction. No active blind decision threshold exists for the broad hadronic bracket.
- **Affine event-record continuation**: Ref. [5] contains a conditional event-record stitch theorem. It consumes affine event supports under the event-manifold receipts and keeps conditioned $H^3$ frame balls as separately typed metadata. It certifies observer-visible tokens crossing real chart or partition interfaces after sector/gauge transport; it does not derive particle species, masses, gauge charges, scattering amplitudes, or geodesic motion.

- **Continuation templates**: black-hole combs, PBH burst templates, dark-sector response laws, the baryogenesis source branch beyond its finite anomaly/current theorem, and critical-string lifts are not recovered-core tests. The natural hypercharge attachment of the $\mathbb{Z}_6$ gauge/deck phase has $k_R = 0$; a nonzero baryon source requires a distinct anomalous record attachment and a CP-odd quotient generator.

#### 5.5.5 Scope Boundaries

The recovered core does not derive the following:

- charged-lepton absolute masses on the available corpus, full flavor-labeled neutrino closure, CKM/PMNS closure, a source-derived flavor-orbit selector, physical quark mass rows, or a general Yukawa hierarchy;
- hadron masses and resonances, which require nonperturbative production computation;
- dark-sector response laws, baryogenesis beyond the finite anomaly/current theorem and gauge/deck no-go, strong-CP proposals, proton-spin fractions, and late-stage spectroscopy templates;
- inflation-replacement, CMB low-$\ell$/parity kernels, $H_0/S_8$ growth modifications, and cosmological dark/anomaly Boltzmann kernels beyond the contract stated in Section 5.3.16;
- a primordial curvature spectrum or physical TT/TE/EE spectra from the screen-spectrum continuation unless the operator-tilt, scalar-release-energy, finite covariant source, physical mode, radial-null, source-dilation or tomography, forward-residual, source-provenance, pooled-reducer, transfer, frozen-likelihood, and no-data-use receipts are present;
- Page-curve/island closure, physical black-hole evaporation claims, PBH/LIGO comb claims, QNM/ringdown predictions, or critical-superstring completion;
- a theorem selecting equal-sized primitive observer carriers, fixed primitive observer capacity, or an isomorphic local cell algebra from $P_\star$ and $N_{\mathrm{CRC}}$ alone;
- a unique microscopic representative or a full fermionic/super-Tannakian gauge reconstruction.

$N_\star = \log D_\star$ is the stable correctable public-record closure coordinate for the cosmological branch. Its source packet, exact finite-size selector, horizon identification, and weak/Higgs common-load carrier are separately typed obligations. $P$ is the output of an incomplete Phase-II outer/inner declared map and is not a Phase-I axiom. A contradiction in a Phase-III continuation retracts that continuation. A contradiction in D10 challenges the quantitative-closure branch. A contradiction in Phase I challenges the recovered-core claim set. If their physical identifications are supplied, $P_\star$ and $N_{\mathrm{CRC}}$ fix the canonical geometric cell area and the total equal-area cell count $K_{\mathrm{cell}} = 4N_{\mathrm{CRC}}/P_\star$ on that chart. They do not select a primitive observer algebra, a fixed observer capacity, or a unique $k$-cell carrier block.

#### 5.5.6 Comparison with other unification approaches

Unified models attempting to tie together QFT, gravity, and SM structure tend to encounter a repeatable set of conceptual difficulties. This subsection examines how the observer-patch holography framework addresses these common pitfalls.

1. **Subsystem factorization in gauge theory and gravity.**

In gauge theories and gravity, the Hilbert space does not cleanly split as "inside $\otimes$ outside" across a cut. This infects entanglement entropy definitions, area terms, edge modes, and observable identification. Many unification attempts handwave this or patch it with conventions.

*Resolution:* The framework builds from a net of von Neumann algebras on patches plus overlap consistency. It does not start from naïve tensor factorization. The gauge-as-gluing + regulator package yields edge-center completion: a canonical block decomposition on collars where the center captures superselection data at the cut, and the state becomes (exactly or approximately) Markov across the collar. The entropy split S($\rho$_C) = S_bulk + $\langle$L_C$\rangle$ follows from having a center with sector labels. This replaces the ad hoc "add an area term" move.

**2. Modular Hamiltonian nonlocality.**

Many entanglement-based gravity derivations depend on modular Hamiltonians that look like local stress-tensor charges (true only in special states/regions). In generic QFT states, modular Hamiltonians are nonlocal, making "first law of entanglement $\Rightarrow$ Einstein equation" arguments fragile.

*Resolution:* The Markov collar condition does heavy lifting: approximate Markov implies approximate modular additivity, with the defect controlled by conditional mutual information. This makes "modular locality" a controlled approximation. It is not treated as an assumption. On the extracted geometric subnet of Theorem 4.2, the controlled tangent-half-space comparison then locks modular flow to geometric dilations with rigid $2\pi$ normalization.

**3. Lorentz invariance as a derived output.**

Discrete microscopic models generally break Lorentz symmetry, and many unified proposals simply postulate Lorentz invariance in the IR.

*Resolution:* Lorentz kinematics are tied to geometric modular flow on caps. On the extracted geometric subnet, the support-visible BW scaling theorem gives cap-pair extraction, support-readable modular covariance, cap-normal BW framing, and modular flow as conformal transformations on $S^2$. The compact cap-normal theorem then identifies a sky direction with $q(\Omega) = (1, \Omega)$, represents $C(\mathbf{c}, \alpha)$ by $n_C = (\cot\alpha, \csc\alpha\, \mathbf{c})$, proves the signed incidence formula, and gives $n_{gC} = \Lambda_g n_C$. Hence $\mathrm{Conf}^+(S^2) \cong \mathrm{PSL}(2,\mathbb{C}) \cong \mathrm{SO}^+(3,1)$. The associated observer-frame chart is $H^3 \simeq \mathrm{SO}^+(3,1)/\mathrm{SO}(3)$, so the observer-facing spatial dimension on that branch is exactly three. A cap gives an $H^3$ plane/half-space, not a preferred observer point or populated bulk. No external spacetime symmetry axiom is added. The step from this frame kinematics to an actual 3+1 event spacetime is the conditional event-manifold packet (Theorem 4.3e): an event base of signature $(-{+}{+}{+})$ over which $H^3$ is strictly the frame fiber, valid exactly on the population/separation/frame/cocycle receipt branch, with countermodels showing $\dim H^3 = 3$ alone promotes nothing.

**4. Dynamics beyond kinematics.**

Many approaches produce emergent geometry/kinematics but stall at dynamics: why Einstein's equations (with the right coefficient) rather than some other geometric PDE?

*Resolution:* The framework combines the MaxEnt-selected fixed-cap generalized-entropy stationarity theorem, the derived K_C = $2\pi$B_C structure on the extracted geometric cap branch, the internal null modular bridge identifying the half-line generator with the local null-stress charge, and the internal small-ball bridge from the geometric cap generator. The E0 dependency discharge states the scaling-limit regularity, bounded-interval kernel, remainder-control, and tensor-upgrade conditions needed by the Einstein branch. It does not rely only on "assume a UV CFT."

**5. Gauge symmetry origin and compactness.**

Most unification stories pick a gauge group and work out consequences. Emergent-gauge approaches sometimes produce noncompact groups or uncontrolled redundancies.

*Resolution:* Gauge symmetry is recast as redundancy in overlap identifications (gauge-as-gluing). The visible fixed-cutoff carriers generate a rigid tensor subcategory with the ordinary forgetful fiber.

When the explicit compact-gauge refinement receipt supplies coherent surjective group pullbacks, block-algebra embeddings, and tensor realizations, the refinement-limit category is constructed and Tannaka-Krein / Doplicher-Roberts reconstruction yields a compact group G on that certified bosonic branch. "Gauge symmetry" names the gluing redundancy at the conceptual level. "Compact group" is the mathematical form compatible with finite-dimensional objectwise fibers; the finite-state refinement maps alone do not produce it.

**6. Classical carrier modes versus quantum particles.**

Writing a symmetry group or a classical Einstein equation is not enough to produce a kinetic term, a physical phase, a quantization, or a particle pole. Conversely, a displayed free quadratic pole does not by itself establish an interacting asymptotic particle.

*Resolution:* Definition 6.18 separates a classical carrier-mode receipt from a quantum- particle receipt. The stated Maxwell, pure-Yang–Mills, and pure-Einstein actions yield transverse or TT classical modes and action-level zero hard-mass parameters on their declared backgrounds and phases. Higgs/Stueckelberg completions, media, confinement, higher-derivative terms, and extra fields remain possible outside those branches; a photon, gluon, or graviton particle is claimed only when the physical Hilbert-space, positive-residue pole, and asymptotic/phase gates pass.

**7. Global consistency, anomalies, and loop patching.**

Building physics from local patches hits loop/holonomy problems: consistent gluing on a tree but obstructions around loops. These obstructions are often anomalies or global topological constraints.

*Resolution:* This is elevated to a first-class organizing principle: gluing data on overlaps define cocycles; central defects define a Čech obstruction class $[z]$, while noncentral defects define a full 2-group/crossed-module orbit together with the separate question whether it admits a strict $G$-valued 1-cocycle representative. Vanishing of $[z]$, or noncentral associator strictifiability, removes the triangle or higher associator defect; global endpoint-only consistency additionally requires trivial represented holonomy for at least one allowed strict representative. Anomalies become one form of failure to glue rather than a mysterious quantum pathology, without conflating the associator and ordinary loop obstructions.

**8. Charge quantization without a GUT.**

Without embedding into a simple GUT group, explaining charge quantization (why all isolated color singlets are integer charged) is awkward. Standard lore requires grand unification or monopoles.

*Resolution:* The framework leans on global group structure (the $\mathrm{Z}_6$ quotient) and derives congruence/selection rules for allowed representations/hypercharges. This gives a structural explanation for integer-charged color singlets without introducing the proton-decay channel of simple-GUT models.

**9. Coupling unification usually forces proton decay.**

Traditional simple-group unification introduces leptoquark gauge bosons (X, Y) mediating proton decay. Experiment keeps pushing limits up, pressuring minimal GUTs.

*Resolution:* The retained D10 discussion is geometric/entropic only at the calibration level: shared edge diffusion data, heat-kernel weights, printed running/matching/threshold/scheme conventions, and extra calibration assumptions can mimic unification-style running without embedding into a simple Lie group. If the reconstructed gauge group genuinely factorizes as a product (sector factorization selector), there are no mixed generators playing the X/Y role. "Unification-like couplings" and "group unification" therefore come apart.

**10. Cosmological constant locality.**

The cosmological constant problem is a graveyard of unified theories: local QFT estimates are enormous, and tiny observed $\Lambda$ seems to demand absurd fine tuning.

*Resolution:* From null modular data, T_ab is reconstructed only up to $\phi$g_ab. Local consistency conditions and null focusing are blind to vacuum-energy shifts, so the Einstein equation is fixed only up to Λg_ab. The dimensionless Λ-capacity relation is global, tied to the static-patch capacity $\log \dim H_{\rm tot}$, while the SI curvature value also uses the selected scale certificate. This resolves the conceptual tension: local microphysics *cannot* fix Λ by structural information-theoretic reasons.

**11. UV infinities and nonrenormalizability.**

Unified programs struggle to give sharp, finite microscopic definitions. Formal continuum structures, infinite entropies, and regularization dependence abound.

*Resolution:* The regulator construction uses local patch algebras that are type-I and finite-dimensional, with a MaxEnt branch whose generator is quasi-local and obeys a Lieb-Robinson bound. So the fixed-cutoff UV branch is interacting in the ordinary finite-range sense, and the fundamental degrees of freedom are finite and live on the screen. The framework does *not* claim a unique microscopic UV completion: physical uniqueness is only modulo gauge or implementation hiding together with inert ancillary stabilization. The genuinely noncentral topological branch is also closed at fixed cutoff by the higher-gauge crossed-module collar theorem, while the support-visible BW scaling branch is closed by theorem and the realized zero-obstruction bosonic compact-gauge branch is carried by its declared theorem stack.

**12. Predictivity vs. parameter explosion.**

Unified models often explode in parameters, sectors, or vacua, becoming hard to test directly because everything depends on choices.

*Resolution:* The framework compresses freedom into a "pixel area" (resolution) parameter and a total Hilbert space capacity (size) parameter, then derives structure from consistency (Lorentz form, the conditional Einstein branch, receipt-certified compact-group reconstruction, charge-quantization patterns, and action-level carrier modes only on explicitly selected phases). The explicit MAR selector then picks the SM factors and sector factorization on the admissible class discussed in Section 6.2. Quantum-particle poles remain a separate receipt rather than an exact-zero output of symmetry.

**Structural pattern.** The framework treats locality, Lorentz invariance, gauge symmetry, and gravity as consequences of consistency conditions among overlapping descriptions together with information-theoretic properties of states. Modular rigidity then supplies the familiar symmetry and dynamical structures.

**Engineering deliverables.** Certain problems can be stated as explicit closure tasks:

- Λ's dimensionless capacity relation is structurally explained as a global capacity parameter; the input-free prediction is the self-closure fixed-point statement plus the selected OPH scale certificate for SI display
- A full microphysical derivation of geometric modular action is required

These are shared challenges across unification approaches. The framework provides an explicit map of where they live and what would resolve them.

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# 6 Consensus, Defects, and Implementation Hiding

The fixed-cutoff consensus package admits an equivalent finite patch-net formulation. Local patch descriptions agree on overlaps, local recovery-derived repair rules remove mismatches, and the physical output is the schedule-independent quotient normal form together with the induced terminal expectation functionals on the declared physical observable algebras. This section records the fixed-point, defect, and gauge-quotient statements in that language.

The consensus lane also carries a simple finite-candidate law-selection model: once one equips candidate repair rules with a fitness functional, replicator dynamics gives a clean toy picture in which more successful reconciliation laws dominate a competing pool. That selection picture is useful for the overall synthesis picture, but in the consensus paper it is a finite-candidate monotonicity result, not part of the recovered-core gravity/gauge theorem surface, not a universality claim, and not a literal cosmological dynamics claim.

On its declared fixed-cutoff branch, *Reality as a Consensus Protocol: The Fixed-Point Computation That Implements Physics* [3] imports four core D1 surfaces into this synthesis surface:

1. **Constraint-code firewall.** A bare finite overlap net is a finite constraint code: its codewords are exactly the globally consistent states $C = \Phi^{-1}(0)$. It is not automatically a QECC/topological code, and its graph min-cut does not determine code distance. Distance/min-cut, Knill–Laflamme resilience, spectral convergence, BFT liveness, and hardware speedup enter only through separate certificates.

2. **Asynchronous confluence.** For the declared accepted repair law, the local-fit contract makes $\Phi$ a Lyapunov functional, hence gives termination on the finite patch net. The fixed-cutoff union-collar gluing package supplies the local diamond on the physical quotient, and only that confluence condition together with repair completeness yields a unique schedule-independent normal form from a fixed initial quotient state. Same-boundary uniqueness requires the additional unique consistent extension condition in the preserved boundary/sector fiber.

3. **Cycle obstruction and higher-gauge defects.** On the abelian branch, global consistency holds exactly when cycle holonomy vanishes; on the genuinely noncentral branch, the crossed-module gluing orbit
$$q_\Sigma \in \check{H}^2(N_\Sigma, H_\Sigma \to G_\Sigma)$$
labels the full fixed-cutoff gluing orbit. It can admit multiple strict representatives related by $H_\Sigma$-valued edge changes and need not determine one ordinary $G_\Sigma$-valued 1-cocycle class. Associator strictifiability is the separate condition that this orbit admit a representative $(g^{\mathrm{str}}, 1)$; strict endpoint-only transport additionally requires at least one allowed strict representative with trivial represented loop holonomy.

4. **Gauge quotient and observable-level confluence.** The repair law descends to the overlap-invariant quotient, and the induced terminal state on the declared physical observable algebras is unique there even when microscopic representatives differ by gauge or sector relabelings inside one quotient-local glued state.

5. **Record algebra and stability.** On the declared fixed-cutoff observer-accessible operator surface, central record projectors carry the Born/Lüders rules directly, while approximate record projectors inherit explicit $(\varepsilon, \delta_{\mathrm{rec}})$ stability bounds on that same event surface. This is a theorem about the stated algebraic record interface, not a requirement that OPH first rebuild every mathematical ingredient from operational records alone.

The consensus paper is equally explicit about the imported repair-law data and what sits outside the fixed-cutoff theorem stack. The declared repair step includes the touched-overlap local-fit contract and the union-collar gluing package on the physical quotient; the branch conditions above that step are repair completeness and, on the Petz branch, the stated support/CPTP clause. On that same fixed-cutoff surface, exact normal-form computation is finite-state and decidable, automatic approximate stability is only collar-local through the splice and record estimates, and long-run noisy approximate consensus is theorem-grade only after a fair-block contraction certificate is supplied for the chosen exported patch-net family. That certificate boundary is not a dependency for the support-visible BW theorem, local Einstein branch, receipt-conditional compact-gauge reconstruction, or realized Standard Model branch.

# 7 Particle-Spectrum Branch

The particle branch follows the same logical order as *Deriving the Particle Zoo from Observer Consistency* [5]. One first fixes the compact-gauge classification basis, applies Minimal Admissible Realization (MAR) to the realized one-Higgs branch to identify the Standard Model quotient, and only then asks how transport-stable overlap data organize the observed particle families.

**Structural carrier and particle-gate package.** On the realized compact-gauge branch, the structural carrier roles are fixed by the compact gauge and MAR-admissibility chain:

$$\frac{\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)}{\mathbb{Z}_6},\qquad N_c=3,\qquad N_g=3.$$

On a declared trace-balanced block carrier $V=C\oplus W$, with $\dim C=3$, $\dim W=2$, and block hypercharges $(-1/3,1/2)$, the selected exterior package

$$\Lambda^2V\oplus\Lambda^4V=Q\oplus u^c\oplus e^c\oplus d^c\oplus L$$

is an exact one-generation Standard Model representation witness. It has the three one-Higgs invariant lines, cancels the gauge and mixed anomalies, and contains four weak doublets per generation. Hence every additive isomorphism-invariant load normalized by $L_P(\mathbb{C})=P$ gives $L_P(\mathrm{Hom}_{SU(2)}(W,M_1))=4P$. A positive unital map between one-dimensional order-unit load lines is unique once the physical order units are identified. The witness does not select its block carrier from screen currents, select the non-vacuum exterior package or $H=W$, remove the omitted $\Lambda^0V$ singlet and other light sectors, or attach the $A_5$ face module to physical families. Those are explicit physical current, determinant, spin-lift, deck-descent, matter selection, no-extra-sector, and family-attachment gates.

**Theorem 7.1** (Conditional icosahedral-screen forcing theorem)**.** *Assume that Euler's twelve-unit-defect rule and the source-derived icosahedral orbit selector produce the twelve-port orbit; that the port-current inner-action, determinant-balance, spin-lift, and axis/center deck-descent receipts produce the trace-balanced $3+2$ block carrier with hypercharges $(-1/3,1/2)$; that minimal admissibility selects $M_1=\Lambda^2V\oplus\Lambda^4V$, identifies $H=W$, and excludes extra light sectors; and that the $A_5$ multiplicity is physically attached to three families with the selecting symmetry hidden, broken, or forgotten on the Yukawa surface. On a receipt-certified compact-gauge continuum/QFT landing, the resulting low-energy chiral package is the Standard Model quotient*

$$\frac{\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)}{\mathbb{Z}_6}$$

*with the exact Standard Model hypercharge lattice, one Higgs doublet, three colors, three attached generations, perturbative anomaly cancellation, even Witten parity, and no simple-GUT $X/Y$ gauge channel.*

*Proof.* The screen receipts produce the declared carrier and deck data. Exterior branching gives the five chiral multiplets and the three one-Higgs invariant lines. The anomaly and center calculations fix the hypercharge lattice and $\mathbb{Z}_6$ quotient. The compact-gauge/MAR theorem fixes the product group and color triplet, while the stated sector and family receipts remove the otherwise available singlet, extra-sector, and family-attachment ambiguities. The continuum/QFT landing is the final physical promotion gate. □

This theorem closes the logical implication, not its premise production. Each named screen-to-current, carrier, sector, family, and continuum receipt remains an independently removable condition and is tracked in the proof ledger.

This does not determine quantum-particle masses. The compact paper separately proves that explicit Maxwell, perturbative pure-Yang–Mills, and pure-Einstein quadratic actions have transverse or TT classical massless modes on their stated backgrounds and phases. Photon, gluon, or graviton particle language additionally requires a positive-energy physical quantization, positive-residue two-point pole, and the appropriate asymptotic/deconfinement receipt.

**Proof route.** The proof route is the one developed in *Recovering Relativity and the Standard Model from Observer Overlap Consistency* [2] and *Deriving the Particle Zoo from Observer Consistency* [5]. Under the explicit compact-gauge refinement receipt, coherent bosonic pullback ladder, symmetry, and forgetful-fiber conditions of the compact paper, the surviving transportable edge-sector category reconstructs a compact internal symmetry group. MAR then selects the realized admissible branch, anomaly freedom fixes the hypercharge lattice, the minimal coupled carrier fixes $N_c = 3$, and the same one-Higgs quark branch fixes $N_g = 3$. The particle branch builds on that structural skeleton and does not bypass it.

| Sector | Claim tier | What is fixed here | Boundary to stronger claim |
|---|---|---|---|
| Carrier roles and classical modes | structural group/content theorem plus conditional quadratic-action theorem | realized quotient, $N_c = 3$, $N_g = 3$; two Maxwell transverse modes, $2\dim G$ perturbative Yang–Mills modes, and two Einstein TT modes on stated branches | these are symmetry-protected massless classical kernels. A quantum zero-mass particle row requires the physical Hilbert-space, pole-residue, and phase gates |
| Electroweak bosons | exact selected-carrier chart + conditional quotient-transport implication + measured-reference inverse adapter | no nonzero source-only physical mass emitted | the selected chart, conditional value law, and inverse adapter have different provenance. The finite-carrier certificate is not emitted, the adapter is a separate reference-fitted comparison map, and no physical complex-pole pair is promoted |
| Fine-structure endpoint | source witness + mixed diagnostic + empirical closure + measurement | four distinct coordinates, detailed in the endpoint audit | the external-data empirical interval misses the measured endpoint. A source-only theorem requires a no-target-leak source-derived hadronic spectral payload |
| Higgs/top stage | conditional downstream split theorem on the declared running, matching, and threshold surface + compare-only exact sidecar | no nonzero source-only physical mass emitted | promotion inherits the source-root, physical-scale, transport, running, matching, rigidity, provenance, uncertainty, and complex-pole gates. The auxiliary direct-top codomain is comparison-only |

| Sector | Claim tier | What is fixed here | Boundary to stronger claim |
|---|---|---|---|
| Quark family | common-scale reciprocal-ray falsification + exact generic interface | no nonzero source-only physical mass emitted | reciprocal-ray closure fails across the audited common scales. Its sub-percent residuals use a target-anchored mixed-convention chart, and a lower-order ablation fits that chart better. The simulator has no Yukawa coupling or Yukawa information. Generation-blind flavor-singlet inputs produce no physical flavor-orbit selector |
| Charged leptons | continuation branch + exact witness + face-carrier theorem | exact $A_5/C_3$ face orbit; conditional contraction theorem; engineered finite model of the declared charged face-quotient schema | no nonzero source-only physical mass is emitted. The bounded model supplies local register state, event readback, and a central accepted/rejected record, closing fixed-cutoff schema existence. Its dimensions, automaton, grading, clock, and response are authored inputs; no frozen receipt excludes target dependence. A separate nature/pole bridge assumes both the physical Yukawa response and singularity readout. Physical source selection, family/Yukawa attachment, interacting kernel, cofinal refinement, and pole-scheme gates are open |
| Neutrinos | rejected target-informed weighted-cycle candidate + diagnostic sidecars | no nonzero source-only physical mass emitted; source-level PMNS, physical ordering, Majorana phases, and absolute masses are unformed | the candidate fails the NuFIT 6.1 correlated profile; its family kernel is a template, its charged basis is open and nearly degenerate, and its shared-basis recovery is tautological |
| Hadrons | backend-gated nonperturbative continuation | stable-channel and readout architecture are defined | source-only masses require one production backend export bundle, then executed unquenching, runtime receipt, and production systematics |

**Claim-tier boundary.** Structural carrier roles and the realized Standard Model branch are theorem-bearing outputs; classical propagation is conditional on the displayed action/background/phase receipts, and quantum-particle interpretation is a stronger open gate. The electroweak quantitative branch separates an exact selected-carrier chart, a conditional value law, and a reference-fitted inverse adapter. The quotient-transport assumptions imply the value law exactly, but the finite carrier does not emit their certificate. The color-balanced $(\sqrt{N_c}/2, N_c)$ amplitude/loop construction defines a distinct alternative model rather than the complete transport law. The hierarchy theorem fixes the dimensionless ratio $v/E_\star$; an independently physical $E_\star$ and a pole receipt govern any mass in GeV. Full $W/Z$ promotion also requires a strict source root, a frozen RG/matching/scheme packet, branch rigidity, target-independent precommitment, and uncertainty control. The Higgs/top stage is conditional on the declared downstream surface and inherits the same source-root, transport, scale, scheme, rigidity, provenance, uncertainty, and complex-pole gates. Measured electroweak quantities are comparison data. The executed empirical hadron closure integrates external spectral input and misses the measured Thomson endpoint by a certified same-scheme interval. That payload tests the map; it does not source-close the Ward-projected hadronic spectral transport needed for a source-only theorem.

**The $\alpha_U(P)$ source proof record.** The electroweak hierarchy used by the clock-scale branch depends first on a source readout of the unified diffusion coupling

$$\alpha_U(P_\star).$$

For a trial pixel $P$, define

$$M_U(P) = E_\star e^{-2\pi} P^{1/6}, \qquad E_{\mathrm{cell}}(P) = \frac{E_\star}{\sqrt{P}}.$$

On the realized color branch,

$$N_c = 3, \qquad \beta_{\mathrm{EW}} := N_c + 1 = 4,$$

and the transmutation scale for a candidate coupling $a$ is

$$v(P,a) = E_{\mathrm{cell}}(P) \exp\left[-\frac{2\pi}{\beta_{\mathrm{EW}} a}\right].$$

The exterior representation theorem fixes the weak-doublet multiplicity $3+1=4$, which also passes Witten parity on the realized branch. Its use as the D10 transmutation coefficient $\beta_{\mathrm{EW}} = 4$ requires the common screen/electroweak load-carrier identification and remains declared. The SU(2) one-loop coefficient of the realized one-Higgs branch is 19/6 and the declared unification packet's coefficient is 1, and neither equals 4. The integer is exact structural mathematics; the physical load attachment is the open receipt. It is not derived by dividing the 24-slot register by six. The source running family is

$$\alpha_i^{-1}(\mu; P, a) = a^{-1} + \frac{b_i}{2\pi} \log\left(\frac{M_U(P)}{\mu}\right), \qquad i = 1,2,3.$$

The beta-coefficient packet $(b_1, b_2, b_3)$, matching convention, threshold packet, and renormalization convention are part of the source proof record. They are declared before any comparison with measured electroweak or gravitational data.

The hypercharge coupling is

$$\alpha_Y(\mu; P, a) = \frac{3}{5}\alpha_1(\mu; P, a).$$

The source $Z$-scale is a fixed point of the source equations, not an inserted measured mass:

$$\mu_Z(P,a) = \frac{v(P,a)}{2}\sqrt{4\pi\alpha_2(\mu_Z; P, a) + 4\pi\alpha_Y(\mu_Z; P, a)}.$$

For $G = \mathrm{SU}(2), \mathrm{SU}(3)$, define the finite representation heat-kernel sums

$$Z_G(t) = \sum_R d_R e^{-tC_2(R)}, \qquad \bar{\ell}_G(t) = \frac{1}{Z_G(t)} \sum_R d_R e^{-tC_2(R)} \log d_R.$$

The source heat-kernel parameters are

$$t_2(P,a) = 4\pi^2 \alpha_2(\mu_Z(P,a); P, a), \qquad t_3(P,a) = 4\pi^2 \alpha_3(\mu_Z(P,a); P, a).$$

The unified-coupling residual is

$$\Phi_U(P,a) := \bar{\ell}_{\mathrm{SU}(2)}(t_2(P,a)) + \bar{\ell}_{\mathrm{SU}(3)}(t_3(P,a)) - \frac{P}{4}.$$

The source value is the zero

$$\Phi_U(P, \alpha_U(P)) = 0.$$

**Definition 7.2** (No-hidden-free-variable admissibility)**.** *The unified-coupling proof record is admissible only when the representation cutoffs, beta packet, matching prescription, threshold packet, and renormalization convention are frozen before comparison with measured electroweak or gravitational data:*

$$N_2, \quad N_3, \quad (b_1, b_2, b_3), \quad \textit{matching prescription}, \quad \textit{threshold packet}, \quad \textit{renormalization convention.}$$

*None of these may be selected by minimizing the error against*

$$M_W, \quad M_Z, \quad \alpha_s(M_Z), \quad \sin^2\theta_W, \quad v_{\text{exp}}, \quad G_{\text{exp}}.$$

*Tuning any of these objects after comparison with observed electroweak or gravitational data makes the hierarchy row a calibration.*

**Theorem 7.3** (No-measured-input unified-coupling proof record)**.** *Let*

$$\mathcal{R}_U = (P_\star, N_2, N_3, I_U, \Phi_U, K_U, a_U^{(0)}, r_U, m_U, \mathrm{DAG}_U)$$

*be an admissible unified-coupling proof record. Here $N_2, N_3$ are the finite representation cutoffs for* $\mathrm{SU}(2)$ *and* $\mathrm{SU}(3)$*, $I_U \subset \mathbb{R}_+$ is the declared physical interval for $\alpha_U$, $\Phi_U$ is the heat-kernel closure residual, $K_U$ is an interval-Newton or Krawczyk operator, $a_U^{(0)}$ is the displayed candidate, $r_U$ is the residual bound, $m_U$ is a lower derivative bound, and* $\mathrm{DAG}_U$ *is the dependency graph.*

*Assume*

$$K_U(I_U) \subset \operatorname{int}(I_U),$$
$$|\Phi_U(P_\star, a_U^{(0)})| \le r_U,$$
$$|\partial_a \Phi_U(P_\star, a)| \ge m_U > 0 \qquad (a \in I_U),$$

*and assume that* $\mathrm{DAG}_U$ *contains no directed path from measured electroweak masses, measured low-energy gauge couplings, measured $G$, Planck area, Planck mass, measured $\Lambda$, or any equivalent gravity-calibrated or electroweak-calibrated scale. Then there is a unique source value*

$$\alpha_U(P_\star) \in I_U$$

*such that*

$$\Phi_U(P_\star, \alpha_U(P_\star)) = 0,$$

*and*

$$|\alpha_U(P_\star) - a_U^{(0)}| \le \frac{r_U}{m_U}.$$

*Consequently, the electroweak hierarchy*

$$\frac{v}{E_\star} = P_\star^{-1/2} \exp\left[-\frac{2\pi}{(N_c+1)\alpha_U(P_\star)}\right]$$

*is no-measured-input on this proof record.*

*Proof.* The residual $\Phi_U(P_\star, a)$ is built from the source-side pixel $P_\star$, declared representation cutoffs, source transmutation law, source running law, and source $Z$-scale fixed-point equation. The dependency condition excludes measured electroweak data and all gravity-calibrated scales. The interval-Newton/Krawczyk inclusion proves existence and uniqueness of a zero of $\Phi_U(P_\star, a)$ inside $I_U$. The residual estimate and derivative lower bound give

$$|\alpha_U(P_\star) - a_U^{(0)}| \le \frac{|\Phi_U(P_\star, a_U^{(0)})|}{m_U} \le \frac{r_U}{m_U}.$$

Substitution into the declared transmutation law gives the displayed hierarchy. The dependency graph contains no measured $G$, Planck unit, measured $M_Z$, measured $M_W$, measured $\alpha_s(M_Z)$, or measured low-energy electroweak coupling ancestor, so the emitted hierarchy is source-side on $\mathcal{R}_U$. □

The theorem fixes a dimensionless ratio. It supplies neither an independently physical $E_\star$ nor a mass in GeV. On the populated branch $P_\star$ is the unique root of the declared local self-read map, so the downstream source expressions inherit that root. Their physical endpoint interpretation requires the corresponding transport and pole constructions.

**Lemma 7.4** (Exponential sensitivity of the hierarchy row)**.** *On the transmutation branch*

$$\frac{v}{E_\star} = P_\star^{-1/2} \exp\left[-\frac{2\pi}{4\alpha_U}\right],$$

*and on a clock row whose dominant scale dependence enters through $m_e^2 \propto v^2$ in the leading hyperfine scaling $\varepsilon_{\mathrm{Cs}} \sim \mathcal{C}_{\mathrm{Cs}}\, \alpha_\star^4\, m_e^2/(m_p E_\star)$, the induced gravity readout satisfies approximately*

$$\frac{\partial \ln G}{\partial \alpha_U} = \frac{2\pi}{\alpha_U^2}.$$

*At $\alpha_U \simeq 0.0411$,*

$$\frac{\partial \ln G}{\partial \alpha_U} \simeq 3.7 \times 10^3.$$

*Therefore a certified relative gravity precision $\eta_G$ requires*

$$|\delta\alpha_U| \lesssim \frac{\eta_G}{3.7 \times 10^3}.$$

*Proof.* For $\beta_{\mathrm{EW}} = 4$,

$$\ln(v/E_\star) = -\frac{1}{2}\ln P_\star - \frac{2\pi}{4\alpha_U}.$$

Thus

$$\frac{\partial \ln(v/E_\star)}{\partial \alpha_U} = \frac{2\pi}{4\alpha_U^2} = \frac{\pi}{2\alpha_U^2}.$$

The clock gap carries the square of the electroweak scale: in the leading hyperfine scaling $\varepsilon_{\mathrm{Cs}} \propto m_e^2 \propto v^2$ at fixed Yukawa, fine-structure, and strong-sector ratios, so

$$\frac{\partial \ln \varepsilon_{\mathrm{Cs}}}{\partial \alpha_U} = 2 \cdot \frac{\pi}{2\alpha_U^2} = \frac{\pi}{\alpha_U^2}.$$

Since the gravity row is quadratic in the clock scale,

$$G \propto \varepsilon_{\mathrm{Cs}}^2,$$

one obtains

$$\frac{\partial \ln G}{\partial \alpha_U} = 2 \cdot \frac{\pi}{\alpha_U^2} = \frac{2\pi}{\alpha_U^2}.$$

□

**Remark 7.5** (Digits policy)**.** *The exponential branch allows precision only to the extent that the source proof records carry interval bounds. The number of printed digits in $G_{\rm SI}$ may not exceed the precision certified by $\mathcal{R}_U$, $\mathcal{R}_{\rm QCD}$, $\mathcal{R}_{\rm flav}$, and $\mathcal{R}_{\rm Cs}$. A display such as*

$$\varepsilon_{\rm Cs} = 2\pi\nu_{\rm Cs}\sqrt{\hbar G_{\rm ref}/c^5}$$

*is a calibration checksum unless the full source stack emits the same interval with a no-G dependency graph.*

**No-$G$ clock hierarchy for the gravity scale.** The numerical gravity row uses the dimensionless cesium clock gap

$$\varepsilon_{\rm Cs} := \frac{\hbar\omega_{\rm Cs}}{E_\star}, \qquad E_\star := \frac{\hbar c}{\ell_\star}, \qquad \omega_{\rm Cs} = 2\pi\nu_{\rm Cs}.$$

The size of this gap is a hierarchy problem for the OPH scale branch. The gap is represented as a factorized source-side readout. The source-only scale branch is accepted as no-$G$ only when the gap factorizes through source-side dimensionless physics:

$$\mathcal{R}_\gamma = \mathcal{R}_U + \mathcal{R}_\alpha + \mathcal{R}_e^{\rm abs} + \mathcal{R}^{\rm 133Cs}_{\rm QCD/nuc} + \mathcal{R}^{\rm 133Cs}_{\rm atom}.$$

Here $\mathcal{R}_U$ emits $\alpha_U(P_\star)$ and the electroweak scale, $\mathcal{R}_\alpha$ emits the electromagnetic coupling used by the atomic Hamiltonian, $\mathcal{R}_e^{\rm abs}$ emits the electron absolute mass ratio, $\mathcal{R}^{\rm 133Cs}_{\rm QCD/nuc}$ emits the cesium nuclear source packet, and $\mathcal{R}^{\rm 133Cs}_{\rm atom}$ emits the hyperfine spectral gap from those dimensionless data.

At leading OPH source level, the electroweak hierarchy is generated by the declared transmutation law

$$\frac{v}{E_\star} = P_\star^{-1/2}\exp\left[-\frac{2\pi}{(N_c+1)\alpha_U(P_\star)}\right], \qquad N_c = 3.$$

Thus the weak-to-UV ratio is generated from the dimensionless source data $P_\star$, $N_c$, and $\alpha_U(P_\star)$. The cesium gap then has the schematic source form

$$\varepsilon_{\rm Cs} = \mathcal{H}_{\rm Cs}\left(\alpha_\star, \frac{v}{E_\star}, \frac{\Lambda_{\rm QCD}}{E_\star}, y_e, y_q, \text{nuclear data}, \text{atomic corrections}\right),$$

where all arguments are dimensionless source outputs on the declared branch. In leading hyperfine scaling this contains the familiar structure

$$\varepsilon_{\rm Cs} \sim \mathcal{C}_{\rm Cs}\,\alpha_\star^4\frac{m_e^2}{m_p E_\star},$$

with relativistic, many-body, recoil, QED, finite-nuclear-size, and nuclear-moment corrections absorbed into $\mathcal{C}_{\rm Cs}$.

**Definition 7.6** (Cesium source packet)**.** *A source-only cesium packet is*

$$\mathcal{D}^{\rm src}_{\rm Cs} = \left(\alpha_\star, \frac{m_e}{E_\star}, \frac{\Lambda_{\rm QCD}}{E_\star}, \{y_q\}, \mathcal{N}_{133}, \mathcal{B}^{\rm Cs}_{\rm atom}\right).$$

$\mathcal{N}_{133}$ *contains the source-side nuclear data for* $^{133}$Cs, *including nuclear spin, magnetic moment, magnetization distribution, charge radius, recoil data, and finite-size data.* $\mathcal{B}^{\rm Cs}_{\rm atom}$ *contains the atomic many-body, relativistic, QED, recoil, and finite-nucleus correction package. Every object in $\mathcal{D}^{\rm src}_{\rm Cs}$ is dimensionless and has no dependency path from measured $G$, Planck area, Planck mass, Planck time, measured $\Lambda$, or an algebraically equivalent gravity-calibrated scale.*

**Definition 7.7** (OPH cesium Hamiltonian)**.** *Given $\mathcal{D}_{\mathrm{Cs}}^{\mathrm{src}}$, define the dimensionless cesium Hamiltonian*

$$\widehat{H}_{\mathrm{Cs}}^{\mathrm{OPH}} = \widehat{H}_{\mathrm{DC}} + \widehat{H}_{\mathrm{Breit}} + \widehat{H}_{\mathrm{QED}} + \widehat{H}_{\mathrm{recoil}} + \widehat{H}_{\mathrm{nuc}} + \widehat{H}_{\mathrm{manybody}}.$$

*The hats indicate that every term is measured in units of $E_\star$, not in SI units and not in Planck units inferred from measured $G$.*

**Theorem 7.8** (No-$G$ OPH clock-hierarchy theorem)**.** *Assume the local pixel proof selects $P_\star$. Assume the unified-coupling proof record $\mathcal{R}_U$ of Theorem 7.3 emits $\alpha_U(P_\star)$ and*

$$\frac{v}{E_\star} = P_\star^{-1/2} \exp\left[-\frac{2\pi}{(N_c+1)\alpha_U(P_\star)}\right].$$

*Assume the electromagnetic endpoint branch $\mathcal{R}_\alpha$ emits $\alpha_\star$, the charged-lepton absolute-scale branch $\mathcal{R}_e^{\mathrm{abs}}$ emits*

$$\frac{m_e}{E_\star} = \frac{y_e}{\sqrt{2}} \frac{v}{E_\star},$$

*the QCD/nuclear branch $\mathcal{R}_{\mathrm{QCD/nuc}}^{133\mathrm{Cs}}$ emits $\mathcal{N}_{133}$, and the atomic branch $\mathcal{R}_{\mathrm{atom}}^{133\mathrm{Cs}}$ supplies isolated spectral enclosures*

$$J_{F=3}, \qquad J_{F=4}$$

*for the source-only Hamiltonian*

$$\widehat{H}_{\mathrm{Cs}}^{\mathrm{OPH}} = \widehat{H}_{\mathrm{DC}} + \widehat{H}_{\mathrm{Breit}} + \widehat{H}_{\mathrm{QED}} + \widehat{H}_{\mathrm{recoil}} + \widehat{H}_{\mathrm{nuc}} + \widehat{H}_{\mathrm{manybody}}$$

*with disjoint intervals and certified hyperfine gap*

$$\varepsilon_{\mathrm{Cs}} = \lambda_{F=4}(\widehat{H}_{\mathrm{Cs}}^{\mathrm{OPH}}) - \lambda_{F=3}(\widehat{H}_{\mathrm{Cs}}^{\mathrm{OPH}}).$$

*Assume further that the combined dependency graph*

$$\mathrm{DAG}_{\mathrm{Cs}} = \mathrm{DAG}\left(\mathcal{R}_U, \mathcal{R}_\alpha, \mathcal{R}_e^{\mathrm{abs}}, \mathcal{R}_{\mathrm{QCD/nuc}}^{133\mathrm{Cs}}, \mathcal{R}_{\mathrm{atom}}^{133\mathrm{Cs}}\right)$$

*contains no directed path from*

$$G_{\mathrm{exp}}, \qquad \ell_P^2 = \frac{\hbar G}{c^3}, \qquad m_P = \sqrt{\frac{\hbar c}{G}}, \qquad t_P = \sqrt{\frac{\hbar G}{c^5}}, \qquad \Lambda_{\mathrm{exp}}, \qquad \frac{3\pi c^3}{\hbar G},$$

*or from any algebraically equivalent gravity-calibrated scale. Then $\varepsilon_{\mathrm{Cs}}$ is a no-$G$ hierarchy readout. Consequently*

$$\gamma_\star = \frac{\ell_\star \nu_{\mathrm{Cs}}}{c} = \frac{\varepsilon_{\mathrm{Cs}}}{2\pi}$$

*is non-circular, and*

$$G_{\mathrm{SI}} = \frac{c^5}{4\pi^2 \hbar \nu_{\mathrm{Cs}}^2} \varepsilon_{\mathrm{Cs}}^2$$

*has no measured-$G$ ancestor.*

*Proof.* $\mathcal{R}_U$ emits the small weak ratio from $P_\star$, $N_c$, and $\alpha_U(P_\star)$, all of which are dimensionless branch data. $\mathcal{R}_\alpha$ emits the electromagnetic coupling used by the atomic Hamiltonian. $\mathcal{R}_e^{\mathrm{abs}}$ emits the electron mass ratio. $\mathcal{R}^{133\mathrm{Cs}}_{\mathrm{QCD/nuc}}$ emits the cesium nuclear source packet. Therefore every coefficient of $\widehat{H}^{\mathrm{OPH}}_{\mathrm{Cs}}$ has no measured-$G$ ancestor. The atomic branch gives isolated spectral enclosures for the two hyperfine levels, so the gap

$$\varepsilon_{\mathrm{Cs}} = \lambda_{F=4}(\widehat{H}^{\mathrm{OPH}}_{\mathrm{Cs}}) - \lambda_{F=3}(\widehat{H}^{\mathrm{OPH}}_{\mathrm{Cs}})$$

is well-defined and inherits the no-$G$ dependency condition of the Hamiltonian.

Finally,

$$\varepsilon_{\mathrm{Cs}} = \frac{\hbar(2\pi\nu_{\mathrm{Cs}})}{\hbar c/\ell_\star} = \frac{2\pi\nu_{\mathrm{Cs}}\ell_\star}{c},$$

so

$$\gamma_\star = \frac{\ell_\star\nu_{\mathrm{Cs}}}{c} = \frac{\varepsilon_{\mathrm{Cs}}}{2\pi}.$$

Substitution into $G_{\mathrm{SI}} = c^3\ell_\star^2/\hbar$ gives the displayed expression for $G_{\mathrm{SI}}$. The only non-display parent of the expression is the source-side $\varepsilon_{\mathrm{Cs}}$ readout, so the gravity row is non-circular under the stated hypotheses. □

**Remark 7.9** (Clock-certificate boundary)**.** *The theorem above is not discharged by printing*

$$\varepsilon_{\mathrm{Cs}} = 3.1139305134\ldots \times 10^{-33}.$$

*That decimal is source-predictive only if it is emitted by*

$$\mathcal{R}_U + \mathcal{R}_\alpha + \mathcal{R}_e^{\mathrm{abs}} + \mathcal{R}^{133\mathrm{Cs}}_{\mathrm{QCD/nuc}} + \mathcal{R}^{133\mathrm{Cs}}_{\mathrm{atom}}.$$

*If instead it is computed as*

$$\varepsilon^{\mathrm{cal}}_{\mathrm{Cs}} = 2\pi\nu_{\mathrm{Cs}}\sqrt{\frac{\hbar G_{\mathrm{exp}}}{c^5}},$$

*then it is a gravity-side calibration checksum, not a source-side clock prediction. The paper surface supplies the algebraic certificate shape and the $\mathcal{R}_U$ numerical witness. The full source-only $\mathcal{R}_{\mathrm{Cs}}$ branch requires the source-only electromagnetic endpoint, charged-lepton absolute-scale gate, source-side cesium nuclear packet, and atomic spectral enclosure.*

**Remark 7.10** (Hadronic and nuclear certificate boundary)**.** *The unresolved load-bearing pieces of the no-G SI gravity row are the hadronic/same-scheme endpoint payload in $\mathcal{R}_\alpha$ and the source-side cesium QCD/nuclear packet in $\mathcal{R}^{133\mathrm{Cs}}_{\mathrm{QCD/nuc}}$. The displayed endpoint and $\varepsilon_{\mathrm{Cs}}$ rows therefore fix those payloads for checksum and comparison purposes. They are not counted as source-only OPH emissions.*

*The source-only completion route is a Ward-projected QCD/nuclear spectral certificate from a dedicated OPH optical-compute backend, or an equivalent nonperturbative backend with manifest provenance and systematics. Ordinary CPU/GPU runs in this paper are not the declared source certificate for that hadronic payload.*

**Direct public-record closure for total capacity.** The global counterpart of the local $P$-closure begins with a supplied positive integer capacity-carrier dimension

$$D = \dim \mathcal{H}_{\mathrm{cap},r,D}, \qquad N = \log D.$$

The official universe-level readback equation is

$$\boxed{N = \log M_0(\mathfrak{U}_N)}.$$

Here the argument fixes the type. In every occurrence, $M_0$ is a multiplicative code size. Once the complete terminal fiber scalarizes, the universe-level notation means

$$M_0(\mathfrak{U}_N) := \widehat{F}_{r,0}(e^N).$$

Thus $N$ is the logarithmic capacity read back from the trial universe, while $M_0(\mathfrak{U}_N)$ is the number of correctable public records. The equation is the logarithmic form of stable direct capacity closure, not a new producer or an appeal to a measured target. The carrier type is frozen in the branch contract. Boundary, total, edge-center, and other Hilbert spaces are not interchangeable. Let $\widetilde{\Omega}_{r,D}$ be the terminal physical quotient fiber reached from the source-derived trial universe $\mathfrak{U}_{r,D}$. Its members are repair-normal and pass the declared local, obstruction, and provenance gates. Membership contains no capacity-equals-$D$ predicate.

For $q \in \widetilde{\Omega}_{r,D}$, observer $O$ has a finite commutative record algebra $\mathcal{R}_O(q)$ with atom set $X_O(q) = \operatorname{At}\mathcal{R}_O(q)$. Each shared interface $e$ has an atom set $X_e(q)$ and a source-derived record-atom readout

$$r_{Oe} : X_O(q) \longrightarrow X_e(q).$$

This atom-level construction is used because the generic visible maps on full observer algebras need not be unital $^*$-homomorphisms in the direction required by an algebra equalizer. The compatible public global sections are

$$X_{\mathrm{pub}}(q) = \{(x_O)_O : r_{Oe}(x_O) = r_{O'e}(x_{O'}) \text{ on every shared interface}\}, \qquad \mathcal{R}_{\mathrm{pub}}(q) = C(X_{\mathrm{pub}}(q)).$$

**Theorem 7.11** (Public global sections)**.** *For every finite observer record diagram, $X_{\mathrm{pub}}(q)$ is finite and $C(X_{\mathrm{pub}}(q))$ is a finite commutative unital $C^*$-algebra. Every isomorphism of observer record diagrams induces a bijection of public global sections and a $^*$-isomorphism of their function algebras.*

*Proof.* The public sections form the subset of the finite product $\prod_O X_O(q)$ satisfying finitely many equalities. A diagram isomorphism transports the atom sets and interface maps, hence compatible tuples bijectively; pullback transports their function algebras. $\square$

Only endogenously writable records count. Define

$$X_{\mathrm{reach}}(q) = \{x \in X_{\mathrm{pub}}(q) : x \text{ terminates an admissible endogenous semantic history}\}.$$

The history may use declared source data, accepted repair, semantic continuation, and quotient-visible writes. It may not use a target record, supplied-capacity metadata, measured $\Lambda$, or the electroweak bridge. The publicness policy is a frozen nonempty family

$$\mathfrak{P}(q) \subseteq 2^{\mathcal{O}(q)} \setminus \{\varnothing\}$$

of authorized observer subsets. Collective, universal-local, and quorum publicness are distinct theorem branches.

For every $x \in X_{\mathrm{reach}}(q)$, authorized set $A \in \mathfrak{P}(q)$, and allowed same-interface semantic continuation $\kappa$, the source construction must emit a joint stochastic kernel

$$K_{A,\kappa}(y \mid x), \qquad y \in Y_{A,\kappa},$$

whose observer marginals agree with the local checkpoint packets. The joint kernel is an independent receipt. It cannot be manufactured from local marginals.

**Theorem 7.12** (Local marginals do not determine public capacity)**.** *There are two binary-input joint checkpoint channels with identical observer-by-observer marginals and different public capacities.*

*Proof.* Let $U$ be uniform. The channel $(Y_1, Y_2) = (U, U \oplus x)$ has uniform input-independent marginals, while the joint parity recovers $x$. Independent uniform outputs have the same marginals and carry no information about $x$. Thus the joint capacities are two and one, respectively. $\square$

For a channel $K$, write $S_K(x) = \{y : K(y \mid x) > 0\}$. A code $C \subseteq X_{\text{reach}}(q)$ is zero-error correctable for $K$ when a decoder $d_K$ satisfies

$$K(d_K^{-1}(x) \mid x) = 1 \qquad (x \in C).$$

**Theorem 7.13** (Support criterion and compound zero-error capacity)**.** *A code is zero-error correctable for $K$ exactly when the supports $S_K(x)$, $x \in C$, are pairwise disjoint. Let $\mathfrak{K}(q)$ contain every authorized checkpoint channel and every declared finite composition. Define the compound confusability graph $G_q$ on $X_{\text{reach}}(q)$ by joining $x \neq x'$ whenever their output supports overlap for at least one $K \in \mathfrak{K}(q)$. Then the exact public-record capacity is*

$$\boxed{M_0(q) = \alpha(G_q)},$$

*where $\alpha$ is the graph independence number.*

*Proof.* A successful decoder cannot assign one output to two inputs, which proves the support criterion in one direction. Pairwise disjoint supports admit the decoder that returns the unique supporting input. A code valid for every declared channel is therefore exactly an independent set in the union of the channel confusability graphs. $\square$

For exact indefinite continuation, replace each kernel by its support relation $R_K = \{(x, y) : K(y \mid x) > 0\}$. Boolean relation composition generates at most $2^{|X_{\text{reach}}|^2}$ relations, so the support semigroup and its compound confusability graph close after finitely many steps.

**Theorem 7.14** (Reversible fast branch)**.** *If every authorized continuation is a known injective deterministic map on $X_{\text{reach}}(q)$, then*

$$M_0(q) = |X_{\text{reach}}(q)|.$$

*Proof.* Injectivity keeps every pair of inputs distinguishable for every continuation, so $G_q$ has no edges. $\square$

Checkpoint invariance is not the capacity definition. A cyclic permutation of $m$ labels has a one-dimensional fixed function algebra, while its inverse decodes all $m$ labels. Thus a fixed-projector count can undercount the correctable capacity arbitrarily.

For a frozen finite channel family or continuation horizon and tolerance $\varepsilon$, let $M_\varepsilon(q)$ be the largest code for which every declared channel has a decoder with worst-input error at most $\varepsilon$.

**Theorem 7.15** (Approximate-capacity stability and zero-error discontinuity)**.** *If corresponding channel rows obey*

$$\sup_{K,x} d_{\text{TV}}(K(\cdot \mid x), \widehat{K}(\cdot \mid x)) \leq \delta,$$

*then*

$$M_{\varepsilon+\delta}(\widehat{\mathfrak{K}}) \geq M_\varepsilon(\mathfrak{K}).$$

*Zero-error capacity is discontinuous at every identity channel on more than one symbol: adding arbitrarily small full-support noise collapses $M_0$ to one.*

*Proof.* Total-variation control changes the probability of each correct-decoding event by at most $\delta$. For the second statement, $(1-\delta)I+\delta U$, with $U$ full support, makes every pair of inputs confusable for every $\delta > 0$. □

The capacity carrier is an exact finite Hilbert space $\mathcal{H}_{\mathrm{cap},r,D}$ of dimension $D$. Each reachable public record class has a nonzero projection $P_x$ on this carrier, and distinct records have orthogonal projections.

**Theorem 7.16** (Carrier bound and saturation rigidity)**.** *Every correctable public code $C$ satisfies*

$$|C| \le D, \qquad \log M_\varepsilon(q) \le \log D = N.$$

*If $M_0(q) = D$, the code projections are rank one and sum to the identity.*

*Proof.* The ranks of nonzero orthogonal projections add and their sum has rank at most $D$. Equality with $D$ codewords forces every rank to be one and the sum to have full rank. □

The primary finite readback remains set-valued:

$$\boxed{\mathfrak{F}_{r,\varepsilon}(D) = \{M_\varepsilon(q) : q \in \widetilde{\Omega}_{r,D}\}.}$$

A scalar $\widehat{F}_{r,\varepsilon}(D)$ exists exactly when the nonempty terminal fiber has one common defined value. The evaluator must distinguish an empty readback, an ambiguous readback, and a singleton scalar readback. Existential closure means $D \in \mathfrak{F}_{r,0}(D)$; selector-free stable closure means

$$\boxed{\mathfrak{F}_{r,0}(D) = \{D\}.}$$

Equivalently, the unclosed kernel

$$K_r(D, m) = |\{q \in \widetilde{\Omega}_{r,D} : M_0(q) = m\}|$$

has support only at $m = D$. One saturated branch is insufficient when the terminal fiber is ambiguous.

The full capacity packet consists of observer lineage, record atoms, interface maps, reachable public sections, publicness policy, global kernels, and carrier projections. An isomorphism preserving that packet transports every exact confusability edge, approximate decoder error, and projection rank. Hence $M_\varepsilon$ is implementation invariant under packet isomorphism.

Once scalarized, the active map $f(D) = \widehat{F}_{r,0}(D)$ is deflationary. It cannot attract a positive fixed point from below. If $f$ is total on a declared finite admissible chain, monotone, and deflationary, iteration from its top element stabilizes at the greatest fixed point. This is a conditional order theorem, not a uniqueness or cutoff-independence theorem. The identity family $f(D) = D$ fixes every dimension; an erasure family $f(D) = 1$ fixes only the bottom. Both satisfy monotonicity and deflation. A physical principle must therefore select a nontrivial cutoff-independent zero of the capacity slack.

Capacity extension and regulator refinement are distinct. For $D \le D'$, an injection of reachable records that reflects confusability,

$$e_{DD'}(x) \sim_{G_{D'}} e_{DD'}(x') \Longrightarrow x \sim_{G_D} x',$$

maps every coarse independent set to a fine independent set and gives $M_0(D) \le M_0(D')$. At fixed $D$, refinement injections with the same property make $M_{0,r}(D)$ a nondecreasing integer bounded by $D$, hence eventually constant along each cofinal sequence.

For regions $A, B$ sewn along public seam $S$, exact public records form the fiber product

$$X_{A\cup B} = X_A \times_{X_S} X_B, \qquad |X_{A\cup B}| = \sum_{z\in X_S} |r_A^{-1}(z)|\,|r_B^{-1}(z)|.$$

This gives an exact CSP, tensor-network, or model-counting route on the reversible branch. A cosmological conclusion additionally requires an exact transfer recurrence, exact fiber-product recursion, or a seam theorem with a certified subleading term.

Define the finite-size slack

$$\boxed{s_r(D) = \log D - \log M_{0,r}(D).}$$

If $\limsup_{D\to\infty} \log M_0(D)/\log D < 1$, closure is excluded at all sufficiently large dimensions. Unit asymptotic density is insufficient: $M(D) = D$ and $M(D) = D - 1$ have the same limiting density but opposite fixed-point behavior. Full physical $N$-closure requires a source-derived finite-size law proving

$$s(D_\star) = 0, \qquad s(D) > 0 \quad (D \neq D_\star)$$

on the declared physical domain, or an equally explicit physical selector.

Two independent physical identifications follow only after stable direct closure. The first hypothesis identifies the same capacity carrier with the de Sitter horizon record:

$$N_\star = \log D_\star = \frac{A_{\mathrm{dS}}}{4\ell_\star^2}, \qquad A_{\mathrm{dS}} = \frac{12\pi}{\Lambda} \qquad \Longrightarrow \qquad \boxed{\Lambda \ell_\star^2 = \frac{3\pi}{N_\star}}.$$

The second hypothesis identifies the screen load with the electroweak load by a positive, unital, refinement-natural map

$$\Xi_r : \mathcal{L}_{\mathrm{screen},r} \longrightarrow \mathcal{L}_{\mathrm{EW},r}.$$

Given the independent screen-sieve and electroweak source laws,

$$\Gamma_{\mathrm{scr}} = \frac{P}{12}\log\frac{N_\star}{\pi}, \qquad \log\frac{E_{\mathrm{cell}}}{v} = \frac{\pi}{2\alpha_U(P)}, \qquad \Xi_r(\Gamma_{\mathrm{scr}}) = \log\frac{E_{\mathrm{cell}}}{v},$$

it gives the conditional Higgs/electroweak residual and coordinate

$$\boxed{R_{\mathrm{EW}} = \alpha_U(P)\log\frac{N_\star}{\pi} - \frac{6\pi}{P} = 0, \qquad N_{\mathrm{bridge}} = \pi \exp\left[\frac{6\pi}{P\alpha_U(P)}\right].}$$

Neither physical identification constructs the public-record map. The independently frozen operational-resolution residual

$$R_\rho = \log M_0 - \frac{\pi}{\rho_{\mathrm{op}}^2}$$

is likewise a downstream test and never a producer definition.

The finite global-section, correctable-code, compound-capacity, approximate-stability, carrier-bound, scalarization, order, extension, refinement, and sewing implications are theorems. Physical closure requires source-derived record-atom restriction maps, endogenous public-record reachability, the frozen publicness policy, a globally coupled checkpoint family, a faithful capacity-carrier representation, a whole-fiber scalar producer, confusability-reflecting extension and refinement maps, the finite-size slack law with one physical zero, the horizon–record identification, and the common screen/electroweak load-carrier identification. No QCD or hadronic backend is a dependency of this program.

**Outer/inner self-consistency for the pixel ratio.** This synthesis paper is the canonical place where the local pixel ratio $P$ is defined, so the compact and particle papers can cite one derivation rather than duplicate it. The construction has six steps.

1. **Outer pixel ratio.** The outer-side local screen datum is the dimensionless area ratio

$$P := \frac{a_{\rm cell}}{\ell_\star^2}.$$

Here $\ell_\star^2 = 3\pi/B_\star$ is supplied by the selected scale certificate. After that scale is emitted, it is displayed in SI language as the Planck area.

2. **Exact equilibrium benchmark.** The total/bulk/edge hierarchy has one exact self-similar balance point. Writing

$$x(C) := \frac{S_{\rm gen}(C)}{S_{\rm bulk}(C)} = 1 + \frac{\langle L_C \rangle}{S_{\rm bulk}(C)},$$

the self-similar balance condition

$$\frac{S_{\rm gen}(C)}{S_{\rm bulk}(C)} = \frac{S_{\rm bulk}(C)}{\langle L_C \rangle}$$

gives $x^2 - x - 1 = 0$ and hence $x = \varphi = (1+\sqrt{5})/2$. The realized branch must sit near, but not exactly at, this point because exact equilibrium is too symmetric to support durable records, structure, and dynamics.

3. **Outer detuning variable.** On the declared closure surface we parametrize the outer-side detuning by

$$\alpha_{\rm ext}(P) := \frac{P-\varphi}{\sqrt{\pi}}, \qquad \text{equivalently} \qquad P = \varphi + \alpha_{\rm ext}(P)\sqrt{\pi}.$$

4. **Declared quantitative anchor from the same pixel.** Let $I \subset \mathbb{R}_+$ be the physical pixel interval and let

$$F_{D10} : I \longrightarrow \mathcal{D}_{D10}$$

denote the declared D10 source map. For any trial $P \in I$, the forward D10 lane emits

$$P \longmapsto \alpha_U(P) \longmapsto (t_U(P), t_{\rm tr}(P)) \longmapsto (t_2(P), t_3(P), v(P)) \longmapsto A_Z(P) = \alpha_{\rm em}^{-1}(m_Z^2; P).$$

5. **Ward-projected Thomson transport.** Let $T_Q$ be the required Ward-projected $\mathrm{U}(1)_Q$ transport operator from the $m_Z$-anchor to the Thomson limit. The complete source-only operator is not emitted. A comparison branch may insert the measured endpoint $\alpha^{-1}(0) = 137.035999177(21)$, while the separate empirical branch integrates external hadronic spectral data and returns 136.2636589431 on $[136.1583832834, 136.3690975240]$. For either declared branch, define

$$A_T(P) := T_Q\big(A_Z(P), F_{D10}(P)\big) = \alpha_{\rm Th}^{-1}(P).$$

The inner coupling used in the outer/inner closure is the coupling, not the inverse coupling:

$$\alpha_{\rm in}(P) := \frac{1}{A_T(P)}.$$

6. **Closure.** The realized pixel ratio is the fixed point for which the outer detuning equals the inner observation coupling:

$$\alpha_{\text{ext}}(P) = \alpha_{\text{in}}(P).$$

Equivalently,

$$H(P) := P - \varphi - \frac{\sqrt{\pi}}{A_T(P)} = 0, \qquad \text{or} \qquad P = \varphi + \frac{\sqrt{\pi}}{A_T(P)} = \varphi + \alpha_{\text{in}}(P)\sqrt{\pi}.$$

Writing

$$G(P) := \varphi + \frac{\sqrt{\pi}}{A_T(P)},$$

the closure problem is the fixed-point condition $G(P) = P$. On any physical interval where this induced map is a self-map and a contraction, the closure root is locally unique. Equivalently, the solver can use

$$H(P) := P - \varphi - \frac{\sqrt{\pi}}{A_T(P)}$$

and accept the unique zero of $H$ on the declared branch interval. Observer-facing data may localize that interval. The measured endpoint cannot replace the closure solve.

The closure interpretation assigns one screen cell two coupled roles. On the outer side it is one cell of the screen whose displacement above the exact self-similar equilibrium $\varphi$ sets the geometric detuning. On the inner side the same cell emits the smallest electromagnetic observation step available on the particle branch. The pixel ratio $P$ is the size of that cell in units of the scale-certificate area $\ell_\star^2$, and the realized $P$ is the value at which those outer and inner readings agree. The fixed-point equation is the mathematical form of that consistency condition. On the measured-endpoint comparison branch, the root is

$$P_C \simeq 1.6309682094, \qquad \alpha^{-1}(0) = 137.035999177(21).$$

This is a comparison coordinate inferred from the measured endpoint. It does not define the pixel used by the theory. The source solve's unique root is the closure point, and the difference between the two is the loop residual: $3.0 \times 10^{-4}$ relative on the source chain and $2.5 \times 10^{-6}$ relative (about $1.6 \times 10^4$ measurement sigma) for the certified self-consistent gauge-width fixed point $\alpha^{-1} = 137.035660136946577\ldots$, a certified fixed point of the declared map with the Ward-projected hadronic transport open. Distinct coordinates describe the audit. The certified source/root witness is $136.994835177413\ldots$. The mixed-provenance no-hadron diagnostic is $137.0359595136\ldots$; it mixes the inner value at $P_{\text{fwd}}$ with $\alpha_U$ at the comparison pixel, is no fixed point, and is excluded from physical output. The external-data empirical closure is $136.2636589431$ on $[136.1583832834, 136.3690975240]$, and the measured endpoint is $137.035999177(21)$. The empirical interval requires a same-scheme correction in $[0.6485541111, 0.8547920666]$ inverse-alpha units. A zero-momentum laboratory measurement sees the dressed $\mathrm{U}(1)_Q$ current after charged-lepton vacuum polarization, confined-quark/hadron spectral transport, and same-scheme endpoint matching. A source-only fixed point requires the missing source-derived hadronic spectral transport and its dependency certificate.

The claim table records the source-audit trunk, mixed diagnostic, empirical endpoint, measurement, and interval disclosures. Downstream quantities inherit the provenance of whichever declared pixel branch they use. A separate hardware note reports an optical-cavity check of the same fixed-point geometry; this is treated as corroborating engineering evidence. The same closure equation also admits an inverse observational use case in which observers infer $P$ and total screen capacity from inside the world.

One source-only route studies the closure directly at zero momentum. It requires a self-contained hadronic transport law rather than the external empirical compilation used by the audit.

**Non-hadron output surface.** For quick orientation, the non-hadron output surface consists of receipt-gated classical carrier modes with no promoted quantum mass rows, the exact electroweak selected-carrier chart, the conditional value-law implication, the distinct reference-fitted inverse adapter, the conditional downstream Higgs/top coordinate, a same-family charged witness, and the common-scale rejection of the reciprocal-ray quark texture. The generic quark interface has six scalar coordinates, and no source-derived flavor-orbit selector is emitted. The rejected weighted-cycle neutrino candidate is retained as a comparison and falsification record, outside the exact-output surface. This synthesis paper uses those facts as lane markers. Ref. [5] carries the exact values, supporting surfaces, diagnostic sidecars, and per-sector caveats.

**Extended derivation.** The structural-to-family route and the full per-sector claim-tier audit are developed in *Deriving the Particle Zoo from Observer Consistency* [5].

# 8 Screen Microphysics, Records, and Observer Continuation

The screen-microphysics lane is the engineering side of the paper corpus. It asks whether some microscopic model might exist in principle and gives one explicit regulated carrier architecture: a federation of finite observer patches with echosahedral multi-port interfaces, recurrent toroidal subchannels, exposed overlap data, records, repair instruments, and observer-facing interfaces all made concrete at fixed cutoff. A spherical screen is used there only as an observer-facing regulator chart for support-visible cuts, not as a claim that the universe is a literal spherical quantum computer.

This lane has five theorem-bearing components. First, the reference architecture itself is explicit. Second, the regulated patch-net embedding theorem is closed on its fixed-cutoff object/local-interface surface, and the edge-sector thermal/Casimir branch carries an explicit fixed-cutoff law with one compact lift above that theorem surface. Third, the fixed-cutoff measurement/Born-rule package is closed on its declared operator and record-event surface. Fourth, the fixed-cutoff Bell/CHSH package is closed on the declared two-wing surface. Fifth, the checkpoint/restoration observer package is closed. The downstream boundaries on this lane are packet-level quotient closure where one wants an autonomous exported dynamics, compact-group/Peter–Weyl bookkeeping on the D10/compact surfaces, fair-block contraction certificates for long-run noisy approximate consensus on exported packet nets, and model-family universality. The support-visible BW / geometric-modular / local-Einstein closure lives on the companion recovered-core papers rather than on this fixed-cutoff engineering lane. A reconstruction of Hilbert, $C^*$- or von Neumann algebra structure, Born probabilities, trace structure, and entropy from operational records alone is a separate question outside this synthesis lane.

**Reference-architecture takeaway.** The synthesis-level point is that the regulated screen-side architecture supplies a concrete habitat in which local observables, overlap observables, record registers, and synchronization maps can all be written without pretending to identify a unique final UV completion. Hardware evidence is not imported by this synthesis theorem surface unless it appears in a public, hash-stable OPH evidence bundle. The OPH-FPE simulator repository is the corresponding finite-run software surface for emitted receipts and replayable OPH experiment bundles [10].

**Imported theorem from Ref. [4] (Fixed-cutoff record algebra and Born-Lüders package).** For each completed compare/write/verify slice of the regulated microphysics, the declared pointer and overlap-sector projectors generate a finite commutative central record algebra

$$\mathcal{Z}_{\mathrm{rec}}.$$

Every observer-accessible event $E$ in that algebra has probability

$$\mathbb{P}(E) = \mathrm{Tr}(\rho P_E),$$

and conditioning on $E$ gives the operational post-measurement state

$$\rho|_E = \frac{P_E \rho P_E}{\mathrm{Tr}(\rho P_E)}.$$

If a practical readout instead uses projectors $Q_a$ with commuting central reference projectors $\widehat{Q}_a$ on the same declared slots and

$$\delta_{\mathrm{rec}} = \max_a \|Q_a - \widehat{Q}_a\|,$$

then

$$\|[Q_a, Q_b]\| \leq 4\,\delta_{\mathrm{rec}},$$

and any accessible-state perturbation $\|\widetilde{\rho} - \rho\|_1 \leq \varepsilon$ changes each declared elementary record-event probability by at most $\varepsilon + \delta_{\mathrm{rec}}$.

The charged-family audit contains one engineered witness for this distinction. Inside a bounded finite patch, eight local register graphs and eight declared paths contain 6,467 matrix units. A noncentral event projection is read into a central two-valued accepted/rejected record, and sixty $A_5$ charts verify the declared equivariance. The evidence bundle proves that the stipulated fixed-cutoff schema is nonempty. Its register sizes, path automaton, signs, clock, and response are authored, while inert ancillary stabilization does not supply a cofinal physical refinement. The witness therefore connects event readback to a public record at fixed cutoff; it does not select the charged source, recovery law, family attachment, or physical pole.

**Imported theorem from Ref. [4] (Fixed-cutoff Bell / CHSH package).** On the declared fixed-cutoff physical observable algebra, fix commuting left/right wing subalgebras, central binary setting registers, binary projective readouts on each wing, and a physical source state on the joint wing algebra. Then the compare slice carries the joint law

$$p(a, b \mid x, y) = \mathrm{Tr}\big(\rho_{LR}\, P^{(L)}_{a|x} P^{(R)}_{b|y}\big),$$

the local marginals are independent of the remote setting, the Bell correlators satisfy

$$E(x, y) = \mathrm{Tr}\big(\rho_{LR}\, A_x B_y\big),$$

and the CHSH combination obeys

$$\big|E(0,0) + E(0,1) + E(1,0) - E(1,1)\big| \leq 2\sqrt{2}.$$

If the source family contains an explicit two-qubit branch with the stated Pauli readouts, the same fixed-cutoff surface saturates $2\sqrt{2}$ exactly on that branch. The source-state input is explicit: the theorem derives the Bell law and the Tsirelson bound from the stated fixed-cutoff operator surface, while Bell-pair preparation and two-qubit factor structure are stated branch conditions.

**Imported theorem from Ref. [4] (Checkpoint/restoration and observer backup).** For an observer patch $P_O$, let the checkpoint data consist of the observer-facing record algebra, the accessible local state, the future update schedule, and the externally visible overlap interface data. Exact restoration reproduces the full future law of observer-accessible events, while an $\varepsilon$-accurate restoration changes that future law by at most $\varepsilon$ in total variation.

**Observer-facing conclusion.** The theorem-level content is modest but sharp: the paper corpus supplies a fixed-cutoff measurement interface, a fixed-cutoff checkpoint/restoration package, and an operational observer-identity criterion on observer-accessible event algebras. Stronger substrate-selection or strange-loop closure claims are outside the recovered theorem package.

**Extended derivation.** The fixed-cutoff edge-law, measurement, Born-rule, Bell/CHSH, and checkpoint/restoration packages are developed in *Federated Echosahedral Screen Microphysics: Patch Hardware, Records, and Observer Synchronization in OPH* [4].

# 9 Worldsheet/String Branch

The synthesis paper carries the string/worldsheet branch on the same theorem boundary used across the paper corpus: a genuine theorem-level bridge from the edge-sector theorem surface to the two-dimensional Yang–Mills partition function, followed by a controlled large-$N_{\text{edge}}$ effective-description theorem on the stated branch. This two-dimensional heat-kernel partition statement and declared-branch large-edge effective description sit beside the compact paper's conditional support-visible compact-gauge four-dimensional Yang–Mills form and repair-gap theorems, whose continuum identification and gap transport require the separately named receipts.

**Imported theorem (Heat-kernel edge-sector bridge).** Import the fixed-cutoff same-overlap edge-law package from *Federated Echosahedral Screen Microphysics* [4]. Under the same overlap-gauge realization and local-Gibbs/MaxEnt edge branch, together with the edge-Hamiltonian and one-sided-algebra hypotheses (EH-1)/(EH-2) of Theorem 6.20 (the sector probabilities are the one-sided $d_R$ convention; the closed partition below is the two-sided $d_R^2$ convention), the edge-sector weights satisfy

$$p_R(t) \propto d_R e^{-tC_2(R)}.$$

Consequently the edge partition function matches the standard two-dimensional Yang-Mills heat-kernel form. The compact paper carries OS/Wightman-strength continuum reconstruction for the four-dimensional Euclidean Yang–Mills form and the OPH mass-gap claim separately on the support-visible compact-gauge branch.

**Imported proof route.** The microphysics source surface carries the fixed-cutoff thermal/Casimir law on the declared overlap interface. Peter–Weyl decomposition [47] then identifies the resulting sum with the standard heat-kernel expression for two-dimensional Yang-Mills on the compact-group refinement lift.

**Compact-carrier theorem.** Writing the closed edge partition function as

$$Z_{\text{edge}}(t) = \sum_R d_R^2 e^{-tC_2(R)},$$

the compact carrier states the exact bridge itself as a named theorem: Peter–Weyl identifies it with the compact-group heat kernel at the identity,

$$Z_{\mathrm{edge}}(t) = K_t(1).$$

The Chapman–Kolmogorov gluing law for $K_t$ is the precise sense in which collar sewing matches the two-dimensional Yang–Mills heat-kernel surface before any large-$N_{\mathrm{edge}}$ continuation is invoked. The compact carrier is a partition-identity theorem on its stated branch. The four-dimensional OPH claim identifies the compact-gauge branch with the Euclidean Yang–Mills form and applies the repair-dynamics theorem on that same support-visible branch.

**Controlled large-$N_{\mathrm{edge}}$ branch.** Fix a distinct large-$N_{\mathrm{edge}}$ realization, with $N_{\mathrm{edge}} \neq N_c = 3$, and the fixed-$\tau$ variable

$$\tau = tN_{\mathrm{edge}}$$

on a compact window $I$. If the resulting edge free energy satisfies the compact paper's large-$N_{\mathrm{edge}}$ criterion, with remainder control on $I$, then the Gross-Taylor rewriting [46] is a controlled theorem-level worldsheet effective description of the edge dynamics on that branch. Critical-superstring claims, worldsheet CFT closure, anomaly cancellation, and full massless-spectrum matching are outside the recovered core theorem package.

**Higher-gauge continuation slot.** The genuinely noncentral crossed-module gluing data on cuts provide the natural continuation-level slot for $B$-field / gerbe data in any string-style reorganization of the OPH edge sector. This is a structural analogy only; it is not an open/closed-string boundary formalism or a critical-string theorem.

# 10 Cross-Lane Theorem Boundary and Continuation Directions

The paper corpus has the following theorem boundary:

1. The fixed-cutoff collar, higher-gauge, consensus, record, Bell/CHSH, checkpoint, and restoration packages are theorem-bearing on their stated finite-regulator surfaces.
2. Physical UV uniqueness is quotient-level and stable under inert ancillas. OPH fixes terminal values on declared physical observables, not a unique microscopic representative.
3. The Lorentz and null-modular chain is closed on the support-visible refinement/scaling branch by the finite cap-normal BW scaling theorem on the geometric subnet: fixed-collar Markov replacement with carried $r_{\mathrm{FR}}$, $\delta^{\mathrm{M}}$, and regularized modular remainders, support-readable modular covariance, support-order faithfulness, BW framing, held-out oriented cross-ratio rigidity, and independently normalized geometric $2\pi$-KMS convergence make the cap modular automorphism geometric. The compact cap-normal theorem then identifies $S^2$ with projective future null rays $q(\Omega) = (1, \Omega)$, represents oriented round caps by $n_C = (\cot\alpha, \csc\alpha\,\mathbf{c})$, proves the signed incidence formula, and obtains $n_{gC} = \Lambda_g n_C$ and $H^3 \simeq \mathrm{SO}^+(3,1)/\mathrm{SO}(3)$. This is not a finite-cell Lorentz-invariance claim, a one-shot upgrade from small CMI to exact Markov geometry, a proof that bare finite consensus produces the cap-normal certificate, or a populated/neutral-bulk claim. The absolute Einstein implication additionally requires one source-derived common-domain tower carrying the event, stress, entropy, uniform-asymptotic, universal-coupling, vacuum-reference, and independent-scale premises. Construction and certification of that tower are work in progress.

4. Record-conditioned observer-frame estimation in $H^3$ is a separate conditional theorem: calibrated modular cap responses select a frame value only when they factor through frame-local $H^3$ data on a compact domain and the finite cap frame has $\alpha > 0$. Exact data identify one frame value; finite noisy data produce a frame ball, and a unique finite value requires a positive residual gap $\Delta_{\mathrm{loc}}$. This does not locate an event, populate the event base, derive particle species or stress-energy, or construct chart-blind neutral bulk.

5. The cosmological constant branch has a closed conditional implication layer and an open execution/identification surface. The descended semantic checkpoint packet defines exact public capacity as the independence number of the compound confusability graph at finite integer dimension. Under a faithful carrier representation and whole-fiber scalarization it is deflationary; under a confusability-reflecting capacity embedding, top-down iteration selects its greatest fixed point, while fixed-capacity refinement injections imply eventual exact stabilization. The record-atom maps, reachability, publicness policy, global coupling, carrier representation, whole-fiber agreement, and finite-size slack law are source constructions in progress. The horizon–record identification then gives $\Lambda_{\mathrm{CRC}}\ell_\star^2 = 3\pi/N_{\mathrm{CRC}}$, and the common screen/electroweak load-carrier map separately gives the electroweak bridge. A diagonal normal-form count constructs neither the off-diagonal map nor deterministic closure. The operational resolution experiment is an independent test rather than the producer.

6. Compact gauge reconstruction uses the theorem-produced combined transportability criterion and fixed-cutoff tensor-generated bosonic sector category. On a cofinal tail carrying the explicit compact-gauge refinement receipt, the bosonic refinement ladder and compatible forgetful fibers reconstruct a compact group from sectors with strictifiable central or higher-associator defects and at least one allowed trivial-holonomy strict representative. That is the classification stage. The receipt-certified MAR-admissible witness data plus the explicit one-Higgs matter package then select the realized Standard Model quotient, exact hypercharge, $N_c = 3$, and $N_g = 3$.

7. The particle branch is sector-split. Electroweak $W/Z$ has an exact selected-carrier chart, a conditional value law whose finite-carrier certificate is not emitted, and a separate reference-fitted inverse adapter. Higgs/top is conditional on its declared downstream surface. The reciprocal-ray quark candidate fails on dimensionless common-scale Yukawas, while the exact generic interface has six scalar coordinates and no source-derived flavor-orbit selector. Its sub-percent residuals use a target-anchored mixed-convention chart. The particle simulator contains no Yukawa coupling and returns a null calibration result. The weighted-cycle neutrino candidate is rejected, charged-lepton source landing is open despite the engineered fixed-cutoff record witness, and source-only hadron masses require a production OPH backend. The empirical hadron endpoint is an external-data validation surface rather than that backend.

8. The string/worldsheet branch is a controlled heat-kernel continuation branch. The critical-string selector is a separate gate on top of that edge-language branch.

**Local unification boundary.** At the synthesis level, one local unification surface is explicit and one uniform theorem tier for $c$, $G$, $W$, $Z$, and $H$ is absent. The declared local input $P$ governs the electroweak/Higgs trunk and fixes the matching between cell area and edge entropy. The gravity normalization is emitted by the selected scale certificate, while the invariant causal speed $c$ is structural from the Lorentz branch and gains its SI label only on the local readout package.

| Quantity | Best value on the paper surface | Common paper-surface chain | Caveat |
|---|---|---|---|
| $c_\star$ | 299792458 m/s by SI definition | structural Lorentz/BW branch $\to$ common invariant null cone/speed $\to$ SI unit convention | the common cone and dimensionless speed equality are structural; the decimal magnitude is not predicted |
| $G$ | $6.674299995910528 \times 10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}}$ on the stated local extension surface | $\gamma_\star = \ell_\star \nu_{\mathrm{Cs}}/c$, $B_\star = 3\pi/\ell_\star^2$, $a_{\mathrm{cell}} = P\ell_\star^2$, $\bar{\ell}_{\mathrm{shared}} = P/4$, $G_{\mathrm{SI}} = c^3\ell_\star^2/\hbar$ | the pixel fixes the shared cell/edge identity and cancels in the Newton area law; the strict classical-regime clause is explicit |
| $M_W$ | no nonzero source-only physical mass emitted | selected carrier $\to$ exact chart; quotient-transport assumptions $\Rightarrow$ conditional value law; inverse target adapter separate | the finite-carrier certificate is not emitted, and no pole mass is promoted |
| $M_Z$ | no nonzero source-only physical mass emitted | same chart, conditional law, and separate adapter | same source-certificate and pole-mass boundary |
| $M_H$ | no nonzero source-only physical mass emitted | double-criticality family from the gauge sector $\to$ frozen boundary-scale candidate $(m_H, m_t) = (125.77, 172.63)$ GeV | inherits the boundary-scale selection theorem plus the source-root, physical-scale, running, matching, rigidity, provenance, uncertainty, and complex-pole gates |

**Conditional electroweak audit coordinates.** The strictest target-free $W/Z$ branch is the source-audit zero-selector law, $(80.330, 91.119)$ GeV, with no target input and a single discrete two-law choice. This is a running/chart coordinate, not a physical pole-mass prediction. The source does not close the renormalized-vev, tadpole, field-content, threshold, matching, running, complex-pole, or theory-covariance map needed to compare it with measured resonance parameters, so its numerical proximity carries no mass evidence. Under the complex-pole conversion of the PDG-2026 reference pair, the chart $W$ coordinate differs from the converted $W$ pole 80.3340 GeV by 0.5 propagated experimental standard deviations and the chart $Z$ coordinate from its converted pole by about 17; both figures are convention-conversion diagnostics without an OPH theory covariance, and the physical readout contract is open. The selected carrier, conditional value law, and reference-fitted inverse adapter give the downstream audit coordinates

$$\begin{aligned} \text{selected carrier} \qquad & (80.3862916924, 91.1829044467)\ \text{GeV}, \\ \text{conditional value law} \qquad & (80.3770000154, 91.1879780779)\ \text{GeV}, \\ \text{reference-fitted inverse adapter} \qquad & (80.3625, 91.1879)\ \text{GeV}. \end{aligned}$$

The Higgs/top pair is read on the double-criticality branch, whose frozen boundary-scale candidate $E_\star e^{-\pi} P^{-1/6}$ gives $(m_H, m_t) = (125.77, 172.63)$ GeV at two loops, with $m_H = 125.72$ GeV on the fit-free curve at the measured top; the declared calibration surface $(125.1995304097, 172.3523553288)$ GeV is a target-anchored fit. These coordinates are excluded from the particle-output ledger.

The local finite package supplies the strict classical-regime clause and the exact selected-carrier chart. The electroweak value law is conditional on the finite quotient-transport certificate. The familiar-unit display split on that same surface is

$$L_{\mathrm{loc}} = \sqrt{a_{\mathrm{cell}}}\,\widehat{L}(P), \qquad t_{\mathrm{loc}} = \frac{\sqrt{a_{\mathrm{cell}}}}{c}\,\widehat{T}(P),$$

$$E_{\mathrm{loc}} = \frac{\hbar c}{\sqrt{a_{\mathrm{cell}}}}\,\widehat{E}(P), \qquad \Theta_{\mathrm{loc}} = \frac{\hbar c}{k_B\sqrt{a_{\mathrm{cell}}}}\,\widehat{\Theta}(P),$$

with dimensionless $\widehat{L}, \widehat{T}, \widehat{E}, \widehat{\Theta}$. The gravity-side shared-edge identity fixes the cancellation of $P$, while the local SI readout comes from $\ell_\star^2$. Meters, seconds, GeV, and Kelvin are downstream display conventions built from the structural $c$ row and the familiar constants $\hbar$, $k_B$.

**Companion papers.** Detailed derivations appear in Refs. [2, 3, 4, 5, 8, 9].

## 11 Conclusion

This paper presents the full framework on one synthesis surface. The five-axiom basis, the derived gravity and gauge branches, the consensus formulation, the particle-spectrum branch, the regulated screen-microphysics lane, the fixed-cutoff measurement and observer package, and the controlled large-$N_{\mathrm{edge}}$ worldsheet effective-description branch appear on one common synthesis surface.

The corpus states its position exactly. It carries zero continuous dials, defines its constants by closure equations, and certifies by interval arithmetic that each declared fixed-point map has exactly one fixed point on its stated numerical domain. Its finite analytic core is establishes the consensus core and the capacity-coupling algebra, and its falsification program is restricted to mature non-cosmological claims with no locked-target registry. Forty-two adversarial findings identify no false theorem in the recovered mathematical core. Fail-closed gates and regression tests reject two implementation defects. The corpus supplies a sharp fixed-cutoff local picture, a derived recovered core for relativity and Standard-Model structure, labeled particle-side quantitative audit surfaces, and a usable microscopic engineering lane. The scope boundaries are equally explicit: the bounded-interval kernel in the null-stress branch, the cosmic record-capacity fixed point for the global branch constant, microscopic representative uniqueness, public validation artifacts for the electroweak rows, strong CP, full flavor/CKM closure, a source-derived flavor-orbit selector, charged-lepton source landing from the declared pixel branch, and source-only hadron masses without a production OPH backend. The quantitative work in progress reduces to four generating objects (the Ward-projected hadronic transport, the capacity readback map $F$, the certified forward electroweak chain, and solver hygiene) plus one source-derived flavor-orbit selector, each a computation or a construction rather than a reinterpretation. Refs. [2, 3, 4, 5] are the depth surfaces for sector-by-sector proofs, quantitative details, and continuation-level material. The local cross-lane numerical surface is explicit enough to summarize concisely: the fixed point $P_\star$ organizes the bosonic mass trunk and the gravity-side shared edge-entropy identity beneath the emitted local $G$ row, while the same familiar-unit package reads meters and seconds from the single ruler $\sqrt{a_{\mathrm{cell}}}$ together with the structural invariant speed $c$, and reads GeV and Kelvin from the inverse local ruler through $\hbar$ and $k_B$.

## A Candidate Microphysics Reference Architecture

The synthesis paper keeps the observer-facing theorem package in the main text, but the concrete federated patch-carrier architecture belongs in an appendix. The theorem-bearing fixed-cutoff source surface for the edge heat-kernel / Casimir leaf is the microphysics companion paper; the appendix records only the synthesis-level restatement imported from that surface. The reference architecture used here is a federation of finite observer patches with echosahedral multi-port interfaces, boundary-fixed patch algebras, boundary-fixed overlap algebras on collars, declared shared readout packets, local repair instruments, and record registers exposed on the observer-facing interface. A spherical screen is a regulator chart for support-visible cuts, not a literal substrate claim.

**Theorem (Regulated patch-net embedding).** On a fixed finite patch federation and declared finite overlap cover, the regulated microphysics construction realizes an explicit finite patch-net layer: patch vertices are genuine finite observer patches, overlap edges are genuine collars or port interfaces, packet fields are actual readouts of declared overlap observables, and allowed repair branches are concrete local channels on the same finite regulator.

**Proof sketch.** All patch and overlap algebras are finite-dimensional at fixed cutoff, and the declared packet fields are readouts of explicit overlap observables in those algebras. The candidate repair menu is implemented by local finite-support channels on the same neighborhoods. Therefore the abstract finite patch-net syntax is embedded into an explicit regulator object rather than introduced as a separate formal layer.

**Theorem (Fixed-cutoff edge heat-kernel branch).** On the thermalized overlap branch of the same regulated architecture, the stationary sector law is

$$\pi_\beta(\alpha) \propto d_\alpha e^{-\beta C_2(\alpha)},$$

and along refinement ladders converging to a compact-group Peter–Weyl decomposition [47] the same weights converge cylinderwise to the heat-kernel/Casimir law used in the OPH edge-sector branch.

**Proof sketch.** The regulated proposal kernel is chosen so that the weighted proposal symmetry $d_\alpha q_{\alpha\beta} = d_\beta q_{\beta\alpha}$ holds. Detailed balance with the Casimir acceptance rule then fixes the stationary distribution in finite dimension, and the compact-group heat-kernel law is the corresponding refinement lift.

**Extended derivation.** The full fixed-cutoff architecture, validation package, and theorem stacks are developed in *Federated Echosahedral Screen Microphysics* [4].

# B Interpretive Epilogue: State-and-Law Habitat

This appendix records the OPH state-and-law habitat available inside the framework. A strange-loop closure theorem is outside this appendix.

The OPH inputs used here are:

1. the patch net $P \mapsto A(P)$ of von Neumann algebras together with isotony and overlap restriction maps;
2. the global state on the inductive-limit algebra, whose restrictions witness a nonempty overlap-consistent local sector;
3. the finite Axiom-3 MaxEnt/refinement branch, which supplies a common finite family of gauge-invariant local constraint observables across cutoffs;
4. on the fixed-cutoff branch, the derived finite type-I presentations and compact boundary gluing groups used elsewhere in the OPH package.

The law slot is encoded by the finite expectation-value coordinates of the retained Axiom-3 constraint family. That choice furnishes a compact convex state-and-law habitat for the framework.

**Theorem (Internal state-and-law habitat theorem).** Fix finitely many screen patches $P_1, \dots, P_N$ in the standard OPH setup. For each $i$, let

$$M_i := A(P_i), \qquad M_{ij} := A(P_i \cap P_j),$$

where $M_i$ is the patch von Neumann algebra and $M_{ij} \subset M_i, M_j$ is the overlap algebra. Let

$$S_i := S(M_i) \subset M_i^*$$

be the full state space of $M_i$, endowed with the weak* topology $\sigma(M_i^*, M_i)$.

Choose a finite family of self-adjoint gauge-invariant local constraint observables

$$\mathcal{O} = \{O_1, \dots, O_m\}$$

from the Axiom-3 MaxEnt branch, with each $O_a$ supported in one of the chosen patches; write $i(a)$ for an index such that $O_a \in M_{i(a)}$.

Define the overlap-consistent state sector

$$X_{\text{ov}} := \left\{ (\omega_1, \dots, \omega_N) \in \prod_{i=1}^{N} S_i : \omega_i|_{M_{ij}} = \omega_j|_{M_{ij}} \text{ for all } i, j \right\}.$$

Define the OPH law-coordinate map

$$c : X_{\text{ov}} \to \mathbb{R}^m, \qquad c(\omega_1, \dots, \omega_N) := \big(\omega_{i(1)}(O_1), \dots, \omega_{i(m)}(O_m)\big).$$

Let

$$X_{\mathcal{O}} := \left\{ (\omega_1, \dots, \omega_N, \ell) \in \left( \prod_{i=1}^{N} S_i \right) \times \mathbb{R}^m : (\omega_1, \dots, \omega_N) \in X_{\text{ov}}, \ \ell = c(\omega_1, \dots, \omega_N) \right\}.$$

Then:

1. the ambient product
$$E := \left( \prod_{i=1}^{N} M_i^* \right) \times \mathbb{R}^m$$
is a Banach space for any product norm;

2. $X_{\mathcal{O}}$ is nonempty and convex;

3. $X_{\mathcal{O}}$ is norm-closed in $E$;

4. $X_{\mathcal{O}}$ is compact in the product topology
$$\tau := \left( \prod_{i=1}^{N} \sigma(M_i^*, M_i) \right) \times \text{Euclidean topology on } \mathbb{R}^m;$$

*Proof.* Each $M_i$ is a von Neumann algebra in the OPH patch net, hence a Banach space, so each dual $M_i^*$ is Banach. A finite product of Banach spaces is Banach, hence so is $E$.

For each $i$, the state space

$$S_i = \{\omega \in M_i^* : \omega \geq 0, \ \omega(1) = 1\}$$

is convex and weak* compact by Banach–Alaoglu, because it is a weak* closed subset of the dual unit ball. For every pair $(i, j)$, restriction along $M_{ij} \subset M_i$ and $M_{ij} \subset M_j$ defines affine weak* continuous maps

$$r_{ij} : S_i \to S(M_{ij}), \qquad r_{ji} : S_j \to S(M_{ij}).$$

Hence $X_{\text{ov}}$ is an intersection of affine equalizer sets, so it is convex and $\tau$-closed inside $\prod_i S_i$. Since $\prod_i S_i$ is compact and convex in the product weak* topology, $X_{\text{ov}}$ is compact and convex as well.

Nonemptiness comes from the OPH global state on the inductive-limit algebra: its restrictions

$$(\omega|_{M_1}, \ldots, \omega|_{M_N})$$

belong to $X_{\text{ov}}$ because the restrictions agree on every overlap by construction.

Each coordinate $\omega_{i(a)} \mapsto \omega_{i(a)}(O_a)$ is affine and weak* continuous, since evaluation at a fixed algebra element is weak* continuous. Thus $c$ is affine and $\tau$-continuous. The graph map

$$G : X_{\text{ov}} \to E, \qquad G(x) := (x, c(x)),$$

is affine and continuous, so its image $X_{\mathcal{O}} = G(X_{\text{ov}})$ is convex and $\tau$-compact.

To see norm-closedness in $E$, let $(x_n, c(x_n)) \to (x, \ell)$ in norm. Norm convergence implies weak* convergence on each coordinate, so $x \in X_{\text{ov}}$ because $X_{\text{ov}}$ is weak* closed. Since $c$ is weak* continuous, $c(x_n) \to c(x)$, while norm convergence in $\mathbb{R}^m$ also gives $c(x_n) \to \ell$. Hence $\ell = c(x)$, so $(x, \ell) \in X_{\mathcal{O}}$.

Finally, $X_{\mathcal{O}}$ is a compact convex subset of the locally convex topological vector space $E$ equipped with $\tau$. □

The additional inputs needed for a strange-loop closure theorem are:

1. a closure map $T$ built from internal operations such as overlap repair, Axiom-3 MaxEnt reprojection, collar recoverability, and record-sector coarse-graining;
2. a proof that the stronger "observer-supporting" or "selected-world" criteria carve out a nonempty $T$-invariant observer-supporting subset of $X_{\mathcal{O}}$;
3. any uniqueness, contraction, or Lyapunov-type stability estimate for that closure map.

The habitat theorem supplies only the ambient Banach/compact-convex setting. It does not define a canonical internal self-map, it does not identify any nonempty invariant observer-supporting subset, and it does not prove uniqueness or stability. In particular, nonemptiness and compact-convexity of the ambient habitat do not by themselves imply the existence of a nonempty invariant observer-supporting sector. This does not erase the exact finite packet branch proved on the consensus surface: there the quotient normal-form map pushes forward to a continuous affine idempotent self-map of the finite packet simplex, with fixed points exactly the packets supported on quotient normal forms. Any further strange-loop reading of the ambient habitat as a closure story for existence is interpretive only and is not part of the recovered theorem package.

## B.1 Additional Problem Closures

Using the phase split of Section 5.5.1, the broader closure claims read as follows. The table also records appendix-tier closure stories inside the official claim ledger. "Declared-branch corollary" means proved under the named branch conditions and cited companion-paper theorem surfaces.

| Claim | Status | Boundary |
|---|---|---|
| Quantum gravity consistency | Conditional theorem under explicit receipts | The Einstein relation follows from one source-derived common-domain tower carrying the geometric modular, null, event, entropy, certified-tail, coupling, vacuum, and scale premises. Construction and certification of such a tower are work in progress. |
| UV completion / microscopic uniqueness | Separate scope boundary | Fixed-cutoff theorem packages exist, but a unique microscopic representative is not identified. The support-visible BW scaling theorem and the receipt-certified realized combined-zero-obstruction bosonic compact-gauge branch (central or higher-associator strictification plus an allowed trivial-holonomy strict representative) are closed on their declared companion-paper theorem surfaces; the separate boundary is microscopic representative uniqueness rather than a missing recovered-core lift. |
| Measurement problem | Separate scope boundary | Ref. [4] closes the fixed-cutoff central-record / Born-Lüders interface, but a full philosophical or continuum-level measurement closure is not derived here. |
| Cosmological principle and horizon homogeneity | Separate scope boundary | MaxEnt symmetry inheritance is suggestive only; an independent derivation of cosmological isotropy and homogeneity sits outside the recovered-core theorems. |
| Observer-relative modular time | Declared-branch corollary | On the $\mathrm{BW}_{S^2}$ geometric branch, observer-relative time is read from the cap modular automorphism parameter of the observer's accessible algebra/state pair. This is the D3 reading only. |
| Global / operational problem of time | Separate scope boundary | OPH does not prove a separate operational-clock theorem or a global solution of the problem of time. |
| Black-hole information paradox | Separate scope boundary | Edge-center sectorization and recoverability motivate an internal resolution template, but the full black-hole-information branch sits outside Phase I. |
| D6 cosmological-capacity closure | Conditional corollary | Conditional on a source-derived public checkpoint packet, stable whole-fiber closure $\mathfrak{F}_{r,0}(D_\star) = \{D_\star\}$, a unique physical zero of the finite-size slack, and horizon–record identification $N_{\mathrm{CRC}} = S_{\mathrm{dS}}$, one gets $\Lambda_{\mathrm{CRC}}\ell_\star^2 = 3\pi/N_{\mathrm{CRC}}$. The static-patch SI display additionally uses the selected OPH scale certificate. The 6.6 percent central-value mismatch with the independently derived electroweak/Higgs bridge is a registered comparison, not a contradiction or significance test before the common-carrier receipt closes and the joint cosmological posterior is propagated. |
| Bare cosmological-constant problem from local null data | Separate scope boundary | The local null-data route determines the Einstein branch only modulo $\Lambda g_{ab}$ and does not derive the screen-capacity identification from bare OPH axioms. |
| Magnetic line and simple-GUT monopole boundary | MAR/tensor global-form corollary | On the physical $\mathbb{Z}_6$ branch, the minimum magnetic line is one electron-Dirac unit with required color-magnetic charge and electromagnetic theta period $2\pi$. This is a global-form discriminator, not a dynamical monopole prediction; the product adjoint removes the simple-GUT X/Y production route, not all 't Hooft lines. |
| Gauge-mediated proton-decay boundary | Product-adjoint corollary | The connected product adjoint has no X/Y generator, so the standard simple-GUT channel is absent; general proton stability is not claimed. |
| Proton spin fraction | Separate scope boundary | The proton-spin claim requires nonperturbative QCD completion beyond Phase I. |
| Dark matter phenomenology | Separate scope boundary | Modular-anomaly response laws are conjectural beyond the recovered gravity chain. |
| Baryon asymmetry scale | Separate scope boundary | Suppression-counting estimates are not a substitute for a derived out-of-equilibrium mechanism. |
| Three generations | MAR-branch corollary | $N_g = 3$ is recovered on the realized Minimal Admissible Realization branch. |
| Downstream flavor structure, Koide, and charged-lepton fits | Separate scope boundary | Companion flavor constructions require extra ansätze and sit outside the recovered core. |
| Edge-to-2D-YM / controlled large-$N_{\mathrm{edge}}$ worldsheet branch | Declared-branch corollary | On the stated overlap-gauge, local-Gibbs/MaxEnt, compact-group, and large-$N_{\mathrm{edge}}$ conditions, the edge-sector heat-kernel bridge and controlled Gross–Taylor worldsheet effective description are theorem-level. This is a two-dimensional partition and worldsheet-effective branch, separate from the four-dimensional compact-gauge theorem. |
| Support-visible compact-gauge Yang–Mills form and mass gap | Conditional theorem under explicit receipts | The finite compact-gauge cylinder system has a proof-bearing projective weak-* / GNS extraction. Identification with the four-dimensional Euclidean Yang–Mills state and the equality $\Delta_{\mathrm{YM}} = \Delta_{\mathrm{rep}}$ additionally require the renormalized Yang–Mills, finite ground-state-transform/cross-fiber, transfer/vacuum, |

## C Observer Continuation and Backup

This appendix uses the fixed-cutoff checkpoint/restoration/error/identity theorem package proved in Ref. [4] and summarizes its algebraic interface in observer language. It does not define a strange-loop closure map on an invariant observer-supporting sector or establish uniqueness and stability for such a map. Nor does it upgrade the fixed-cutoff theorem stack to continuation across redesigned environments: the proved package is same-interface restoration on observer-accessible event algebras with explicit error control.

### C.1 Observer as Algebraic Pattern

Here, an observer is represented by

$$O = (P, \mathcal{A}(P), \rho, R),$$

where $P$ is a screen patch, $\mathcal{A}(P)$ is its local algebra, $\rho$ is the local state, and $R$ is the record algebra. Records are carried exactly by central record projectors and, on practical readout surfaces, by approximately commuting projectors in overlap centers, so they are shareable without violating no-cloning constraints for generic quantum states.

Ref. [4] proves a fixed-cutoff observer-facing measurement interface: a finite central record algebra, Born probabilities for its event projectors, and Lüders conditioning on that same commuting record algebra. This appendix uses that fixed-cutoff measurement package in observer language. If a practical readout uses projectors that are $\delta_{\rm rec}$-close in operator norm to that central reference algebra and the accessible state is perturbed by at most $\varepsilon$ in trace norm, each declared elementary record-event probability shifts by at most $\varepsilon + \delta_{\rm rec}$.

### C.2 Markov Collar Factorization

For a collar tripartition $A$-$B$-$D$, the exact Markov normal form is used only when $I(A : D|B) = 0$ holds literally, or along a controlled fixed-collar family for which the exact-Markov replacement modulus $\delta^{\rm M}_{A:B:D}(\varepsilon) \to 0$. In that exact or controlled limit one obtains

$$\rho_{ABD} = \bigoplus_{\alpha} p_\alpha\, \rho^{(\alpha)}_{Ab^\alpha_L} \otimes \rho^{(\alpha)}_{b^\alpha_R D}.$$

The sector label $\alpha$ is classical center data. This decomposition is the mathematical basis for extracting an interior observer state with a controlled boundary interface. Small CMI by itself supplies a recovered comparison state; it is not silently promoted to an exact Markov state.

### C.3 Checkpoint and Restoration Map

A checkpoint is the tuple

$$\mathcal{C} = (R, \alpha, \rho^{(\alpha)}_{\rm int}),$$

with $\rho^{(\alpha)}_{\rm int}$ the interior state after fixing $\alpha$. Given a compatible target environment state $\sigma^{(\alpha)}_{\rm env}$, a restored state is

$$\rho^{(\alpha)}_{\rm new} = \rho^{(\alpha)}_{\rm int} \otimes \sigma^{(\alpha)}_{\rm env}.$$

For approximate Markov collars, recovery is controlled by the standard bound

$$\|\rho_{ABD} - (\mathrm{id}_A \otimes \mathcal{R}_{B\to BD})(\rho_{AB})\|_1 \le 2\sqrt{1 - e^{-\varepsilon}} \le 2\sqrt{\varepsilon}.$$

Here $\varepsilon$ is measured in nats. This gives quantitative trace-distance control on the recovered comparison state under finite CMI (finite collar error). The exact HJPW splice form instead requires literal exact Markovity or the separate fixed-collar replacement modulus tending to zero, in each case together with the Markov-split alignment hypothesis of Section 2.3 identifying the HJPW factors with the preselected edge factors. Interpreting either control as observer continuation requires additional modeling assumptions beyond the bound itself.

The fixed-cutoff microphysics claim is stronger than the bare recoverability estimate written above. In Ref. [4], exact checkpoint restoration preserves the full future law of observer-accessible events, while an $\varepsilon$-accurate restoration changes that future law by at most $\varepsilon$ in total variation. The stronger strange-loop / substrate-transfer closure story is outside this theorem.

### C.4 Physical Meaning

At fixed cutoff, Ref. [4] proves a checkpoint/restoration theorem and backup corollary for observer-accessible event algebras. The proved package stops at same-interface backup/restoration with explicit future-law error control; stronger continuation and substrate-selection claims are outside it. This appendix states the observer-facing meaning of that theorem and its algebraic prerequisites. The stronger substrate-selection, strange-loop closure, uniqueness, and stability package is separate from the fixed-cutoff theorem.

## D Cosmology, Horizons, and Modular-Anomaly Continuations

This appendix carries the cosmology- and continuation-facing technical statements for the synthesis paper. They do not enlarge the recovered core; they make the structural continuation boundary explicit on the synthesis surface.

### D.1 Observer Registry and Clock-Naturality Certificate Contract

For finite exported observer systems, a theorem-faithful evidence bundle separates executor metadata from observer semantics. It contains:

(i) a semantic history DAG or labeled poset $\mathsf{H}_O = (E_O, \preceq_O, \ell_O)$, with event keys computed from semantic payload, observer token, visible footprint, and semantic parents;

(ii) a provenance envelope containing worker ID, repair iteration, retry count, queue position, timestamp, and packet latency, none of which may enter the semantic event key unless declared as physical input;

(iii) local observer registry groupoids with disjoint patch-observer, cap-observer, and future-observer namespaces, explicit birth-event classes, continuation arrows, split/merge lineage arrows, and overlap functors preserving those data;

(iv) a global registry descent certificate: cocycle agreement, trivial monodromy, no duplicate global observer identifiers, no mixed patch/cap/future namespace collisions, and no reuse of local anchor-patch identifiers as local observer indices;

(v) observer-algebra extraction data

$$\mathsf{ObsAlg}(\widehat{q}_{\mathrm{nf}}) = (\mathcal{A}_O, \omega_O, \mathcal{R}_O, \ldots)$$

whose outputs are state-preserving isomorphic for history-equivalent or implementation-equivalent terminal augmented normal forms;

(vi) a support-cap chart map $J_O$ natural with those observer-algebra isomorphisms; and

(vii) a clock instrument $\Theta_O$ with affine reparameterization and residual bound for state, calibration, and record-readout errors.

**Theorem D.1** (Finite observer-clock certificate soundness)**.** *On a finite exported model, suppose the augmented repair relation on $\widehat{Q} = \{(q, [\mathsf{H}])\}$ terminates, terminal states are exactly the consistent physical states with completed semantic records, every augmented critical pair has a history-coherent join, implementation steps are linearizable to semantic commits or stutters, registry descent passes the groupoid cocycle and monodromy checks, and the observer-algebra/chart/clock maps satisfy the state-preserving and affine-residual certificates above. Then worker count, queue order, retry delivery, scheduler counters, and packet latency that is not declared physical input do not change the observer-readable history class, registry class, or clock law beyond the declared affine clock residual.*

*Proof.* Termination plus the augmented local diamond gives confluence by Newman's lemma for terminal history classes rather than only physical quotient states. Linearizability says every implementation execution projects to the same semantic relation up to stutters, so executor metadata cannot create new semantic events. Registry descent identifies the same global observer groupoid across local presentations. State-preserving observer-algebra extraction and natural support-cap charts transport the modular flow along the same observer identity, while the clock instrument certificate bounds the operational readout after the declared affine reparameterization. □

**Remark D.2** (Falsifiers)**.** *The certificate fails if semantic event identifiers depend on worker IDs, timestamps, queue positions, repair iteration counters, or retries; if duplicate retry delivery creates two visible record commits; if two namespaces collide in the global registry; if registry cocycles or lineage arrows fail on overlaps; if histories from two executors are not labeled-poset isomorphic; or if clock residuals exceed the declared bound. These are public evidence-bundle failures that exceed field-name mismatches.*

## D.2 Cosmological Principle

**Lemma 4.1 (MaxEnt symmetry inheritance).** Let $G$ act on the screen by automorphisms $\alpha_g$. If the constraint set is $G$-invariant and von Neumann entropy is $G$-invariant, then the MaxEnt optimizer $\omega$ can be taken $G$-invariant. If the optimizer is unique, it is automatically $G$-invariant.

**Theorem 4.2 (Isotropy from SO(3)-invariant constraints).** Under SO(3)-invariant constraints and MaxEnt uniqueness, the reference state satisfies

$$\omega \circ \alpha_g = \omega \qquad \forall g \in \mathrm{SO}(3).$$

The corresponding stress tensor has perfect-fluid form:

$$\langle T_{ab} \rangle = \rho\, u_a u_b + p(g_{ab} + u_a u_b).$$

**Theorem 4.3 (Schur-type homogeneity).** Let $(\Sigma, h_{ij})$ be a three-dimensional Riemannian manifold. If at every point $p \in \Sigma$, the curvature tensor is SO(3)-invariant, then

$$R_{ijkl}(p) = K(p)(h_{ik}h_{jl} - h_{il}h_{jk}),$$

and the second Bianchi identity forces $\nabla_m K = 0$, so $K$ is constant.

**Corollary 4.4.** If OPH supplies isotropy for all observers, spatial geometry is a constant-curvature space form: $S^3$, $\mathbb{R}^3$, or $H^3$.

**Theorem 4.5 (FLRW emergence).** With the semiclassical Einstein equation from recoverable generalized entropy / entanglement equilibrium, a perfect-fluid stress tensor from SO(3)-invariant MaxEnt, and positive $\Lambda$ from finite screen capacity, the emergent geometry is FLRW:

$$ds^2 = -dt^2 + a(t)^2 \left( \frac{dr^2}{1 - kr^2} + r^2 d\Omega^2 \right),$$

with Friedmann equations

$$H^2 + \frac{k}{a^2} = \frac{8\pi G}{3}\rho + \frac{\Lambda}{3}, \qquad \dot{H} - \frac{k}{a^2} = -4\pi G(\rho + p).$$

## D.3 Horizon-Problem Control

**Theorem 6.1 (Conditional collar-CMI bound).** For a finite-range Gibbs tripartition $A$-$B$-$D$ with collar width $\delta$, assume the uniform strong conditional matrix-mixing premise of Theorem 2.5. Then

$$I(A : D \mid B) \leq c \cdot |\partial C|_{\mathrm{UV}} \cdot e^{-\delta/\xi},$$

with the explicit constants stated there. Ordinary two-point clustering does not supply the premise. On the exact central-interface branch, $I(A : D \mid B) = 0$.

**Corollary 6.2.** For bounded observable $O$ with $\|O\| \sim 1$,

$$|\Delta\langle O\rangle| \leq 2\sqrt{1 - e^{-\varepsilon}} \leq 2\sqrt{\varepsilon}$$

when $\varepsilon$ is CMI in nats. On the CMB anisotropy target $\delta T/T \lesssim 10^{-5}$, this gives the familiar $\varepsilon$-scale bound used in the horizon-problem bookkeeping:

$$\varepsilon \lesssim 2.5 \times 10^{-11} \text{ nats}, \qquad \frac{\delta}{\xi} \gtrsim \ln\!\left( \frac{c \cdot |\partial C|_{\mathrm{UV}}}{\varepsilon_{\max}} \right) \approx 24.4 + \ln\!\big(c|\partial C|_{\mathrm{UV}}\big).$$

The number 24.4 is only the unit-prefactor value. The boundary count cannot be dropped. With $\xi \approx \sqrt{P}\,\ell_\star \approx 1.28\,\ell_\star$, where $\ell_\star$ is displayed as the usual Planck length after the selected scale certificate, the corresponding collar condition is

$$\delta_{\mathrm{CMB}} \gtrsim 1.28\,\ell_\star \left[24.4 + \ln\!\big(c|\partial C|_{\mathrm{UV}}\big)\right].$$

This is a state-recovery benchmark for a bounded observable, not a derivation of CMB anisotropy. On a continuum family the separate rate condition $\delta/\xi - \log|\partial C|_{\mathrm{UV}} \to +\infty$ is required.

**Theorem 6.3 (Homogeneity as the MaxEnt default).** If MaxEnt constraints are uniform across UV cells with no marked points, the MaxEnt state is invariant under triangulation automorphisms. In the continuum limit this gives SO(3) invariance.

**Reading rule.** The cosmological-principle and horizon-homogeneity package is a branch-local theorem package on the realized symmetric MaxEnt branch. The exact theorem-level content is:

1. SO(3)-invariant constraint data plus MaxEnt uniqueness force an isotropic reference state and a perfect-fluid stress tensor.
2. If the same isotropy condition holds for all observers, the Schur/Bianchi argument forces constant-curvature spatial slices.
3. Combined with the semiclassical Einstein branch and positive $\Lambda$ from finite screen capacity, the metric takes FLRW form.
4. Markov control supplies an explicit patch-overlap anisotropy bound and the collar benchmark used in the CMB homogeneity bookkeeping.

The package does not prove from the bare OPH axioms alone that the realized cosmological branch must satisfy SO(3)-invariant constraint data, that MaxEnt uniqueness holds in every cosmological sector, or that the observed universe saturates the displayed CMB benchmark.

## D.4 Conditional Screen-Spectrum Continuation

**Theorem 6.4 (Conditional OPH screen spectrum).** Let $(\mathcal{S}_r)$ be a cofinal finite spherical OPH screen system with a schedule-independent quotient-normal-form scalar $q_r$, removal of the background and dipole sector, and a target-free positive quadratic repair operator $K_r$. If $K_r \to K$, the source-selected finite scalar release energy satisfies $2E^{\mathrm{src}}_{q,r}/d_r \to A_q$, and local MaxEnt is imposed at fixed expected quadratic release energy, then $q_r$ converges in finite harmonic distributions to the centered Gaussian screen field with covariance $A_q K^{-1}$. Exact per-sample release energy would instead give a microcanonical ellipsoid. If the certified repair-scale measure is $t^{\theta/2}\,dt/\Gamma(1+\theta/2)$, then

$$K_{\mathrm{asy}} = (-\Delta_{S^2})^{1+\theta/2}, \qquad C^q_{\ell,\mathrm{asy}} = A_q[\ell(\ell+1)]^{-1-\theta/2},$$

as the large-$\ell$ asymptotic model. The theorem-grade finite-$\ell$ conformal-shell family uses the normalized gamma-ratio precision

$$\kappa_\ell(\theta) = \frac{\Gamma(\ell+2+\theta/2)}{\Gamma(\ell-\theta/2)},$$

with

$$C^q_\ell = A_q \frac{\Gamma(\ell-\theta/2)}{\Gamma(\ell+2+\theta/2)}.$$

If $q$ is additionally certified as the thin-shell pullback of a homogeneous curvature field with $\Delta^2_\zeta(k) = A_\zeta(k/k_\star)^{-\theta}$, this same operator supplies the exact finite-$\ell$ lift; the pure fractional Laplacian is only the asymptotic scaffold. Suppose the same source construction emits a strongly continuous full-collar survival cocycle with generator density $P_\star/24$, together with the orientation-reversal half-collar identity. Then

$$\theta = \frac{P_\star}{48}, \qquad n_s = 1 - \frac{P_\star}{48}, \qquad \kappa^{\mathrm{edge}}_{\mathrm{rep}} = \frac{P_\star}{48(P_\star - \varphi)}.$$

A finite one-step survival value determines $\theta$ through $-\log u_q(\log b)/\log b$; it does not supply the infinitesimal generator receipt. If a scale-natural source embedding transports this cocycle to the physical covariance through $D_s^{-1} C_\zeta D_s = e^{-\theta s} C_\zeta$, the source family is $\Delta^2_\zeta(k) = A_\zeta(k/k_\star)^{-\theta}$. A single shell has an infinite-dimensional radial kernel, including positive ambiguities. Complete radial cross-covariances provide the independent tomography route.

**Receipt boundary.** The conditional theorem fixes the exact angular family. Its radial theorem gives physical source-dilation and cross-covariance tomography as separate uniqueness routes. A finite OPH source DAG that emits the primitive collar ensemble, full-collar generator density, half-collar identity, conformal precision, and physical dilation-intertwiner or tomography receipt is work in progress. Fawzi–Renner/Markov-collar observable control supplies upper bounds on release observables; it is not an amplitude equality. Finite-width windows require an exact Bessel-kernel error certificate. Physical TT/TE/EE transfer and a frozen likelihood contract are separate continuation tasks.

## D.5 Cosmological-Constant / Screen-Capacity Closure

**Theorem 6.5 (D5–D6 local/global closure stack).** On the gravity lane, assume the local Einstein branch has been recovered only modulo $\Lambda g_{ab}$, stable direct public-record closure at a carrier dimension

$$\mathfrak{F}_{r,0}(D_\star) = \{D_\star\}, \qquad N_{\mathrm{CRC}} = \log D_\star,$$

and the independent horizon–record identification readout

$$N_{\mathrm{CRC}} = S_{\mathrm{dS}},$$

the standard de Sitter entropy relation

$$S_{\mathrm{dS}} = \frac{A_{\mathrm{dS}}}{4G} = \frac{3\pi}{G\Lambda},$$

and the standard de Sitter static-patch formulas

$$r_{\mathrm{dS}} = \sqrt{\frac{3}{\Lambda}}, \qquad t_\Lambda = \frac{r_{\mathrm{dS}}}{c}.$$

Then:

1. local null data determine the Einstein branch only modulo $\Lambda g_{ab}$;
2. with the selected scale certificate, the same branch has the global display
$$G_{ab} + \frac{3\pi}{G N_{\mathrm{CRC}}}\, g_{ab} = 8\pi G\, \langle T_{ab}\rangle;$$
3. the same D6 closure fixes the entropy and dimensionless capacity relations
$$S_{\mathrm{dS}} = N_{\mathrm{CRC}}, \qquad A_{\mathrm{dS}} = 4G N_{\mathrm{CRC}}, \qquad r_{\mathrm{dS}} = \sqrt{\frac{3}{\Lambda}}, \qquad t_\Lambda = \frac{r_{\mathrm{dS}}}{c};$$
4. the observed cosmic age is a downstream FLRW benchmark rather than an additional theorem output.

So the cosmological-constant package is one local/global theorem stack rather than a split local argument plus an unrelated global readout.

**Scope boundary.** The D6 hypotheses are the local Einstein branch, stable whole-fiber public-record closure, horizon–record identification, the de Sitter entropy relation, and the standard static-patch formulas. The local null-data route does not by itself determine the global capacity; that value is fixed only on a stable direct-capacity branch with a physical finite-size selector and horizon–record identification. The conditional capacity closure fixes $\Lambda_{\mathrm{CRC}}\ell_\star^2 = 3\pi/N_{\mathrm{CRC}}$; the SI static-patch scale additionally requires the selected scale certificate.

**Capacity self-closure target.** At finite regulator use the frozen capacity-carrier dimension $D$, with $N = \log D$. For each terminal state in the unclosed fiber, form compatible global sections of the local record-atom diagram, retain the endogenously reachable sections, freeze the authorized publicness policy, and use the source-derived joint checkpoint kernels. Their compound confusability graph $G_q$ gives

$$M_0(q) = \alpha(G_q), \qquad \mathfrak{F}_{r,\varepsilon}(D) = \{M_\varepsilon(q) : q \in \widetilde{\Omega}_{r,D}\}.$$

The scalar active map exists only when the whole nonempty terminal fiber has one common defined value. Fixed checkpoint projectors are not the capacity: a cyclic permutation fixes only constant functions while preserving every record label. Local checkpoint marginals are also insufficient to determine the required joint channel.

A faithful capacity-carrier representation gives $M_\varepsilon(q) \le D$, and stable closure is

$$\mathfrak{F}_{r,0}(D_\star) = \{D_\star\}.$$

Equality at zero error forces a rank-one complete public record basis. Under With a confusability-reflecting embedding, capacity extension is monotone, so a total scalar deflationary map on a declared finite chain reaches its greatest fixed point from the top. Fixed-$D$ refinement then stabilizes exactly. None of these implications supplies cutoff independence or a unique physical zero of

$$s(D) = \log D - \log M_0(D).$$

The finite kernel

$$K_r(D, m) = |\{q \in \widetilde{\Omega}_{r,D} : M_0(q) = m\}|$$

distinguishes one closed state from deterministic whole-fiber closure. A diagonal argmax is an extra selector. After direct active closure, the horizon–record identification yields the D6 area-law display, while the common screen/electroweak load-carrier identification yields the electroweak bridge. The operational scale $\rho_{\rm op}$ is an independent estimator tested through $\log M_0 - \pi/\rho_{\rm op}^2$; it does not define the map. The $\Lambda$-located value remains a measured-side comparison coordinate.

**Capacity normalization.** The branch uses $N_{\rm scr}$ as the de Sitter entropy capacity. The bare horizon ratio is

$$N_{\rm patch} = \left(\frac{r_{\rm dS}}{\ell_P}\right)^2,$$

so

$$N_{\rm scr} = \pi N_{\rm patch} = \frac{3\pi}{\Lambda \ell_P^2}.$$

For the observed late-time scale, $N_{\rm patch} \simeq 1.05 \times 10^{122}$ and $N_{\rm scr} \simeq 3.31 \times 10^{122}$. These are benchmark displays, not high-precision inputs for the Newton row.

## D.6 Black-Hole Structural Package and Continuation Boundary

The AMPS-style information trilemma assumes:

1. an outgoing mode $B$ entangled with an interior partner $A$ (smooth horizon),
2. late radiation $B$ entangled with early radiation $R$ (unitarity), and
3. monogamy of entanglement.

The false assumption in this setting is the naive tensor factorization

$$\mathcal{H} \overset{?}{=} \mathcal{H}_{\rm inside} \otimes \mathcal{H}_{\rm outside} \otimes \mathcal{H}_{\rm radiation}.$$

**Theorem 7.1 (Edge-center black-hole decomposition).** On the fixed-cutoff edge-center collar branch, assume the black-hole collar state lies in the exact Markov class or in the idealized exact-recoverability limit used elsewhere in the paper stack. Then the collar decomposition gives

$$\rho_{A_\delta B_\delta D_\delta} = \bigoplus_\alpha p_\alpha \left( \rho_{A_\delta b_L^\alpha} \otimes \rho_{b_R^\alpha D_\delta} \right).$$

The glue between inside and outside is the edge-sector label $\alpha$ in the center. Given $\alpha$, inside and outside factorize.

**Theorem 7.2 (Hawking/KMS normalization).** On the geometric modular branch where the horizon generator carries the standard $2\pi$ KMS normalization, the outside algebra is in a KMS state at inverse temperature

$$\beta = \frac{2\pi}{\kappa}$$

with respect to the horizon Killing generator.

**Corollary 7.3 (Recoverability-style interior encoding).** On the same fixed-cutoff collar carrier, when $I(A_\delta : D_\delta \mid B_\delta) \le \varepsilon$, there exists a recovery channel $\mathcal{R}_{B_\delta \to A_\delta B_\delta}$ such that

$$\|\rho_{A_\delta B_\delta D_\delta} - (\mathcal{R}_{B_\delta \to A_\delta B_\delta} \otimes \mathrm{id}_{D_\delta})(\rho_{B_\delta D_\delta})\|_1 \le 2\sqrt{1 - e^{-\varepsilon}} \le 2\sqrt{\varepsilon}.$$

Here $\varepsilon$ is in nats. So the interior collar data are encoded in the exterior collar together with the shared cut data, rather than supplied by an independent tensor factor.

**Corollary 7.4 (Discrete area spectrum and Schwarzschild transition identity).** The same edge-center package carries a discrete area operator

$$L_C = \sum_\alpha (\log d_\alpha) P_\alpha, \qquad A_\alpha = 4G_{\mathrm{geom}} \log d_\alpha = 4\ell_\star^2 \ln d_\alpha,$$

and for a Schwarzschild sector transition $d \to d'$ one has

$$\Delta(Mc^2) = k_B T_H \ln(d'/d).$$

On the emission branch $d' = d/k$ with $k > 1$, the emitted line energy magnitude is

$$E_{\mathrm{emit}} = k_B T_H \ln k.$$

**Scope boundary.** The retained structural theorem package is exactly Theorem 7.1, Theorem 7.2, Corollary 7.3, and Corollary 7.4. Integer-transition combs, QNM selectors, Page-type linewidth estimates, PBH burst templates, Kerr/LIGO horizon spectroscopy templates, and any Page-curve or island closure are outside that structural core. Those items require continuation inputs such as an integer-transition selection rule, a semiclassical evaporation-power model, a QNM/transition identification, greybody matching, or an explicit island prescription.

The same no-relabeling rule applies here. A finite capacity register is not an exterior mass, an archive record is not Hawking radiation, a finite repair spectrum is not a GR quasinormal-mode spectrum, and an exact finite recovery threshold is not a Page time unless the corresponding independent physical bridge supplies source-separated readout, calibration, residuals, controls, and fixed validation references.

### D.7 Modular-Anomaly Continuation

**Theorem 9.1 (Modular additivity defect).** Define the collar modular additivity defect by

$$\Delta K_\delta := K_{ABD} - K_{AB} - K_{BD} + K_B.$$

Then

$$\langle \Delta K_\delta \rangle_\omega = -I(A : D \mid B)_\omega.$$

This is a scalar recovery identity. It can enter the modular-anomaly dark-sector continuation only after a separate matching theorem or hypothesis identifies it with an anomalous local modular-energy source. By itself it constructs neither a rank-two tensor nor a dark-matter theorem.

The continuation therefore reserves an anomalous tensor slot only for a finite covariant source packet that supplies source localization, CMI-to-modular-source matching when that route is used, rank-two reconstruction, conservation, normalization, and universal coupling. Once those receipts pass, the modified Einstein equation is

$$G_{ab} + \Lambda g_{ab} = 8\pi G \left( \langle T_{ab} \rangle + \langle T_{ab}^{\mathrm{anom}} \rangle \right),$$

with rest-frame normalization

$$\langle T_{00}^{\mathrm{anom}} \rangle := \frac{15}{8\pi^2} \frac{\delta \langle K_C^{\mathrm{anom}} \rangle}{\ell^4}.$$

On that receipt-certified continuation surface the tensor gravitates and is covariantly conserved. Calling it dark additionally requires the no-electromagnetic-coupling receipt. Central or recoverability data alone do not establish any of these properties.

The dark-sector continuation promotes scalar repair occupation to a canonical pair $(n, \theta)$. Conditional on the proposed action, its dilute homogeneous phase scales as pressureless matter and its cubic condensed phase gives $a_R = \sqrt{a_b a_0}$ in the spherical deep regime. The action supplies a repair current and metric stress rather than a virtual source.

**Reading rule.** The scalar modular-additivity identity is a recovery theorem. The canonical repair pair, baryonic source map, dimensional couplings, complete constitutive law, relativistic refinement limit, abundance, and physical likelihoods are work in progress. Every cosmological quantity on this surface is diagnostic.

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**Data Availability declaration:** The paper stack, source manuscripts, public PDFs, simulation code, and implementation notes are available through the OPH project repository and linked public artifact surfaces. Some hardware and prototype evidence requires a sanitized ancillary audit bundle before public disclosure because raw logs may include device configuration, operational metadata, or private infrastructure details. The intended audit bundle contains source hashes, simulator configs, finite-receipt JSON, verifier code, hardware acceptance records, calibration logs, and selected run traces sufficient to replay the stated implementation claims.

**Code Availability declaration:** The OPH simulator, finite-consensus harnesses, visualization payload schemas, and public paper build sources are maintained in the OPH project repositories. Submission artifacts include a standalone source file and a source bundle so that the manuscript can be rebuilt without relying on private local paths.

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| Lane | Output | Role | Support boundary |
|---|---|---|---|
| Carrier modes | two transverse Maxwell modes; $2\dim G$ perturbative Yang–Mills modes; two Einstein TT modes | symmetry-protected massless classical kernels on the stated action branches | no quantum zero-mass particle row until the physical Hilbert-space, pole-residue, and phase gates pass |
| $W/Z$ | no nonzero source-only physical mass emitted | strict source-audit zero-selector branch; exact selected-carrier chart; conditional quotient-transport implication; reference-fitted inverse adapter | the coordinate pairs have different provenance. The finite-carrier certificate and the source-law selection principle deciding the discrete two-law choice are not emitted, and no physical complex-pole pair is promoted |
| Fine structure | four distinct coordinates | source witness, mixed diagnostic, empirical closure, and measurement | the empirical interval misses the measured endpoint. Source-only promotion requires the source-derived hadronic spectral backend |
| Higgs/top | no nonzero source-only physical mass emitted | double-criticality family from the gauge sector; frozen boundary-scale candidate; declared-surface calibration fit kept separate | promotion inherits the boundary-scale selection theorem plus the source-root, physical-scale, running, matching, rigidity, provenance, uncertainty, and complex-pole gates |
| Charged leptons | no nonzero source-only physical mass emitted | target-anchored diagnostic, exact $A_5/C_3$ face-carrier lemma, and conditional nature/pole transport | the affine map and engineered record model are authored. The nature bridge assumes the physical Yukawa response identity, while the pole bridge assumes the face-quotient Dyson singularity readout; its explicit zero-self-energy kernel is only a free witness. Physical source selection, attachment, interacting kernel, infrared completion, coherent branch, and cofinal refinement are open |
| Quarks | no nonzero source-only physical mass emitted | common-scale reciprocal-ray falsification; exact six-scalar interface; restricted source-spread lower bound | the candidate fails across the audited common scales. Its sub-percent residuals use a target-anchored mixed-convention chart. The simulator contains no Yukawa coupling or Yukawa information, while generation-blind flavor singlets supply no physical flavor-orbit selector |
| Neutrinos | no nonzero source-only physical mass emitted | rejected target-informed template candidate | the correlated profile rejects the candidate; the shared-basis recovery was tautological, the charged basis is open, and no source-only |